

Instructions for Use

# Infinity<sup>®</sup> CentralStation



## **WARNING**

For a full understanding of the performance characteristics of this medical device, the user should carefully read this manual before use of the medical device.

**Infinity CentralStation  
Infinity Telemetry System  
Infinity M300  
Software VF8**

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## Intended Use

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The Infinity CentralStation is intended for use by healthcare professionals for the purpose of centralized monitoring within the hospital or clinical environment of adult, pediatric and neonatal patient monitor data on the Infinity patient monitoring network. The Infinity CentralStation is intended to measure and produce visual and audible alarms for one or more physiological parameters. The Infinity CentralStation with Rest ECG can provide interpretive diagnostic statements and reports for adults and pediatric patients when connected to a monitor with ECG monitoring capability.

Infinity M300 Telemetry System is intended to be used by healthcare professionals for the purpose of monitoring ECG and pulse oximetry parameters of ambulatory and non-ambulatory adult and pediatric patients in healthcare facilities. Infinity M300 is intended to operate with the Infinity CentralStation using wireless communication over the Infinity patient monitoring network.

Infinity Telemetry System is intended for use in an environment where patient care is provided by healthcare professionals (physicians, nurses, technicians) when the professional determines that a device is required to measure and produce visual and audible alarms for adult and pediatric patient populations. Infinity Telemetry System transmits via radio frequency physiological signals for display and/or measurement at the Infinity CentralStation. The Infinity CentralStation Telemetry System with TruST is intended for 12-Lead ECG monitoring with a reduced set of electrodes. Reconstructed leads are intended for real-time assessment of ST segment changes.

### NOTE

The Infinity CentralStation, Infinity Telemetry receiver, and Infinity M300 Central Charger are not intended for use in the patient vicinity.

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.

### CAUTION

All Dräger hardware and screen shots shown in this manual are examples only. Actual product or screens may differ slightly.

## Introduction

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Before using all Dräger devices, read all provided documentation carefully.

### **WARNING**

**Using software not approved by Dräger on the CPU of the Infinity CentralStation may adversely affect operation of the Infinity Network.**

**Diagnosis based on interpretation of 12-lead monitoring results should only be done by qualified personnel. Prior to final interpretation and diagnosis, qualified physicians should review suggested diagnostic statements and all other available information.**

## Trademarks

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## Documentation Features

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### Warnings, Cautions, and Notes

#### Definitions

##### **WARNING**

A warning statement provides important information about a potentially hazardous situation which, if not avoided, could result in death or serious injury.

##### **CAUTION**

A caution statement provides important information about a potentially hazardous situation which, if not avoided may result in minor or moderate injury to the user or patient, or in damage to the equipment or other property.

##### **NOTE**

A note provides additional information intended to avoid inconvenience during operation.

### Applicability

For applicability of discontinued product names, (e.g. Infinity SC 7000, Infinity SC 8000, Infinity SC 9000XL) see the current Infinity Network Software Compatibility Chart.

## Device Symbols

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Marking on various Infinity CentralStation hardware components may include the following:



Consult accompanying documents



Display On/Off button



CPU On/Off LED

**IPX7**

Protected against the effects of immersion

**IPX4**

Protected against splashing water

**IPX1**

Protected against dripping water

**IPX0**

Non-protected against water



Defibrillator-proof equipment, type CF

**IC: 267I-ABTU4**

Canadian certification (The term “IC:” before the radio certification number only signifies that Industry Canada technical specifications were met.)



Observe WEEE (Waste Electrical and Electronic Equipment) disposal requirements (12)



Complies with the European Medical Device Directive 93/42/EEC for medical devices



Danger: Risk of explosion if used in presence of flammable anesthetics



Tested to comply with FCC standards



USB connector

## For Your Safety and That of Your Patient

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|                                                      |    |
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Strictly follow these Instructions for Use

**WARNING**  
To maintain patient safety, adhere to all **WARNINGS** and **CAUTIONS** listed in these Instructions for Use and on equipment labels.

Accessories

**WARNING**  
These Instructions for Use list the accessories that have been tested and are approved for use with the device. Using any other accessories may adversely affect the operation of the device. Contact your local Dräger representative for a current list of approved accessories.

Restriction of Distribution

The Infinity Telemetry System complies with the Radio Equipment and Telecommunications Terminal Equipment Directive (1999/5/EC).

Patient monitoring equipment, however sophisticated, should never be used as a substitute for the human care, attention, and critical judgment that only trained health care professionals can provide.

General WARNINGS and CAUTIONS

The following WARNINGS and CAUTIONS apply to general operation of the devices. WARNINGS and CAUTIONS specific to subsystems or particular features appear with those topics in later sections of the manual.

**WARNING**  
For the devices to be properly grounded, connect the power cord to a hospital-grade outlet.

**CAUTION**  
Federal Law (U.S) restricts these devices to sale by or on the order of a physician.

**CAUTION**  
These devices are for use in health care facilities only and exclusively by persons with specific training and experience in its use.

Patient safety

The design of the equipment, the accompanying literature, and the labeling on the equipment consider that the purchase and use of the equipment is restricted to trained professionals, and that certain inherent characteristics of the equipment are known to the trained operator. Instructions, warning, and caution statements are limited, therefore, largely to the specifics of the Dräger design.

This publication excludes references to various hazards which are obvious to a medical professional and operator of this equipment, to the consequences of product misuse, and to potentially adverse effects in patients with abnormal conditions. Product modification or misuse can be dangerous.

**WARNING**  
To avoid explosions, devices should not be used in the presence of flammable anesthetic mixture including oxygen, ether, nitrous oxide, or cyclopropane.

**WARNING**  
To minimize the risk of patient strangulation, carefully place and secure all patient cables away from the patient’s neck and face.



### **CAUTION**

Follow local regulations for safe disposal of batteries. To prevent fire or explosion, never dispose of batteries in fire.

### **EU Directive 2002/96/EC (WEEE)**

These devices are subject to EU Directive 2002/96/EC (WEEE). They are not registered for use in private households, and may not be disposed of at municipal collection points for waste of electrical and electronic equipment. Dräger Medical has authorized a firm to dispose of these devices in the proper manner. For more detailed information, please contact your local Dräger Medical organization.

### **Note on EMC/ESD risk for the devices function**

Electromedical devices are subject to special precautionary measures concerning EMC and must be installed and put into operation in accordance with the EMC information described on page 327. Additional information is available from DrägerService on request.

Portable and mobile RF communications equipment can affect medical electrical equipment.

### **WARNING**

**Do not operate the devices in areas such as: magnetic resonance imaging (MRI) environments, aircraft, ambulance, home or hyperbaric chambers.**

### **WARNING**



**Connector pins with an electrostatic discharge (ESD) warning sign should not be touched and no connections should be made between these connectors without implementing ESD protective measures. Such precautionary procedures may include anti-static clothing and shoes, the touch of a ground stud before and during connecting the pins or the use of electrically isolating and antistatic gloves. All staff involved in the above should receive instruction in these ESD precautionary procedures.**

### **CAUTION**

To prevent equipment failure, do not use wireless phones and radios within 33 feet (10 m) of the equipment.

### **Site of Operation**

Only use devices (monitor, pods, modules and accessories) in areas that meet the environmental requirements outlined in the section beginning on page 309.

### **WARNING**

**Do not operate medical devices (monitor, pods, modules and accessories) in close proximity to equipment that emits microwave or other high-frequency emissions which may interfere with the medical devices' operation.**

### **WARNING**

**To avoid electric shock, inspect all cables before use. Never use cables that appear cracked, worn, or damaged in any way (doing so may compromise performance or put the patient at risk).**

### CAUTION

When placing devices, make sure adequate ventilation exists and prevent overheating by positioning the devices with at least 2 inches (5 cm) of space around all sides, if possible.

### CAUTION

To avoid short-circuiting and otherwise damaging the devices, do not allow fluids to come in contact with the device when it is connected to a power outlet. If fluids are accidentally spilled on the equipment, remove the affected unit from service as soon as possible and contact the service personnel to verify that patient safety is not compromised.

## Networking

When networking with electrical devices, the operator is responsible for ensuring that the resulting system meets the requirements set forth by the following standards:

- IEC 60601-1 (EN 60601-1)  
Medical electrical equipment  
Part 1: General requirements for safety
- IEC 60601-1-1 (EN 60601-1-1)  
Medical electrical equipment  
Part 1-1: General requirements for safety  
Collateral standard: Safety requirements for medical electrical systems
- IEC 60601-1-2 (EN 60601-1-2)  
Medical electrical equipment  
Part 1-2: General requirements for safety  
Collateral standard: Electromagnetic compatibility; Requirements and tests
- IEC 60601-1-4 (EN 60601-1-4)  
Medical electrical equipment  
Part 1-4: General requirements for safety  
Collateral standard: Programmable electrical medical systems

Strictly follow the Instructions for Use for each networked device.

## Electrical Safety

### WARNING

**Because of the danger of electric shock, never remove the cover of a device while it is in operation or connected to a power outlet.**

### WARNING

**Any connected devices, or combination of devices, not complying with the requirements in these Instructions for Use may adversely affect the operation of the Infinity CentralStation and/or Infinity M300. Before using the Infinity CentralStation and/or Infinity M300 with other devices, consult the accompanying documentation of all connected devices.**

**Only equipment tested and approved by Dräger should be connected to the Infinity Network. Failure to do so may adversely affect operation of the Infinity Network.**

### CAUTION

Electrical connections to equipment not listed in these Instructions for Use should only be made following consultation with the respective manufacturers. Equipment malfunction may result with the risk of patient injury.

### CAUTION

To avoid injuring the patient, do not touch any connector or mounting screw on the device when you are touching the patient. In addition, do not allow the conductive parts of electrodes and cables to contact other conductive parts or ground.

## Defibrillator Precautions

The Infinity M300 and Infinity Telemetry transmitter are protected against high-frequency interference from defibrillators and electrosurgical units and against 50- and 60-Hz power line interference.

### WARNING

**To protect the monitor from damage during defibrillation, for accurate ECG information, and to protect against noise and other interference, use only ECG electrodes and cables specified by Dräger. Other sensors may not provide adequate protection against defibrillation.**

### CAUTION

To prevent burns and electric shock due to rerouting of electrical current through electrodes, do not position the defibrillator pads near any electrodes or sensors.

### CAUTION

Only defibrillate across the chest.

## Electrosurgery

Observe the following precautions during electrosurgery to minimize electrosurgical unit (ESU) interference and maximize operator and patient safety.

### WARNING

**For better performance and to reduce the hazard of burns during surgery, always use accessories designed for ESU environments. Do not use skin temperature probes.**

### WARNING

**To reduce the hazard of burns during electrosurgery, keep the sensor or transducer (ECG, pressure, SpO<sub>2</sub>) and their associated cables away from the surgical site, the ESU return electrode, and earth ground.**

## Maintenance, Modifications, and Repairs

### WARNING

**Operation of the device should be in accordance with part 15.242 sections A, D, E, F, G, and H of the FCC rules.**

Safety requires regular equipment inspection and maintenance. Be sure that devices are used and serviced in accordance with Dräger Service Documentation: "Preventative Maintenance Procedure" and "Preventative Maintenance Schedule."

### CAUTION

Using accessories beyond the life span recommended by the manufacturer may compromise patient safety due to device failure.

This service documentation and a full technical description of the Infinity Network are available from your local Dräger representative.

An effective maintenance schedule should be established for your monitoring equipment and reusable supplies. This should include inspection, as well as general cleaning, on a regular basis. The maintenance schedule must comply with the policies of your hospital infection control department or biomedical department. Check with your biomedical maintenance to be sure preventative maintenance and calibration is complete.

Once a year, check all cables, devices, and accessories for damage, ground resistance, chassis and patient leakage currents, and all alarm functions. Also, ensure that all safety labels are legible. Maintain a record of safety inspections. Safety checks, device verification, calibration, and maintenance should be carried out and verified by properly trained personnel.

**WARNING**

Repair of the device may only be carried out by trained service personnel. Regular annual maintenance (functional and safety test) according to IEC 60601-1 is recommended, regardless of national regulations or laws (for example, accident prevention regulations). Connecting this medical device to other medical devices could result in additional maintenance requirements. Consult the documentation for these other devices or software to identify additional requirements.

Dräger recommends contracting with DrägerService for any repairs. Use only authentic Dräger repair parts during maintenance. Using non-Dräger repair parts may adversely affect the operation of the device (see the “Maintenance” chapter).

**Maintenance Intervals**

| Task                                      | Interval       | By whom                   |
|-------------------------------------------|----------------|---------------------------|
| Device Service and Maintenance            | 2 years        | Authorized technician     |
| Replace Hard Drives (without RAID option) | 4 years        | Authorized technician     |
| Replace Hard Drives (with RAID option)    | When necessary | Authorized technician     |
| M300 Battery                              | 1 year         | Technical Personnel       |
| Clean Optional ICS Air Filters            | 2 months       | Medical or Tech Personnel |

# Infinity CentralStation

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## General Description/Product Option & Accessory Information

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### Infinity CentralStation

Infinity CentralStation provides centralized monitoring and critical care management for bedside and telemetry monitored patients. From the Infinity CentralStation, clinicians can gain access to patient information for patients on the Infinity Network. The Infinity CentralStation displays waveforms, parameters, and alarm status of Infinity bedside monitors, Infinity M300, Infinity Telemetry, and ventilators (via an etCO<sub>2</sub> pod or an MIB interface) for up to 16 patients on a single screen or up to 32 patients using two screens. The standard Infinity CentralStation includes the following features:

- Central Processing Unit (CPU)
- Configurable Main Screen display
- Audio alarm annunciation
- 2-hour storage of 4-waveform Full Disclosure
- 2-hour storage of up to 1000 events per patient
- Export of waveforms
- 24- or 72-hour graphical and tabular trends
- Accessory Kit
- Instructions for Use (paper and online .PDF in application)
- one or two PC displays, or touch screens
- external speakers,
- mouse and keyboard
- optional remote displays
- 2-channel R50-N recorder
- network laser printer
- uninterruptible power supply
- Infinity Network components

#### Options for the Infinity CentralStation include:

- Second video
- RAID 1 for storage disk
- CPU mounting package
- Patient licenses to support 32-patients
- 28-, 48-, or 72-hour Full and Event Disclosure
- Expanded Full Disclosure
- VentCentral™
- Patient Census
- Rest ECG Analysis
- Infinity Telemetry
- Infinity M300
- ST Analysis
- TruST 12-lead
- Infinity Paging Interface for alarm notification via Infinity Gateway Suite
- Infinity Symphony for web-based review of stored data
- Infinity WebViewer and Infinity Pocket WinView for near real-time review of patient data
- HL7 bi-directional interfacing via Infinity Gateway Suite

## Infinity M300

Infinity M300 is a patient worn transceiver that uses the Infinity CentralStation as the primary patient monitoring display and primary alarm source. The Infinity M300 device includes a color display, local alarm alerts and rechargeable internal battery. Infinity M300 use is restricted to one patient at a time. The standard Infinity M300 includes the following features:

- Infinity M300 patient worn device
- Infinity M300 Bedside Charger
- Infinity M300 Central Charger
- Infinity M300 programming kit
- ECG lead sets
- Infinity CentralStation

### Options for the Infinity M300 include:

- Infinity CentralStation options, as indicated on page 22
- Pulse Oximetry software enabled at Infinity CentralStation
- Pulse Oximetry cables and sensors
- Infinity M300 Replacement battery
- Infinity M300 disposable pouches
- Infinity M300 shower pouches
- Infinity M300 bedside clips

## Infinity Telemetry

Infinity Telemetry is a patient worn transmitter that uses the Infinity CentralStation as the primary patient monitoring display and primary alarm source. The Infinity Telemetry device is lightweight, water resistant, and uses disposable 9V batteries. Infinity Telemetry use is restricted to one patient at a time. The standard Telemetry includes the following features:

- Infinity Telemetry Receiver CPU
- Infinity Telemetry Receiver Expansion Kits
- Infinity Telemetry UHF or VHF antenna components
- Infinity Telemetry Transmitters
- Infinity Telemetry programming kit
- ECG lead sets
- Infinity CentralStation

### Options for the Infinity Telemetry include:

- Infinity CentralStation options, as indicated on page 22
- MicroO2+ Pulse Oximeter and connection cable
- VitalConnection cable for Non-invasive Blood pressure
- Non-invasive Blood pressure hoses and cuffs
- SpO2 measurements
- Analog ECG cable
- Pulse Oximetry cables and sensors
- 9V disposable batteries or disposable lithium ion batteries
- Infinity Telemetry disposable pouches

## Infinity Network

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The Infinity Network is a standards-based network providing connectivity and data access among Infinity devices.

The Infinity Network can include *wired* and *wireless* components.

For detailed information about the configuration, installation and operation of components in the Infinity Network, contact your local Dräger representative.

### WARNING

**Loss of communication between the Infinity CentralStation and the bedside monitor is possible. Dräger recommends using the bedside monitor for primary diagnosis and the Infinity CentralStation for patient viewing.**

### NOTE

Please refer to manufacturer's documentation for hazards associated with wireless components.



**WARNING**

- The Infinity CentralStation CPU and its display(s) must always stay on. Never press any of the On/Off buttons to turn the CPU or the display(s) off.
- The Infinity CentralStation CPU must be placed so that it can be heard easily if the external speakers become non-functional. You must keep the Infinity CentralStation external speakers connected at all times.
- KeyboardVideoMouse (KVM) extenders tested to be compatible with the Infinity CentralStation may be used. KVM extenders, as with any peripheral of the Infinity CentralStation, require good judgement and maintenance checks to ensure that all peripherals are connected properly and in working condition. When using a KVM extender, the internal speakers within the Infinity CentralStation CPU may not be heard in the event of a KVM disconnection. When using a KVM extender, it is recommended that the hospital implement a routine maintenance procedure to check the integrity of the KVM extender audio cable connection. In the event a visual alarm is present without an audible alarm, check the Infinity CentralStation speaker volume setting and all speaker connections.

## Local and Remote Patient Monitoring

**Local patients** are patients admitted to the Infinity CentralStation you are currently viewing, where the Infinity CentralStation is the primary monitor.

**Remote patients** are patients on the Infinity network that are available to view on the Infinity CentralStation you are currently viewing, yet the Infinity CentralStation is not the primary monitor.

- **Bedside patients** - Remote control is possible only if the Bed Control and the *Bed Silence* features are enabled at the Infinity CentralStation (see page 61) and remote control is activated at the bedside monitor (refer to monitor documentation).
- **Telemetry patients** - Remote control is possible only if the remote functions are enabled at patient's local Infinity CentralStation (see page 67).

Although you can view patients outside the Infinity CentralStation Monitoring Unit, you cannot silence alarms or make any setup changes for those patients.

## Interacting with the Infinity CentralStation Keys

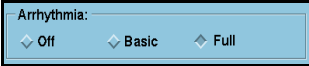
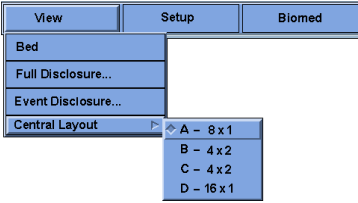
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To interact with the Infinity CentralStation, you can use a keyboard, a mouse, or a touch screen. When the Infinity CentralStation is configured to use the touch screen display, the keyboard and mouse remain active. If the touch screen capability is lost, you can still use the mouse and/or keyboard to interact with the Infinity CentralStation.

The Infinity CentralStation keyboard has the following *quick-access keys*:

- **F1** silences all active alarms at the Infinity CentralStation for 1 minute.
- **F5** displays **Main Screen**.
- **F8** invokes **Help**.
- **F9** generates Shift report for currently assigned beds.
- **F11** initiates a timed recording for all beds assigned to Main Screen. If the Second Video option is enabled, recordings are generated for the display in which the mouse pointer is located.
- **Print Screen** prints the contents of a screen on an optional laser printer.

## User Interface

| User Interface                  | Function/Purpose                                                                           | Remarks/Illustration/Example                                                                                                                                                                                                                                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BUTTONS</b>                  |                                                                                            |                                                                                                                                                                                                                                                                                                                    |
| Action button                   | Executes a function                                                                        | <p>Examples of action buttons are:</p> <ul style="list-style-type: none"> <li>– <b>Audio Pause (1 min)</b>, which silences alarms at the Infinity CentralStation (1 minute).</li> <li>– <b>Audio Paused (in bed view)</b>, which silences the alarms of an individual patient.</li> <li>– <b>Accept</b></li> </ul> |
| Menu button                     | Activates a menu                                                                           |                                                                                                                                                                                                                                  |
| Option button                   | A small rectangle inside the button implies <i>fixed</i> selections.                       |                                                                                                                                                                                                                                  |
| Radio button                    | Available choices are visible. The button next to the current selection appears depressed. |                                                                                                                                                                                                                                  |
| Increment button                | Allows scrolling through predetermined settings                                            |                                                                                                                                                                                                                                  |
| Page button                     | Allows scrolling through screen pages                                                      |                                                                                                                                                                                                                                  |
| <b>MENUS</b>                    |                                                                                            |                                                                                                                                                                                                                                                                                                                    |
| Cascading menu                  | Menu items followed by an arrow indicate additional related submenu selections.            |                                                                                                                                                                                                                                |
| <b>SCROLL BARS</b>              |                                                                                            |                                                                                                                                                                                                                                                                                                                    |
| Horizontal/vertical scroll bars | Allows navigation through data.                                                            | <p>To scroll through information, click on the arrows or drag the slide bar.</p>                                                                                                                                               |

| User Interface       | Function/Purpose                                                                                    | Remarks/Illustration/Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>POP-UPS</b>       |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Confirmation pop-up  | Safeguards against accidental execution of a function that may have significant consequences.       | Example: When trying to change the time, a confirmation pop-up appears because the change may affect other devices on the network.                                                                                                                                                                                                                                                                                                                                                                                                         |
| Informational pop-up | Contains information, warnings about operational errors, or necessary steps that must be performed. | Example: If attempting to transfer data from a bed that is not in 'Standby' mode, an informational pop-up will display the following message: "The source bed for the transfer must be in <i>Standby</i> . Please put this bed in <i>Standby</i> ."                                                                                                                                                                                                                                                                                        |
| Menu-specific pop-up | Provides function-specific information and may accommodate setting changes.                         | The <b>Assign Bed</b> pop-up in the <b>Setup Central Layouts</b> menu allows you to select another bed in the network for display.                                                                                                                                                                                                                                                                                                                                                                                                         |
| Password pop-up      | A Password pop-up restricts access to functions only intended for authorized personnel.             | <ul style="list-style-type: none"> <li>– The <i>clinical password</i> safeguards several setup procedures of the Infinity CentralStation. Often these setup procedures are performed by a Nurse Manager or a Unit Director.</li> <li>– The <i>biomedical password</i> protects functions only intended for use by Service or Biomedical personnel. These functions include setting up the Infinity CentralStation, accessing the system console, configuring locked options, and accessing Diagnostic and Clinical Events logs.</li> </ul> |

## Online Help

---

The Infinity CentralStation **Instructions for Use** are available via the **Help** screen. When active, the **Instructions for Use** occupies the left side of the screen.

### Using Online Help

| Function | Instruction                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open     | Click on <b>Help</b> in the upper-right corner of the Infinity CentralStation menu bar to launch the Infinity CentralStation Instructions for Use.                                                 |
| Search   | Use the <i>Find</i> function  , <i>Bookmarks</i> tab, and navigation tools to search for the wanted instructions. |
| Close    | Close the Instructions for Use by clicking on <b>File</b> and <b>Close</b> in the Help window, or on the X-button in the window's upper-right corner.                                              |

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# Infinity M300

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## About Infinity M300

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### WARNING

**Do not use the Infinity M300 near equipment that produces static or gradient magnetic fields.**

### NOTE

During installation, the Infinity CentralStation must be configured to work with Infinity M300.

Infinity M300 is a patient worn device that uses the Infinity CentralStation as the primary patient monitoring display and primary alarm source.

Before monitoring with the Infinity M300, the patients are admitted and patient data is displayed at the Infinity CentralStation. After admitting the patient to the Infinity CentralStation, the Infinity CentralStation can be used to customize the monitoring settings for specific patients. If the Infinity M300 is not communicating with the Infinity CentralStation over the network, Infinity M300 settings cannot be changed from the Infinity CentralStation.

Patient monitoring is interrupted and patient data is no longer displayed at the Infinity CentralStation under the following conditions:

- Patient discharge
- Battery failure

These conditions are indicated at the Infinity CentralStation by the following messages:

- BED DISCONNECTED (at the Infinity CentralStation display)

After a patient is discharged, the 'BED DISCONNECTED' banner displays at the local Infinity CentralStation.

### NOTE

DISCHARGED (at the Infinity M300 display)

After a patient is discharged, the 'DISCHARGED' banner displays on the Infinity M300.

- OFFLINE (at both the Infinity M300 and Infinity CentralStation displays).

After a network failure, an 'OFFLINE' alarm is received.

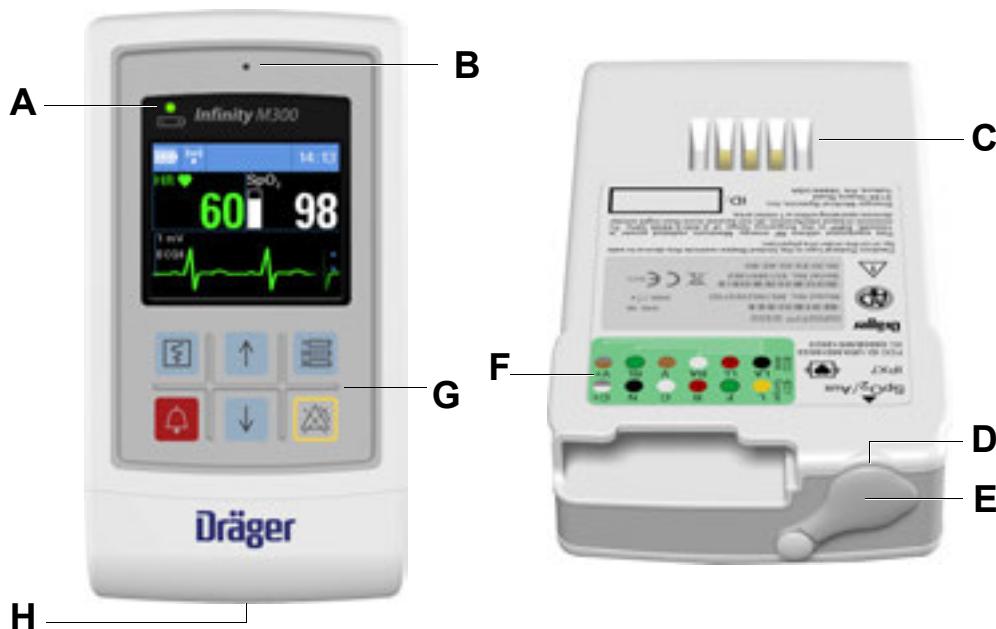
Upon discharging the patient, trends and demographic data are erased from the Infinity M300 memory. Full disclosure and Event disclosure data can still be viewed from the Infinity CentralStation Census application.

The Infinity M300 has a built-in power saving feature that automatically turns the Infinity M300 display off when not in use. This does not interrupt patient monitoring. Parameters, waveforms, alarms, and other information continue to be displayed and announced at the Infinity CentralStation.

The Infinity M300 screen is the secondary display for:

- ECG waves (heart rate), and
- Pulse Oximetry (SpO<sub>2</sub>, pleth waveform)





**A** Battery Status LED Indicator (see page 36)

**B** Audible alarm speaker

**D** SpO2/Aux port

**E** SpO2/Aux port cover

### WARNING

The Infinity M300 speaker is intended for use only in the patient vicinity. It is not intended for primary alarm annunciation. Use the Infinity CentralStation speaker for primary alarm annunciation.

**C** Charger interface

### NOTE

The SpO2 port is also used to run diagnostics on the Infinity M300. If you see the message 'Console Mode' on the Infinity M300 screen, please remove the device from service and contact your Hospital's Biomedical Engineering Dept.

**F** Lead wire illustration

**G** Keys

**H** Battery compartment

## User Controls

The Infinity M300 fixed keys give you quick access to a variety of functions.



### Record key

Press the Record key to:

- generate a timed recording, or
- cancel a timed recording, or
- store and record an event at the Infinity CentralStation.

#### NOTE

This timed recording's duration is 20 seconds and includes 10 seconds of recording delay time.



### Staff Alert key

Press the Staff Alert key to send a STAFF ALERT message from the Infinity M300 to the Infinity CentralStation, along with a *Serious* alarm.

#### NOTE

For Infinity M300 patients admitted to the Infinity CentralStation, the Infinity M300 Staff Alert setting (see page 300) determines if the Infinity M300 Staff Alert key is enabled and how it can be used.



### Up and Down arrow keys

Press either the Up Arrow or Down Arrow key to:

- scroll through available-waveforms from the Infinity M300 monitoring screen, or
- increase or decrease the menu field value (e.g., increase or decrease volume)



### Views key

Press the Views key to:

- Page through *Monitoring*, *Electrode Check*, *Volume (if enabled at the Infinity Central Station)*, and *Demographics* screens, or
- Take Infinity M300 out of Standby mode, or
- Accept/confirm user-selected values.

Press (and hold) the Views key for more than 3 seconds to turn the LCD display On or Off.



### Alarm Paused key

Press (and hold) the Alarm Paused key for more than 3 seconds to:

- Display a confirmation screen and, if you select **Yes**,
- invokes Infinity M300's ALARM PAUSED feature, which makes the Infinity M300 stop making alarms and, therefore, eliminates the Infinity CentralStation alarms associated with that specific Infinity M300/patient. It does not affect Infinity CentralStation alarms in any other way.

If Alarm Paused is ON, pressing the Alarm Paused key again turns Alarm Paused OFF.

#### NOTE

The Infinity CentralStation Alarm Paused setting (see page 300) determines if the Infinity M300 Alarm Paused key is enabled and how it can be used.

## Audio Alerts

If the Infinity M300 speaker is enabled (see page 300) audio alerts will be triggered for the following conditions.

### WARNING

The Infinity M300 speaker is intended for use only in the patient vicinity. It is not intended for primary alarm annunciation. Use the Infinity CentralStation speaker for primary alarm annunciation.

| Condition               | Description              |
|-------------------------|--------------------------|
| Acknowledgement         | 2 brief tones            |
| Error                   | 1 tone                   |
| <i>Find Device</i> tone | 5 beeps alternating tone |
| Alarm conditions        | ADV, SER, LT tones       |

### Adjusting Alarm Tones

You can adjust the Infinity M300 alarm tone pattern and the volume of the alarm.

To adjust the volume of the tones at the Infinity CentralStation, see page 41.

To adjust the volume of the tones at the Infinity M300, see page 34.

To adjust the Infinity M300 alarm tone pattern at the Infinity CentralStation, see page 67.

### Finding the Infinity M300

Infinity M300 generates five alternating *Find Device* tones when you request it at the Infinity CentralStation (see page 79).

## Starting up Infinity M300

To turn on the Infinity M300, simultaneously press the up and down arrows. When you turn Infinity M300 on, a startup screen displays the current software version and possible messages. During this operation Infinity M300 also sounds a tone to confirm that alarm annunciation is functional.

**WARNING**

If no verification tone sounds, do not use the device and contact the Hospital Biomedical Engineering Department.

Contact the Hospital Biomedical Engineering Department upon receiving any of the following messages:

| Startup Message                              | Condition                                                             |
|----------------------------------------------|-----------------------------------------------------------------------|
| High Temperature Reset                       | Reset due to the internal temperature exceeding the acceptable limit. |
| Default Patient Settings Restored            | User settings were restored to factory defaults.                      |
| Software Reset, see Diagnostic Log           | Reset due to a software error.                                        |
| Patient Data Erased (followed by error code) | Trend data or at least one stored event was erased.                   |
| Low Battery Reset                            | An error was detected with the battery.                               |
| Software Watchdog                            | Reset due to software error.                                          |

## Infinity M300 Power

The Infinity M300's rechargeable battery can be recharged during patient use by inserting the Infinity M300 into its *Bedside Charger*. The battery runtime is shortened with continuous use of the display and speaker. The device's Battery Status LED Indicator provides indication to the battery status (see page 38).

**NOTE**

The Infinity M300 enters a deep-discharge state when it is stored away from either the bedside or the central chargers for extended periods. To exit the deep-discharge state, place the Infinity M300 in either the Bedside or Central Charger for at least 30 minutes.

**CAUTION**

- When the Infinity M300 is worn in the shower, an Infinity M300 Shower Pouch must be used to prevent water from entering the battery compartment.
- When the Infinity M300 internal battery requires replacement, please contact the Hospital Biomedical Engineering Department. Battery replacement information is described in service documentation.

**NOTE**

Recycle or dispose of batteries in accordance with the directive EC 91/156/EWG or equivalent country-specific regulation.

## Infinity M300 Bedside Charger

### WARNING

Place the Infinity M300 device to avoid any possibility of the device falling on or injuring patient, or any accessory cables accidentally becoming entangled around patient's neck.

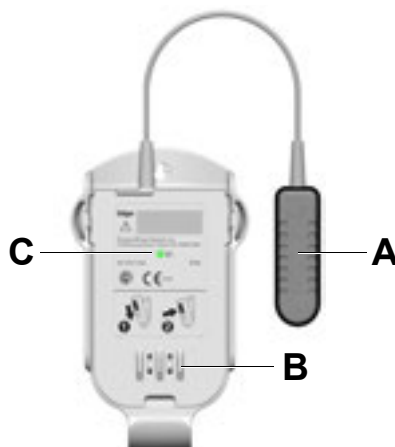
### NOTE

For patients in a bed or chair, do not place the Infinity M300 Bedside Charger in a patient's pocket or pouch. Use the Infinity M300 Bed Clip (see the following illustration) to clip the Bedside Charger to the bed linen instead.



**A**  
**B**

**A** Bed linen  
**B** Infinity M300 Bed Clip



**A** Power supply  
**B** Infinity M300 Bedside Charger  
**C** Green LED

### NOTE

When illuminated, the green LED on the Infinity M300 Bedside Charger indicates that the bedside charger is connected to the AC Mains voltage and is ready to charge the Infinity M300.

When the Infinity M300 screen is enabled, it displays a battery icon that shows an estimate of the remaining available battery power.

When the screen is off, battery condition is indicated using the Battery Status LED Indicator.

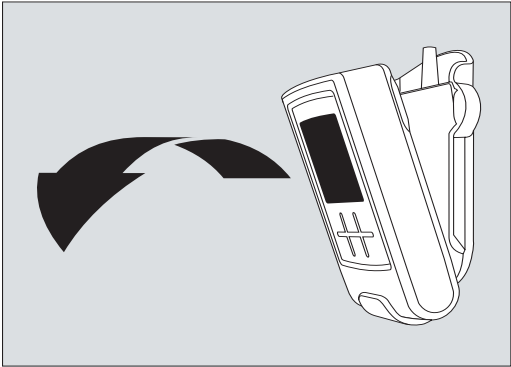
When the remaining battery level is 10%, the *Low Battery* message appears on the Infinity CentralStation and on the Infinity M300 display. When the remaining battery level is 5%, an Advisory alarm is generated at the Infinity CentralStation and the *Recharge Battery* message displays on the Infinity CentralStation and on the Infinity M300 display. These status messages disappear when the Infinity M300 is placed in a Bedside Charger or Central Charger.

When Infinity M300 is removed from the Charger, the green Battery Status LED indicator blinks once every 5 seconds and the battery icon shows available battery power if the Infinity M300 screen is enabled.

**Battery Status LED Indicator Description**

| Behavior                                          | Description                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------|
| Blinking green<br>(approximately every 5 seconds) | Power On<br>Infinity M300 is being used away from the charger.<br>Battery OK  |
| Blinking amber<br>(approximately every 5 seconds) | Power On<br>Infinity M300 is being used away from the charger.<br>Low battery |
| Solid amber                                       | Charging in progress (while in charger)                                       |
| Solid green                                       | Charging complete (while in charger)                                          |

**Removing Infinity M300 from the *Bedside Charger***



Place your fingers over the top of the Infinity M300 device, between the back of the Infinity M300 and the *Bedside Charger*. Gently pry the top of the Infinity M300 device away from the *Bedside Charger* until the Infinity M300 is loose. Lift the Infinity M300 Device out of the charger.

**Placing Infinity M300 into the *Bedside Charger***

Place the bottom of the Infinity M300 device into the *Bedside Charger*. Gently press the top of the Infinity M300 toward the *Bedside Charger* until it snaps into place.

**Infinity M300 Central Charger**

**WARNING**

**Remove the Lead Wire Set from Infinity M300 before placing the Infinity M300 into the Central Charger, or possible electrical shock could occur.**

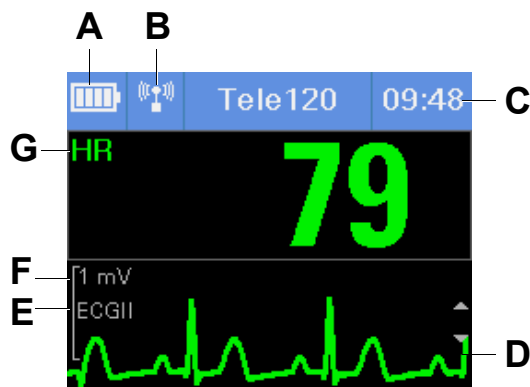
The Infinity M300 should be stored in the *Central Charger*, which accommodates up to 10 devices. When an Infinity M300 is placed on the *Central Charger*, the battery recharges and the patient is discharged from the Infinity CentralStation. This also erases any patient data stored in the Infinity M300's memory (i.e. trends, demographics).

**CAUTION**

**Do not use the Infinity M300 Central Charger near patients (fluids there may damage the charger).**

## Infinity M300 Parameter Displays

### ECG Display



- A The **Battery Charge** icon appears in the Message Area shows the available battery power.
- B The **Wireless Signal Strength** icon appears at the left of the blue Message Area along the top.
- C The **Bed Label** and **Time** also appear in the Message Area.
- D The **Waveform**. If pacer detection is enabled, pacer spikes are indicated on the waveform.
- E The **Lead Label**. The default is Lead II.
- F The **Waveform Scale** next to the waveform is configured at the Infinity CentralStation.
- G The **HR Parameter** box.

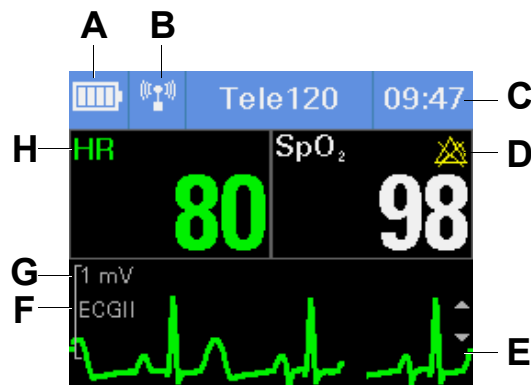
### Waveform Area

The waveform scale (located at the left of the waveform) is configured at the Infinity CentralStation. Use the Infinity M300 **Up Arrow** and **Down Arrow** keys to scroll through available leads.

### ECG/SpO2 Display

#### NOTE

SpO<sub>2</sub> does not display unless the parameter is enabled at the Infinity CentralStation.



- A The **Battery Charge** icon appears in the Message Area and illustrates the available battery power.
- B The **Wireless Signal Strength** icon appears in the Message Area.
- C The **Bed Label** and **Time** also appear in the Message Area.
- D The **Alarms OFF** Icon appears when the alarms are **OFF**.
- E The **Waveform**.
- F The label of the **Lead** currently being displayed.
- G The **Waveform Scale** next to the waveform is configured at the Infinity CentralStation.
- H The **HR Parameter** box appears on the left, and the **SpO<sub>2</sub> Parameter** box is on the right.

## Other Infinity M300 Displays

### Start Up Screen

The **Start Up** screen displays while the Infinity M300 is powering up. A progress bar provides a visual display of the *Start up* process.

### New Patient Screen

Once **Start Up** is completed, the **New Patient** screen appears if a previous patient's data remains in the Infinity M300 memory. To admit a new patient, you must select **Yes** to erase the previous patient's data from memory.

Use the **Up Arrow** or **Down Arrow** keys to toggle between **Yes** and **No** selections.

#### WARNING

**When you admit a patient to the Infinity M300, make sure that the Alarm Limits and Arrhythmia Settings are appropriate for the patient. Upon admit some predefined default settings may not be applied.**

#### NOTE

To admit a patient to the Infinity M300, use the ADT menu option at the Infinity CentralStation to access the Admit screen.

#### NOTE

To discharge a patient from the Infinity M300, you can either use the ADT menu option at the Infinity CentralStation to access the Discharge option, or you can place the Infinity M300 on the *Central Charger*.

When discharging a patient:

- Patient data is erased in Infinity M300 memory.
- The patient is discharged from the Infinity CentralStation.

When an Infinity M300 patient is discharged from the Infinity CentralStation (see page 125), the *Discharge* banner covers the Infinity M300 screen, along with the message, *Press*  *to exit*.

To exit the Infinity M300 **Discharge** screen, press the **Views** key.

### Transfer Screen

To transfer patient information from one Infinity M300 to another (or to a bedside monitor), you must put the Infinity M300 in Standby mode and then initiate the transfer at the destination Infinity CentralStation (see page 126).

During the transfer process the message, *Transferring data*, displays on the destination Infinity CentralStation screen. When the transfer is complete, a Discharge screen appears on the original Infinity CentralStation (not the destination Infinity CentralStation).

### Standby Screen

Standby mode must be initiated at the Infinity CentralStation. When a telemetry patient's monitor is in *Standby* mode, the *Standby* banner displays on the Infinity M300 screen along with the message, *Press*  *to exit*.

When *Standby* mode is selected at the Infinity CentralStation, the Infinity M300 is affected as follows:

- Patient monitoring is suspended,
- Screen is blank except for a *Standby* banner,
- Network and system status messages circulate in the message area.

You can set *Standby* time-out selections at the Infinity CentralStation. Selection options are: **No time out, 5 mins., 10 mins., 20 mins., 30 mins., 1 hr., 2 hr., and 3 hr.**

To return the Infinity M300 to *Monitoring* mode, press the Infinity M300 **Views** key.



## Demographics Screen

The Infinity M300 **Demographics** screen shows patient demographic information. All patient demographic information is received from the Infinity CentralStation and may include the following:

- Name
- ID#
- Bed Label
- Care Unit
- Patient Category

## Volume Setup Screen

If the Infinity M300 **Speaker** is enabled at the Infinity CentralStation (see page 301) you can access the Infinity M300 **Volume Setup** screen and change the device alarm volume. To access the Infinity M300 **Volume Setup** screen press the **Views** key.

To scroll through the available volume settings, press the Infinity M300 **Up Arrow** or **Down Arrow** key. Available selections are: **Off**, **10%**, **20%**, **30%**, **40%**, **50%**, **60%**, **70%**, **80%**, **90%**, and **100%**.

### CAUTION

If speaker is disabled, the **Volume** screen will not be available to the user.

Although the percentage settings are always available when Volume Control is enabled, 'Off' is only available when the M300 patient is being monitored at the ICS. If M300 goes out of range, 'Off' becomes disabled.

### WARNING

**Set alarm volume in accordance with ambient sound level in the room so that alarm volume is audible.**

## Electrode Status Screen

An **Electrode Status** screen provides a status of electrode connection, enabling you to quickly check which electrodes are disconnected without disturbing the patient. The color of the dot indicates which electrodes are connected (or not connected) to the patient.

- Green = connected
- Red = disconnected

To access the **Electrode Status** screen press the **Views** key.

## Infinity M300 Alarm Processing

### CAUTION

Upon startup and after a patient discharge, the Infinity M300 alarms will *not* be active until a numeric value has been received for the corresponding parameter

### Alarm Grade Display

Infinity M300 Monitoring System supports three alarm severity grades:

- life-threatening
- serious
- advisory

Each grade is accompanied by a different pattern and color-coded display, shown as follows:

| Severity Grade   | Color  | Audible Tone                                              |
|------------------|--------|-----------------------------------------------------------|
| Life-threatening | Red    | Two Frequencies (alternating high/low)                    |
| Serious          | Yellow | Two medium tempo, medium pitch notes, followed by a pause |
| Advisory         | Cyan   | Single Frequency, low                                     |

When a parameter is alarming, the parameter box flashes the color representing the severity grade and the Infinity M300 sounds an associated tone.

Alarm Paused

**WARNING**  
No alarms are announced when the Infinity M300 is in *Alarm Paused* mode.

**CAUTION**  
The Alarm Paused key on the Infinity M300 can only be disabled at the Infinity CentralStation.

To initiate *Alarm Paused*, press and hold the **Alarm Paused** key for 3 seconds. A confirmation screen displays. If **YES** is selected (to confirm), an *Alarm Paused* message displays in the Infinity M300 Message area with a timer, which represents the length of pause time remaining.

For a description of alarm behavior at the Infinity CentralStation, see page 157.

Trends

You can review and set up the Trends for an Infinity M300 patient at the Infinity CentralStation (see page 202). Although Infinity M300 does not display trend data, it can store up to 72 hours of trend data for each monitored parameter.

All trend data is deleted when you:

- discharge the patient, or
- update Infinity M300 software.

Setting up Infinity M300 at the Infinity CentralStation

Infinity M300 Setup Screens

| If you want to...                                                         | Refer to...                            | Location |
|---------------------------------------------------------------------------|----------------------------------------|----------|
| Assign Infinity M300 IDs and IP Addresses                                 | Biomed <b>Telemetry Devices</b> screen | page 299 |
| Configure Infinity M300 System Defaults                                   | Biomed <b>M300 Setup</b> screen        | page 299 |
| Configure Infinity M300 monitoring for an individual patient              | Bed View Telemetry setup screens       | page 117 |
| Assign an Infinity M300 to a patient                                      | <b>Admit</b> screen                    | page 120 |
| Set system defaults for Infinity CentralStation telemetry patient windows | Telemetry <b>System Setup</b> screen   | page 67  |
| View Infinity M300 status information and configure device keys           | <b>M300 Setup</b> screen               | page 299 |

## Infinity M300 Setup

The **M300 Setup** screen is accessible for local Infinity M300 patients and displays information received from the Infinity M300. It provides Infinity M300 status information and permits configuration functions.

| Selection                                                                                                                                                                                                                       | Description                                                                                                                                                                                                                              | Settings                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Telemetry ID                                                                                                                                                                                                                    | Displays Infinity M300 identification number                                                                                                                                                                                             | N/A                                                                                                                |
| Battery Level Bar Graph                                                                                                                                                                                                         | Graphically displays remaining battery time<br><br>The Infinity M300 current battery voltage appears next to the Bar Graph.                                                                                                              | N/A                                                                                                                |
| ECG Lead Prep                                                                                                                                                                                                                   | Shows the quality of the ECG Lead Prep. The displayed information is received from Infinity M300 and can be updated dynamically.                                                                                                         | N/A                                                                                                                |
| Alarm Paused <sup>1</sup>                                                                                                                                                                                                       | Determines if the Infinity M300 <b>Alarm Pause</b> key is enabled<br><br>Alarm Paused time is set in Telemetry System Setup (see page 67)                                                                                                | <ul style="list-style-type: none"> <li>– On</li> <li>– Off</li> </ul>                                              |
| Record <sup>1</sup>                                                                                                                                                                                                             | Generates manual timed recordings                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>– Record</li> <li>– Off</li> <li>– Record/Store</li> <li>– Store</li> </ul> |
| Staff Alert <sup>1</sup>                                                                                                                                                                                                        | Determines if Infinity M300 <b>STAFF ALERT</b> key is enabled or disabled.<br><br>When <b>On</b> is selected, you can generate a STAFF ALERT alarm at the Infinity Central Station by pressing the Infinity M300 <b>STAFF ALERT</b> key. | <ul style="list-style-type: none"> <li>– On</li> <li>– Off</li> </ul>                                              |
| Volume Control <sup>1</sup>                                                                                                                                                                                                     | Determines if the Infinity M300 speaker is enabled.                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>– Enabled</li> <li>– Disabled</li> </ul>                                    |
| <b>WARNING</b><br><b>The Infinity M300 speaker is intended for use only in the patient vicinity. It is not intended for primary alarm annunciation. Use the Infinity CentralStation speaker for primary alarm annunciation.</b> |                                                                                                                                                                                                                                          |                                                                                                                    |

| Selection                                                                                                                          | Description                                                          | Settings                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Speaker Volume                                                                                                                     | Sets Infinity M300 audible alarm                                     | <ul style="list-style-type: none"><li>– Off</li><li>– 10 - 100%</li></ul>                                                  |
| <b>WARNING</b><br>Set the alarm volume in accordance with the ambient sound level in the room so that the alarm volume is audible. |                                                                      |                                                                                                                            |
| Accept/Undo                                                                                                                        | Enables you to save or discard the changes made since the last save. | <ul style="list-style-type: none"><li>– Accept saves any changes</li><li>– Undo reverts to the previous settings</li></ul> |
| <sup>1</sup> Unless the default setting (page 299) is Per Patient, these selections are unavailable for selection.                 |                                                                      |                                                                                                                            |

## Recordings

---

### Recording Types

You can generate manual timed recordings from the Infinity M300 by pressing the device Record key or using the Infinity CentralStation. You can also generate manual or continuous timed recordings for an Infinity M300 patient admitted to the Infinity CentralStation from the Infinity CentralStation (page 175).

## Monitoring a Patient Using Infinity M300

---

### ECG Monitoring

To prepare for ECG monitoring:

- 1 Admit the Infinity M300 connected to the patient to the desired Infinity CentralStation patient window (see page 119).
- 2 Apply electrodes to patient (see page 305).
- 3 Connect the appropriate ECG lead wire set to the Infinity M300.
- 4 Attach the lead wires to the electrodes.
- 5 Set the alarm and arrhythmia limits at the Infinity CentralStation (see page 166).
- 6 Check the demographics screen to ensure that you are monitoring the correct patient.

#### NOTE

If communication with a Infinity CentralStation can not be established when trying to admit a patient, contact or call your local Dräger representative.

### SpO<sub>2</sub> Monitoring

To prepare for SpO<sub>2</sub> monitoring:

- 1 Select an appropriate sensor and place the sensor on the patient.
- 2 Turn SpO<sub>2</sub> "**ON**" in the Patient View Setup screen (see page 142).
- 3 Connect the pulse oximetry cable to the Infinity M300.

#### NOTE

Pulse Oximetry must be enabled via a locked option. Please call your local Dräger representative.

- 4 Set alarm limits at the Infinity CentralStation (see page 166).

## Infinity M300 Messages

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### Message Area

The Infinity M300 message area displays alarms, network-generated or locally generated messages. If more than one message must display, the messages circulate in a round-robin manner, with each message appearing for 1 second. These messages include:

- **Alarm messages**, which display in cyan characters on a background color defined by the alarm severity grade (see page 161).
- **Status messages**, which display in black characters on a white background.
- **System information messages**, which display in white text on a blue background.
- **Network messages**, which display in yellow text on a blue background.

See page 97 for a description of **Status message** display at the Infinity CentralStation.

#### NOTE

If communication with a Infinity CentralStation can not be established when trying to admit a patient, please contact or call your local Dräger representative.

## Infinity M300 Status Messages

| Alarm Message in Message Area | Grade | Description                                                                                      | Action                                                                                                                         |
|-------------------------------|-------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| STAFF ALERT                   | SER   | The Infinity M300 <b>STAFF ALERT</b> key was pressed.                                            | Check the patient.<br>This message requires acknowledgement.                                                                   |
| Recharge battery              | ADV   | Infinity M300 battery charge is low.                                                             | Recharge battery or contact the Hospital Biomedical Engineering Dept. to replace Infinity M300 battery.                        |
| HR Alarms Off                 | SER   | Heart rate alarms are turned Off at the Infinity CentralStation.                                 | Pay close attention to the patient.<br>This message remains until HR alarms are turned back ON at the Infinity CentralStation. |
| Alarm Paused: xxx             | ADV   | <b>Alarm Paused</b> key was pressed. The remaining pause time displays with the message.         | Condition remains until time expires or until Alarm Paused key is pressed at the Infinity M300 while Alarm Paused is active.   |
| Offline                       | SER   | The Infinity M300 cannot communicate with the Infinity CentralStation.                           | Check the patient.<br>Ensure that the patient is not out of the designated coverage area.                                      |
| Lead X <b>OFF</b>             | ADV   | ECG lead Disconnected as indicated.                                                              | Check the patient.<br>Reapply electrodes.                                                                                      |
| HR Limits <b>OFF</b>          | N/A   | The Infinity M300 will not alarm for HR Limit violations.                                        | Informational message only.                                                                                                    |
| HR, ASY, VF <b>OFF</b>        | N/A   | The Infinity M300 will not alarm for HR Limit violations, asystole, or ventricular fibrillation. | Informational message only.                                                                                                    |
| Bed Disconnected              | N/A   |                                                                                                  | The patient has been discharged.                                                                                               |

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# Infinity Telemetry Transmitter

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|                                                           |    |
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## Transmitter Components

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### WARNING

**Do not use the Infinity Telemetry transmitters near equipment that produces static or gradient magnetic fields.**

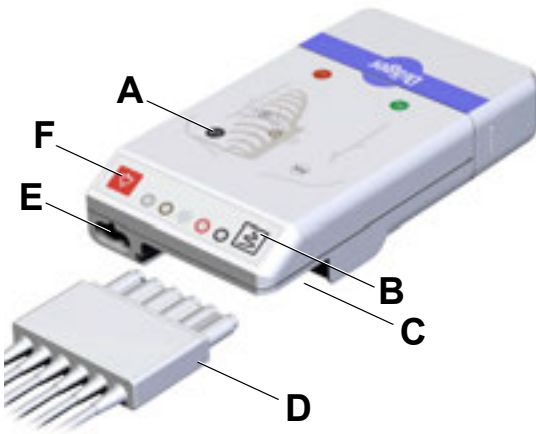
### Lead Connector

The Infinity TruST Telemetry transmitter, which has a 6-pin lead connector, can accommodate a 3-, 5-, or 6-wire ECG lead set.

### Infinity TruST Telemetry Transmitter

### WARNING

**Do not use Infinity Monoleads with Infinity Telemetry. Doing so adversely affects performance and could result in misdiagnosis.**



- A** Lead Placement Diagram
- B** Recording Button
- C** Lead Connector  
Accommodates 3-, 5-, and 6-Wire Lead Set
- D** Lead Block  
6-Wire Lead Set is illustrated
- E** Programming Port /MicroO2+ Interface/Analog  
ECG Cable Port/Vital Connection Cable Inter-  
face
- F** Staff Alert Button

## Programming Port Connector/Interface

The transmitter has a connector that can be used to do the following:

- Program the transmitter.
- Connect to the **MicrO2® pulse oximeter** for monitoring SpO2 and pulse (see page 142).



- Connect a **Telemetry Analog ECG Cable** to an Infinity bedside monitor (via the MultiMed pod) to display ECG waveform (lead II) and possible pacer spikes.



### NOTE

For the best possible ECG signal, you should only use the analog ECG cable with 5- or 6-wire lead sets.

- Connect a **Vital Connection Cable** to an Infinity bedside monitor to display NIBP, SpO2 and PLS (see page 143).

## Transmitter Buttons

### NOTE

The transmitter buttons are only functional if they have been activated (see page 301).

- The red **staff alert** button allows the patient to initiate a serious alarm which sounds at the Infinity CentralStation. The message **Staff Alert** is displayed in the patient window. You can silence this alarm at the Infinity CentralStation with the **Audio Pause (1 min)** button.
- The white **recording** button allows you to begin a recording. Press this button less than 3 seconds for a timed recording; press it longer than 3 seconds for a continuous recording.

If you press both transmitter buttons simultaneously, a 1 mV calibration pulse is superimposed on the patient's Bed View lead display at the Infinity CentralStation.

Transmitter LED Error Patterns (Normal Mode)

Transmitter Self-Test

When you put a battery into a transmitter, it performs a self-test. If the test completes successfully, all of the LEDs flash briefly. If an error is detected, all the LEDs flash initially and then alternate an error-specific pattern.

Transmitter Orientation as you examine LED Patterns

The Transmitter shown here is the Infinity TruST Telemetry transmitter. Error codes display on four of the LEDs



| From                                                                    | To | Description                                                                | Action                                                                                                                   |
|-------------------------------------------------------------------------|----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| One lighted diode “ripples” across display.<br>●○○○○○ ○○○○○<br>— —————▶ |    | Battery is critically low. Transmitter is operating, but not sending data. | Re-insert or replace the battery immediately (see page 54). If the error continues, take the transmitter out of service. |
| Two diodes flash<br>●●○○○ ○○○○○                                         |    | Transmitter unable to tune                                                 | Reprogram the transmitter.                                                                                               |
| Two diodes flash<br>○●●○○ ○○○○○                                         |    | Analog ECG cable connected                                                 | --                                                                                                                       |

For other possible transmitter-related error messages that may appear at the Infinity CentralStation, see page 58.

## Transmitter Operating Modes

### Normal Mode

During *Normal* mode the transmitter is connected to a patient and collects data which it relays to the Infinity CentralStation.

#### NOTE

*Staff* mode lasts for 30 seconds, after which the transmitter returns to *Normal* operating mode.

### Staff Mode

*Staff* mode may be used to troubleshoot the transmitter or indicate a lead-off condition.

To activate Staff mode:

- Press the *Staff Alert Button* for more than 3 seconds.

The LEDs may identify the possible conditions displayed in the following table:

| Condition                                      | LED Reaction                                                                                                             |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Lead(s)-off                                    | The LED(s) corresponding to the detached lead(s) flash(es).                                                              |
| Reference lead-off<br>(or all other leads-off) | All LEDs flash simultaneously<br><br>On transmitters using 3-wire mode only 3 LEDs flash when the reference lead is off. |
| Low battery <b>and</b> Lead-off                | The LED patterns alternate for each condition.                                                                           |

## Transmitter Battery

**WARNING**

To avoid explosion, do not recharge or disassemble a battery, or dispose of it in fire.

**CAUTION**

- Infinity Telemetry Shower Pouch must be used when it is worn in the shower so that water does not get in the battery compartment.
- Always remove the battery before storing the transmitter. If the transmitter will not be in use for an extended time, remove the battery.
- Always use a new, fully charged battery each time you begin monitoring a patient with the transmitter.

**NOTE**

Recycle or dispose of batteries in accordance with the directive EC 91/156/EEG or equivalent country-specific regulation.

| Battery Type                                                                                                                                                 | Nominal Voltage | Typical Life   | Characteristics                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|----------------------------------------------|
| Alkaline/manganese oxide<br>(DURACELL PROCELL model<br>PC1604)                                                                                               | 9 V             | 2 days minimum | General purpose battery.<br>Good shelf life  |
| <b>WARNING</b><br>Use only the brand <b>ULTRALIFE (model U9VL-FP) lithium battery.</b><br>Any other lithium battery may present a risk of fire or explosion. |                 |                |                                              |
| Lithium/manganese dioxide<br>(ULTRALIFE model U9VL-FP)                                                                                                       | 9 V             | 4 days minimum | High energy density.<br>Excellent shelf life |

## Installing or Replacing the Battery

### WARNING

**Do not use the Infinity Telemetry transmitter unless the battery cover is securely in place.**

Each time you begin monitoring a patient with the transmitter, you should install a new transmitter battery.



Also, to avoid interruption in patient monitoring, promptly replace the transmitter battery when:

- the Transmitter Low Battery alarm displays at the Infinity CentralStation (see page 58), or
- the transmitter LED lights flash (see page 53).

To install the transmitter battery:

- 1 Remove the transmitter battery cover.
- 2 Insert a fully-charged battery into the transmitter.  
The transmitter performs a self-test.
- 3 Slide the battery cover back on and snap into place.  
If the battery cover is not closed properly the transmitter will not operate.

## Transmitter Setup

---

The **Transmitter Setup** screen provides transmitter status information and permits configuration and programming functions for *local* telemetry channels.

### Accessing the Transmitter Setup Screen

To access the Transmitter Setup Screen:

- 1 Open the patient's Bed View screen.
- 2 Click on **Setup**.
- 3 Click on **Telemetry**.
- 4 Click on **Transmitter Setup**.

## Received Transmitter Status Information

| Received Transmitter Status section of the Transmitter Setup menu                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Information                                                                                                                                                                                                                                                                                                                                                                                                                 | Description                                                                                                                                                                                             |
| ID                                                                                                                                                                                                                                                                                                                                                                                                                          | The transmitter ID cannot be changed; it is configured by the Hospital Biomedical Engineering Dept. during installation.                                                                                |
| ECG Lead Wires                                                                                                                                                                                                                                                                                                                                                                                                              | 3-, 5-, or 6-wire mode cannot be changed by the user; it is configured by the Hospital Biomedical Engineering Dept. during installation.                                                                |
| Battery                                                                                                                                                                                                                                                                                                                                                                                                                     | A bar graph showing the transmitter battery charge and current voltage. The color of the graph also indicates the status of the remaining battery charge.                                               |
| ECG Lead Prep                                                                                                                                                                                                                                                                                                                                                                                                               | <p>The color of the dot indicates the quality of lead prep.</p> <p>green = good</p> <p>yellow = acceptable</p> <p>red = poor</p> <p>*L* = Lead Off</p> <p>*N* = No signal detected from transmitter</p> |
| <ul style="list-style-type: none"> <li>– ECG Lead Prep <b>RL</b> is not displayed as this is a reference lead only.</li> <li>– An Infinity TruST Telemetry transmitter programmed for 6-wire monitoring displays: RA, LA, LL, C, and C+.</li> <li>– An Infinity Telemetry transmitter programmed for 5-Wire monitoring displays: RA, LA, LL, and C. If programmed for 3-wire monitoring, it displays: RA and LA.</li> </ul> |                                                                                                                                                                                                         |
| Update Status                                                                                                                                                                                                                                                                                                                                                                                                               | Allows you to update the screen with the latest status information.                                                                                                                                     |
| Frequency                                                                                                                                                                                                                                                                                                                                                                                                                   | Cannot be changed by the user; it is configured by the Hospital Biomedical Engineering Dept. during installation.                                                                                       |



## Transmitter Setup Screen Functions

### Customizing the Transmitter Buttons

#### NOTE

You can only activate/deactivate the transmitter buttons for an individual patient if *Per Patient* mode was selected during setup (see page 303).

To customize the transmitter buttons:

- 1 Open the **Transmitter Setup** screen (see page 55).
- 2 Click on the **On/Off** toggle button next to **Transmitter Record:**  
and/or  
Click on the **On/Off** toggle button next to **Transmitter Staff Alert:**.
- 3 Click on **Accept** to save or **Undo** to keep previous settings.

## Attaching the Transmitter to the Patient

---

To attach the transmitter to the patient:

- 1 Verify that the transmitter is working properly (check the LEDs).
- 2 Apply electrodes to patient. (see page 305).
- 3 Connect the appropriate ECG lead wire set to the transmitter.
- 4 Attach the lead wires to the electrodes.
- 5 Admit the patient to the Infinity CentralStation patient window for which the transmitter is selected.

## Transmitter Messages

| Status Messages                                                                                  |                          |       |                                                    |                                                                                                                        |
|--------------------------------------------------------------------------------------------------|--------------------------|-------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Alarm Message                                                                                    |                          | Grade | Description                                        | Action                                                                                                                 |
| Parameter Area                                                                                   | Status Area              |       |                                                    |                                                                                                                        |
| HR: <value><br>ARR: <value><br>PVC/min: <value><br>SpO2: <value><br>PLS: <value><br>STx: <value> | Staff Alert              | SER   | The transmitter staff alert button was pressed.    | Check the patient.<br><br>This message requires acknowledgement.                                                       |
| HR: *N*<br>ARR: <blank><br>PVC/min: <blank><br>SpO2: <blank><br>PLS: <blank><br>STx: <blank>     | Transmitter no signal    | ADV   | The receiver cannot detect the transmitter.        | Be sure patient is within telemetry antenna range.<br><br>Check cable connections.<br><br>Replace transmitter battery. |
| HR: *I*<br>ARR: <blank><br>PVC/min: <blank><br>SpO2: <blank><br>PLS: <blank><br>STx: <blank>     | Transmitter ID incorrect | ADV   | The transmitter ID and programmed ID do not match. | Reprogram transmitter with correct ID.<br><br>Make sure that transmitters do not have duplicate IDs.                   |
| HR: <blank><br>ARR: <blank><br>PVC/min: <blank><br>SpO2: <blank><br>PLS: <blank><br>STx: <blank> | Transmitter low battery  | ADV   | Transmitter's battery charge is low.               | Replace transmitter battery immediately.<br><br>The message remains until battery is replaced.                         |

# Infinity CentralStation Setup

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**Accessing the Setup Screens** . . . . . 60

Password-Protected Screens . . . . . 60

**Setup - Central Screen** . . . . . 61

**Setup - Recorder Screen** . . . . . 63

**Setup - Central Layouts Screen** . . . . . 64

## Accessing the Setup Screens

To access the Setup screen:

- 1 Click on **Setup** in the Main Screen menu bar.
- 2 Click on one of the drop down menu selections.

| Selection          | Reference              |
|--------------------|------------------------|
| Central...         | page 61                |
| Recorders...       | page 63                |
| Central Layouts... | page 64                |
| Telemetry          | page 65 and<br>page 81 |

## Password-Protected Screens

Access to the following setup functions is controlled with assignment of a required password (see page 291).

| Screen                                    | Required Password                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Setup > Telemetry > System Setup          | Clinical                                                                                                |
| Setup > Telemetry > Patient View Defaults | Clinical                                                                                                |
| Setup > Telemetry > Alarm Limits Defaults | Clinical                                                                                                |
| Setup > Telemetry > Arrhythmia Defaults   | Clinical                                                                                                |
| Setup > Telemetry > ST Defaults           | Clinical                                                                                                |
| Setup > Central                           | Clinical/Biomed                                                                                         |
| Setup > Recorder                          | Clinical/Biomed                                                                                         |
| Setup > Central Layout                    | Clinical (optional)<br>Password access to this function can be set during configuration (see page 291). |

If a password is required, a pop-up window appears when you click on the **Setup** menu item.

To enter a password:

- 1 Type the password in the pop-up text entry box.

### NOTE

You can only enter the password if the mouse pointer is on the text entry box.

- 2 Click on **Accept**. If the password is incorrect, an error message displays.

## Setup - Central Screen

To open the Setup - Central screen, refer to page 60.

| Selection                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                         | Available Settings                                                                                                                                                | Default   |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Sweep Speed                                                                                                    | Determines how fast the erase bar sweeps across the screen to update the waveforms                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>– 25 mm/sec</li> <li>– 50 mm/sec</li> </ul>                                                                                | 25 mm/sec |
| Display Limits                                                                                                 | Determines whether or not alarm limits appear in the Infinity CentralStation parameter areas                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF</li> </ul>                                                                                             | OFF       |
| Alarm Volume                                                                                                   | Determines alarm tone volume                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>– 10% - 100%</li> </ul>                                                                                                    | 50%       |
| <b>WARNING</b><br><b>Always set the alarm volume so it can be heard during the busiest periods of the day.</b> |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                   |           |
| Attention/Error Volume                                                                                         | Determines the attention and error tone volume                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>– 10% - 100%</li> </ul>                                                                                                    | 50%       |
| Display Timeout                                                                                                | Determines how long the screen will display without user interaction.                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>– 1 min</li> <li>– 3 min</li> <li>– 5 min</li> <li>– No Timeout</li> </ul>                                                 | 3 min     |
| Audio Paused Enable                                                                                            | <p>Determines whether or not you can silence bedside alarms from the Infinity CentralStation.</p> <p>Remote silence is only possible at the Infinity CentralStation if also activated at the bedside network device.</p>                                                                                                                                            | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF</li> </ul>                                                                                             | ON        |
| Bed Control Enable                                                                                             | <p>Determines whether or not you can control certain bedside monitor functions from the Infinity CentralStation.</p> <p>Remote control is only possible at the Infinity CentralStation if also activated at the network device.</p> <p>If this function is disabled, the Infinity CentralStation can still silence bedside alarms if Audio Paused Enable is ON.</p> | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF - The alarm and arrhythmia setup entries and Relearn buttons are unavailable for selection.</li> </ul> | ON        |

| Selection                                                                                                                                                                                                           | Description                                                                                                                                                                                                                         | Available Settings                                                                                                                                                         | Default                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>WARNING</b><br>A confirmation pop-up warns that changing the time or date at the Infinity CentralStation also changes the time at all devices connected to the network. Changes cannot be undone once confirmed. |                                                                                                                                                                                                                                     |                                                                                                                                                                            |                           |
| Hour: Minute                                                                                                                                                                                                        | Sets current time                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>– Hour</li> <li>– Minute</li> </ul>                                                                                                 | --                        |
| Day - Month                                                                                                                                                                                                         | Sets current date                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>– Day</li> <li>– Month</li> </ul>                                                                                                   | --                        |
| Year                                                                                                                                                                                                                | Sets current year                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>– 1995 - 2099</li> </ul>                                                                                                            | --                        |
| Report Trend Order                                                                                                                                                                                                  | Displays System Trend Order Setup screen, which allows selection and sorting of parameters for a Shift (page 183) or Graphical Trend Report (page 188).<br><br>To configure Trend Order Setup for a specific patient, see page 196. | <ul style="list-style-type: none"> <li>– Enable Auto Scale (page 188). Click the selection box to activate.</li> <li>– Trends Per Page. Click 1, 2, 3, 4, or 5.</li> </ul> | Disabled<br><br><br><br>3 |
| Batch Shift Report                                                                                                                                                                                                  | Permits assignment of report length for batch shift reports<br><br>When you press F9, a Shift Report is generated for all currently active beds.                                                                                    | <ul style="list-style-type: none"> <li>– 2 Hour</li> <li>– 4 Hour</li> <li>– 8 Hour</li> <li>– 12 Hour</li> <li>– 24 Hour</li> </ul>                                       | --                        |
| Accept/Undo                                                                                                                                                                                                         | Enables you to save or discard the changes made since the last save.                                                                                                                                                                | <ul style="list-style-type: none"> <li>– Accept saves any changes</li> <li>– Undo reverts to the previous settings</li> </ul>                                              | --                        |

## Setup - Recorder Screen

To open the **Setup - Recorder** screen, refer to page 60.

| Selection                                                                                                                                                                                                                                                       | Description                                                                        | Available Settings                                                                                                                                                | Default |                           |                         |         |         |        |             |        |                |        |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------|-------------------------|---------|---------|--------|-------------|--------|----------------|--------|--|
| Primary Recorder                                                                                                                                                                                                                                                | Selects preferred recorder                                                         |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| Secondary Recorder                                                                                                                                                                                                                                              | Determines recorder to be used when Primary Recorder is unavailable                |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| When you click on Assign...in the Assign Recorder pop-up, a list of available recorders displays.<br>– Click on the desired recorder.<br>– Click on Accept in the pop-up to choose the selected recorder, or click Cancel.                                      |                                                                                    |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| Speed                                                                                                                                                                                                                                                           | Sets recording speed                                                               | – 50, 25, 12.5, 6.25, 1 mm/sec                                                                                                                                    | 25 mm/s |                           |                         |         |         |        |             |        |                |        |  |
| Alternate Speed                                                                                                                                                                                                                                                 | Determines recording speed when R 50 Alternate Speed key is pressed (see page 174) | – 50, 25, 12.5, 6.25, 1 mm/sec                                                                                                                                    | 50 mm/s |                           |                         |         |         |        |             |        |                |        |  |
| Manual Recording Duration                                                                                                                                                                                                                                       | Sets time length of a Manual Recording                                             | – 5/6, 10, 15, 20 sec                                                                                                                                             | 20 sec  |                           |                         |         |         |        |             |        |                |        |  |
| Manual Recording Delay                                                                                                                                                                                                                                          | Determines amount of pre-event data included in Manual Recording                   | – 5, 10, 15 sec<br>Available selections depend on your recording speed setting. Inactive settings for a particular recording speed are unavailable for selection. | 10 sec  |                           |                         |         |         |        |             |        |                |        |  |
| <table><tr><th>Manual Recording Duration</th><th>Available Delay Setting</th></tr><tr><td>5/6 sec</td><td>– 5 sec</td></tr><tr><td>10 sec</td><td>– 5, 10 sec</td></tr><tr><td>15 sec</td><td>– 5, 10 15 sec</td></tr><tr><td>20 sec</td><td></td></tr></table> |                                                                                    |                                                                                                                                                                   |         | Manual Recording Duration | Available Delay Setting | 5/6 sec | – 5 sec | 10 sec | – 5, 10 sec | 15 sec | – 5, 10 15 sec | 20 sec |  |
| Manual Recording Duration                                                                                                                                                                                                                                       | Available Delay Setting                                                            |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| 5/6 sec                                                                                                                                                                                                                                                         | – 5 sec                                                                            |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| 10 sec                                                                                                                                                                                                                                                          | – 5, 10 sec                                                                        |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| 15 sec                                                                                                                                                                                                                                                          | – 5, 10 15 sec                                                                     |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| 20 sec                                                                                                                                                                                                                                                          |                                                                                    |                                                                                                                                                                   |         |                           |                         |         |         |        |             |        |                |        |  |
| Accept/Undo                                                                                                                                                                                                                                                     | Enables you to save or discard the changes made since the last save.               | – Accept saves any changes<br>– Undo reverts to the previous settings                                                                                             | --      |                           |                         |         |         |        |             |        |                |        |  |

## Setup - Central Layouts Screen

To open the **Setup - Central Layouts** screen, refer to page 60.

| Selection             | Description                                                                                          | Available Settings                                                                                                                                                                                                                              | Default               |
|-----------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Select Central Layout | Permits configuration and storage of desired layout                                                  | A, B, C, D                                                                                                                                                                                                                                      | --                    |
| Central Layout Name   | Text box for naming the layout                                                                       | 12 character max.                                                                                                                                                                                                                               | --                    |
| Central Layout Mode   | Sets the number of beds and waveforms per bed that will display                                      | <ul style="list-style-type: none"> <li>– 2 Beds x 4 waves each</li> <li>– 4 Beds x 2 waves each</li> <li>– 4 beds x 4 waves each</li> <li>– 8 Beds x 2 waves each</li> <li>– 8 Beds x 1 waves each</li> <li>– 16 Beds x 1 waves each</li> </ul> | 2 Beds x 4 waves each |
| Notes                 | Permits display of note area (see page 102)                                                          | <ul style="list-style-type: none"> <li>– OFF</li> <li>– ON</li> </ul>                                                                                                                                                                           | OFF                   |
| Assign...             | Activates the Assign Bed screen, which lists all of the beds being advertised in the monitoring unit | <ul style="list-style-type: none"> <li>– Select a bed from the list</li> <li>– Click on Accept</li> </ul>                                                                                                                                       | --                    |
| Accept/Undo           | Enables you to save or discard the changes made since the last save.                                 | <ul style="list-style-type: none"> <li>– Accept saves any changes</li> <li>– Undo reverts to the previous settings</li> </ul>                                                                                                                   | --                    |



# Infinity M300 System Setup

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|                                                               |    |
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## Overview

The Telemetry setup screens allow you to set *system defaults* for Infinity CentralStation telemetry channels (see page 68). These defaults are activated at telemetry patient admission.

### Accessing the Telemetry Setup Screens

- Select **Telemetry** from the **Setup** drop-down menu.

| Submenu Selection                          | Required Password | Category Selection | Reference |
|--------------------------------------------|-------------------|--------------------|-----------|
| System...                                  | Clinical          |                    | page 67   |
| Patient View Defaults                      | Clinical          |                    | page 68   |
| ST Defaults<br>(Requires option, page 294) | Clinical          |                    | page 71   |
| Alarm Limits Defaults...                   | Clinical          | Adult              | page 74   |
|                                            |                   | Pediatric          | page 74   |
| Arrhythmia Defaults...                     | Clinical          | Adult              | page 76   |
|                                            |                   | Pediatric          | page 76   |
| Find Device                                | Clinical          |                    | page 79   |

### Changing the System Default Patient Category

You can select a system default patient category that is initially used by all telemetry channels using the **System** screen setting. The system default patient category setting is **Adult**.

When you select **Alarm Limits Defaults** or **Arrhythmia Defaults**, an additional submenu permits selection of **Adult** or **Pediatric** patient category screens. Each of these screens includes the patient category in the screen name. The **Alarm Limits Defaults** and **Arrhythmia Defaults** screens have a selection that permits you to quickly change to the alternate category screen (see page 76).

To configure the patient category for an individual Telemetry patient, see page 121.

## Infinity M300 System Setup Screen

To access the Infinity M300 **System Setup** screen, see page 66.

| Selection                  | Description                                                                                                                                                                                                                                                                                               | Settings                                                                                                        | Default  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------|
| Alarm Pause Time           | Determines duration of Alarm Paused Time                                                                                                                                                                                                                                                                  | 1, 2, 3, 4, 5 min                                                                                               | 3 min.   |
| Alarm Tone Pattern         | Determines alarm pattern used by the Infinity M300 device only.                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>– Infinity</li> <li>– High Frequency</li> <li>– Low Frequency</li> </ul> | Infinity |
| Weight Units               | Sets weight units in <b>Admit</b> screen.                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>– kg</li> <li>– lbs</li> </ul>                                           | lbs.     |
| Height Units               | Sets height units in <b>Admit</b> screen.                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>– cm</li> <li>– inches</li> </ul>                                        | inches   |
| Remote Audio Pause Enabled | <p>Determines whether or not you can silence alarms for telemetry patients from other monitors in the monitoring unit.</p> <p>When <b>OFF</b>, the <b>Audio Paused</b> button in Bed View is unavailable for selection and the corresponding icon will not appear in the Main Screen during an alarm.</p> | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF</li> </ul>                                           | ON       |
| Remote Control Enabled     | Determines if you can control telemetry patients' monitoring at other monitors in the monitoring unit.                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF</li> </ul>                                           | ON       |
| Patient Category           | Sets default patient category.                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>– Adult</li> <li>– Pediatric</li> </ul>                                  | Adult    |
| Pacer Detect Mode          | Sets pacer detection mode.                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>– Basic</li> <li>– Advanced</li> </ul>                                   | Basic    |
| Beds x - x                 | Toggle button for selecting beds 1 - 16 or beds 17 - 32.                                                                                                                                                                                                                                                  | –                                                                                                               |          |
| Label                      | <p>Permits assignment of <i>Bed Labels</i> to each Infinity CentralStation <b>Channel</b> (or patient window).</p> <p>You can select up to 32 bed labels that will identify Infinity CentralStation Main Screen patient windows.</p>                                                                      | – 7 characters max                                                                                              |          |
| Alarm Group                | Assigns an <b>Alarm Group</b> to each Infinity CentralStation <b>Channel</b> (or patient window).                                                                                                                                                                                                         | – 3 characters max                                                                                              |          |

| Selection   | Description                                                          | Settings                                                                                                                   | Default |
|-------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------|
| Accept/Undo | Enables you to save or discard the changes made since the last save. | <ul style="list-style-type: none"><li>– Accept saves any changes</li><li>– Undo reverts to the previous settings</li></ul> |         |

## Infinity M300 Patient View Defaults Setup Screen

The Main Screen **Default Patient View Setup** screen allows you to define system-wide, default, patient-monitoring settings. It is similar to the Bed View **Patient View Setup**, which allows you to customize a *particular* patient's monitoring (see page 290).

The **Default Patient View Setup** screen for Telemetry patients provides selections to configure SpO2 and ECG monitoring. When you click on the button labeled **SpO2**, the setup selections for that parameter display. When you click on the button labeled **ECG**, the setup selections for that parameter display.

| Selection     | Required Action                                                                                                         | Available Settings                                                                                                                                                                                                                                                                  |
|---------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Channel Setup |                                                                                                                         |                                                                                                                                                                                                                                                                                     |
| Parameter     | Click in Channel <b>1</b> through <b>6 Parameter</b> column.<br><br>For 6-wire monitoring 8 channel rows are available. | HR, ARR, ST, SpO2 + PLS, None                                                                                                                                                                                                                                                       |
| Waveform      | Click in the Channel's ' <b>Waveform</b> ' column.                                                                      | <ul style="list-style-type: none"><li>– 3-wire mode - I, II, III</li><li>– 5-wire mode - I, II, III, aVR, aVL, aVF, V</li><li>– 6-wire mode - I, II, III, aVR, aVL, aVF, V, V+</li></ul>                                                                                            |
| Gain/Scale    | Click in the Channel's <b>Gain/Scale</b> column                                                                         | 0.25, 0.5, 1, 2, 4, 8 mV                                                                                                                                                                                                                                                            |
|               | <b>ECG QRS Processing Settings</b>                                                                                      | The Infinity CentralStation uses an AAMI-compliant regular QRS threshold when you select a channel size of 1, 2, 4, or 8 mV/cm. If you select a channel size of 0.25 or 0.5 mV/cm, the Infinity CentralStation lowers the detection threshold, and the AAMI requirement is not met. |
|               | <b>Display Scale mV/cm</b>                                                                                              |                                                                                                                                                                                                                                                                                     |
|               | 8                                                                                                                       |                                                                                                                                                                                                                                                                                     |
|               | 4                                                                                                                       |                                                                                                                                                                                                                                                                                     |
|               | 2                                                                                                                       |                                                                                                                                                                                                                                                                                     |
|               | 1                                                                                                                       |                                                                                                                                                                                                                                                                                     |
|               | 0.5                                                                                                                     |                                                                                                                                                                                                                                                                                     |
|               | 0.25                                                                                                                    |                                                                                                                                                                                                                                                                                     |
|               |                                                                                                                         | Gain                                                                                                                                                                                                                                                                                |
|               |                                                                                                                         | QRS ≥ 0.5 mV                                                                                                                                                                                                                                                                        |
|               |                                                                                                                         | QRS ≥ 0.5 mV                                                                                                                                                                                                                                                                        |
|               |                                                                                                                         | QRS ≥ 0.5 mV                                                                                                                                                                                                                                                                        |
|               |                                                                                                                         | QRS ≥ 0.5 mV                                                                                                                                                                                                                                                                        |
|               |                                                                                                                         | QRS ≥ 0.15 mV                                                                                                                                                                                                                                                                       |
|               |                                                                                                                         | QRS ≥ 0.15 mV                                                                                                                                                                                                                                                                       |

| Selection                                                                                                                                                                                                                                                                                                                                                                                                            | Required Action                                                                                                                                                                                                                                                | Available Settings                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ECG Monitoring Setup</b>                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                 |
| QRS Processing                                                                                                                                                                                                                                                                                                                                                                                                       | Select leads for ECG and ARR monitoring.                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>– ECG1 &amp; ECG2</li> <li>– ECG1</li> </ul>                                                                                                             |
| <b>WARNING</b><br>High amplitude (>0.15mV) P- and T-waves of long duration may register as integral QRS complexes. To ensure that the system accurately detects low heart rate in these cases, select the lead with the highest R-wave (relative to the T- and/or P-wave) for ECG1. If the system continues to misinterpret P- or T-waves, reposition the electrodes or use other modalities to monitor the patient. |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                 |
| ECG 1<br>ECG 2                                                                                                                                                                                                                                                                                                                                                                                                       | Top and second lead assigned to ECG 1 and ECG 2 display                                                                                                                                                                                                        |                                                                                                                                                                                                 |
| ECG Color                                                                                                                                                                                                                                                                                                                                                                                                            | Assign color of ECG associated waveforms to telemetry channel for an individual patient.                                                                                                                                                                       | <ul style="list-style-type: none"> <li>– Red</li> <li>– White</li> <li>– Yellow</li> <li>– Green (Default)</li> <li>– Light Blue</li> <li>– Blue</li> <li>– Purple</li> <li>– Orange</li> </ul> |
| ECG Lead Wires                                                                                                                                                                                                                                                                                                                                                                                                       | Select device lead wire mode                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>– 3</li> <li>– 5 (Default)</li> <li>– 6</li> </ul>                                                                                                       |
| Pacemaker<br>Detection                                                                                                                                                                                                                                                                                                                                                                                               | Set the pacer detection function.<br><br>When <b>On</b> , pacer spikes display on a patient's waveform whenever a pacemaker pulse is detected.<br><br>The <b>Fusion</b> selection is only available if the <b>Pacer Detect Mode is Advanced</b> (see page 67). | <ul style="list-style-type: none"> <li>– ON (Default)</li> <li>– OFF</li> <li>– FUSION</li> </ul>                                                                                               |
| <b>WARNING</b><br>Special surveillance is necessary for patients with pacemakers who are monitored with Pacemaker Detection set to Fusion mode because more pacer spikes may register as integral QRS complexes.                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                 |

| Selection                                                                                                                                                                                             | Required Action                                                                                                                                                               | Available Settings                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TruST 12-lead<br>(Selection only appears when <b>ECG Wire Mode</b> setting is <b>6 wire</b> and TruST locked option is enabled.)                                                                      | Activate/deactivate TruST monitoring (see page 117).<br>Precordial leads are fixed as V2 and V5.                                                                              | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF (Default)</li> </ul>                                                                                                                 |
| <b>SpO2 Monitoring Setup</b>                                                                                                                                                                          |                                                                                                                                                                               |                                                                                                                                                                                                 |
| SPO2 Monitoring                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>– Enable/disable SpO2 monitoring</li> <li>– Selection appears unavailable for selection if an SpO2 option is not available.</li> </ul> | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF (Default)</li> </ul>                                                                                                                 |
| <b>NOTE</b> <ul style="list-style-type: none"> <li>– You can only activate SpO2 Monitoring for local telemetry patients.</li> <li>– If SpO2 Monitoring is OFF, the parameter box is blank.</li> </ul> |                                                                                                                                                                               |                                                                                                                                                                                                 |
| SpO2 Color                                                                                                                                                                                            | Assign color of SpO2 associated waveforms to telemetry channel for an individual patient.                                                                                     | <ul style="list-style-type: none"> <li>– Red</li> <li>– White (Default)</li> <li>– Yellow</li> <li>– Green</li> <li>– Light Blue</li> <li>– Blue</li> <li>– Purple</li> <li>– Orange</li> </ul> |
| Sensor Type                                                                                                                                                                                           | Displays the device sensor type.                                                                                                                                              | <ul style="list-style-type: none"> <li>– Nellcor</li> <li>– Masimo</li> </ul>                                                                                                                   |
| <b>NOTE</b> <p>As of release VF8.4, the sensor type is fixed and determined at the time of purchase and cannot be changed via the Sensor Type button.</p>                                             |                                                                                                                                                                               |                                                                                                                                                                                                 |
| Accept/Undo                                                                                                                                                                                           | Enables you to save or discard the changes made since the last save.                                                                                                          | <ul style="list-style-type: none"> <li>– Accept saves any changes</li> <li>– Undo reverts to the previous settings</li> </ul>                                                                   |

## Infinity M300 ST Defaults Setup

The **Default ST Setup** screen is similar to the **ST Setup** screen in Bed View, which allows you to customize a *particular* patient's ST monitoring. The **Default ST Setup** screen allows you to define system-wide ST monitoring default settings, except ST alarm default settings which must be configured in the **Setup - Alarm Limits** screen (see page 74). Also, ST monitoring default settings are for 3- or 5-wire monitoring only. All 6-wire monitoring must be configured at the individual patient's Bed View **ST Setup** screen (see page 155).

If ST monitoring is enabled, the default **ST Setup** settings are activated whenever a telemetry patient is admitted. They can also be activated on a Bed View **ST Setup** screen that is configured for 3-/5-wire monitoring.

The following table displays the available selections within the **ST Setup** screen.

For additional information on ST monitoring, see page 149.

| ST Setup Screen Selections: 6-Lead, TruST OFF                                      |                                                                                                     |                                                                       |         |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------|
| Selection                                                                          | Description                                                                                         | Available Settings                                                    | Default |
| ST Lead 1                                                                          | Sets ECG lead vector for ST Lead 1 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | I       |
| ST Lead 2                                                                          | Sets ECG lead vector for ST Lead 2 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | II      |
| ST Lead 3                                                                          | Sets ECG lead vector for ST Lead 3 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | III     |
| ST Lead 4                                                                          | Sets ECG lead vector for ST Lead 4 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | aVR     |
| ST Lead 5                                                                          | Sets ECG lead vector for ST Lead 5 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | aVL     |
| ST Lead 6                                                                          | Sets ECG lead vector for ST Lead 6 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | aVF     |
| ST Lead 7                                                                          | Sets ECG lead vector for ST Lead 7 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | V       |
| ST Lead 8                                                                          | Sets ECG lead vector for ST Lead 8 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, V+, None                                | V+      |
| ST Units                                                                           | Sets units for ST measurements                                                                      | mm<br>mV                                                              | mm      |
| Event Duration                                                                     | Determines how long an ST event has to remain outside the set ST alarm limit before an alarm sounds | Traditional Telemetry:<br>15, 30, 45, 60, 75, 90,<br>105, 120 seconds | 60 sec  |
|                                                                                    |                                                                                                     | M300: 60 seconds                                                      | 60 sec  |
| ST Measurement Point                                                               | Determines whether the ST measuring point is 60 or 80 ms after QRS offset                           | 60 ms<br>80 ms                                                        | 80 ms   |
| – Click on <b>Accept</b> to save changes or <b>Undo</b> to keep previous settings. |                                                                                                     |                                                                       |         |

| ST Setup Screen Selections: 6-Lead, TruST ON |                                                   |                                                                          |         |
|----------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|---------|
| Selection                                    | Description                                       | Available Settings                                                       | Default |
| ST Lead 1                                    | Sets ECG lead vector for ST Lead 1 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | I       |
| ST Lead 2                                    | Sets ECG lead vector for ST Lead 2 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | II      |
| ST Lead 3                                    | Sets ECG lead vector for ST Lead 3 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | III     |
| ST Lead 4                                    | Sets ECG lead vector for ST Lead 4 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | aVR     |
| ST Lead 5                                    | Sets ECG lead vector for ST Lead 5 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | aVL     |
| ST Lead 6                                    | Sets ECG lead vector for ST Lead 6 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | aVF     |
| ST Lead 7                                    | Sets ECG lead vector for ST Lead 7 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | V2      |
| ST Lead 8                                    | Sets ECG lead vector for ST Lead 8 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | V5      |
| ST Lead 9                                    | Sets ECG lead vector for ST Lead 9 parameter box  | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | dV1     |
| ST Lead 10                                   | Sets ECG lead vector for ST Lead 10 parameter box | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None | dV3     |



| <b>ST Setup Screen Selections: 6-Lead, TruST ON</b> |                                                                                                     |                                                                                                                           |        |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|
| ST Lead 11                                          | Sets ECG lead vector for ST Lead 11 parameter box                                                   | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None                                                  | dV4    |
| ST Lead 12                                          | Sets ECG lead vector for ST Lead 12 parameter box                                                   | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None                                                  | dV6    |
| ST Lead 13                                          | Sets ECG lead vector for ST Lead 13 parameter box                                                   | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None                                                  | STVM   |
| ST Lead 14                                          | Sets ECG lead vector for ST Lead 14 parameter box                                                   | I, II, III, aVR, aVL, aVF, V2, V5, dV1, dV3, dV4, dV6, STVM, STCVM, None                                                  | STCVM  |
| ST Units                                            | Sets units for ST measurements                                                                      | mm<br>mV                                                                                                                  | mm     |
| Event Duration                                      | Determines how long an ST event has to remain outside the set ST alarm limit before an alarm sounds | Traditional Telemetry:<br>15, 30, 45, 60 75, 90, 105, 120 seconds                                                         | 60 sec |
|                                                     |                                                                                                     | M300: 60 seconds                                                                                                          | 60 sec |
| ST Measurement Point                                | Determines whether the ST measuring point is 60 or 80 ms after QRS offset                           | 60 ms<br>80 ms                                                                                                            | 80 ms  |
| Accept/Undo                                         | Enables you to save or discard the changes made since the last save.                                | <ul style="list-style-type: none"> <li>Accept saves any changes</li> <li>Undo reverts to the previous settings</li> </ul> | --     |

## Infinity M300 Alarm Limits Defaults Screen

The **Default Settings - Alarm Limits** screen allows you to define system-wide, telemetry alarm default settings. These default settings are activated upon patient admittance or by clicking **Restore System Defaults** on a particular patient’s Bed View screen (see page 155).

The patient category is part of the screen title, and parameter limits are based on that category (see page 75).

The **Default Settings - Alarm Limits** screen is similar to the **Setup - Alarm Limits** screen in Bed View in which you can customize alarm limits for a *particular* patient.

The following table displays the available selections within the **Alarm Limits** screen.

| Button/Function             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arrhythmia                  | Opens <b>Default Settings - Arrhythmia</b> screen (see page 76)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Parameter alarm limit setup | <div>To setup the alarm limits:<div><div>1</div><div>Click on the table row you wish to set up.<br/>This activates the configuration mode for that row, from which you can set the limits within each column.</div><div>2</div><div>Click on the arrows or toggle button so that the desired setting appears.</div></div><div><div><div>SpO2</div><div>OK</div><div>OFF</div><div>100</div><div><div>▲</div><div>▼</div></div><div><div>90</div><div><div>▲</div><div>▼</div></div><div>REC</div></div></div><div><div>3</div><div>Click on <b>OK</b>.</div><div>4</div><div>Repeat steps 1 - 3 for each parameter.</div></div></div></div> |
| →Pediatric...               | – Changes the patient category from <b>Adult</b> to <b>Pediatric</b> or vice versa.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| →Adult...                   | – Changes all of the parameters and limits to the defaults in the selected patient category.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

The following table displays the **Alarm Limits Parameter Settings**.

| Parameter                                                                                      | Heading       | Adult Default     | Pediatric Default | Setting Range                         |
|------------------------------------------------------------------------------------------------|---------------|-------------------|-------------------|---------------------------------------|
| HR                                                                                             | Upper Limit   | 120 beats/min     | 150 beats/min     | 25 - 300 beats/min                    |
|                                                                                                | Lower Limit   | 45 beats/min      | 50 beats/min      | 20 - 295 beats/min                    |
|                                                                                                | Alarm         | On                | On                | On/Off                                |
|                                                                                                | Alarm Archive | Rec/Store         | Rec/Store         | Rec, Store, Rec/Store, Off            |
| SpO2                                                                                           | Upper Limit   | 100%              | 100%              | 21 - 100%                             |
|                                                                                                | Lower Limit   | 90%               | 90%               | 20 - 99%                              |
|                                                                                                | Alarm         | Off               | On                | On/Off                                |
|                                                                                                | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| PLS                                                                                            | Upper Limit   | 120 beats/min     | 150 beats/min     | 35 - 240 beats/min                    |
|                                                                                                | Lower Limit   | 45 beats/min      | 50 beats/min      | 30 - 235 beats/min                    |
|                                                                                                | Alarm         | Off               | On                | On/Off                                |
|                                                                                                | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| ST<br>(I, II, III, aVF, aVL, aVR, V, V+, V1, V2, V3, V4, V5, V6, dV1, dV2, dV3, dV4, dV5, dV6) | Upper Limit   | +2 mm<br>+0.20 mV | +2 mm<br>+0.20 mV | -14.9 to +15.0 mm<br>-1.49 to +1.5 mV |
|                                                                                                | Lower Limit   | -1mm<br>-0.10 mV  | -1mm<br>-0.10 mV  | -15.0 to +14.9 mm<br>-1.5 to +1.49 mV |
|                                                                                                | Alarm         | Off               | Off               | On/Off                                |
|                                                                                                | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| STVM<br>STCVM                                                                                  | Upper Limit   | +2 mm<br>+0.2 mV  | +2 mm<br>+0.2 mV  | 0.1 to 45 mm<br>0.01 to 4.5 mV        |
|                                                                                                | Lower Limit   | 0                 | 0                 | 0.0 to 44.9 mm<br>0.0 to 4.49 mV      |
|                                                                                                | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| PVC/min                                                                                        | Upper Limit   | 10 beats/min      | 10 beats/min      | 1 - 60 beats/min                      |
|                                                                                                | Alarm         | Off               | Off               | On/Off                                |
|                                                                                                | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |

## Infinity M300 Arrhythmia Defaults Screen

This screen allows you to define system-wide default settings for arrhythmia event categories. These defaults are passed to the Infinity M300 upon patient admission or by clicking **Restore System Defaults** on a particular patient's Bed View screen (see page 155).

The patient category is part of the screen title, and event defaults are based on that category (see page 77).

The **Default Settings - Arrhythmia** screen is similar to the **Setup - Arrhythmia** screen in Bed View in which you can customize each event category for a *particular* patient (see page 133).

### Available Functions on the Arrhythmia Screen

The following table displays the available selections within the **Arrhythmia screen**.

| Button/Function                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm Limits...                  | Opens <b>Default Settings - Alarm Limits</b> screen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Select Arrhythmia event settings | <p>To setup the arrhythmia events:</p> <ol style="list-style-type: none"><li>Click on the table row of the event you wish to set up. This activates the configuration mode for that row during which you can set limits within each column.</li></ol> <div><div>VT</div><div>OK</div><div>LT</div><div><div></div><div></div></div><div>120</div><div><div></div><div></div></div><div>10</div><div><div></div><div></div></div><div>REC/STORE</div></div> <ol style="list-style-type: none"><li>Click on the arrows or toggle button so that the desired setting appears.</li><li>Click on <b>OK</b>.</li><li>Repeat steps 1 to 3 for each parameter.</li><li>Click on <b>Accept</b> to save or <b>Undo</b> to cancel changes.</li></ol> |
| Set arrhythmia 'monitoring mode' | <p>To select the monitoring mode for arrhythmia, click Select.</p> <ul style="list-style-type: none"><li>Off</li><li>Basic (Default)</li><li>Full</li></ul> <p>In order to set the arrhythmia monitoring mode to Off, you must enter the clinical password in the <i>Password</i> pop-up.</p> <p>Full arrhythmia monitoring requires that the locked option is enabled.</p>                                                                                                                                                                                                                                                                                                                                                               |
| →Pediatric...                    | <ul style="list-style-type: none"><li>Changes the patient category from <b>Adult</b> to <b>Pediatric</b> or vice versa.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| →Adult...                        | <ul style="list-style-type: none"><li>Changes all of the parameters and limits to the defaults in the selected patient category.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Events available for arrhythmia monitoring are determined by the arrhythmia and alarm limit settings.

| Mode            | Available Events                                                    |
|-----------------|---------------------------------------------------------------------|
| Full            | ASY, VF, VT, ARTF, RUN, AIVR, CPT, BGM, PAUS, TACH, BRDY, SVT, BRDY |
| Basic (Default) | ASY, VF, VT, ARTF                                                   |
| Off             |                                                                     |

The following table displays the available selections within the **Arrhythmia Event Settings** screen.

| Parameter | Heading       | Adult Default Setting                                       | Pediatric Default Setting                            |
|-----------|---------------|-------------------------------------------------------------|------------------------------------------------------|
| ASY       | Count         |                                                             |                                                      |
|           | Rate          |                                                             |                                                      |
|           | Alarm Grade   | LT                                                          |                                                      |
|           | Alarm/Archive | Rec, Rec/Store                                              |                                                      |
| VF        | Count         |                                                             |                                                      |
|           | Rate          |                                                             |                                                      |
|           | Alarm Grade   | LT                                                          |                                                      |
|           | Alarm/Archive | Rec, Rec/Store                                              |                                                      |
| VT        | Count         | 5 - 15 Default: $\geq 10$                                   | 5 - 15 Default: $\geq 10$                            |
|           | Rate          | 100 - 200<br>Default: $\geq 120$                            | 100 - 200<br>Default: $\geq 160$                     |
|           | Alarm Grade   | LT (default), SER, ADV, OFF                                 |                                                      |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                  |                                                      |
| ARTF      | Count         |                                                             |                                                      |
|           | Rate          |                                                             |                                                      |
|           | Alarm Grade   | LT, SER, ADV, OFF (default)                                 |                                                      |
|           | Alarm/Archive | OFF                                                         |                                                      |
| RUN       | Count         | Not adjustable; upper value is VT count - 1. Default: 3 - 4 |                                                      |
|           | Rate          | Not adjustable; same as VT rate. Default: $\geq 120$        | Not adjustable; same as VT rate. Default: $\geq 160$ |
|           | Alarm Grade   | LT, SER (default), ADV, OFF                                 |                                                      |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                  |                                                      |

| Parameter | Heading       | Adult Default Setting                                              | Pediatric Default Setting                                                    |
|-----------|---------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| AIVR      | Count         | Not adjustable. Default: $\geq 3$                                  |                                                                              |
|           | Rate          | Not adjustable, upper value is VT rate - 1. Default: $\leq 119$    | Not adjustable, upper value is VT rate - 1. Default: $\leq 159$              |
|           | Alarm Grade   | LT, SER, ADV (default), OFF                                        | Rec, Store, Rec/Store, Off                                                   |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| CPT       | Count         |                                                                    |                                                                              |
|           | Rate          |                                                                    |                                                                              |
|           | Alarm Grade   | LT, SER, ADV (default), OFF                                        |                                                                              |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| BMG       | Count         |                                                                    |                                                                              |
|           | Rate          |                                                                    |                                                                              |
|           | Alarm Grade   | LT, SER, ADV (default), OFF                                        |                                                                              |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| PAUSE     | Count         |                                                                    |                                                                              |
|           | Rate          | 1.0, 1.5, 2.0, 2.5 (default), 3.0, 3.5                             | 1.0, 1.5, 2.0 (default), 2.5, 3.0, 3.5                                       |
|           | Alarm Grade   | LT, SER, ADV, OFF (default)                                        |                                                                              |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| TACH      | Count         | 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15. Default: $\geq 8$           |                                                                              |
|           | Rate          | 100 - 200<br>Default: $\geq 130$                                   | 100 - 200<br>Default: $\geq 180$                                             |
|           | Alarm Grade   | LT, SER, ADV, OFF (default)                                        |                                                                              |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| BRDY      | Count         | $\geq 8$ bpm                                                       |                                                                              |
|           | Rate          | 30 - 70<br>Default: $\leq 50$                                      | 30 - 105<br>Default: $\leq 60$                                               |
|           | Alarm Grade   | LT, SER, ADV, OFF (default)                                        | LT, SER (default), ADV, OFF                                                  |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |
| SVT       | Count         | 3, 4, 5, 6, 7, 8, 9, 10<br>Default: $\geq 3$                       |                                                                              |
|           | Rate          | 120, 130, 140, 150, 160, 170, 180, 190, 200<br>Default: $\geq 150$ | 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220<br>Default: $\geq 220$ |
|           | Alarm Grade   | LT, SER (default), ADV, OFF                                        |                                                                              |
|           | Alarm/Archive | Rec, Store, Rec/Store, Off                                         |                                                                              |

## Infinity M300 Find Device

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You can initiate a *Find Device* series of 5 alternating tones that can help you to locate an Infinity M300 in the event it is misplaced.

To locate a device:

- 1** Select **Telemetry** from the Main Screen **Setup** drop-down menu.
- 2** Select **Find Device**.  
Scroll down to/select the Infinity M300 ID you are trying to find.

### NOTE

Devices admitted to the Infinity CentralStation will not appear in the list as selections.

- 3** Click on the device you wish to locate.
- 4** Click on **Locate**.  
The Infinity CentralStation sends a command to the selected Infinity M300 to sound a *Find Device* tone (see page 35).

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# Infinity Telemetry System Setup

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## Overview

Set *system defaults* for Infinity CentralStation telemetry channels using the Telemetry setup screens. These defaults are activated at telemetry patient discharge.

### Accessing the Telemetry Setup Screens

- Select **Telemetry** from the **Setup** drop-down menu.

| Telemetry Submenu Selections                |                   |                     |           |
|---------------------------------------------|-------------------|---------------------|-----------|
| Submenu Selection                           | Password Required | Category Selections | Reference |
| System...                                   | Clinical          |                     | page 83   |
| Patient View Defaults                       | Clinical          |                     | page 84   |
| ST Defaults (requires option, see page 294) | Clinical          |                     | page 86   |
| Recordings                                  | Clinical          |                     | page 87   |
| Alarm Limits Defaults...                    | Clinical          | Adult               | page 88   |
|                                             |                   | Pediatric           |           |
| Arrhythmia Defaults...                      | Clinical          | Adult               | page 90   |
|                                             |                   | Pediatric           |           |

### Changing the System Default Patient Category

You can select a system default patient category that is initially used by all telemetry channels using a **System** screen setting (see page 83). The system default patient category setting is **Adult**. If after making changes you want to return all settings to the defaults, see page 291.

When you select **Alarm Limits Defaults** or **Arrhythmia Defaults**, an additional submenu permits selection of **Adult** or **Pediatric** patient category screens. Each of these screens includes the patient category in the screen name. The **Alarm Limits Defaults** and **Arrhythmia Defaults** screens have a selection that permits you to quickly change to the alternate category screen (see page 88).

To configure the patient category for an individual Telemetry patient see page 121.

# Infinity Telemetry System Setup Screen

To access the **Telemetry System Setup** screen, see page 82.

The following table displays the System Setup screen selections.

| Selection                   | Description                                                                                                                                                                                                                                                                                              | Settings                                                                                                                  | Defaults |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------|
| Alarm Paused Time           | Determines duration of <b>Alarm Paused Time</b>                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>1, 2, 3, 4, 5 min</li> <li>Disable</li> <li>No Time Out</li> </ul>                 | 3 min    |
| Weight Units                | Sets weight units in <b>Admit</b> screen                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>kg</li> <li>lbs</li> </ul>                                                         | lbs      |
| Height Units                | Sets height units in <b>Admit</b> screen                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>cm</li> <li>in</li> </ul>                                                          | in       |
| Remote Audio Paused Enabled | <p>Determines whether or not you can silence alarms for telemetry patients from other monitors in the monitoring unit</p> <p>When <b>OFF</b>, the <b>Audio Paused</b> button in Bed View is unavailable for selection and the corresponding icon will not appear in the Main Screen during an alarm.</p> | <ul style="list-style-type: none"> <li>ON</li> <li>OFF</li> </ul>                                                         | ON       |
| Remote Control Enabled      | Determines if you can control telemetry patients' monitoring at other monitors in the monitoring unit                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>ON</li> <li>OFF</li> </ul>                                                         | ON       |
| Default Patient Category    | Sets default patient category                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Adult</li> <li>Pediatric</li> </ul>                                                | Adult    |
| Pacer Detect Mode           | Sets pacer detection mode                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Basic</li> <li>Advanced</li> </ul>                                                 | Basic    |
| Bed Label                   | <p>Permits assignment of <i>Bed Labels</i> to each Infinity CentralStation <b>Channel</b> (or patient window).</p> <p>You can select up to 32 bed labels that will identify Infinity CentralStation Main Screen patient windows.</p>                                                                     | <ul style="list-style-type: none"> <li>7 character max.</li> </ul>                                                        |          |
| Alarm Group                 | Assigns an <b>Alarm Group</b> to each Infinity CentralStation <b>Channel</b> (or patient window)                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>3 character max.</li> </ul>                                                        |          |
| Accept/Undo                 | Enables you to save or discard the changes made since the last save.                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Accept saves any changes</li> <li>Undo reverts to the previous settings</li> </ul> |          |

## Infinity Telemetry Patient View Defaults Setup Screen

If a password is required to access this screen, see page 60.

The Main Screen **Default Patient View Setup** screen allows you to define system-wide, default patient-monitoring settings. It is similar to the

**Patient View Setup** screen in Bed View, which allows you to customize a specific patient's monitoring.

The following table displays the Patient View Setup screen selections.

| Selection                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required Action                                                                            | Available Settings                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Channel Setup</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                            |                                                                                                 |
| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                 | Click in the Channel's <b>Parameter</b> column.                                            | Depends on monitoring selections                                                                |
| Waveform                                                                                                                                                                                                                                                                                                                                                                                                                                  | Click in the Channel's <b>Waveform</b> column                                              | <div><div>– 3-wire mode - II</div><div>– 5-wire mode - I, II, III, aVR, aVL, aVF, V</div></div> |
| <b>Monitoring Setup</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                            |                                                                                                 |
| QRS Processing                                                                                                                                                                                                                                                                                                                                                                                                                            | Select leads for ECG and ARR monitoring.                                                   | <div><div>– ECG1 &amp; ECG2</div><div>– ECG1</div></div>                                        |
| <div><div><b>WARNING</b></div><div>High amplitude (&gt;0.15mV) P- and T-waves of long duration may register as integral QRS complexes. So that the system accurately detects low heart rate in these cases, select the lead with the highest R-wave (relative to the T- and/or P-wave) for ECG1. If the system continues to misinterpret P- or T-waves, reposition electrodes or use other modalities to monitor the patient.</div></div> |                                                                                            |                                                                                                 |
| Gain/Scale                                                                                                                                                                                                                                                                                                                                                                                                                                | Click in the Channel's <b>Gain/Scale</b> column                                            | 0.25, 0.5, 1, 2, 4, 8 mV                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ECG QRS Processing Settings</b>                                                         |                                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Display Scale mV/cm</b>                                                                 | <b>Gain</b>                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 8                                                                                          | QRS ≥ 0.5 mV                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4                                                                                          | QRS ≥ 0.5 mV                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2                                                                                          | QRS ≥ 0.5 mV                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                          | QRS ≥ 0.5 mV                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.5                                                                                        | QRS ≥ 0.15 mV                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.25                                                                                       | QRS ≥ 0.15 mV                                                                                   |
| Parameter Color                                                                                                                                                                                                                                                                                                                                                                                                                           | Assign system-wide color of SpO2, NIBP, and ECG associated waveforms to telemetry channels | red, white, yellow, green (default), light blue, blue, purple, and orange                       |

| Selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Required Action                                                                                                                         | Available Settings                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Pacemaker Detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Set the pacer detection function. When <b>ON</b> , pacer spikes display on a patient's waveform whenever a pacemaker pulse is detected. | On, Off, Fusion                                                                                                               |
| <b>WARNING</b> <ul style="list-style-type: none"> <li>– When Pacemaker Detection is set on Fusion mode pacer spikes may register as integral QRS complexes. Special surveillance is necessary for patients with pacemakers who are monitored with this mode enabled.</li> <li>– Fusion mode pacer detection is not intended for use with large-voltage, unipolar pacemakers. It is intended for use only with biphasic pacemakers. Please observe the following:</li> <li>– Before selecting Fusion mode be certain that the patient has a biphasic pacemaker (external or implanted) and that it is accurately programmed as appropriate for that patient.</li> <li>– Do not select Fusion mode if you are not certain what type of pacemaker is in use, or how it is programmed.</li> <li>– Select Pacer Fusion mode only to suppress repeated false asystole and/or false low heart rate alarms.</li> <li>– Selection of Fusion mode may increase the risk of falsely counting pacer spikes as QRS complexes, and may cause cardiac arrest to be undetected. Therefore, special surveillance of any pacemaker patient monitored with this mode is strongly recommended.</li> </ul> |                                                                                                                                         |                                                                                                                               |
| <b>NOTE</b> <ul style="list-style-type: none"> <li>– Display the lead with the least interference and the highest R-wave in the patient's waveform channel 1.</li> <li>– Verify that the HR calculation is accurate in comparison to the ECG waveform.</li> <li>– Verify that the ♥<sup>P</sup> symbol is displayed for each paced beat.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                         |                                                                                                                               |
| SpO2 Monitoring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Enable/disable SpO2 monitoring.                                                                                                         | On, Off                                                                                                                       |
| <b>NOTE</b> <ul style="list-style-type: none"> <li>– This selection is only available for local telemetry patients.</li> <li>– If SpO2 Monitoring is OFF, SpO2 and PLS parameter boxes are blank.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                         |                                                                                                                               |
| Accept/Undo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enables you to save or discard the changes made since the last save.                                                                    | <ul style="list-style-type: none"> <li>– Accept saves any changes</li> <li>– Undo reverts to the previous settings</li> </ul> |

## Infinity Telemetry ST Defaults Setup

The **Default ST Setup** screen is similar to the **ST Setup** screen in Bed View, which allows you to customize a *particular* patient's ST monitoring. The **Default ST Setup** screen allows you to define system-wide ST monitoring default settings, except ST alarm default settings which must be configured in the **Setup - Alarm Limits** screen (see page 88).

The default **ST Setup** settings are activated whenever a telemetry patient is discharged. They can also be activated on a Bed View **ST Setup** screen that is configured for 5- or 6-wire monitoring. For more information on ST monitoring, see page 149.

The following tables display the ST Setup screen selections:

| ST Setup Screen Selections: 5-Lead |                                                                                                     |                                                                                                                           |         |
|------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------|
| Selection                          | Description                                                                                         | Available Settings                                                                                                        | Default |
| ST Lead 1                          | Sets ECG lead vector for ST Lead 1 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | I       |
| ST Lead 2                          | Sets ECG lead vector for ST Lead 2 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | II      |
| ST Lead 3                          | Sets ECG lead vector for ST Lead 3 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | III     |
| ST Lead 4                          | Sets ECG lead vector for ST Lead 4 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | aVR     |
| ST Lead 5                          | Sets ECG lead vector for ST Lead 5 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | aVL     |
| ST Lead 6                          | Sets ECG lead vector for ST Lead 6 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | aVF     |
| ST Lead 7                          | Sets ECG lead vector for ST Lead 7 parameter box                                                    | I, II, III, aVR, aVL, aVF, V, None                                                                                        | V       |
| ST Units                           | Sets units for ST measurements                                                                      | mm<br>mV                                                                                                                  | mm      |
| Event Duration                     | Determines how long an ST event has to remain outside the set ST alarm limit before an alarm sounds | 15, 30, 45, 60 75, 90, 105, 120 seconds                                                                                   | 60 sec  |
| ST Measurement Point               | Determines whether the ST measuring point is 60 or 80 ms after QRS offset                           | 60 ms<br>80 ms                                                                                                            | 80 ms   |
| Accept/Undo                        | Enables you to save or discard the changes made since the last save.                                | <ul style="list-style-type: none"> <li>Accept saves any changes</li> <li>Undo reverts to the previous settings</li> </ul> | --      |

## Infinity Telemetry Recordings Setup Screen

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If a password is required to access this screen, refer to page 60.

### Recordings Setup Screen Functions

Use the **Recordings Setup** screen to set various telemetry-specific recording properties, independent of the basic Infinity CentralStation recorder settings.

The following table displays the Recordings Setup screen selections:

| Selection                 | Description                                                                       | Available Settings                                                                                                            | Default |
|---------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| Waveform Selection        | Determines whether waveforms to be printed are selected automatically or manually | Auto - Automatically selects top-most displayed waveform for printing<br><br>Manual - Prints selected Waveform 1 and 2        | Auto    |
| Waveform 1                | Selects first waveform for R 50 recording                                         | I, II, III, aVF, aVL, aVR, V                                                                                                  | II      |
| Waveform 2                | Selects second waveform for R 50 recording                                        | I, II, III, aVF, aVL, aVR, V                                                                                                  | V       |
| Alarm Condition Waveform: | Selects top waveform for R 50 recording                                           | On, Off                                                                                                                       | On      |
| Accept/Undo               | Enables you to save or discard the changes made since the last save.              | <ul style="list-style-type: none"> <li>– Accept saves any changes</li> <li>– Undo reverts to the previous settings</li> </ul> | --      |

## Infinity Telemetry Alarm Limits Defaults Screen

Use the **Default Settings - Alarm Limits** screen to define system-wide, telemetry alarm default settings. These default settings are activated at telemetry patient discharge or by clicking **Restore System Defaults** on a particular patient's Bed View screen (see page 166).

The patient category is part of the screen title, and parameter limits are based on that category (see page 88).

The **Default Settings - Alarm Limits** screen is similar to the **Setup - Alarm Limits** screen in Bed View in which you can customize alarm limits for a *particular* patient.

The following table displays the available functions on the Default Settings - Alarm Limits screen:

| Button/Function             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arrhythmia                  | Opens <b>Default Settings - Arrhythmia</b> screen (see page 90)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Parameter alarm limit setup | <div>To set up the alarm limits:<ol style="list-style-type: none"><li>Click on the table row you wish to set up. This activates the configuration mode for that row during which you can set limits within each column.</li></ol><div><div>SpO2</div><div>OK</div><div>OFF</div><div>100</div><div><div></div><div></div></div><div>90</div><div><div></div><div></div></div><div>REC</div></div><ol style="list-style-type: none"><li>Click on the arrows or toggle button so that the desired setting appears.</li><li>Click on <b>OK</b>.</li><li>Repeat steps 1 to 3 for each parameter.</li></ol></div> |
| →Pediatric...               | – Changes the patient category from <b>Adult</b> to <b>Pediatric</b> or vice versa.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| →Adult...                   | – Changes all of the parameters and limits to the defaults in the selected patient category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

The following table displays the Alarm Limits Parameter settings:

| Alarm Limits Parameter Settings |               |               |                   |                            |
|---------------------------------|---------------|---------------|-------------------|----------------------------|
| Parameter                       | Heading       | Adult Default | Pediatric default | Setting Range              |
| HR                              | Upper Limit   | 120 beats/min | 150 beats/min     | 25 - 300 beats/min         |
|                                 | Lower Limit   | 45 beats/min  | 50 beats/min      | 20 - 295 beats/min         |
|                                 | Alarm         | On            | On                | On/Off                     |
|                                 | Alarm Archive | Rec/Store     | Rec/Store         | Rec, Store, Rec/Store, Off |



| Alarm Limits Parameter Settings                                                                                                               |               |                   |                   |                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|-------------------|---------------------------------------|
| Parameter                                                                                                                                     | Heading       | Adult Default     | Pediatric default | Setting Range                         |
| SpO2                                                                                                                                          | Upper Limit   | 100%              | 100%              | 21 - 100%                             |
|                                                                                                                                               | Lower Limit   | 90%               | 90%               | 20 - 99%                              |
|                                                                                                                                               | Alarm         | On                | On                | On/Off                                |
|                                                                                                                                               | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| PLS                                                                                                                                           | Upper Limit   | 120 beats/min     | 150 beats/min     | 35 - 240 beats/min                    |
|                                                                                                                                               | Lower Limit   | 45 beats/min      | 50 beats/min      | 30 - 235 beats/min                    |
|                                                                                                                                               | Alarm         | Off               | Off               | On/Off                                |
|                                                                                                                                               | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| ST (I, II, III, aVF, aVL, aVR, V, V+, V1, V2, V3, V4, V5, V6, dV1, dV2, dV3, dV4, dV5, dV6)                                                   | Upper Limit   | +2 mm<br>+0.20 mV | +2 mm<br>+0.20 mV | -14.9 to +15.0 mm<br>-1.49 to +1.5 mV |
|                                                                                                                                               | Lower Limit   | -1mm<br>-0.10 mV  | -1mm<br>-0.10 mV  | -15.0 to +14.9 mm<br>-1.5 to +1.49 mV |
|                                                                                                                                               | Alarm         | Off               | Off               | On/Off                                |
|                                                                                                                                               | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| STVM<br>STCVM                                                                                                                                 | Upper Limit   | +2 mm<br>+0.2 mV  | +2 mm<br>+0.2 mV  | 0.1 to 45 mm<br>0.01 to 4.5 mV        |
|                                                                                                                                               | Lower Limit   | 0                 | 0                 | 0.0 to 44.9 mm<br>0.0 to 4.49 mV      |
|                                                                                                                                               | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
|                                                                                                                                               |               |                   |                   |                                       |
| PVC/min                                                                                                                                       | Upper Limit   | 10 beats/min      | 10 beats/min      | 1 - 60 beats/min                      |
|                                                                                                                                               | Alarm         | Off               | Off               | On/Off                                |
|                                                                                                                                               | Alarm Archive | Off               | Off               | Rec, Store, Rec/Store, Off            |
| <b>NOTE</b><br>NIBP alarm limits must be set up at the bedside monitor.<br>For details, see the bedside monitor <i>Instructions for Use</i> . |               |                   |                   |                                       |

## Infinity Telemetry Arrhythmia Defaults Screen

This screen allows you to define system-wide default settings for arrhythmia event categories. These defaults are activated at telemetry patient discharge or by clicking **Restore System Defaults** on a particular patient's Bed View screen (see page 133).

The patient category is part of the screen title, and event defaults are based on that category (see page 91).

The **Default Settings - Arrhythmia** screen is similar to the **Setup - Arrhythmia** screen in Bed View in which you can customize each event category for a *particular* patient (see page 133).

| Available Functions on the Arrhythmia Screen |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Button/Function                              | Description/Selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Alarm Limits...                              | Opens <b>Default Settings - Alarm Limits</b> screen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Select Arrhythmia event settings             | <p>To setup the arrhythmia events:</p> <div><div><div>VT</div><div>OK</div><div>LT</div><div><div>▲</div><div>▼</div></div><div>120</div><div><div>▲</div><div>▼</div></div><div>10</div><div><div>▲</div><div>▼</div></div><div>REC/STORE</div></div><p><b>1</b> Click on the table row of the event you wish to set up. This activates the configuration mode for that row during which you can set limits within each column.</p><p><b>2</b> Click on the arrows or toggle button so that the desired setting appears.</p><p><b>3</b> Click on <b>OK</b>.</p><p><b>4</b> Repeat steps 1 to 3 for each parameter.</p><p><b>5</b> Click on <b>Accept</b> to save or <b>Undo</b> to cancel changes</p></div> |
| Set arrhythmia 'monitoring mode'             | <p>To select the arrhythmia monitoring mode, click on the appropriate radio button.</p> <ul style="list-style-type: none"><li>– Off</li><li>– Basic (Default)</li><li>– Full</li></ul> <p>In order to set the arrhythmia monitoring mode to Off, you must enter the clinical password in the <i>Password</i> pop-up.</p>                                                                                                                                                                                                                                                                                                                                                                                     |

Events available for arrhythmia monitoring are determined by the monitoring mode.

| Mode            | Available Events                                              |
|-----------------|---------------------------------------------------------------|
| Full            | ASY, VF, VT, ARTF, RUN, AIVR, CPT, BGM, PAUS, TACH, BRDY, SVT |
| Basic (Default) | ASY, VF, ARTF, VT                                             |
| Off             | –                                                             |

| Arrhythmia Event Settings |               |                                                                 |                                                                 |
|---------------------------|---------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| Parameter                 | Heading       | Adult Default Setting                                           | Pediatric Default Setting                                       |
| ASY                       | Count         | –                                                               | –                                                               |
|                           | Rate          | –                                                               | –                                                               |
|                           | Alarm Grade   | LT                                                              |                                                                 |
|                           | Alarm/Archive | Rec, Rec/Store                                                  |                                                                 |
| VF                        | Count         | –                                                               | –                                                               |
|                           | Rate          | –                                                               | –                                                               |
|                           | Alarm Grade   | LT                                                              |                                                                 |
|                           | Alarm/Archive | Rec, Rec/Store                                                  |                                                                 |
| VT                        | Count         | 5 - 15 Default: $\geq 10$                                       | 5 - 15 - Default: $\geq 10$                                     |
|                           | Rate          | 100 - 200 Default: $\geq 120$                                   | 100 - 200 - Default: $\geq 120$                                 |
|                           | Alarm Grade   | LT (default), SER, ADV, OFF                                     |                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                      |                                                                 |
| ARTF                      | Count         | –                                                               | –                                                               |
|                           | Rate          | –                                                               | –                                                               |
|                           | Alarm Grade   | LT, SER, ADV, OFF (default)                                     |                                                                 |
|                           | Alarm/Archive | OFF                                                             |                                                                 |
| RUN                       | Count         | Not adjustable; upper value is VT count - 1. Default: 3 - 4     |                                                                 |
|                           | Rate          | Not adjustable; same as VT rate. Default: $\geq 120$            | Not adjustable; same as VT rate. Default: $\geq 160$            |
|                           | Alarm Grade   | LT, SER (default), ADV, OFF                                     |                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                      |                                                                 |
| AIVR                      | Count         | Not adjustable. Default: $\geq 3$                               |                                                                 |
|                           | Rate          | Not adjustable, upper value is VT rate - 1. Default: $\leq 119$ | Not adjustable, upper value is VT rate - 1. Default: $\leq 159$ |
|                           | Alarm Grade   | LT, SER, ADV (default), OFF                                     |                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                      |                                                                 |
|                           |               |                                                                 |                                                                 |

| Arrhythmia Event Settings |               |                                                                       |                                                                                 |
|---------------------------|---------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Parameter                 | Heading       | Adult Default Setting                                                 | Pediatric Default Setting                                                       |
| CPT                       | Count         | —                                                                     | —                                                                               |
|                           | Rate          | —                                                                     | —                                                                               |
|                           | Alarm Grade   | LT, SER, ADV (default), OFF                                           |                                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
| BGM                       | Count         | —                                                                     | —                                                                               |
|                           | Rate          | —                                                                     | —                                                                               |
|                           | Alarm Grade   | LT, SER, ADV (default), OFF                                           |                                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
| PAUSE                     | Count         | —                                                                     | —                                                                               |
|                           | Rate          | 1.0, 1.5, 2.0, 2.5 (default),<br>3.0, 3.5                             | 1.0, 1.5, 2.0 (default), 2.5,<br>3.0, 3.5                                       |
|                           | Alarm Grade   | LT, SER, ADV, OFF (default)                                           |                                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
| TACH                      | Count         | 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15<br>- Default: $\geq 8$          |                                                                                 |
|                           | Rate          | 00 - 200 - Default: $\geq 130$                                        | 100 - 200 - Default: $\geq 180$                                                 |
|                           | Alarm Grade   | LT, SER, ADV, OFF (default)                                           |                                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
| BRDY                      | Count         | —                                                                     | —                                                                               |
|                           | Rate          | 30 - 70    Default: $\leq 50$                                         | 30 - 105    Default: $\leq 60$                                                  |
|                           | Alarm Grade   | LT, SER, ADV, OFF (default)                                           | LT, SER (default), ADV, OFF                                                     |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
| SVT                       | Count         | 3, 4, 5, 6, 7, 8, 9, 10<br>Default: $\geq 3$                          |                                                                                 |
|                           | Rate          | 120, 130, 140, 150, 160, 170,<br>180, 190, 200<br>Default: $\geq 150$ | 120, 130, 140, 150, 160, 170, 180,<br>190, 200, 210, 220<br>Default: $\geq 220$ |
|                           | Alarm Grade   | LT, SER, ADV, OFF (default)                                           |                                                                                 |
|                           | Alarm/Archive | Rec, Store, Rec/Store, Off                                            |                                                                                 |
|                           |               |                                                                       |                                                                                 |

# Main Screen

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## Overview

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The Infinity CentralStation Main Screen displays waveforms, parameters, status messages, banners, and alarm messages for telemetry and bedside patients on the Infinity Network. It can be configured in various central layouts.

## Main Screen Layout

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The Infinity CentralStation displays monitoring data for patients who are assigned to a window of the current Main Screen. The Main Screen layout can be customized with the **Setup Central Layout** screen (see page 64). For a description of screen layouts, see page 95.

## Split vs. Full Main Screen

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The following table shows how much patient data can be visible in the Split and Full Main Screen layouts. If the NOTES feature (see page 102) is enabled (see page 64), less waveform data displays.

| Main Screen layout<br>(rows x columns) | Split Screen   | Full Screen    | Amount of Waveform<br>Data (approximate) |
|----------------------------------------|----------------|----------------|------------------------------------------|
| 2 x 4                                  | Not applicable | Yes            | 8 - 10 seconds                           |
| 4 x 2                                  | Not applicable | Yes            | 8 - 10 seconds                           |
| 8 x 1                                  | Not applicable | Yes            | 8 - 10 seconds                           |
| 4 x 4                                  | Yes            | Not applicable | 4 seconds                                |
| 8 x 2                                  | Yes            | Not applicable | 4 seconds                                |
| 16 x 1                                 | Yes            | Not applicable | 4 seconds                                |

## Selecting a Main Screen Layout

To select a layout for the Main Screen:

- 1 Click on **View** in the Main Screen menu bar.
- 2 Click on the **Central Layout** menu selection. A submenu with available *Main Screen layout* selections appears.
- 3 Click on the desired layout. A pop-up window displays.
- 4 Click on **Yes** in the pop-up to select the new layout or on **No** to exit the menu and keep the current layout.

## Main Screen Setup

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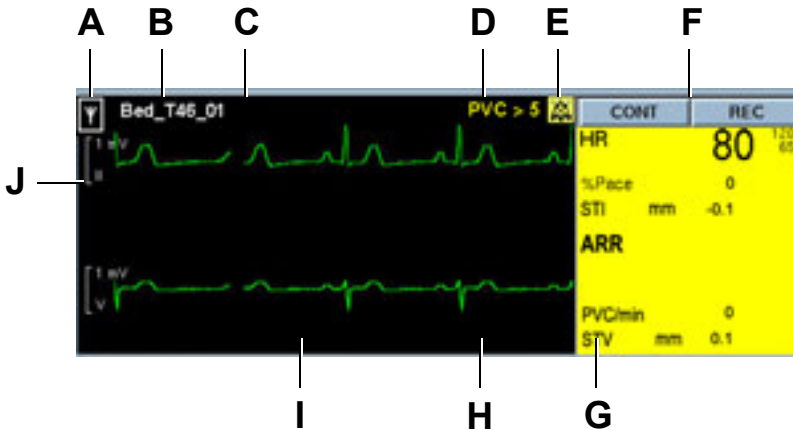
### Menu Bar

| Selection           | Allows you...                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| View                | To access the Full/Event Disclosure applications (see page 209)                                                                                                              |
| Setup               | To configure Infinity CentralStation monitoring<br>To configure Main Screen layouts (see page 64)<br>To access the Telemetry setup menus (see page 65)                       |
| Biomed              | To observe the status of all devices on the network<br>To configure the Infinity CentralStation and telemetry receivers<br>To enable options<br>To access the system console |
| Help                | To open the Infinity CentralStation Instructions for Use on the screen (see page 29)                                                                                         |
| Audio Pause (1 min) | To silence active alarms at the Infinity CentralStation for one minute                                                                                                       |

## Waveform Area - Patient Window

Depending on the selected Main Screen layout, each patient window may have up to four waveforms. All patient waveforms are updated from left to right.

In addition to parameter waveforms, the patient window includes other information.



- A** Monitoring icon: Manual configuration mode, telemetry, or wireless monitoring (page 98)
- B** Bed Label for Infinity M300 or Infinity Telemetry. The bed label is followed by the device ID
- C** Patient Name
- D** Message area for the display of: Bedside alarms, banner, status, Standby, or discharge messages
- E** Icons from bedside monitors
- F** Keys to generate a *Timed* or *Continuous* recording (page 175)
- G** Icon area for the display of the MIB Ventilator Alarm
- H** Area for the display of the alarm messages from the ventilator
- I** Area for the display of the Wave Paused and Standby messages
- J** Lead Label and Scale



| Banner and Status Messages |           |         |                              |                                                                                                      |
|----------------------------|-----------|---------|------------------------------|------------------------------------------------------------------------------------------------------|
| Banner                     | Telemetry | Monitor | Appearance                   | Description                                                                                          |
| Alarm Paused               | Yes       | Yes     | Black text/yellow background | Patient's alarm function is disabled                                                                 |
| All Alarms OFF             | Yes       | Yes     | Black text/yellow background | Patient's alarm function is disabled                                                                 |
| HR Alarms OFF <sup>1</sup> | Yes       | Yes     | Black text/yellow background | Patient's alarm function is disabled                                                                 |
| HR Limits OFF              | Yes       | Yes     | Black text/yellow background | Patient's alarm function is disabled                                                                 |
| HR, ASY, VF OFF            | Yes       | Yes     | Black text/yellow background | Patient's alarm function is disabled                                                                 |
| Bed Disconnected           | No        | Yes     | Black text/white background  | Bedside monitor disconnected for patient transport                                                   |
| Bedside Offline            | Yes       | Yes     | Yellow text                  | Network communication interrupted                                                                    |
| CODE <sup>2</sup>          | No        | Yes     | Black text /red background   | Code fixed key was pressed at bedside monitor                                                        |
| Discharged                 | Yes       | Yes     | Black text/white background  | Patient discharged or transferred at bedside monitor                                                 |
| Duplicate Address          | No        | Yes     | Yellow text                  | CPS/IDS detects a duplicate address on network                                                       |
| Offline                    | No        | Yes     | Yellow text                  | Network communication interrupted                                                                    |
| Pacer Fusion               | Yes       | Yes     | Black text/white background  | Pacemaker Fusion mode activated for local telemetry patient                                          |
| Pacer Off                  | Yes       | Yes     | Black text/white background  | Pacemaker detection deactivated                                                                      |
| Pacer On                   | Yes       | Yes     | Black text/white background  | Pacemaker detection activated                                                                        |
| Pacer On/Fusion            | Yes       | Yes     | Black text/white background  | Pacemaker Fusion mode activated for a remote bed that cannot distinguish between On and Fusion modes |
| STANDBY <sup>3</sup>       | Yes       | Yes     | Black text/white background  | Patient monitor in <i>STANDBY</i> mode                                                               |

### Banner and Status Messages

<sup>1</sup> If an alarm occurs while this banner is displayed, the alarm message replaces the banner during the alarm.

<sup>2</sup> If a code condition is activated at the bedside during an alarm, the banner, CODE, replaces the alarm message.

<sup>3</sup> When a patient monitor is put into Standby mode, only the patient name and bed label displays. When a telemetry patient is put in to Standby mode the ECG waveform continues to display along with patient name, lead labels and scales. A selectable Standby label also displays (see page 113).

For both bedside and telemetry patients, the framed Standby message displays prominently over the center of the top waveform area.

### Icons

| Icon                                                                                                                          | Location                                                                                                                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Audio Paused<br>                              | Upper right corner of top waveform channel<br><br>The icon only appears if the <i>Audio Paused</i> function is enabled for bedside patients (see page 61) or the <i>Remote Silence</i> function is enabled for telemetry patients (see page 67). | Appears when the patient has an active non-silenced alarm or a latching alarm message (a message corresponding to an alarm whose condition is no longer valid but that has not yet been acknowledged).<br><br>Click on icon to clear a latching alarm message or silence the current alarm for 1 minute (for a telemetry patient the alarm is silenced at the Infinity CentralStation; for a bedside patient it is silenced at the bedside monitor and at the Infinity CentralStation). |
| All Alarms OFF<br>                          | Next to parameter value                                                                                                                                                                                                                          | Appears in place of alarm limits when the parameter's alarms are disabled                                                                                                                                                                                                                                                                                                                                                                                                               |
| Alarm Volume Zero<br>MIB Alarm Tone Off<br> | Lower-right corner of patient's top waveform channel in orange                                                                                                                                                                                   | Appears when you click on the <b>MIB Alarm Tone</b> icon.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MIB Alarm Tone ON<br>                       | Lower-right corner of patient's top waveform channel                                                                                                                                                                                             | Appears in a patient's top waveform channel in Main Screen when the MIB alarm tone is enabled.                                                                                                                                                                                                                                                                                                                                                                                          |

| Icons                                                                             |                                               |                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------|
| Icon                                                                              | Location                                      | Description                                                                       |
|  | Left of bed label                             | Identifies a telemetry patient                                                    |
|  | Left of bed label                             | Identifies that the waveform configuration mode is active for the bedside patient |
|  | Left of bed label                             | Identifies a patient being monitored by a wireless monitor                        |
|  | Upper right of Event Disclosure Review Screen | When checked, denotes event for inclusion in Shift Report                         |

### Markers

Pacer spikes for all leads are blue. They start at the detection point in the waveform and end at the ECG scale bar height.

Breath detection spikes for impedance respiration display as white lines vertically centered on the waveform at detection point and extend 1 cm.

### Stopping Waveforms

You can stop all patient waveforms by clicking anywhere in the waveform area. Stopping a patient's waveforms in Main Screen does not stop them in Bed View.

When you stop the waveforms:

- The banner **WAVE(S) PAUSED** displays below the topmost waveform.
- The **CONT** and **REC** buttons appear unavailable for selection in the parameter area. You can request a manual recording, but not a print screen.
- Parameter values and banners continue to be updated and the *Bed Silence* icon and alarm message display when there is an alarm.

To restart the waveforms, click inside the waveform area. All previously displayed waveforms are erased and replaced by new waveforms.

### Waveform Color

The Infinity CentralStation supports the display of all real time and stored waveforms and parameter values using the bedside currently selected colors or telemetry system default settings. For information about changing waveform color at a bedside monitor, see the monitor's Instructions for Use. For information about changing waveform color for a specific telemetry patient, see page 115.

### Telemetry Waveform Dropout Display (Infinity Telemetry only)

You can choose to display a dropout on a telemetry waveform in a clearly identifiable negative *Square-wave* configuration to indicate a non-physiological event should RF dropouts occur when using Infinity

Telemetry. Square-wave display must be configured via a system Biomed setting during installation.

### Parameter Areas

The parameter areas to the right of each waveform may consist of several parameter fields.

The parameter areas allow you to access each patient's Bed View (see the chapter starting on page 107).

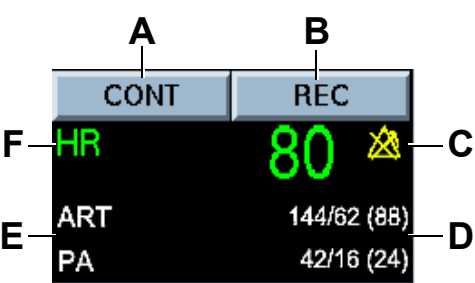
When there is an alarm, the background of the respective parameter area on the Infinity Central-Station flashes in the color associated with the highest grade alarm for that bed.

A parameter area appears blank if no parameter data is available for a displayed patient due to one of the following conditions:

- patient monitor is in *Standby* or the patient was discharged,
- monitor was disconnected for patient transport,
- monitor/transmitter stopped communicating with network.

#### NOTE

The parameter area does not flash or change color during *Offline*, *Bedside Offline*, or *Duplicate Address* conditions.



- A Generates a continuous recording
- B Generates a timed recording
- C Displays alarms off icon when alarms are turned off, or alarm limits when alarms are turned on
- D Secondary parameter values
- E Secondary parameter labels
- F Primary parameter label

#### NOTE

Parameter units are only displayed if they appear at the bedside monitor. For telemetry patients, the telemetry units are always displayed.

## Primary Vs. Secondary Parameters

The Infinity CentralStation displays primary and secondary parameters. The primary parameter has an associated waveform displayed in the waveform area, but secondary parameters do not.

The number of a patient's displayed primary and secondary parameters depends on the selected Main Screen layout.

| Layout | Primary | Secondary   |
|--------|---------|-------------|
| 16 x 1 | One     | up to two   |
| 8 x 1  | One     | up to two   |
| 8 x 2  | Two     | up to four  |
| 4 x 2  | Two     | up to four  |
| 4 x 4  | Four    | up to eight |
| 2 x 4  | Four    | up to eight |

Each parameter field may contain:

- one single-valued primary parameter and up to two secondary parameters, or
- one multi-valued parameter set (e.g. pressures)

The *Primary Parameter* is distinguished by its size. The *Primary Parameter* value(s) and the label are the color of the associated waveform. If the *Primary Parameter* has no waveform displayed, the values and the label appear white.

Up to two *Secondary Parameters*, not associated with waveforms, can be displayed per *Primary Parameter*. *Secondary Parameters* are white and appear smaller.

*Secondary Parameters* consisting of triple-valued pressures are displayed in a single row, in the format SYS/DIA (MEAN).

If the next parameter box at the bedside monitor consists of a “set” of parameter values for ST or the MIB device, only the topmost parameter value will be used as a *Secondary Parameter*.

## Allocation of Parameters - Bedside Monitors

The bedside monitors determine how parameters are allocated to the parameter areas of the Infinity CentralStation. If more parameter slots are available than parameters to fill them, the fields that are not allocated are blank.

If no parameters are available for a bed assigned to Main Screen because the monitor is in “Standby” mode, the patient has been discharged, or the monitor has been removed for transport, the entire waveform and parameter area for that bed appears blank. A message banner in the waveform area explains the reason for the absence of data.

Parameters from the Delta family of bedside monitors (Delta/Kappa/Delta XXL) are assigned in descending order. The top waveform/parameter of the bedside monitor becomes the first *Primary Parameter*/waveform at the Infinity CentralStation. Parameters continue to be assigned in consecutive order (*Primary Parameters* first, then *Secondary Parameters*) from the bedside to available parameter fields at the Infinity CentralStation. The Infinity CentralStation supports two icons specific to these monitors.

- The  symbol in the parameter box alerts users that the O2 lower alarm limit has been set to a value less than 21% at the MultiGas Module.
- Parameters that originate from devices that are not part of the Infinity Network, such as MIB devices, and duplicate parameters originating at the Delta/Delta XL/Kappa/GammaX XL bedside monitors are marked with an asterisk. The same is true when parameters are derived from two different sources within the Infinity Network (such as the monitor and a connected pod).

### NOTE

If a single-value *Primary Parameter* has minor parameter(s) associated with it (e.g., SpO2 has PLS as a minor parameter), the Infinity CentralStation designates these minor parameter(s) as secondary.

Parameters from a Gamma/Gamma XL/Vista bedside monitor are assigned using the top waveform and corresponding parameter as the first *primary waveform/parameter* at the Infinity CentralStation. The second waveform and associated parameter fill the next *Primary/Secondary* fields. The remaining parameters are assigned top-to-bottom, left-to-right, from the bedside parameter boxes to available parameter fields at the Infinity CentralStation.

### Allocation of Parameters - Telemetry Channels

The waveforms selected on the system **Patient View Setup** screen (see page 68) determine the display order of parameters and waveforms in Main Screen. Any changes in the waveform and parameter assignments in Bed View are automatically reflected in Main Screen.

### Notes Area

If the NOTES function is enabled (see page 64), an area for entering patient notes displays before each patient window.

Notes are only stored locally at the Infinity CentralStation, they are not stored at the bedside monitor. Stored notes are deleted under one of the following conditions:

- when you reconnect a bedside monitor to that network location,
- when you discharge the patient,
- when you remove the patient from Main Screen,
- when you select a Main Screen layout that does not include the patient.

The notes area is relative to the number of waveforms in a waveform area. Each notes area can contain up to 300 characters and show a scroll bar.

### Status Area

The status area runs along the bottom of the screen. It displays:

- Alarm and Status messages (to the left).  
These messages may originate from the Infinity CentralStation, network recorders, or bedside monitors. With Dual Display, the status messages appear on both displays. Some status messages appear briefly and indicate a one-time occurrence. Other status messages appear for as long as the cause of the message exists.
- Host Label (near the center).  
Identifies the Infinity CentralStation.
- Time and Date (to the right).  
Displays the current day and time.
- Surveillance alarm messages (to the left).  
See page 161.

## Manual Waveform - Configuration Mode

To select different waveforms (not parameters) for a bedside patient in Main Screen:

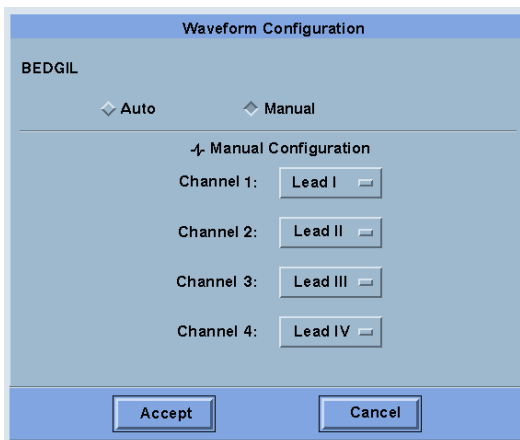
### NOTE

Once you configure a bedside patient's waveform setup via the manual configuration mode, the new setup remains intact for that channel even if you discharge the patient and admit a different one to that channel. For a new patient you can manually change waveforms again or switch to 'Automatic' mode.

### NOTE

The waveform scale information displayed underneath the manual waveform pop-up is not redrawn after the pop-up is closed. However, this does *not* affect the waveform or patient monitoring. To completely redraw the waveform and associated scale, click the left mouse button twice.

- 1 Move the mouse over the patient's waveform channel.
- 2 Press the *right* mouse button.  
The **Waveform Configuration** pop-up displays.



- 3 Click on the **Manual** radio button.
- 4 Click on the option button next to the desired channel. A list of available waveforms appears.
- 5 Set the desired waveform.
- 6 Repeat steps 3 and 4 for additional channels.
- 7 When you are finished, click on **Accept** to keep the changes or **Cancel** to return to previous settings.

## Main Screen - Status Area Messages

---

The following table lists status messages that may display on the Main Screen. For recorder-specific status messages, see page 171.

| Device/Network Status Messages                                                                                                         |           |                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status Message                                                                                                                         | Tone      | Description                                                                                                                                          |
| Central Offline                                                                                                                        | Serious   | Infinity CentralStation is not communicating with network due to incorrect configuration information.                                                |
| Reverting to Default Settings for yyy Data                                                                                             | Serious   | Error during a setup file read operation                                                                                                             |
| Recorder Setup Data Not Saved<br>Central Setup Data Not Saved<br>Central Configuration Data not Saved<br>Locked Options Data Not Saved | Serious   | Error during a setup file write operation. The file will be created at the next restart.                                                             |
| Central Configuration Incomplete                                                                                                       | Serious   | All labels identifying Infinity CentralStation are not entered.                                                                                      |
| Standalone                                                                                                                             | Attention | The patient is not connected to the network.                                                                                                         |
| <xxx> Remote Control Failed                                                                                                            | Attention | Remote control action failed for named device.                                                                                                       |
| Notes Access Error                                                                                                                     | Attention | Notes cannot be accessed.                                                                                                                            |
| Central Duplicate Address                                                                                                              | Serious   | Infinity CentralStation has the same address as another network device.                                                                              |
| File Access Error                                                                                                                      | Attention | File cannot be accessed.                                                                                                                             |
| Data Export Finished                                                                                                                   | Attention | The data has arrived at the MegaCare system.                                                                                                         |
| Waveform Export Completed                                                                                                              | None      | The data was sent to the third party device.                                                                                                         |
| Waveform Export Failed                                                                                                                 | Attention | The data export to the third party device failed.                                                                                                    |
| Hardware Watchdog Failure                                                                                                              | None      | Watchdog timer failure during start-up.<br><br>Call the Hospital Biomedical Engineering Dept.                                                        |
| No Bed Assigned                                                                                                                        | Attention | No bed has been assigned to the selected parameter area in Main Screen. Select the correct parameter area or assign a patient to the available slot. |



| Device/Network Status Messages                                                       |             |                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status Message                                                                       | Tone        | Description                                                                                                                                                                         |
| Copy Logs Started                                                                    | None        | The copying of the logs has started. Wait until the procedure is finished.                                                                                                          |
| Copy Log Failed                                                                      | Advisory    | The logs could not be copied. Ensure that the USB media is connected. Try to copy the log again (see page 298).                                                                     |
| Patient Not found in HIS/CIS                                                         | None        | There was no HIS CIS data available for the patient. Check if you have selected the right patient.                                                                                  |
| Too Many Beds in this Monitoring Unit                                                | None        | You have tried to add more than 32 beds in this monitoring unit. Before adding new beds to this monitoring units, you need to delete some beds. The maximum of beds per unit is 32. |
| Copy Logs Completed - Remove Disk                                                    | Acknowledge | USB media can be removed.                                                                                                                                                           |
| Parameter (YYY) not found                                                            | None        | The selected parameter is not detected. Make sure the necessary hardware is connected properly.                                                                                     |
| <xxx> stands for the device/host label; yyy stands for the specific setup data file. |             |                                                                                                                                                                                     |

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## Patient Setup - Bed View

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|                                                      |     |
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## General Bed View Setup

In Bed View you can:

- remotely view a patient's monitor,
- remotely view a bedside patient's ventilation data (with VentCentral option),
- print the Bed View screen, and
- print a simultaneous ECG Report (see page 181).

The Bed View screen constantly updates patient data, immediately reflecting changes in configuration or signal availability from the patient monitor.

You can configure a bedside patient's setup at the Infinity CentralStation and override bedside monitor setup when the bedside monitor is configured appropriately and the Infinity CentralStation **Bed Control Enable** feature is **On** (see page 61).

### NOTE

For bedside monitor configuration instructions, refer to the monitor Instructions for Use.

Available Infinity CentralStation functions differ depending on whether a patient is local or remote to the Infinity CentralStation Monitoring Unit.

| Function                                               | WITHIN<br>monitoring unit | OUTSIDE<br>monitoring unit |
|--------------------------------------------------------|---------------------------|----------------------------|
| Relearn ECG                                            | Yes                       | --                         |
| Request timed recording                                | Yes                       | --                         |
| Silence alarm                                          | Yes                       | --                         |
| Set up alarm limits                                    | Yes                       | --                         |
| Set up arrhythmia events                               | Yes                       | --                         |
| Admit patient                                          | Yes                       | --                         |
| Edit patient demographic information                   | Yes                       | --                         |
| ST Setup and measuring points                          | Yes                       | --                         |
| VentCentral review (bedside monitor patients)          | Yes                       | Yes                        |
| Bed View Print screen request                          | Yes                       | Yes                        |
| Bed View Print Simultaneous ECG Report request         | Yes                       | Yes                        |
| Stop waveforms                                         | Yes                       | Yes                        |
| Review trend graphs and tables                         | Yes                       | Yes                        |
| Review alarm limits                                    | Yes                       | Yes                        |
| Review arrhythmia event setup                          | Yes                       | Yes                        |
| Review patient's demographic information               | Yes                       | Yes                        |
| Review Full/Event Disclosure Data (with Client option) | Yes                       | Yes                        |
| Request reports                                        | Yes                       | Yes                        |

## Accessing a Patient's Bed View Screen

---

### Viewing a Patient Displayed in Main Screen

To view a patient that is displayed in the Main Screen:

- Click on the patient's parameter area in the Main Screen patient window.

When the Bed View screen opens, the last viewed screen mode will display. Bed View mode descriptions and selection instructions are on page 115.

### Viewing a Patient Not Displayed in Main Screen

To view a patient that is not displayed on the Main Screen:

- 1 Click on **View** on the Main Screen menu bar.
- 2 Click on the **Bed...** menu selection.  
A screen with two list boxes appears.
- 3 Click on a care unit in the box labeled, **Select Care Unit**.  
  
A list of patients in that care unit displays in the box labeled **Select Bed**.
- 4 Use the scroll bars to scroll through the available patients, and click on the desired patient.
- 5 Click **Continue**.

## Bed View Screen Layout

---

A patient's Bed View screen occupies the left half of the Infinity CentralStation display, leaving patients in the Main Screen still visible. It displays parameters that originate at a bedside monitor or patient worn device.

## Waveform Assignment

### Bedside Monitor Patients

The Infinity CentralStation Bed View screen can display bedside monitor main screen layouts.

### Telemetry Patients

You can assign specific parameters and waveforms to each available Bed View waveform channel. The availability of displayed waveforms depends on the ECG configuration mode (3-, 5-, or 6-wire) of the transmitter assigned to a particular channel.

## Bed View Menu Bar

| Selection           | Allows you to...                                                                                                                                                                                                                                                                                                     |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Review              | – access patient Ventilator Settings Review screen (page 237)                                                                                                                                                                                                                                                        |
| Setup               | <ul style="list-style-type: none"> <li>– access patient arrhythmia setup menu (page 129)</li> <li>– access patient alarm limits setup menu (page 157)</li> <li>– access Telemetry setup menus (page 117) with Telemetry option</li> <li>– access VentCentral functions (page 237) with VentCentral option</li> </ul> |
| ADT                 | <ul style="list-style-type: none"> <li>– admit /discharge/transfer patient (page 119)</li> <li>– edit patient demographics (page 121)</li> </ul>                                                                                                                                                                     |
| Main Screen         | – access Main Screen (page 93)                                                                                                                                                                                                                                                                                       |
| Help                | – open the Infinity CentralStation <b>Instructions for Use</b> on screen                                                                                                                                                                                                                                             |
| Audio Pause (1 min) | – silence all active alarms for 1 minute                                                                                                                                                                                                                                                                             |

## Bed View Submenu Selections

The Bed View submenu selections are below the main menu bar, and differ for *Telemetry* and *bed-side* patients.

### NOTE

If any selection appears unavailable for selection, remote control is not possible or the function is not available for the patient category.

| Bedside     | Telemetry   | Description/Selection                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relearn     | Relearn     | <ul style="list-style-type: none"> <li>– <b>Relearn ECG</b> initiates a <i>Relearn</i> of the normal ECG waveform template for processing arrhythmias and ST. If arrhythmia monitoring is <b>Off</b>, this selection is unavailable for selection and cannot be executed.</li> <li>– <b>Relearn RESP</b> initiates a <i>Relearn</i> of the breath detection threshold (only available for bedside patients).</li> </ul> |
| Record      | Record      | – Starts a timed recording                                                                                                                                                                                                                                                                                                                                                                                              |
| Print       | Print       | <ul style="list-style-type: none"> <li>– <b>Bed View</b> initiates print request of Bed View screen.</li> <li>– <b>Simultaneous ECG report</b> Initiates print request of that report (see page 181).</li> <li>– If <b>Print</b> button is unavailable for selection, no printer is configured.</li> </ul>                                                                                                              |
| Audio Pause | Audio Pause | <ul style="list-style-type: none"> <li>– Silences all active alarms at the monitor and/or Infinity CentralStation for 1 minute. For bedside monitor patients, <b>Bed Silence Enable</b> must be <b>On</b> (see page 61).</li> </ul>                                                                                                                                                                                     |

| Bedside | Telemetry   | Description/Selection                                                                                                                                                                                                                                                                                                                                                               |
|---------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| View    | View        | <ul style="list-style-type: none"> <li>– <b>Bed View</b> (see page 115)</li> <li>– <b>All Leads</b> - displays all available leads from the bedside or telemetry patient (see page 115)</li> <li>– The <b>All Leads</b> screen mode supports the Hexaxial (Cabrera) display format (see page 151)</li> <li>– <b>Ventilator</b> - (with VentCentral option, see page 239)</li> </ul> |
| --      | Alarm Pause | – Suppresses any further alarms for a predefined period                                                                                                                                                                                                                                                                                                                             |
| --      | Standby     | <ul style="list-style-type: none"> <li>– Opens pull-down menu of STANDBY text labels (see page 113) which once selected display in the upper right of the waveform channel</li> <li>– Suspends monitoring</li> <li>– Displays “<b>STANDBY</b>” prominently in the center of the top waveform</li> </ul>                                                                             |
| --      | Save Event  | <ul style="list-style-type: none"> <li>– Allows manual storage of an 18-second waveform. <b>Save Event</b> is also available for patients not assigned to Main Screen but located in the same monitoring unit (if the remote control function is enabled).</li> </ul>                                                                                                               |

## Information Area

The information area is directly above the top waveform. It is visible in Bed View so alarm and bedside messages can be seen.

### NOTE

If the patient category 'Neonate' is selected and displayed at the bedside monitor, the category label 'Neonate' appears in the center of the information area.

| Message Type                                   | Display Color                   | Description                                                                                              |
|------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Banners</b>                                 |                                 |                                                                                                          |
| <sup>1</sup> CODE<br>(Delta/Delta XL/Kappa)    | Black text on red background    | The <b>Code</b> key was pressed at the bedside monitor                                                   |
| ALARM PAUSED<br>(Delta/Delta XL and telemetry) | Black text on yellow background | Alarms are turned off for the selected patient for a selected period of time. (Countdown timer displays) |
| All Alarms OFF                                 |                                 | Alarms are turned off indefinitely. (No countdown timer displayed)                                       |
| <sup>2, 3</sup> HR ALARMS OFF                  |                                 | HR alarms are off                                                                                        |
| BED DISCONNECTED                               | Black text on white background  | Bedside monitor is disconnected                                                                          |
| (yyy) Out of Range (LOW)                       |                                 | Appears when a parameter falls below the measuring range                                                 |
| (yyy) Out of Range (HIGH)                      |                                 | Appears when a parameter exceeds the measuring range                                                     |
| STANDBY (secondary)                            |                                 | Patient monitor is in Standby mode                                                                       |

| Message Type                                                                                                                         | Display Color                      | Description                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------|
| Banners                                                                                                                              |                                    |                                                                                                  |
| DISCHARGE                                                                                                                            | White text on black background     | The patient was discharged at the bedside monitor.                                               |
| Monitor Internal Battery Low                                                                                                         |                                    | The wireless bedside monitor internal battery is low.                                            |
| Internal Battery Depleted                                                                                                            |                                    | The wireless bedside monitor internal battery is depleted. The patient's waveform area is blank. |
| Local Messages                                                                                                                       |                                    |                                                                                                  |
| Alarm Messages                                                                                                                       | Color corresponding to alarm grade | See pages 129, 139, 149, and 237 for messages.                                                   |
| Parameter-specific local messages                                                                                                    | Black text on white background     | See pages 129, 139, 149, and 237 for messages.                                                   |
| STAFF ALERT                                                                                                                          |                                    | Infinity Telemetry or Infinity M300 STAFF ALERT key was pressed.                                 |
| TRANSMITTER FAILURE                                                                                                                  |                                    | Technical error                                                                                  |
| TRANSMITTER NO SIGNAL                                                                                                                |                                    | No signal from the transmitter                                                                   |
| MIB Alarm Messages                                                                                                                   | Orange                             | See page 237                                                                                     |
| <sup>1</sup> If a code condition is activated at the bedside during an alarm, the banner CODE replaces the alarm message.            |                                    |                                                                                                  |
| <sup>2</sup> If an alarm occurs while this banner is displayed, the alarm message replaces the banner for the duration of the alarm. |                                    |                                                                                                  |
| <sup>3</sup> This banner appears if no local bedside messages are displayed.                                                         |                                    |                                                                                                  |



## Waveform Area

Each waveform in Bed View consists of approximately 4 seconds of data. Waveform colors for both telemetry and bedside patients are described on page 117.

### NOTE

Waveforms in Bed View are always updated at a 25 mm/s sweep speed, regardless of the selected bedside monitor setting.

The waveform area does not display a waveform when the:

- bedside monitor/transmitter is disconnected,
- patient has been transferred or discharged,
- bedside monitor is in *Standby* mode,
- bedside monitor or Infinity Telemetry receiver is offline, or
- wireless bedside monitor battery is depleted.

## Telemetry Signal Strength

In a telemetry system there are conditions during which signal losses or “dropouts” can occur. When there is a signal loss the Infinity CentralStation displays a blank in the telemetry patient’s waveform. However, during installation of Infinity Telemetry, you can configure the Infinity CentralStation to display a “dropout” as a negative square-wave (see page 100). To change how the Infinity CentralStation displays dropouts, contact Dräger Service.

## Putting Monitoring Into Standby Mode

### WARNING

You **must** take the patient out of Standby to resume monitoring when the patient returns to the care unit.

*Standby* mode allows you to suspend patient monitoring when necessary. When patient monitoring is in *Standby* mode, you can still view all of the patient’s stored data such as trends and configuration settings from any Infinity CentralStation with the client option.

When selected, *Standby* mode will display the message, **Standby** prominently in the center of the waveform.

## Bedside Monitor Patients

Patients being monitored at a bedside must be put in *Standby* at that bedside’s monitor.

## Infinity M300 and Infinity Telemetry Patients

*Standby* is initiated at the Infinity CentralStation for local telemetry patients or for remote telemetry patients if their Infinity CentralStation server has enabled remote control.

For telemetry patients you can select a secondary Standby label that indicates a Standby status.

### NOTE

Monitoring of a telemetry patient assigned to a different Infinity CentralStation within the same monitoring unit can also be put in *Standby* mode provided the remote control functions are enabled at both devices (see page 67).

To select a Standby label for a telemetry patient:

- 1 Open the telemetry patient’s Bed View screen.
- 2 Click on **Standby** to display a pull-down of the following available Standby labels:

|                 |                   |
|-----------------|-------------------|
| <b>Standby</b>  | <b>Shower</b>     |
| <b>Off Unit</b> | <b>Procedure</b>  |
| <b>Cath Lab</b> | <b>Ultrasound</b> |
| <b>X-Ray</b>    | <b>MRI</b>        |
| <b>Surgery</b>  | <b>CT</b>         |

- 3 Click on the desired label.

### NOTE

- To cancel Standby and resume monitoring, click on Standby again, or click the VIEWS button on the Infinity M300.
- Standby labels can be customized by your local Dräger service representative.

## Effect of Standby Mode at the Infinity CentralStation

- All alarms become *inactive*.
- Waveforms continue to display, if available.
- Main Screen parameter labels remain visible but values do not display.
- The message **STANDBY** displays prominently in the patient waveform area. For a telemetry patient, a selected label displays in the upper right of the waveform channel.

## Parameter Areas in Bed View

---

Each waveform channel in Bed View has an associated parameter area, which displays information. The amount of information may differ slightly between *bedside* and *telemetry* patients.

| Information Types                                                                                                                                                                   | Bedside Patient                                          | Telemetry Patient                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| Parameter values                                                                                                                                                                    | Varies according to parameters monitored at bedside      | HR, ARR, PVC/min, ST, NIBP, SPO <sub>2</sub> , and PLS                         |
| Alarm limits                                                                                                                                                                        | Flow (Insp/exp)                                          | Always display                                                                 |
|                                                                                                    | All Alarms OFF symbol displays when alarms are disabled. |                                                                                |
| Blinking green heart symbol<br>( <i>only for Traditional/Infinity Telemetry</i> )<br><br>For pacemaker patients the letter 'P' also appears next to the green blinking heart symbol | Always displays                                          | Displays for local patients when the function is activated during System setup |
| +++                                                                                                                                                                                 | The measurement value exceeds the measuring range.       |                                                                                |
| - - -                                                                                                                                                                               | The measurement value is below the measuring range.      |                                                                                |

Parameter label and unit defaults

| Bedside                                           |                                                                      | Telemetry              |                 |                |
|---------------------------------------------------|----------------------------------------------------------------------|------------------------|-----------------|----------------|
| Parameter Labels/Units                            | Color                                                                | Parameter Labels/Units | Parameter Color | Waveform Color |
| HR                                                | Matching the color of the associated waveform at the bedside monitor | HR                     | Green           | Green          |
| ARR, PVC/min                                      |                                                                      | ARR, PVC/min           | Green           | Green          |
| STx                                               |                                                                      | STx                    | Green           | Green          |
| SpO2 and PLS                                      |                                                                      | SpO2 + PLS             | White           | Blue           |
| NIBP<br>(for Traditional/Infinity Telemetry only) |                                                                      | NPB                    | White           | --             |
| Ventilator                                        | Blue                                                                 | --                     | --              | --             |
| Alarm Limits                                      | Color                                                                | Alarm Limits           | Color           | Color          |
| HR                                                | Gray (provided the limits are displayed at the bed)                  | HR                     | Gray            | Gray           |
| ARR, PVC/min                                      |                                                                      | ARR, PVC/min           | Gray            | Gray           |
| STx                                               |                                                                      | STx                    | Gray            | Gray           |
| SpO2 and PLS                                      |                                                                      | SpO2 and PLS           | Gray            | Gray           |
| Symbols                                           | Color                                                                | Symbols                | Color           | Color          |
| No symbols are displayed                          |                                                                      | Blinking hearts        | Green           | Green          |

## Bed View Screen Display

The Bed View screen has three different displays:

- *Bed View* is a view of a local or remote patient.
- *All Leads* displays all available leads of an individual patient (telemetry or bedside).
- *Ventilator* (see page 240) displays the ventilation status of an individual bedside patient.

### Selecting the Bed View

To select a Bed View:

- 1 Click on the patient's parameter box in the Main Screen to activate the Bed View mode that was last active.
- 2 Click on **View**.
- 3 Click on **Bed View**, **All Leads**, or **Ventilator**.

## Bed View Review Screens

If applicable options are enabled, the Bed View **Review** screens allow the following:

| If you want to...                 | From the patient's Bed View screen...                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access Trends                     | <b>1</b> Click on <b>Review</b> in the Bed View menu bar.<br><b>2</b> Click on <b>Trend Graphs...</b> or <b>Trend Table...</b> (see page 201).         |
| Review Full/Event Disclosure Data | <b>1</b> Click on <b>Review</b> in the Bed View menu bar.<br><b>2</b> Click on <b>Full Disclosure...</b> or <b>Event Disclosure...</b> (see page 209). |
| Review Ventilator Settings        | <b>1</b> Click on <b>Review</b> in the Bed View menu bar.<br><b>2</b> Click on <b>Ventilator Settings Review...</b> (see page 237).                    |

## Alarm Limits and Arrhythmia Setup

The Bed View application provides Alarm Limits and Arrhythmia setup tables for customizing a patient's monitoring attributes.

### Bedside Patients

For bedside patients, the Alarm Limits and Arrhythmia setup tables reflect those at the corresponding bedside monitor. When a bedside monitor acknowledges that it is being remotely controlled, a message is displayed in the local message area at the bedside monitor. Once the bed has successfully processed the change, the setup tables at the bedside monitor *and* at the Infinity CentralStation reflect the change. If the bedside monitor is unable to receive or process a change in setup, the following happens:

- The table entry returns to the previously saved value.
- A corresponding status message is displayed in the status area of the Infinity CentralStation (see page 104).
- An attention tone sounds.

### Telemetry Patients

Configuration of a telemetry patient's alarm setup is similar to that of a bedside patient except for the addition of SpO<sub>2</sub>, PLS, and ST parameters for the telemetry patient.

For Telemetry patients you can select **Alarm Limits** and **Arrhythmia** setup screens according to patient category (see page 66).

### Configuring Alarm Limits

For information regarding Alarm Setup, see page 157.

### Configuring Arrhythmia Setup

Refer to page 129 for setup information.

## Telemetry Setup Screens

When the Telemetry option is enabled, the Bed View **Setup** pull-down menu includes a **Telemetry** menu item which has several submenu selections. Settings made in these submenu screens for individual patients temporarily supersede the system defaults (see page 65) until you discharge the patient or restore system defaults.

You can only perform these setup functions for telemetry patients who are local (see page 26) or admitted to another Infinity CentralStation within the same monitoring unit.

### Accessing the Telemetry Setup Screens

To access the Telemetry Setup screens:

- 1 Select **Telemetry** from the Bed View **Setup** menu.
- 2 Select **Patient View, ST, or M300 Setup/Transmitter Setup** from the submenu.

## Infinity TruST 12-Lead Monitoring Option

### NOTE

Infinity TruST is intended for 12-lead ECG monitoring with a reduced electrode set. Reconstructed leads are intended for real-time assessment of ST segment changes.

Infinity TruST is a 12-lead ECG obtained with only six conventionally placed electrodes. Its function is based on the knowledge that, in general, the signal from a measured lead provides information common to other leads. When this information is appropriately combined, the signal of leads not otherwise configured can be interpolated. This type of lead derivation has a high correlation with measured leads.

TRUST electrodes are placed in accordance with current clinical practice. Like a conventional 6-lead ECG system, waveforms from eight leads (typically, Leads I, II, III, aVR, aVL, aVF, V2, and V5) can be viewed on the Infinity CentralStation. These leads are processed and displayed in the same manner as a conventional system. However, unlike a conventional system, TRUST processes and displays four additional lead waveforms. These derived

leads are viewable in the same fashion as the conventional leads, but are designated as dV1, dV3, dV4, and dV6.

Electrode configuration includes limb leads and the selection of any two precordial leads. Leads V2 and V5 are typically used as these often provide the highest correlation for the remaining precordial leads. Alternate lead configurations can be selected based on signal amplitudes and clinical requirements. See page 305 and page 49 for details regarding proper electrode placement.

### WARNING

- There may be instances where the QRS morphology in one of the four derived leads differs from that of an equivalent conventional lead. In these instances, always refer to the conventional lead.
- Do not select derived leads for ECG processing.

Use of TruST Reference ECG

When **TruST 12-lead** is activated, the Infinity CentralStation uses generic coefficients that are based on a general population. Performance in this mode meets the established correlation to conventional leads.

The following table shows the correlation between TruST *generic* derived leads and conventionally measured leads for various lead configurations (a value of 1.0 being a perfect match). This table displays the recommended lead configurations and respective correlations.

| Correlation | Precordial Lead Placement |
|-------------|---------------------------|
| 0.956       | V2, V5                    |
| 0.953       | IPPV                      |
| 0.953       | V1, V5                    |
| 0.952       | V3, V5                    |
| 0.950       | V1, V6                    |

The performance above is based on a minimum 0.3 mV amplitude and QRS duration < 180 milliseconds on patients with a body surface area (BSA) of 1.5 – 2.5 m<sup>2</sup>. Performance on patients outside these ranges may be less.

Patient View Setup Chest Lead Selection

When the Infinity TruST 12-Lead Monitoring option is enabled (see page 299) and the **TruST 12-Lead:** setting is **On**, recommended lead placements for **Set V** and **Set V+** are V2 and V5. Available selections are V1 through V6 and the lead selections must match the actual electrode placement on the patient's body.

With the two actual selected chest leads, the Infinity CentralStation calculates and reconstructs the four remaining chest lead measurements designated as dV1 through dV6. Chest lead selection on the **Patient View Setup Screen** will affect the patient's related screen displays (e.g. **ST Setup**, **ST Points**, **Trend Graphs**, **Trend Table**, **All Leads**).

**NOTE**  
When using Infinity M300, the precordial leads are always V2, V5. They cannot be changed.

## Admit/Discharge/Transfer

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## Overview

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On the Infinity CentralStation you can view, enter, and edit demographic information for any patient located within the Infinity CentralStation monitoring

unit. Although entering demographics is not essential for monitoring, this data provides helpful information for easy patient identification.

## Admitting a Patient

---

### NOTE

Be sure to enter a patient's demographic information accurately.

You can edit and view demographics of patients at the Infinity CentralStation where they were originally admitted. If remote control is enabled you can view and change demographics for patients within the same monitoring unit even though they are not currently displayed in a Main Screen patient window.

You can view, but *cannot* make changes to, demographics for patients outside the monitoring unit. In this case, the **Admit** screen appears unavailable for selection.

### NOTE

- If the patient was removed for patient transport or the Infinity M300 monitor is not communicating with the Infinity network, you cannot view, enter, or edit demographic data. The **Admit** screen for the viewed patient appears blank.
- The Admit screen is unavailable for selection for patients outside the monitoring unit.

### Admitting a patient from the Bed View

To admit a patient from the Bed View:

- 1 Click on **ADT** in the Bed View menu bar.
- 2 Click on **Admit**.

### Admitting a patient from the Main Screen

To admit a patient from the Main Screen:

- 1 Click on the Bed Label/Patient Name in the patient's waveform area.
- 2 Click on **Admit**.



## Entering Patient Demographics

To open the **Admit** screen, see page 120.

### Demographic Data Selections of the Admit Screen

| Selection                                                                              | Possible Settings                                                                                                                | Special Consideration...                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                                                                   | 1 to 25 characters                                                                                                               | --                                                                                                                                                                                                                                                        |
| ID#                                                                                    | 1 to 12 characters                                                                                                               | --                                                                                                                                                                                                                                                        |
| Birth Date                                                                             | Day-Month-Year selection fields                                                                                                  | Click on the up/down arrow buttons to scroll to the appropriate settings for each category.                                                                                                                                                               |
| Physician                                                                              | 1 to 12 characters                                                                                                               | --                                                                                                                                                                                                                                                        |
| Admit Date                                                                             | Day-Month-Year selection fields                                                                                                  | Click on the up/down arrow buttons to scroll to the appropriate settings for each category.                                                                                                                                                               |
| Height                                                                                 | <ul style="list-style-type: none"> <li>– 20 to 240 cm</li> <li>or</li> <li>– 8 to 100 inches</li> </ul>                          | Click on the up/down arrow buttons to scroll to the appropriate settings.                                                                                                                                                                                 |
| Weight                                                                                 | <ul style="list-style-type: none"> <li>– 0.0 to 350.0 kg</li> <li>– 0 to 722 lb</li> </ul>                                       | <p>The unit of measure is determined by the bedside monitor or, in case of telemetry patients, by the setting selected under System setup (see page 71).</p> <p>Click on the up/down arrow buttons to scroll to the appropriate settings.</p>             |
| Category                                                                               | <i>Bedside monitor:</i><br>Adult, Pediatric, Neonate                                                                             | Obtained from bedside monitor<br>Cannot be modified                                                                                                                                                                                                       |
|                                                                                        | <i>Telemetry patient:</i><br>Adult, Pediatric                                                                                    | Select patient category from drop-down list.                                                                                                                                                                                                              |
| Gender                                                                                 | Male, Female, or Unknown                                                                                                         | --                                                                                                                                                                                                                                                        |
| Pacer/ICD/PCD:                                                                         | <ul style="list-style-type: none"> <li>– PACER</li> <li>– ICD</li> <li>– PCD</li> <li>– EXT</li> <li>– NONE (default)</li> </ul> | <p>20-character maximum limit.</p> <p>The Pacer/ICD/PCD identifier displays on all patient reports. If a bedside does not support this function the Pacer/ICD/PCD label displays on the report with a place to write the patient's pacer information.</p> |
| Telemetry ID:                                                                          | Lists available devices from a configuration file populated during Infinity M300 Setup (see page 299).                           | Click on a device in the list.                                                                                                                                                                                                                            |
| Get HIS/CIS Demographics                                                               | Allows you to retrieve a patient's demographics from the hospital's admission system.                                            | You must have the correct patient ID.                                                                                                                                                                                                                     |
| – Click on <b>Accept</b> to save changes or <b>Undo</b> to keep the original settings. |                                                                                                                                  |                                                                                                                                                                                                                                                           |

## Changing Patient Category

When you change **Category** in the **ADMIT** screen, a pop-up displays with the message:

### CAUTION

Alarm settings, arrhythmia settings, and QRS processing will be changed to the defaults for this category <selected category>.

You must press **Confirm** in the pop-up to accept the change.

Whenever you successfully change a patient category, the following occur.

- Label and QRS-sensitivity settings are changed.
- All Alarm Limit and Arrhythmia settings are changed.
- If arrhythmia is enabled, a RELEARN is initiated.

## Admitting Wireless Monitor Patients

---

The Infinity CentralStation treats *wired* and *wireless* monitors similarly at admission.

### NOTE

- For detailed information about the configuration and operation of wireless components in the Infinity Network, refer to the Dräger publication, “Infinity OneNet Network Planning and Installation Handbook” or “Infinity Network Planning, Design, and Installation Handbook, *Third Edition*.”
- See specific monitor documentation for information about using the monitor with the Infinity CentralStation in a wireless network configuration.

## Moving Between Wired and Wireless Communication

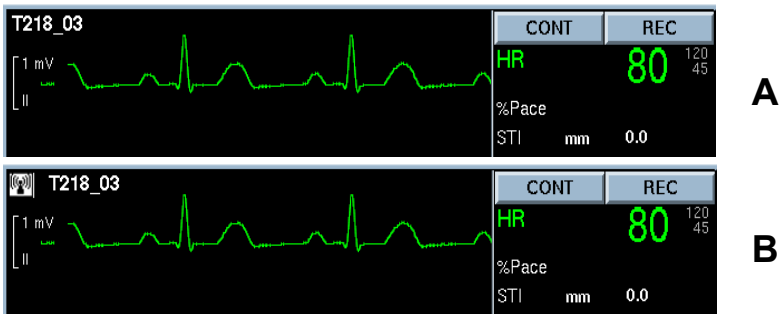
When a wireless bedside monitor is docked at an Infinity Docking Station, the monitor automatically accepts the Infinity Docking Station's Care Unit and Bed Label assignments and communicates with the Infinity Network. When the bedside monitor is removed from the Infinity Docking Station, the wireless network maintains communication.

### Assigning a Wireless Monitor to a Patient Window

If a wireless bedside monitor is removed from the Infinity Docking Station (IDS) and it is configured to keep the IDS bed label, the Infinity CentralStation

patient window remains assigned to the wireless monitor and displays a *wireless icon* before the bed label. When the wireless monitor returns to the IDS, the wireless icon disappears from the patient window bed label.

If you dock another bedside monitor on the IDS associated with the wireless monitor, the new monitor can select the same bed label and becomes available for assignment on the Assign Bed screen in the Main Screen (see page 63).



- A** Waveform of patient on monitor communicating to the network via IDS
- B** After monitor is removed from IDS and begins wireless communication, the waveform is preceded by icon.

### Collecting Full/Event Disclosure Data

If a patient is currently admitted to Full/Event disclosure and is using a wireless monitor, all data collection will continue regardless of bed label change.

#### NOTE

There will be short breaks in Full/Event Disclosure data while the network resumes communication with the patient monitor.

You must re-assign an Infinity CentralStation patient window in the following situations:

- If a wireless monitor bed label is changed to one that is not currently assigned to a Infinity CentralStation patient window, or
- If you dock another bedside monitor on a IDS whose bed label is currently associated with a wireless monitor.

## Wireless Transmission Interruption

If a wireless monitor loses contact with all access points and wireless transmission is interrupted, the Infinity CentralStation generates an *Offline* alert.

## Admitting Under Special Conditions

---

| Circumstance                                                                                | Effect on Admit Functions                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bedside monitor is powering up.                                                             | Patient admission is not possible.                                                                                                                                                                                               |
| Infinity M300/CPU/receiver is offline.                                                      | Patient admission is not possible.                                                                                                                                                                                               |
| Bedside monitor, bedside monitor IDS, or Infinity M300 cannot communicate with the network. | The Infinity CentralStation exits the <b>Admit</b> screen without accepting any changes, displays a blank Bed View screen, and generates a serious alarm.<br><br>Patient admission is not possible.                              |
| The patient monitor is either put into or taken out of <i>Standby</i> mode.                 | No effect; the demographic information can be viewed and edited.                                                                                                                                                                 |
| The bedside monitor is removed for transport (Pick and Go).                                 | The Infinity CentralStation exits the Admit screen without accepting changes and displays a blank Bed View screen. The banner BED DISCONNECTED appears in Bed View and in Main Screen.<br><br>Patient admission is not possible. |
| The bedside monitor returns from transport.                                                 | You can view, enter, and/or edit the patient's demographic information immediately.                                                                                                                                              |

## Discharging a Patient

---

- **Bedside patients** must be discharged at the bedside monitor.
- **Telemetry patients** must be discharged at the Infinity CentralStation where they were originally admitted. Remote discharges are not possible.
- The Pacer Identifier label reverts to *None*.
- A *Bed Disconnected* banner displays in Main Screen.
- (Infinity M300 only) If the patient's **SpO2 Monitoring** setting was **ON**, an SpO2 license becomes available.

When you discharge a telemetry patient:

- Demographics are deleted.
- All patient setup data such as alarm limits, arrhythmia setup, etc. are replaced by system defaults.
- Trends are deleted.
- All pending and active recordings are cancelled.

### Discharging a Telemetry Patient

To discharge a Telemetry patient:

- 1 Open the patient's Bed View screen.
- 2 Click on **ADT** in the Bed View menu bar.
- 3 Click on **Discharge**. This activates the **Admit** screen with the discharge confirmation pop-up.
- 4 Click on **Yes** inside the pop-up to discharge the patient or on **No** to cancel the discharge.

For information about how a patient discharge affects the Infinity M300, see page 40.

## The Patient Transfer Function

---

### NOTE

- After a successful transfer, the source monitor automatically discharges the patient.
- If an error occurs during the transfer, the data remains unchanged at the original location.

You can transfer a patient's trend and demographic data from one location to another. For example, you can transfer data from a telemetry patient at the Infinity CentralStation to any bedside monitor on the Infinity network, or vice versa, or you can transfer telemetry patients' data from one Infinity CentralStation to another. All transfers must be initiated from the destination device (the monitor/Infinity CentralStation to which you wish to transfer the data).

## Special Data Transfer Conditions

During a patient transfer from a telemetry patient to a Gamma/Gamma XL/Vista monitor, only the data from ST leads I and II is sent to the monitor. ST values may be offset by up to 3 minutes due to different trend update intervals between devices.

## Transferring Data to the Infinity CentralStation

To transfer data to the Infinity CentralStation:

- 1 Put the *source* device (bedside monitor or Infinity CentralStation Telemetry patient window or Infinity M300) in *Standby* mode. Only beds that are in *Standby* mode are candidates for transfer.
- 2 Open the patient's Bed View screen at the *destination* Infinity CentralStation.
- 3 Click on **ADT** in the Bed View menu bar.
- 4 Click on **Transfer**.  
The **Transfer Data From** screen displays.
- 5 Click on the desired care unit in the **Select Care Unit**: section of the screen.  
A list of care unit beds displays in the **Select Bed**: window.
- 6 Click on the desired bed from the selection list.

If the selected bed's monitor is not in *Standby* mode an information pop-up displays the following message:

***The source bed for the transfer must be in Standby. Please put this bed in Standby.***

- 7 Click on **Continue** to close the pop-up.

A **Confirmation** pop-up displays a message similar to the following:

***Before proceeding with Transfer...  
You have requested to transfer data from Bed\_xx to Tel\_xx.  
This action will replace any stored data at Tel\_xx with data from Bed\_xx.  
Data stored at Bed\_xx will also be deleted once the transfer is complete.  
Do you wish to begin the transfer?***

- 8 Click on **Yes** to transfer, or click **No** to cancel.

## Transferring Data From the Infinity CentralStation

### Transferring data for Bedside Patients

To transfer data from the Infinity CentralStation for Bedside patients:

- 1 At the Infinity CentralStation select the patient whose data you wish to transfer and open the patient's Bed View screen.
- 2 Click on **Standby** in the Bed View menu bar.
- 3 Go to the *destination* monitor and start the transfer (refer to the monitor documentation).

### Transferring data for Telemetry Patients

To transfer data from the Infinity CentralStation for Telemetry patients:

- Go to the *destination* Infinity CentralStation to start the transfer. See "Transferring Data to the Infinity CentralStation" on page 126 for additional information.

## Transfer Messages

| Transfer Message             | Description                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transfer not allowed         | A transfer of neonatal patient data was attempted.                                                                                                                                       |
| Patient Data Transfer Failed | The data transfer was unsuccessful. Partial data received by the destination device is discarded. The source monitor remains in <i>Standby</i> mode ready for a restart of the transfer. |
| Transfer in Progress         | The data is currently being transferred. You cannot make any changes to the source/destination device during this time.                                                                  |
| Transfer Completed           | The data transfer was successful. All data is permanently deleted from the source monitor, which remains in <i>Standby</i> mode.                                                         |

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# Arrhythmia

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## Overview

With the Infinity CentralStation, Infinity Telemetry, and Infinity M300 systems, arrhythmia monitoring is available for adult and pediatric patients. The mode you select (Full, Basic, or OFF) determines the events processed. Arrhythmia monitoring is not available for neonates.

The system matches incoming beats against beats previously recorded and stored in a reference template. Through this process, the system can verify an arrhythmia event's occurrence, classify it, and draw clinically useful conclusions based on the frequency and morphology of the signal. The system considers all beats questionable if a baseline shift exceeds specified limits.

### WARNING

**Electrical artifacts of non-cardiac origin, such as seizure, may prevent detection of certain arrhythmias. Do not rely solely on ECG with seizure-prone patients.**

### NOTE

QRS classification and arrhythmia detection are intentionally disabled for patients with QRS complexes less than 0.25 mV in amplitude when the display scale is set to 0.5 or 0.25 mV/cm. An artifact condition (ARTF) may occur when the ECG signal does not meet these minimums. Try selecting a lead with larger amplitude QRS complexes. If this is not possible, you can continue to monitor HR, and turn OFF ARR Monitoring for patients whose QRS complexes do not meet these minimums.

The system uses the results of QRS processing for arrhythmia analysis. During multiple-lead arrhythmia processing, each lead's QRS complexes are measured and compared against its learned dominant normal beat. The system classifies beats based on information acquired from all available leads.

## Pacemaker and TENS Precautions

Keep pacemaker patients under close observation. Certain difficulties are inherent to ECG monitoring and require special attention. Use care not to misinterpret ECG waveforms of patients with pacemakers.

### WARNING

#### Pacemaker Performance

**In areas of uncertain performance, the monitor has been designed to err in the direction of false positive alarms rather than false negative alarms.**

**In paced patients, QRS complexes may not be counted that result in false low rate alarms under the following circumstances:**

- Fused beats and asynchronous pacers when coupling intervals are +10 to -90 msec.
- 700 mV pacer pulses followed by QRS complexes less than 0.5 mV.
- Asynchronous pacer pulses with overshoot.
- Misinterpretation of pacemaker pulses as R-waves

**Even though the Infinity CentralStation passed the Association for the Advancement of Medical Instrumentation (AAMI) pacer pulse rejection test, it cannot anticipate every waveform characteristic. The system may count heart rates inaccurately and misinterpret rate-dependent arrhythmias in some paced patients. Do not rely entirely on the displayed heart rate to assess a paced patient's condition. Always observe closely and monitor carefully the patient's vital signs.**

### For Patients without a Pacemaker

For patients without a pacemaker, do the following:

- Disable pacer detection (see page 69)
- Be aware that transcutaneous electrical nerve stimulation (TENS) signals are similar to pacemaker spikes and may be labeled as such.

### WARNING

**Valid QRS complexes following mislabeled TENS signals could be rejected, resulting in false asystole or low heart rate alarms. Follow the directions for pacemaker patients. If TENS signals continue to be interpreted as pacer spikes, disable pacemaker detection.**

## About the Arrhythmia Template

---

The system creates a reference template based on its identification of the patient's dominant QRS pattern. It then classifies individual beats by comparing them with the learned reference template. In the third and final phase of arrhythmia processing, the monitor compares sequences of valid beats with the template.

In most situations, the learning phase takes about 30 to 40 seconds. If the system detects more than 100 QRS complexes and less than 16 matching beats, the system displays a message stating that it is unable to learn. While the system is in the learning phase, all arrhythmia alarms and trend collection are suspended; LRN appears in the parameter box; and a message stating that the system 'relearning' displays in the local message area.

## Relearning the Template

---

During the *Relearn* process a beat classification template is created against which each detected beat is compared and classified.

An automatic *Relearn* is initiated in the following conditions:

- Arrhythmia monitoring is enabled for the first time.
- A patient's monitor (or Infinity CentralStation) comes out of *Standby* mode.
- The processing mode (ECG1 or ECG 1&2) changes.
- Lead assignment changes.
- TRUST monitoring is enabled.
- The telemetry **Patient Category** changes.

During the *learning* process no arrhythmia events are reported, and the parameter area displays the message *LRN*.

### NOTE

If you assign more than one lead, and if only one of those leads is available, the algorithm will learn from that lead only.

Dräger recommends that you initiate a manual *Relearn* of the patient's normal template in the following instances:

- Eight hours after the last *Relearn* occurred,
- After the patient's electrodes are repositioned, replaced, or disturbed, or
- If the message CANNOT LEARN <lead> or CANNOT LEARN is displayed.

### Initiating a Manual Relearning

To relearn a patients' QRS rhythm:

- 1 Open the patient's Bed View screen.
- 2 Click on **Relearn** in the Bed View menu bar.
- 3 Click on **Relearn ECG**.

Once the *Relearn* successfully completes, the assigned leads do not require relearning, even if they fall off and are reconnected.

If the message *CANNOT LEARN* is displayed in the patient's waveform area, try to *Relearn* once more and confirm the quality of the signal. If the message persists, reposition or properly replace the electrodes, then try again.

### NOTE

If the ST option is enabled, the ST algorithm also performs a Relearn on the ST template whenever a Relearn of the arrhythmia template is initiated.

## Customizing a Patient's Arrhythmia Setup

### Accessing the Arrhythmia Setup Table

To access the Setup Table for Arrhythmia:

- 1 Open the patient's Bed View screen and click on **Setup** in the Bed View menu bar.

### 2 Click on Arrhythmia....

|   |                                                                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Column heading is 'Alarm Archive' for telemetry patients admitted to Event Disclosure and 'Record' for telemetry patients not admitted to Event Disclosure. |
| 2 | Opens Alarm Limits Setup Table (page 167)                                                                                                                   |
| 3 | Selects Arrhythmia monitoring mode                                                                                                                          |
| 4 | Restores values to system settings                                                                                                                          |
| 5 | Row in 'configuration mode'                                                                                                                                 |

### Available Functions on the Arrhythmia Setup Table

| Heading/Function  | Description                                                                                                                                                                                                                                                             | Available Settings                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm Grade       | <ol style="list-style-type: none"> <li>1 Click on the '<b>Alarm Grade</b>' column of the event you wish to configure and scroll to the desired setting.</li> <li>2 Click <b>OK</b>.</li> </ol>                                                                          | <ul style="list-style-type: none"> <li>– L-T</li> <li>– SER</li> <li>– ADV</li> <li>– OFF</li> </ul>                                                                                                     |
| Rate<br><br>Count | <p>Determines when an event call is triggered.</p> <ol style="list-style-type: none"> <li>1 Click on the '<b>Rate</b>' or '<b>Count</b>' column of the event you wish to configure and scroll to the desired setting.</li> <li>2 Click <b>OK</b>.</li> </ol>            | <p>See page 135.</p> <p>For arrhythmia settings based on the <b>Rate</b> and <b>Count</b> settings of other calls, there may be no scrolling arrows (e.g. <b>RUN</b> is independent on VT settings).</p> |
| Record            | <p>Automatically generates a timed recording for each validated alarm condition.</p> <ol style="list-style-type: none"> <li>1 Click on the 'Record' column of the event you wish to configure and scroll to the desired setting.</li> <li>2 Click <b>OK</b>.</li> </ol> | <ul style="list-style-type: none"> <li>– ON</li> <li>– OFF</li> </ul> <p>This function cannot be <b>OFF</b> for <b>ASY</b> and <b>VF</b>.</p>                                                            |

| Heading/Function                                                                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Available Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm/Archive                                                                                                                                                                                                                                                                                                                                                                | <p>Depending on the setting, can automatically store waveform and generate a timed recording for each validated alarm condition.</p> <p>Each stored event is 18 seconds long, and includes 9 seconds of data collected before and after the event. (Alarms do <i>not</i> have to be turned on for events to be stored.)</p> <ol style="list-style-type: none"> <li>1 Click on the '<b>Alarm Archive</b>' column of the event you wish to configure and scroll to the desired setting.</li> <li>2 Click <b>OK</b>.</li> </ol> | <ul style="list-style-type: none"> <li>– REC</li> <li>– REC/STORE</li> <li>– STORE</li> <li>– OFF</li> </ul> <p>This function cannot be <i>OFF</i> for ASY and VF.</p>                                                                                                                                                                                                                                                                                                                  |
| Arrhythmia                                                                                                                                                                                                                                                                                                                                                                   | <p>Determines which arrhythmia events are monitored</p> <ul style="list-style-type: none"> <li>– Click on the appropriate radio button.</li> </ul> <div data-bbox="337 710 763 791" data-label="Image"> </div> <p><b>NOTE:</b> To select OFF for telemetry patients, a password is required.</p>                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>– Basic - ASY, VF, VT, ART</li> <li>– Full - ASY, TACH, BRDY, PAUS, VT, RUN, AIVR, CPT, BGM, ARTF, SVT</li> </ul> <p>(‘Full’ arrhythmia capability at the Infinity Central-Station for bedside patients is only active if the bedside monitor has that capability.)</p> <ul style="list-style-type: none"> <li>– OFF</li> </ul> <p>(If you select ‘OFF’, arrhythmia setup table and arrhythmia field in patient’s parameter area are blank.)</p> |
| <p><b>WARNING</b></p> <p>If arrhythmia monitoring is Off, event alarms are automatically turned off. The exceptions are ASY and VF events, which always generate an alarm since they are part of ECG monitoring. However, these events will not cause an alarm if the HR alarm function is also OFF. Therefore, it is strongly recommended that HR alarms are always ON.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Available Bedside Event Settings

For available event settings for telemetry patients, see page 77.

### NOTE

Some arrhythmia settings are based on the Rate and Count settings for other calls (e.g. RUN is dependent on the VT settings).

For available monitor bedside settings, see the appropriate monitor Instructions for Use.

| Event | Count (bpm)                                                         | Rate (bpm)                                                                 | Alarm Setting                 | Record Setting          | Store Setting           |
|-------|---------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------|-------------------------|-------------------------|
| ASY   | --                                                                  | --                                                                         | Not adjustable<br>Default: LT | On, Off                 | On, Off                 |
| VF    | --                                                                  | --                                                                         |                               | Default: On             | Default: On             |
| VT    | 5 - 15<br>Default: $\geq 10$                                        | 100 - 200<br>Default: $\geq 120$                                           | LT, Off, ADV, SER             | On, Off<br>Default: Off | On, Off<br>Default: Off |
| ARTF  | --                                                                  | --                                                                         |                               |                         |                         |
| RUN   | Not adjustable;<br>upper value is VT<br>count -1.<br>Default: 3 - 9 | Not adjustable;<br>same as VT rate.<br>Default: $\geq 120$                 | LT, Off, ADV, SER             | On, Off<br>Default: On  | On, Off<br>Default: On  |
| AIVR  | Not adjustable<br>Default: $\geq 3$                                 | Not adjustable,<br>upper value is VT<br>rate -1 bpm<br>Default: $\geq 119$ |                               | On, Off<br>Default: Off | On, Off<br>Default: Off |
| CPT   | --                                                                  | --                                                                         | LT, Off, ADV, SER             | On, Off<br>Default: On  | On, Off<br>Default: On  |
| BGM   | --                                                                  | --                                                                         |                               | On, Off<br>Default: On  | On, Off<br>Default: On  |
| SVT   | 3 - 10<br>Default: $\geq 3$                                         | 120 - 200<br>Default: $\geq 150$                                           | LT, Off, ADV, SER             | On, Off<br>Default: On  | On, Off<br>Default: Off |
| TACH  | 5 - 15<br>Default: $\geq 8$                                         | 100 - 200<br>Default: $\geq 130$                                           |                               | On, Off<br>Default: Off | On, Off<br>Default: Off |
| BRDY  | Not adjustable<br>Default: $\geq 8$                                 | 30 - 105<br>Default: $\geq 50$                                             | LT, Off, ADV, SER             | On, Off<br>Default: Off | On, Off<br>Default: Off |

| Event                             | Count (bpm)                  | Rate (bpm)                | Alarm Setting                  | Record Setting          | Store Setting           |
|-----------------------------------|------------------------------|---------------------------|--------------------------------|-------------------------|-------------------------|
| PAUS                              | --                           | 1.0 - 3.5<br>Default: 2.5 | LT, Off, ADV, SER              | On, Off<br>Default: Off | On, Off<br>Default: Off |
| PVC/min                           | 0 - 50<br>Default: $\geq 12$ | --                        | Not adjustable<br>Default: SER | On, Off<br>Default: On  | On, Off<br>Default: On  |
| To set alarm limits, see page 167 |                              |                           |                                |                         |                         |

#### NOTE

HR, ASY, and BRDY alarms may be active simultaneously. The HR alarm function provides adjustable alarm limits for heart rates less than 30 beats/min.

## Arrhythmia Beat and Rhythm Classification

| Category            | Classification                                                                                                                |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASY                 | Asystole<br>indicated if 4 seconds pass without the detection of a valid QRS complex                                          |
| VF                  | Ventricular Fibrillation<br>indicated if the source device identifies a sinusoidal waveform with fibrillation characteristics |
| VT <sup>1</sup>     | Ventricular Tachycardia<br>“N” or more consecutive PVCs have been detected with a beat-to-beat rate $\geq$ the VT rate.       |
| RUN <sup>1</sup>    | Ventricular run<br>a series of 3 to “N – 1” consecutive PVCs with a beat-to-beat rate $\geq$ the VT rate                      |
| AIVR                | Accelerated idioventricular rhythm<br>a series of 3 or more PVCs with a rate less than the VT rate                            |
| CPT                 | Ventricular Couplet<br>a sequence of beats with the pattern: normal, PVC, PVC, normal                                         |
| BGM                 | Ventricular Bigeminy<br>a sequence of beats with the pattern: normal, PVC, normal, PVC, normal                                |
| TACH <sup>1,2</sup> | Sinus Tachycardia<br>“N” or more consecutive normal beats with a beat-to-beat rate $\geq$ to the set TACH rate                |



| Category                                                                                                 | Classification                                                                                                                     |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| BRDY <sup>2</sup>                                                                                        | Sinus bradycardia<br>8 or more consecutive normal beats are detected with an average rate $\leq$ to the set sinus bradycardia rate |
| PAUS                                                                                                     | Pause<br>a sequence of 2 normal or PVC beats with an interval $\geq$ to pause rate value in seconds ( $\pm$ 100ms)                 |
| SVT                                                                                                      | Supra ventricular Tachycardia<br>"N" or more consecutive normal beats with a beat-to-beat rate $\geq$ the SVT rate                 |
| ARTF                                                                                                     | Artifact<br>indicated if more than 50% of the beats in the last minute are questionable                                            |
| <sup>1</sup> "N" represents the value in the Arrhythmia setup table 'Count' column.                      |                                                                                                                                    |
| <sup>2</sup> Arrival of a VPB or other abnormal beat breaks the analysis sequence and analysis restarts. |                                                                                                                                    |

Beat classification refers to the analysis of *individual* beats. If the new beat's features do not match those of the reference template, the new beat is classified as abnormal, paced, or questionable.

The monitor uses all detected beats to calculate the heart rate, eliminating questionable beats from arrhythmia classifications.

The following table describes available beat classifications:

| Label | Event and Beat Classification                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASY   | Asystole<br>4 seconds pass without the detection of a valid QRS complex                                                                                                     |
| VF    | Ventricular Fibrillation<br>The monitor identifies a sinusoidal waveform with fibrillation characteristics <sup>1</sup>                                                     |
| VT    | Ventricular Tachycardia<br>N or more PVC's are detected in a time interval<br>$T = (60 * (N - 1)) / R$ , where N is defined as the VT count and R is defined as the VT rate |
| RUN   | Ventricular Run<br>Series of 3 to N-1 consecutive PVCs with a beat-to-beat rate $\geq$ the VT rate <sup>1</sup>                                                             |
| AIVR  | Accelerated Idioventricular Rhythm<br>Series of 3 or more PVCs with a rate less than the VT rate                                                                            |
| SVT   | Supraventricular Tachycardia<br>N or more consecutive normal beats, with a beat-to-beat rate greater than or equal to the SVT setting                                       |
| CPT   | Ventricular Couplet<br>Sequence of beats with the pattern: normal, PVC, PVC, normal                                                                                         |
| BGM   | Ventricular bigeminy<br>Sequence of beats with the pattern: normal, PVC, normal, PVC, normal                                                                                |

| Label                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Event and Beat Classification                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TACH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sinus Tachycardia<br>N or more consecutive normal beats, with a beat-to-beat rate $\geq$ TACH rate setting <sup>2,3</sup>                                                                                                                                                                                                                                                                                                         |
| BRDY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sinus bradycardia<br>8 or more consecutive normal beats, with an average rate $\leq$ sinus bradycardia rate setting <sup>2</sup><br><br><b>NOTE</b><br><ul style="list-style-type: none"> <li>– When in neonatal mode, bradycardia is a low heart rate alarm.</li> <li>– Brady Alarm (neonate only), which is a life threatening alarm, can be configured independently of the low HR alarm, which is a serious alarm.</li> </ul> |
| PAUS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Pause<br>Sequence of two beats classified as normal or PVC, with interval $\geq$ pause rate value in seconds ( $\pm 100$ ms)                                                                                                                                                                                                                                                                                                      |
| ARTF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Artifact<br>More than 50% of beats in the last minute classified as questionable                                                                                                                                                                                                                                                                                                                                                  |
| <p><sup>1</sup>Certain ventricular tachycardias have sinusoidal waveforms closely resembling those of ventricular fibrillation. Because of the similarity of these waveforms, the monitor may classify such types of ventricular tachycardia as ventricular fibrillation, the more serious of the two conditions.</p> <p><sup>2</sup>“N” is the event count set in the Arrhythmia setup table’s count column.</p> <p><sup>3</sup>A PVC or other abnormal beat breaks the analysis sequence and restarts analysis.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Telemetry Pulse Oximetry and NIBP Monitoring

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## Overview

---

Infinity Telemetry and Infinity M300 offer pulse oximetry monitoring.

The Infinity M300 contains an internal pulse oximeter. To enable pulse oximetry monitoring through Infinity M300, a license option must be enabled.

Infinity Telemetry connects to the MicrO2+ Pulse Oximeter to provide pulse oximetry monitoring. Infinity Telemetry also connects to Infinity monitors using the Vital Connection cable for both pulse oximetry and NIBP monitoring (see page 142).

SpO2 and PLS measurements and their associated trend values are labeled identically regardless of the source (Infinity monitor/Vital Connection Cable, MicrO2/MicrO2+, or bedside monitor).

## Pulse Oximetry

---

A pulse oximeter determines the percentage of functional hemoglobin saturated with oxygen (% SpO2) in the patient's arterial blood. A light sensor on the patient's finger measures the absorption levels of red and infrared light.

Since oxyhemoglobin and deoxyhemoglobin absorb different amounts of red light, but nearly the same amount of infrared light, the monitor uses the difference between the two measurements to calculate the percentage of saturated hemoglobin. Because light absorption varies with blood volume and blood volume varies with the pulse rate, a pulse rate (PLS) can also be derived.

### WARNING

**Bright sunlight can interfere with pulse oximetry measurements, causing erratic or missing values. When the sensor is likely to be exposed to direct sunlight, cover it with an opaque material.**

### WARNING

**Elevated levels of carboxyhemoglobin or methemoglobin in monitored patients can result in inaccurate pulse oximetry readings**

### WARNING

**SpO2 measurements are particularly sensitive to pulsations in the artery and the arteriole. Therefore, measurements may not be accurate if a patient is experiencing shock, hypothermia, anemia, or has received medications that reduce artery blood flow**

### WARNING

**Infrared remote control devices, such as those used with TVs and VCRs that are aimed directly at the SpO2 sensor can interfere with accurate pulse oximetry measurements.**

### WARNING

**Check the sensor at least every four hours. Move the sensor if there is any sign of skin irritation or impaired blood circulation.**

### WARNING

**Significant levels of indocyanine green, methylene blue, or other intravascular dyes can interfere with accurate pulse oximetry measurements**

**WARNING**

**Dyes, nail polishes and other substances, may absorb an abnormal amount of red light, which can effect the accuracy of the measurement. Be sure to apply the sensor to a site free of any artificial pigments.**

**NOTE**

- A functional tester cannot be used to assess the accuracy of a pulse Oximeter probe or a pulse Oximeter monitor. Because pulse oximeter equipment measurements are statistically distributed, only about two-thirds of pulse oximeter equipment measurements can be expected to fall within  $\pm$  Arms of the value measured by a CO-oximeter.
- If there is independent demonstration that a particular calibration curve is accurate for the combination of a pulse oximeter monitor and a pulse oximeter probe, then a functional tester can measure the contribution of a monitor to the total error of a monitor/probe system. The functional tester can then measure how accurately a particular pulse oximeter monitor is reproducing that calibration curve.

## Sensor Application

---

Infinity M300 supports Dräger, Nellcor, and Masimo SpO<sub>2</sub> sensors, and Infinity Telemetry supports Dräger and Nellcor SpO<sub>2</sub> sensors. If you connect a sensor that is different from the sensor type configured for the system, the Infinity CentralStation displays an error message.

To apply the sensor:

- 1 Select a sensor that is best suited for your patient.

**NOTE**

For a complete list of accessories available with this product, contact your local Dräger Medical representative to obtain a catalogue.

- 2 Clean reusable sensors before and after each use.
- 3 Position the sensor correctly and attach it to your patient (see sensor manufacturer's recommendations).
- 4 Connect the sensor to the intermediate cable.

- 5 Inspect the sensor application site frequently. If the sensor is too tight, it may damage the tissue and impede blood flow. If the sensor is damaged, do not use it.

**CAUTION**

Read the instructions provided with the sensor to select the best application technique and to review all safety related information.

## Turning Pulse Oximetry Monitoring On/Off

---

To turn on SpO<sub>2</sub> monitoring:

- 1 Open the patient's **Bed View** screen.
- 2 Click on **Setup** in the Bed View menu bar.
- 3 Click on **Telemetry**. (If the selection does not appear, the Telemetry option is not enabled).
- 4 Click on **Patient View**. The Patient View Setup screen displays.
- 5 (*Infinity M300 only*) Click on the SpO<sub>2</sub> button.
- 6 Click on **SpO<sub>2</sub>** monitoring.
- 7 Click on the **ON/OFF** toggle button next to **SpO<sub>2</sub> Monitoring**.
- 8 If you choose to monitor SpO<sub>2</sub>, select settings for **SpO<sub>2</sub> Color**.

9 (*Infinity M300 only*) Select settings for **SpO<sub>2</sub> Sensor**.

10 Click on **Accept** to save the changes, or click **Undo** to keep the previous settings.

When SpO<sub>2</sub> monitoring is set as a system default, but turned off for an individual patient, the parameter box SpO<sub>2</sub> and PLS fields are blank.

### Display of Pulse Oximeter Information

When pulse oximetry monitoring is enabled, the Bed View at the Infinity CentralStation also displays:

- SpO<sub>2</sub> value and alarm limits (or All Alarms OFF icon if alarms are off)
- **PLS** and alarm limits

## Vital Connection Cable

---

### CAUTION

Do not allow fluids to come in contact with the Vital Connection Cable or its transmitter interface. If fluids are accidentally spilled on equipment, remove affected device from service as soon as possible. Contact the Hospital Biomedical Engineering Dept. to ensure that there is no compromise in electrical safety.

### NOTE

Use of the Vital Connection cable requires software version VF5 or higher in the Infinity CentralStation and Infinity Kappa, Delta, Delta XL, Gamma X XL, and Vista XL patient monitors.

The Vital Connection Cable links the Infinity Telemetry Transmitter with a compatible Infinity bedside monitor so that a telemetry patient's SpO<sub>2</sub> and NIBP measurements can be transmitted to the Infinity CentralStation.



## Getting an NIBP or SpO2 measurement

To get an NIBP or SpO2 measurement:

- 1 Turn on Pulse Oximetry monitoring (see page 142).
- 2 Set up the **Patient View** screen *Channels* (see page 84) so that **NIBP** and **SPO2 + PLS** parameters display in the patient window.
- 3 Plug the appropriate end of the Vital Connection Cable into the local telemetry patient's transmitter programming port.
- 4 Attach the other end of the Vital Connection Cable into the X-8 port of a compatible bedside monitor on the Infinity network. Be sure that the Vital Connection LED is lit. This determines an active connection.
- 5 For details about obtaining SpO2 and NIBP measurements, refer to the appropriate Infinity patient monitor *Instructions for Use*.



### NOTE

If you use the Vital Connection Cable to spot-check SpO2, you must disable the SpO2 alarm to avoid a reoccurring Advisory alarm caused by disconnecting the Vital Connection cable. If the SpO2 alarm is not disabled, an Advisory alarm will continue every minute.

The Vital Connection LED indicates connection status to both associated devices (bedside or transmitter) as follows:

| LED appearance   | Status                                                                                                                 |
|------------------|------------------------------------------------------------------------------------------------------------------------|
| Not lit          | There is no physical connection to or communication with either bedside or transmitter.                                |
| Continuous light | The Vital Connection cable is physically connected and communication is established with both bedside and transmitter. |

A Vital Connection interface failure or disconnection will send the following message to the Infinity CentralStation.

Ext. Device NO SIGNAL

As soon as the patient's NIBP measurement is available the Infinity CentralStation displays the data in a parameter box as illustrated. The measured value updates whenever you take a new measurement. The latest measurement displays until you discharge the patient or the measurement is older than 24 hours.



A Systolic measurement <sup>1</sup>

B Diastolic measurement <sup>1</sup>

C Time of last measurement

D Mean measurement<sup>1</sup>

E Units of measurement

<sup>1</sup>When alarms are OFF the *Alarm Off* icon displays

Vital Connection Cable measurements display on the Bed View and Main Screen of the Infinity CentralStation, and are stored in Tabular and Graphic Trends.



## Alarm Functions

---

For specific Infinity CentralStation **SpO<sub>2</sub>** and **PLS** alarm setup instructions, see page 157.

## Alarm Messages

---

### SpO<sub>2</sub> Alarm Messages via Infinity Telemetry and Infinity M300

| Alarm Messages                                            |                                      |             |                                                 |                                                                                 |
|-----------------------------------------------------------|--------------------------------------|-------------|-------------------------------------------------|---------------------------------------------------------------------------------|
| Parameter Area                                            | Status Information Area              | Alarm Grade | Description                                     | Action                                                                          |
| PLS: <value>                                              | PLS > UL                             | SER         | Parameter value exceeds the upper alarm limit.  | Check the patient.<br>Reset the alarm limits.                                   |
| PLS: <value>                                              | PLS < LL                             | SER         | Parameter value is below the lower alarm limit. | Check the patient.<br>Reset the alarm limits.                                   |
| PLS: +++                                                  | Pulse rate out of range              | SER         | The pulse rate is outside the measuring range.  | Check the patient.                                                              |
| SpO <sub>2</sub> : <value>                                | SpO <sub>2</sub> > UL                | SER         | Parameter value exceeds the upper alarm limit.  | Check the patient.<br>Reset the alarm limits.                                   |
| SpO <sub>2</sub> : <value>                                | SpO <sub>2</sub> < LL                | SER         | Parameter value is below the lower alarm limit. | Check the patient.<br>Reset the alarm limits.                                   |
| Messages unique to MicroO <sub>2</sub>                    |                                      |             |                                                 |                                                                                 |
| SpO <sub>2</sub> : *U*<br>PLS: *U*<br>P1                  | MicroO <sub>2</sub> sensor unplugged | ADV         | Sensor is disconnected from the oximeter.       | Check all cable connections.<br><br>Replace sensor, if necessary.               |
| SpO <sub>2</sub> : *A*<br>PLS: *A*<br>P2<br>(at oximeter) | MicroO <sub>2</sub> transparent      | ADV         | Sensor cannot detect any signals.               | Check the patient.<br><br>Check sensor and reapply or replace it, if necessary. |

| Alarm Messages                                                                  |                         |                      |                                   |                                                                                                                |
|---------------------------------------------------------------------------------|-------------------------|----------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------|
| Parameter Area                                                                  | Status Information Area | Alarm Grade          | Description                       | Action                                                                                                         |
| SpO2: *A*<br>PLS: *A*<br>P3<br>(at oximeter)                                    | Micro2 opaque           | ADV                  | Sensor does not detect any light. | Make sure the LED on the oximeter is not blocked.                                                              |
| SpO2: *A*<br>PLS: *A*<br>P4<br>(at oximeter)                                    | Micro2 artifact         | ADV                  | Motion artifact                   | Ensure that the patient remains as still as possible.<br><br>Check sensor; reapply or replace it if necessary. |
| SpO2: *A*<br>PLS: *A*<br>P5<br>(at oximeter)                                    | Micro2 regulation error | ADV                  | Too much ambient light            | Cover the sensor with an opaque material.                                                                      |
| SpO2: *A*<br>PLS: *A*<br>P6<br>(at oximeter)                                    | Micro2 weak signal *    | ADV/SER <sup>1</sup> | Pulse is weak or nonexistent.     | Check the patient.<br><br>Check sensor/oximeter and reapply or replace it, if necessary.                       |
| SpO2: *N*<br>PLS: *N*                                                           | Micro2 no signal        | ADV                  | Oximeter signal is not available. | Check connections and Micro2 battery.                                                                          |
| <sup>1</sup> After the 30-second advisory alarm, this alarm is upgraded to SER. |                         |                      |                                   |                                                                                                                |

| SpO2 Alarm Messages via Vital Connection Cable |                         |             |                                                                                                                                                                                                                                       |                                               |
|------------------------------------------------|-------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Alarm Message                                  |                         |             |                                                                                                                                                                                                                                       |                                               |
| Parameter Area                                 | Status Information Area | Alarm Grade | Description                                                                                                                                                                                                                           | Action                                        |
| SpO2: <value>                                  | SpO2 > UL               | SER         | Parameter value exceeds the upper alarm limit.                                                                                                                                                                                        | Check the patient.<br>Reset the alarm limits. |
| SpO2: <value>                                  | SpO2 < LL               | SER         | Parameter value is below the lower alarm limit.                                                                                                                                                                                       | Check the patient.<br>Reset the alarm limits. |
| SpO2: +++<br>SpO2: - - -                       | out of range            | SER         | Parameter rate is outside the measuring range                                                                                                                                                                                         | Check the patient.<br>Reset the alarm limits. |
| PLS: <value>                                   | PLS > UL                | SER         | Parameter value exceeds the upper alarm limit.                                                                                                                                                                                        | Check the patient.<br>Reset the alarm limits. |
| PLS: <value>                                   | PLS < LL                | SER         | Parameter value is below the lower alarm limit.                                                                                                                                                                                       | Check the patient.<br>Reset the alarm limits. |
| PLS: +++<br>PLS: - - -                         | Pulse rate out of range | SER         | The pulse rate is outside the measuring range.<br><br><b>PLS: +++</b> will also display at the Infinity CentralStation if the measured PLS at the bedside exceeds 255. The actual PLS value displays at the bedside monitor, however. | Check the patient.<br>Reset the alarm limits. |
| SpO2: * * *<br>PLS: * * *                      | SpO2<br>Technical Alarm | ADV         | SpO2 technical Alarm                                                                                                                                                                                                                  | See bedside monitor<br>Instructions for Use   |

| NIBP Alarm Messages via Vital Connection Cable |                         |             |                                         |                                                                                                                            |
|------------------------------------------------|-------------------------|-------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Alarm Message                                  |                         |             |                                         |                                                                                                                            |
| Parameter area                                 | Status/Information area | Alarm Grade | Description                             | Action                                                                                                                     |
| S <value><br>S + + +<br>S - - -                | SYS Alarm               | SER         | NIBP alarm at bedside monitor           | Check the patient and treat if necessary.<br><br>Reset the alarm limits                                                    |
| D <value><br>D + + +<br>D - - -                | DIA Alarm               | SER         | NIBP alarm at bedside monitor           | Check the patient and treat if necessary.<br><br>Reset the alarm limits                                                    |
| M <value><br>M + + +<br>M - - -                | MEAN Alarm              | SER         | NIBP alarm at bedside monitor           | Check the patient and treat if necessary.<br><br>Reset the alarm limits                                                    |
| S * * *<br>D * * *<br>M * * *                  | NIBP Technical Alarm    | ADV         | Technical NIBP alarm at bedside monitor | This message displays for any bedside-generate NIBP Advisory alarm. For details, see bedside monitor Instructions for Use. |

## Telemetry ST Analysis Option

---

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## Overview

ST deviation measurements are performed on averaged ECG segments. An averaged ECG segment is 900 ms long and is calculated from a minimum of 4 normal QRS complexes having occurred in the previous 15 seconds. To reduce the effects of

artifact, only normal beats are included in the average (if no normal beats are available, no complex or corresponding parameter box values display).

ST parameters are included in the trend function, and you can configure the ST alarm functions as for any other parameter (see page 157).

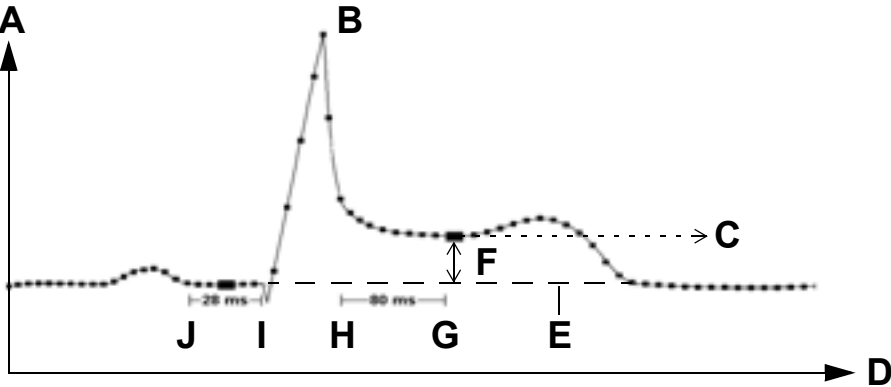
## ST Segment Morphology

ST segment deviations are defined in terms of displacement above or below the isoelectric level. The measurement compares the *isoelectric point* to the ST deviation measurement point.

The isoelectric point defines the point of zero voltage (no electrical activity, 0 mm) with a default position of 28 ms before the onset of the QRS complex on the horizontal (time) axis. The ST deviation point

occurs in the ST segment between the QRS offset (J point) and the T-wave, at a default position of 80 milliseconds after the QRS offset.

The value of the ST measurement point is compared to the isoelectric point value, and the difference between the two measurements is the ST deviation. The onset and offset points for the QRS complex are determined by the QRS detection algorithm.



A mV axis

B Fiducial point

C ST level

D Time axis

E Isoelectric level

F ST Deviation

G ST Measurement point. Default ST measurement point is equal to 80 ms after QRS offset

H QRS offset

I QRS onset

J Isoelectric point. Default ST measurement point is equal to 28 ms before QRS onset

## Parameter Display

---

When ST monitoring is enabled the Infinity CentralStation parameter area lists the ECG leads selected by the user as data sources for ST analysis. The ST values displayed depend on the transmitter lead configuration mode.

## Relearning ST

---

The Infinity CentralStation initiates a learning phase automatically under the following circumstances:

- Whenever you enable ST for the first time
- After you change leads
- When the monitor, Infinity M300, or Infinity CentralStation comes out of *Standby* mode
- After a device restart

To relearn a patient's QRS rhythm:

- 1 Open the patient's Bed View screen.
- 2 Click on **Relearn** in the Bed View menu bar.

### 3 Click on **Relearn ECG**.

During the learning phase, the ST fields within the parameter area do not display any parameter values, and the message **ECG RELEARNING** appears in the information area of the Bed View screen. A vertical dotted line on the ST trend graphs marks the time of a learning phase, and the label LRN appears to the right of the trend window when you move the trend cursor on the dotted line. Each learning phase of a patient's QRS is entered in the clinical events log.

## ST Points Setup Screen

---

The **ST Points Setup** screen displays the current averaged complex for each lead with associated deviation measurement values and times.

### NOTE

Infinity M300 does not support Hexaxial (Cabrera) display.

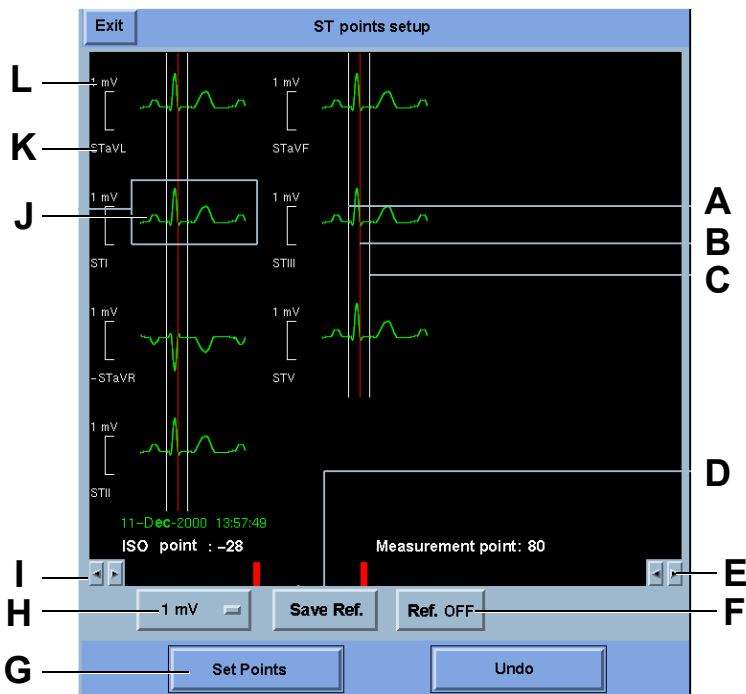
### Hexaxial (Cabrera) Lead Support

In Cabrera display mode (see page 291), the aVR lead is presented in inverted polarity as -aVR. The order of the lead presentation in Cabrera is: aVL, I -aVR, II aVF, III V1-V6.

## Accessing the ST Points Setup Screen

To access the ST Points Setup screen:

- 1 Open patient's Bed View screen.
- 2 Click on **Setup** in the Bed View menu bar.
- 3 Click on **ST Points**.



A ISO point

B J-Point

C ST point

D Stores new reference complex

E Moves the reference point back and forth

F Displays/removes latest reference complex

G Accepts ST setup changes

H Changes the display size (amplitude)

I Moves the ISO point back and forth

J Averaged complex

K Lead label

L Amplitude



## Display Order of ST Complexes

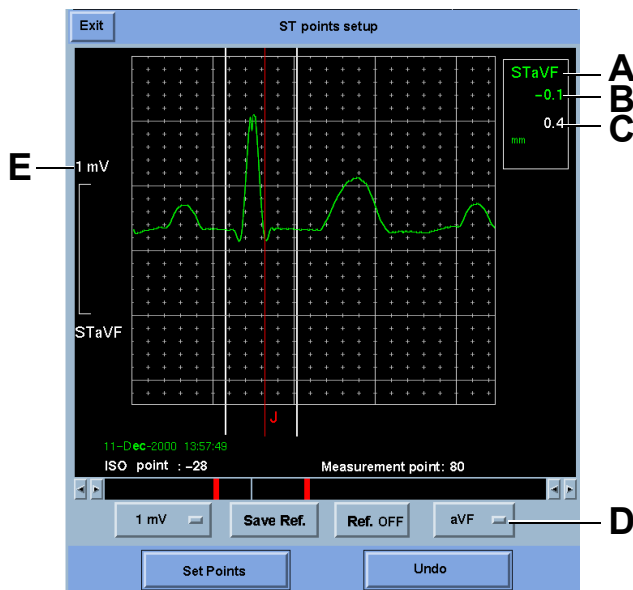
| 3/5-wire<br>1/7 lead |     | 6-wire<br>12-lead |                     |
|----------------------|-----|-------------------|---------------------|
| I                    | aVF | I                 | aVF V3, dV3         |
| II                   | aVL | II                | aVL V4, dV4         |
| III                  | V   | III               | V, V1, dV1 V5, dV5  |
| aVR                  | V   | aVR               | V+, V2, dV2 V6, dV6 |

The averaged complexes and corresponding ST deviation measurements and times are updated as each new complex is computed. If an averaged complex is not available, the reserved display area appears blank. You can change the gain of the complexes and adjust the measurement points.

The current ST measurement point, the QRS offset (J-point), and isoelectric point are labeled and identified by vertical lines. If you make any adjustments to the measurement point, the ST deviation is recomputed and displayed below the current value.

### Zoom Mode

- To examine an individual ST complex in greater detail and view the current ST measurement point selections, click on the desired complex in the **ST Points Setup** screen.



- A Lead label
- B Reference deviation
- C Current deviation value
- D Lead selection
- E Amplitude

### NOTE

A portion of the complex may be clipped if you do not select an appropriate gain.

## Available Functions on ST Points Setup Screen

| Function                                                                                    | Description                                                                                                                                                                                                                                                                                                                                      | Available Settings                                                                                         |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Adjust Display Size                                                                         | Click on the option button and select from available settings.                                                                                                                                                                                                                                                                                   | 0.25, 0.5, 1, 2, 4, and 8 mV/cm                                                                            |
| Save Ref.                                                                                   | Permits comparison of reference ST complex with another, more recent complex<br>Click on <b>Save Ref.</b>                                                                                                                                                                                                                                        | Not applicable                                                                                             |
| Ref. ON/OFF                                                                                 | Allows visual comparison of a stored reference QRS complex to current complex<br>Click on <b>Ref. On/Off</b> toggle button.<br>(Current complex is green and the reference complex is purple.)                                                                                                                                                   | Not applicable                                                                                             |
| Change ISO point                                                                            | Click on the arrow buttons in the lower left of the screen and scroll to the desired setting.<br><br>The number next to <b>ISO point</b> is the time before QRS onset.                                                                                                                                                                           | The setting changes in 4 ms increments each time you click on the arrows.<br><br>Default: <b>-28 ms.</b>   |
| Change ST Deviation Measurement Point                                                       | Click on the arrow buttons in the lower right of the screen and scroll to the desired setting.<br><br>The ST deviation point is set from the QRS offset to the end of the displayed ECG complex (including the T-wave).<br><br>When you change the <b>Measurement Point</b> , the ST deviation is recomputed and the current value is displayed. | The setting changes in 4 ms increments each time you click on the arrows.<br><br>Default: <b>60/80 ms.</b> |
| – Click on <b>Set Points</b> to change settings or <b>Undo</b> to keep previous selections. |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                            |

A marker is put into the ST trend graphs and an entry is stored in the clinical events log. If you move the trend cursor over the marker, the label **CHG** appears.

## Telemetry ST Setup Screen

### NOTE

You can change ST settings for telemetry patients at the Infinity CentralStation, but changes for bedside patients must be done at the bedside monitor.

Use the telemetry ST Setup screen to customize a telemetry patient's ST setup. For bedside patients, please refer to the specific monitor documentation.

### Accessing the Telemetry ST Setup Screen

A patient's ST Setup Screen differs according to transmitter monitoring mode.

To access the Telemetry ST Setup screen:

- 1 Open the patient's Bed View screen.
- 2 Click on **Setup** in the Bed View menu bar.
- 3 Click on **Telemetry**.
- 4 Click on **ST**.

| Transceiver Bed View ST Setup Selections                                                                                                              |                                                                                                      |                                                 |                                                                                                          |                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Selection                                                                                                                                             | 3/5-wire monitoring                                                                                  | 6-wire monitoring                               | TRUST 12-Lead Monitoring                                                                                 | Description                                                                                                                                        |
| ST Lead 1 - 7<br><b>NOTE:</b> In 3-wire monitoring mode, <b>ST Lead 1</b> is set to <b>Lead II</b> and <b>ST Leads 2 - 7</b> are set to <b>NONE</b> . | None, I, II, III, aVR, aVL, aVF, V                                                                   | None, I, II, III, aVR, aVL, aVF, V, V+          | None, I, II, III, aVR, aVL, aVF, 2 chest leads (V1 - V6), 4 reconstructed leads (dV1 - dV6), STVM, STCVM | Sets displayed lead order in the Bed View parameter boxes<br><br>Click on the desired <b>ST Lead x</b> option button and scroll to a lead setting. |
| ST Lead 9 - 14                                                                                                                                        | Not applicable                                                                                       | Settings appear gray and are set to <b>None</b> | None, I, II, III, aVR, aVL, aVF, 2 chest leads (V1 - V6), 4 reconstructed leads (dV1 - dV6), STVM, STCVM | Sets displayed lead order in the Bed View parameter boxes<br><br>Click on the desired <b>ST Lead x</b> option button and scroll to a lead setting. |
| Event Duration                                                                                                                                        | <b>15, 30, 45, 60, 75, 90, 105, and 120 seconds</b><br>For Infinity M300 this is fixed at 60 seconds |                                                 |                                                                                                          | Specifies the time an ST alarm condition must persist before it is classified as a valid alarm                                                     |
| Restore System Defaults                                                                                                                               | See Chapter 5 for details on ST default settings.                                                    | Not applicable                                  | Not applicable                                                                                           | Click on <b>Restore System Defaults</b> .                                                                                                          |
| – Click on <b>Accept</b> to change settings or <b>Undo</b> to keep previous selections.                                                               |                                                                                                      |                                                 |                                                                                                          |                                                                                                                                                    |

## ST Alarms

An ST alarm occurs when the ST deviation measurement for any lead violates the set alarm limits for the user-selected duration (see page 155). An ST alarm is cleared if the majority of intervals fol-

lowing the alarm are within the specified limits. ST alarms are subject to the same alarm guidelines as any other parameter. Refer to page 157 for information on setting up the various alarm functions.

## ST Alarm Messages

The following alarm messages pertain specifically to ST parameters, and are displayed in the status area in Main Screen and in the top waveform channel in Bed View. In addition, the parameter areas

display an abbreviated message in the appropriate parameter field. Please note that the 'x' next to the ST label (e.g. in the message STx:\*\*\* ) is a placeholder for the actual ST lead.

| Message        |                         |                |                                          |                                                |
|----------------|-------------------------|----------------|------------------------------------------|------------------------------------------------|
| Parameter Area | Status/Information Area | Alarm Grade    | Description                              | Action                                         |
| STx: <value>   | STx > UL                | SER            | Value exceeds upper alarm limit.         | Check the patient.<br>Reset alarm limits.      |
| STx: <value>   | STx < LL                | SER            | Value is below lower alarm limit.        | Check the patient.<br>Reset alarm limits.      |
| STx: +++       | STx OUT OF RANGE (HIGH) | SER            | Value exceeds upper measuring range.     | Check the patient.                             |
| STx - - -      | STx OUT OF RANGE (LOW)  | SER            | Value exceeds lower measuring range.     | Check the patient.                             |
| STx: <blank>   | ST x CANNOT LEARN       | Not applicable | Algorithm cannot determine ST values.    | Check the electrodes<br>Relearn                |
| STx: <blank>   | ECG RELEARNING          | Not applicable | ARR/ST morphologies are being relearned. | Wait until the relearning process is finished. |

# Alarms

---

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## Overview

The Infinity CentralStation is the primary alarm annunciator for telemetry patients; for bedside patients the bedside monitor is the primary alarm annunciator.

For **Telemetry patients**, the Infinity CentralStation issues visual and audible alarm signals for the following:

- HR, SpO2, PLS, ST, NIBP (Infinity Telemetry only) and arrhythmia event parameters and associated technical conditions,
- Physiological conditions that are detected by the algorithms,
- Technical network events, and
- Infinity CentralStation or telemetry system error conditions.

For patients connected to **Bedside Monitors** and **MIB-connected devices**, all alarm signals are sent to the Infinity CentralStation for display and broadcast.

**WARNING**

**For primary monitoring and diagnosis of bedside patients, use the bedside monitor. Use the Infinity CentralStation only for remote assessment of a patient’s status.**

The Infinity CentralStation can report surveillance alarm messages for life-threatening and active serious alarms of any bedside monitor within the monitoring unit even though it is not assigned to its current Main Screen.

For information about silencing surveillance alarms, see "Silencing Surveillance Alarms" on page 162.

## Alarm Validation - Infinity Telemetry

The Infinity CentralStation generates an alarm when a parameter remains outside alarm limits longer than an assigned delay time. If the limit violation ceases within the delay time, no alarm occurs.

| Parameter               | Upper Limit Alarm Delay | Lower Limit Alarm Delay |
|-------------------------|-------------------------|-------------------------|
| HR                      | 2 seconds               | no delay                |
| SpO2/SpO2*              | 4 seconds               | 4 seconds               |
| ΔSpO2                   | no delay                | N/A                     |
| PLS/PLS*                | 4 seconds               | no delay                |
| ST (telemetry patients) | user-defined (page 155) | user-defined (page 155) |

## Alarm Validation - Infinity M300

### NOTE

The Alarm delay time from an Infinity M300 device to the Infinity CentralStation is less than 4 seconds.

| Parameter        | Upper Limit Alarm Delay | Lower Limit Alarm Delay |
|------------------|-------------------------|-------------------------|
| HR               | 6 seconds               | 6 seconds               |
| SpO <sub>2</sub> | 6 seconds               | 10 seconds              |
| PLS              | 6 seconds               | 10 seconds              |
| ST               | 60 seconds              | 60 seconds              |

### NOTE

If communication is interrupted between the network and the Infinity M300 device, the Infinity M300 continues monitoring and automatically sets its alarm volume to 100%. Any alarms that are active at the time the Infinity M300 lost connection and all future alarms are annunciated by the Infinity M300 until communication is reestablished with the Infinity CentralStation.

If there is an active alarm while Infinity M300 has lost connection with the Infinity CentralStation, the Infinity M300 screen will not shut off despite the Infinity CentralStation Display Shut Off Timer setting. You can not shut off the Infinity M300 screen manually unless the alarm is paused.

For bedside monitor alarm delays, see the applicable monitor *Instructions for Use*.

## Alarm Grades

All alarms are classified according to three alarm grades:

- Life-threatening
- Serious
- Advisory

Each alarm grade has its own unique tone and color scheme (see page 161). Depending on the

seriousness of the alarm condition, alarms are either *Latching* or *Non-latching*.

### NOTE

If several alarms are simultaneously active for a single patient, the Infinity CentralStation reports the highest grade *active* alarm (alarm that has not been silenced and whose condition remains valid).

Latching alarms

**Latching Alarms** indicate either life-threatening or serious conditions such as an asystole or a ventricular fibrillation. Life-threatening alarm audible *and* visual indicators continue (even if the alarm condition is no longer valid) until the alarm is manually acknowledged.

Serious alarms behave similarly, except that only the visual alarm indicators continue until you acknowledge the alarm.

**NOTE**  
A latching alarm is cleared if another life-threatening or serious alarm occurs for the same parameter.

Non-latching Alarms

**Non-latching Alarms** indicate advisory conditions and continue for as long as the event exists. The visual and audible alarm signals stop automatically

when the alarm condition ends. You can also silence *non-latching* alarms manually, as described on page "Silencing Alarms" on page 163.

| Visual and Audible Alarm Signals |                                                                                    |                                                             |                                                                  |
|----------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------|
| Alarm Grade and                  | Life-threatening (e.g. asystole, ventricular fibrillation)                         | Serious (limit violation)                                   | Advisory (e.g. network events, bedside events such as lead-off). |
| Priority                         | Highest priority                                                                   | Medium priority                                             | Lowest priority                                                  |
| Alarm Indication in Main Screen  | Parameter area flashes red.                                                        | Parameter area flashes yellow.                              | Parameter area flashes white.                                    |
| Alarm Indication in Bed View     | Parameter area of respective parameter flashes red.                                | Parameter area of respective parameter flashes yellow.      | Parameter area of respective parameter flashes white.            |
| Alarm Tone                       | Two-tone, high pitched, rapid tone                                                 | Two, medium-pitch, short tones followed by a pause          | Short, single, low-pitch tone every 10 seconds.                  |
| Message Location in Main Screen  | Message displays in red in top waveform channel.                                   | Message displays in yellow in top waveform channel.         | Message displays in white in top waveform channel.               |
| Message Location in Bed View     | Message displays in red in information area.                                       | Message displays in yellow in information area.             | Message displays in white in information area                    |
| Latching behavior                | Message displays in parameter area and tone continues until alarm is acknowledged. | Message stays in waveform area until alarm is acknowledged. | Not applicable.                                                  |



## The Surveillance Feature

---

This feature provides alarm messages for life-threatening and serious alarms associated with bedside monitors that are not currently assigned to the Infinity CentralStation Main Screen but are located within the same monitoring unit.

### Display of Surveillance Alarm Messages

Surveillance alarms are displayed at the bottom-left of the Main Screen status area. A surveillance alarm message includes the bed label and alarm grade.

If the surveillance alarm is for a wireless bedside monitor an antenna icon precedes the alarm message.

 **Serious Alarm at <BED2>**

Message background colors vary with alarm grade and status.

| Alarm Grade      | Status                                                                             | Background Color |
|------------------|------------------------------------------------------------------------------------|------------------|
| Life-threatening | Alarm condition is valid and the alarm has not been silenced at the bed.           | Flashing red     |
|                  | Alarm condition is no longer valid but the alarm has not been silenced at the bed. | Flashing red     |
|                  | Alarm condition is valid and the alarm has been silenced at the bed.               | Solid red        |
| Serious          | Alarm condition is valid and the alarm has not been silenced at the bed.           | Flashing yellow  |
|                  | Alarm condition is valid and the alarm has been silenced at the bed.               | Solid yellow     |

### Multiple Alarm Situations/Priorities

If several beds issue surveillance alarms, the Infinity CentralStation cycles to the next message in the list once every 3 seconds.

Audible Alarm Signals

The Infinity CentralStation sounds the latest alarm condition with the highest priority. Audible alarm signals accompanying surveillance alarms act the same as other alarms.

Silencing Surveillance Alarms

A surveillance alarm continues until it is either acknowledged or the alarm condition ceases. You can silence the audible alarm signal of a surveillance alarm locally by clicking on **Alarm Silence** in the Main Screen menu bar or by pressing **F1**.

| Special Surveillance Alarm Conditions                                                                                       |                                                |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Condition                                                                                                                   | Effect on surveillance alarm                   |
| Bedside CPS/IDS goes offline or fails while a surveillance alarm for that bed is reported at the Infinity CentralStation.   | Surveillance alarm is canceled at the bedside. |
| A bedside monitor goes offline or fails while a surveillance alarm for that bed is reported at the Infinity CentralStation. | Surveillance alarm is canceled at the bedside. |
| The bedside CPS/IDS or monitor changes to stand-alone mode.                                                                 | Surveillance alarm is canceled at the bedside. |
| The bedside CPS/IDS or monitor detects that it has the same IP address as another device on the network.                    | Surveillance alarm is canceled at the bedside. |
| Monitor is removed from the network.                                                                                        | Surveillance alarm is canceled.                |

## MIB Ventilator Alarms

For MIB-connected ventilator events, the Infinity CentralStation provides alarm messages which are accompanied by a serious alarm tone. However, not all ventilator messages are reported at the Infinity CentralStation. MIB ventilator alarms are described in detail in "VentCentral Option" on page 237.

The most recent MIB ventilator alarm message appears in the patient's top waveform channel in orange on the Main Screen.

In Bed View, the MIB ventilator alarm message appears in the information area in orange.

MIB ventilator alarm messages are *non-latching*. If a patient experiences multiple alarm conditions simultaneously, the messages rotate. Unlike other Bed View parameters, the parameter boxes of MIB ventilator parameters do not blink while in alarm.

## Alarm Groups

Use an alarm group to configure monitors and telemetry patients to display their highest grade active alarm at each device in the alarm group. If multiple beds in the alarm group detect an alarm, each corresponding alarm message briefly rotates along the bottom of the screen on each device. To assign a telemetry channel to an alarm group,

see page 67. For bedside patients, you must set the alarm group at the bedside monitor. (Refer to the monitor *Instructions for Use*.)

### CAUTION

Care must be taken when assigning patients to alarm groups. Assigning a patient to an incorrect alarm group could result in alarms going unheard or unanswered by the intended caregiver.

## Silencing Alarms

The Audio Pause (1 min) function clears all audible and visual alarm indicators for life-threatening and serious *latched* alarms (see page 160). This function also suspends audible signals for life-threaten-

ing and serious alarms associated with conditions that are still valid for one minute. There are two ways to silence alarms.

|                           |                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Audio Pause (1 min) | Silences active alarms for all local patients at the Infinity CentralStation for 1-minute.                                                                                                                                                                                                       |
| Audio Pause               | Silences active alarms for an individual bedside patient at the Infinity CentralStation and at the bedside monitor for one minute, when the Bed Audio Paused function is enabled at the Infinity CentralStation (see page 61).<br><br>Silences alarms for an individual local telemetry patient. |

#### NOTE

- The Infinity CentralStation's *Audio Pause (1 min)* function also deactivates one-time alarms. The **Audio Pause (1 min)** button does not remain selected if it has deactivated a one-time alarm. Also, you cannot deactivate the *Audio Pause (1 min)* state by clicking on the **Audio Pause (1 min)** button again.
- If a new alarm occurs while an alarm is silenced, the Infinity CentralStation announces the new alarm. Silencing the new alarm resets the timer for all previously silenced alarms for another minute.

#### NOTE

- The Infinity M300's Audio Paused timer is 2 minutes.
- When the Infinity M300 is already in the Audio Pause state, pressing the Audio Pause button again at the Infinity CentralStation causes the Audio Pause state to be cancelled.

After the *Audio Pause (1 min)* period expires, the alarm tone resumes if the alarm condition still exists. Visual alarm indicators are also reactivated.

## Silencing Local Alarms

To silence all active local alarms (telemetry, bedside, and MIB alarms) at the Infinity CentralStation for one minute:

- Click on **Audio Pause (1 min)** in the Main Screen menu bar, **or**
- Press the **F1** key from the keyboard.

A local **Audio Pause (1 min)** has the following consequences:

- The alarm tone stops for all active alarms at the Infinity CentralStation for one minute.
- The affected parameter area continues to flash in the color of the highest grade alarm.

- Alarm messages and the yellow **Audio Pause** icon displayed next to the message remain displayed.
- The orange ventilator MIB Alarm Tone icon remains displayed (for an MIB alarm).
- The **Audio Pause (1 min)** button remains selected.

After the **Audio Pause (1 min)** period expires, the alarm tone resumes if the alarm condition is still valid.

In contrast to a local **Audio Pause (1 min)**, which silences **all** active alarms at the Infinity CentralStation only, a Bed Audio Pause silences all alarms for an individual patient at the Infinity CentralStation and at the bedside.

### From Main Screen

- For Bedside Monitor alarms, click on the yellow **Audio Pause** icon, which appears in the patient's waveform channel during an alarm. If the icon appears unavailable for selection for a bedside patient, this indicates that the required **Audio Pause** function has not yet been enabled at the Infinity CentralStation (see page 61), or that the patient is outside the monitoring unit (in which case you cannot silence the alarm).
- Press the **F1** key from the keyboard or click the yellow **Audio Pause (1 min)** icon in the Main screen to silence ventilator and MIB alarm tones.

**NOTE**

The crossed orange bell-shaped MIB Alarm Tone button (see illustration) appears in the lower-right corner of patient's top waveform channel.

**It is important to understand** that, when this button is selected and set to 'Off,' audible ventilator or MIB alarm annunciation are disabled at the Infinity CentralStation until the button is selected again to set it back to 'On.' This button should not be used in an attempt to silence ventilator or MIB alarms.

You should use the keyboard's F1 key or the **Audio Pause (1 min)** icon to silence the alarm.

**From Bed View**

- Click on the **Audio Pause** button in the Bed View menu bar.

**NOTE**

You cannot silence MIB ventilator alarms in Bed View. The button appears unavailable for selection.

**NOTE**

Bed Audio Paused does not affect the *Audio Paused* state of other patients or any other local alarms that may be present at the Infinity CentralStation. It only silences the alarms at the selected patient.

Silencing a bedside alarm has the following consequences:

- silences all active alarms for an *individual* bedside patient at the bedside monitor **and** at the Infinity CentralStation. This function is only available for bedside patients within the monitoring unit of the Infinity CentralStation server whose Bed Silence function is enabled (see page 61).
- silences all active alarms for an individual local telemetry patient at the Infinity CentralStation.
- freezes the visual alarm indicators in the color of the highest grade active alarm.

After the *Audio Paused* period expires, the alarm tone resumes if the alarm condition is still valid. Also, the visual alarm indication is reactivated.

## Configuring a Patient's Alarm Setup

The **Setup - Alarm Limits** screen appearance varies depending on the parameters monitored. The following illustration shows an example of the Setup - Alarm Limits screen.

To adjust alarm tone, see page 61.

Exit

Setup - Alarm Limits

| Parameters | All Alarms | Upper Limit | Current Value | Lower Limit | Alarm Archive |
|------------|------------|-------------|---------------|-------------|---------------|
| HR         | ON         | 120         |               | 45          | REC           |
| SpO2       | OK OFF     | 100         |               | 90          | REC           |
| PLS        |            | 120         |               | 45          | REC           |
| STI        |            | 2.0         |               | -2.0        | REC           |
| STII       |            | 2.0         |               | -2.0        | REC           |
| STIII      |            | 2.0         |               | -2.0        | REC           |
| STaVF      |            | 2.0         |               | -2.0        | REC           |
| STaVL      |            | 2.0         |               | -2.0        | REC           |
| STaVR      |            | 2.0         |               | -2.0        | REC           |
| STV        |            | 2.0         |               | -2.0        | REC           |
| PVC/min    |            | 10          |               |             | REC           |
|            |            |             |               |             | REC           |

D

Auto Set

Arrhythmia...

B

C

Restore System Defaults

- A Parameter row in Configuration mode
- B Opens Arrhythmia Setup screen
- C Restores system default settings (only available on telemetry patient screen)
- D Resets alarm limits to automatically calculated values (see page 168)

## Opening the Alarm Limits Setup Screen

---

To open the Setup screen:

- 1 Open the patient's Bed View screen.
- 2 Click on Setup in the Bed View menu bar.
- 3 Click on **Alarm Limits...**

## Pausing Alarms

### Telemetry Patient

To pause alarms for Telemetry patients:

- 1 Open the patient's Bed View screen.
- 2 Click on **Alarm Paused** in the Bed View menu bar.

After you click on **Alarm Paused**, all current *Active* alarms clear and additional alarm events are suppressed for a predefined time (see page 67 and

page 88). During the *Alarm Paused* time period, the banner ALARM PAUSED and time remaining for the pause displays. For Infinity Telemetry only, the user can set the Alarm Pause function to "No Time Out." In this case the banner will say ALL ALARMS OFF instead.

### Bedside Patient

To pause alarms for Bedside patients:

- 1 Open patient's **Alarm Limits Setup** screen (see page 88).
- 2 Click on the parameter whose alarm you wish to turn on/off.  
  
The parameter row changes to *Configuration Mode*.
- 3 Set the **On/Off** toggle button in the **All Alarms** column.
- 4 Click on **OK**.
- 5 Repeat steps 2 through 4 for additional parameter alarms.

#### NOTE

When a parameter alarm is *Off*, all parameter settings in the table row display in yellow and a crossed bell icon appears in the *All Alarms* column.

## Changing a Parameter's Alarm Limits

To change Alarm Limits for a single parameter:

- 1 Open patient's **Alarm Limits Setup** screen (see page 167).
- 2 Click on the parameter whose alarm limits you wish to change. The parameter row changes to *Configuration Mode*.
- 3 Click on the up/down arrows in the **Upper Limit** and **Lower Limit** columns to set limits.
- 4 Click on **OK**.
- 5 Repeat steps 2 through 4 for each parameter.

| Alarm Limit Ranges for Telemetry Patients |                  |                   |         |                   |                |
|-------------------------------------------|------------------|-------------------|---------|-------------------|----------------|
| Parameter                                 | Patient Category | Upper Limit Range | Default | Lower Limit Range | Default        |
| HR                                        | Adult            | 20 – 300 bpm      | 120 bpm | 15 – 295 bpm      | 45 bpm         |
|                                           | Pediatric        | 50 – 150 bpm      | 150 bpm | 50 – 150 bpm      | 50 bpm         |
| SpO <sub>2</sub>                          | Adult            | 21 – 100%         | 100%    | 20 – 99%          | 90%            |
|                                           | Pediatric        | 21 – 100%         | 100%    | 20 – 99%          | 90%            |
| PLS                                       | Adult            | 35 – 250 bpm      | 120 bpm | 30 – 245 bpm      | 45 bpm         |
|                                           | Pediatric        | 50 – 150 bpm      | 150 bpm | 50 – 150 bpm      | 50 bpm         |
| ST                                        | Adult            | -1.49 – +1.5 mV   | +2.0 mV | -1.5 – +1.49 mV   | -2.0 mV        |
|                                           | Pediatric        | -14.9 – +15.0 mm  | 20.0 mm | -15 – +14.9 mm    | -20.0 mm       |
| PVC/min                                   | Adult            | 1 – 60            | 10      | Not applicable    | Not applicable |
|                                           | Pediatric        | 1 – 60            | 10      | Not applicable    | Not applicable |

### Alarm Limit Ranges for Bedside Patients

Refer to the bedside monitor *Instructions for Use*.

| Automatic Alarm Limits Selection (Auto Set) |                                                                               |                                                                                |
|---------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Parameter                                   | Upper Limit                                                                   | Lower Limit                                                                    |
| SpO <sub>2</sub>                            | 105% of current value                                                         | 95% of current value                                                           |
| PVC/min                                     | N/A                                                                           | N/A                                                                            |
| HR, PLS                                     | 125% of current value                                                         | 80% of current value                                                           |
| ST                                          | Current value <i>plus</i> absolute value of upper alarm limit default setting | Current value <i>minus</i> absolute value of lower alarm limit default setting |

#### NOTE

If there is no current value for the parameter, the alarm limits do not change.



## Initiating an Auto Set

As soon as you click on **Auto Set**, the alarm limits for all monitored parameters are replaced. There is no confirmation process.

To use the Auto Set function:

- 1 Open patient's **Alarm Limits Setup** screen (see page 167).
- 2 Click on **Auto Set**.

### NOTE

If a parameter value is outside the limit range during the **Auto Set**, alarm limits are not changed for that parameter.

The automatically calculated alarm limits can be manually replaced at any time (see page 167).

## Turning Alarm Recordings ON/OFF

To configure timed recordings, see page 63.

To turn Alarm Recordings On/Off:

- 1 Open patient's **Alarm Limits Setup** screen (see page 167).
- 2 Click on the table row of the parameter whose alarm recording function you wish to change. The parameter row changes to *Configuration Mode*.
- 3 If the last column heading is '**Record**', select **ON** or **OFF** by clicking on the toggle button. If the last column heading is '**Alarm Archive**', click on the desired selection.
- 4 Click on **OK**.
- 5 Repeat steps 2 through 4 for each parameter.

## Special Conditions

| What happens if...                                                                                       | Effect on alarm recording function                                                                                                                           |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an alarm is reactivated (for example when an alarm condition is valid beyond the Alarm Silence state.)   | A new alarm recording is <i>not</i> generated.                                                                                                               |
| an alarm is silenced during an alarm recording or pending recording                                      | All alarm progress and pending recordings are printed completely.                                                                                            |
| a parameter's alarm function is turned off                                                               | Alarm recording is still generated <i>provided</i> the alarm recording function is turned on.                                                                |
| an <i>alarm waveform</i> is activated in the Telemetry Recording Setup screen (see page 74) <sup>1</sup> | The waveform associated with the alarm parameter is printed in the bottom channel of the strip recording (instead of the waveform assigned to that channel). |
| <sup>1</sup> This function pertains only to telemetry patients.                                          |                                                                                                                                                              |

Turning Event Storage On/Off

When a parameter’s storage and alarm functions are turned on, waveforms are stored automatically at the bedside for each validated alarm condition.

To turn Event Storage On/Off:

- 1 Open patient’s **Alarm Limits Setup** screen (see page 167).
- 2 Click on the table row of the parameter whose alarm recording function you wish to set up. The parameter row changes to *Configuration* mode.

- 3 Click on the arrow buttons in the **Alarm Archive** column and scroll to **STORE**.
- 4 Click on the **OK** button.
- 5 Repeat steps 2 through 4 for each parameter.

Special Conditions

| What happens if...                                                                                      | Effect on alarm storage function                                                                                                     |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| an alarm is reactivated (for example, when an alarm condition is valid beyond the Alarm Silence state.) | A new alarm waveform is <b>not</b> stored.                                                                                           |
| the <i>alarm waveform</i> is activated in the Telemetry Recording Setup menu <sup>1</sup>               | The waveform associated with the alarm parameter is stored in the bottom channel (instead of the waveform assigned to that channel). |
| <sup>1</sup> This function pertains only to telemetry patients.                                         |                                                                                                                                      |

## Recordings / Reports

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## Overview

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### WARNING

**If you require a diagnostic quality recording you must use a Rest ECG Report (see page 295). Using any recording other than Rest ECG Report may lead to misdiagnosis.**

From the Infinity CentralStation, you can request several types of recordings on the R 50-N recorder.

### NOTE

The R50-N recorders should always be installed and configured as part of the Infinity M300 system.

- An optional *network* laser printer

### NOTE

All Telemetry ECG recordings and reports are filtered at a bandwidth of 0.5 - 40 Hz. For the filtering bandwidth of a bedside patient's ECG recording/report, see the bedside monitor Instructions for Use.

When you request a recording for a *bedside patient* at the Infinity CentralStation, the request is sent back to the bedside monitor for printing on a recorder assigned at that monitor. For telemetry patients, recordings are printed at the recorder assigned to the local Infinity CentralStation (see page 63).

If an assigned recorder is unavailable at the time of a recording request, the request (or recorded data) is stored and executed as soon as the recorder becomes available.

If the Full/Event Disclosure and/or VentCentral options are enabled, you can also print a variety of reports on a laser printer.

Status messages in the Main Screen status area indicate when a request is accepted, canceled, or rejected. See page 104 for a list of Status messages.

## Laser Printers

---

A laser printer can print Infinity CentralStation screens and reports. Refer to the manufacturer's operating instructions for specific information regarding the use of your particular type of laser printer.

## Processing of Strip Recordings

---

A recording request can generate an *active* or *pending recording*.

### Active Recordings

If a recorder is available and no pending recordings of higher priority exist, the recording begins printing immediately.

If a recording is interrupted and another recorder is available, the complete recording is rerouted and printed at the available recorder. If no recorder is available, the recording or request becomes a *pending recording* and is added to the top of the *pending recording* list.

### Requesting a Remote Recording

Other Infinity Central Stations within the same monitoring unit can request recordings of local patients. These requests use the recording attributes of the remote device.

### Pending Recordings

If no recorder is available to process a print request, the data or recording request and is stored until it can be printed.

#### NOTE

The actual data of a *timed* recording is stored, then printed. For a *continuous* recording only the request is stored; the data is that which is current at the actual print time.

*Pending recordings* are stored on a first-in, first-out basis. As soon as the buffer is full, the oldest request is deleted. The buffer can store the following number of recordings for each patient:

- two manually requested *timed* recordings
- one *continuous* recording
- one automatic alarm or event recording.

## Recorder Setup

---

The R 50-N recorders has two buttons:

- The ***mm/s*** button on the recorder's front panel (**Alternate Speed** on older recorders)  
The **Alternate Speed** button allows you to change the recording speed while a recording is in progress. The recorder stops briefly and then restarts automatically at the new recording speed.
- The **Stop** button stops a recording in progress.

## Replacing the Recorder Paper

To replace the recorder paper:

- 1 Open the paper door and remove the empty paper roll and any paper remaining in the printing mechanism.
- 2 Place a new paper roll with printed side facing up into the spool holder. Unroll a few inches of paper from the bottom.
- 3 Align the paper roll with the paper guides, and close the door. (If not aligned properly, the paper may jam.)
- 4 Generate a *timed* recording to verify that the recorder is connected properly, and the paper is loaded correctly.

## Setting up a Recording

For general recording setup attributes, see page 63. For telemetry-specific recording setup information, see page 74.

## Recording Types

| Recording Type           | Priority | Description                                                                                                                                | Alarm Recording Override                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous recording     | 1        | A strip recording that runs until interrupted manually (generated from Infinity CentralStation).                                           | A continuous recording request will cancel any alarm recording in progress.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Alarm recording          | 2        | A timed recording generated automatically in response to a limit violation for a parameter whose recording function is enabled.            | <ul style="list-style-type: none"> <li>– If a second alarm recording is requested within 5 seconds of the first, the second request is ignored if the alarm grade is an equal or a lower alarm grade (e.g. if the first is a life-threatening and the second is a serious alarm).</li> <li>– If the second alarm recording is a higher alarm grade, the first recording is canceled and the second is printed.</li> <li>– If the second request occurs 5 seconds after the first, both recordings are either printed or are stored and printed later.</li> </ul> |
| Timed recording (Manual) | 3        | A strip recording that consists of delay and real-time data and runs for a specified time (generated from or Infinity CentralStation).     | Not applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Event recording          | 4        | A timed recording that is generated automatically in response to an occurrence of an arrhythmia event whose recording function is enabled. | Not applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Timed Recordings

Timed recordings are strip recordings of a specified duration. They contain *delay* data originating *before* the recording is initiated and *real-time* data acquired *after* the recording started.

### NOTE

If the patient was monitored for less time than the specified *delay*, the amount of delay data is adjusted accordingly. If no delay data is available at the recording request time, that part of the printed waveform is blank.

Timed recordings consist of up to two waveforms. For a waveform to be printed, it must first be displayed. Pressure waveforms are recorded in either *Standard* or *Pressure Overlap* mode, depending on their configuration at the bedside monitor (see Instructions for Use for specific bedside monitor). The size of the printed waveforms is identical to the screen display.

## Manual vs. Automatic Recordings

Timed recordings are either requested manually or generated automatically when a parameter or arrhythmia event occurs whose recording function is enabled. For detailed information on how to enable/disable an event's/parameter's automatic recording function, see page 134 and page 169.

An alarm strip recording is the same as one manually requested except that it has a different *recording type* in the header.

## Telemetry Recordings

All alarm-generated telemetry strip recordings ignore *Manual* timed settings and use the following fixed settings.

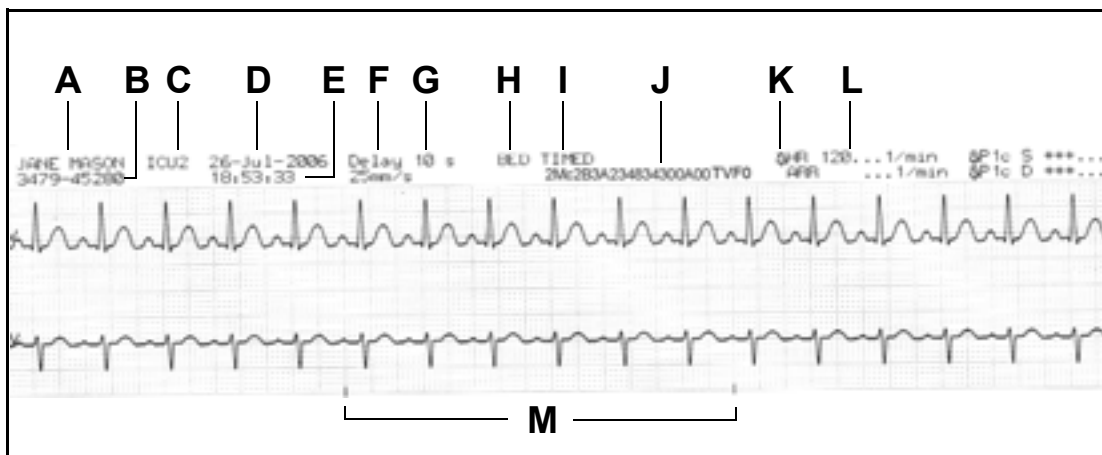
- Duration: 20 seconds
- Delay: 10 seconds
- Speed: 25 mm/sec

## Header Information

The header along the top of the recording contains a diagnostic code, information about the patient, the recorder settings, the monitor, and the monitored parameters. It also shows the values and *Alarm Silence* indicators that are valid at the time of the recording request.



## Timed recording with header



- A** Patient Name
- B** Patient ID
- C** Bed Label
- D** Date of recording request
- E** Time the recording was printed
- F** Selected delay
- G** Selected recording speed
- H** Recorder label
- I** Recording type: CENTRAL TIMED, CENTRAL CONT., alarm string for alarm recording, arrhythmia label for event recording
- J** Diagnostic Code (see page 178)
- K** Alarm Silence indicator
- L** Parameter value at the time of the request
- M** 3-Second interval markers

**Diagnostic Code1**

| <b>Digit</b> | <b>Description</b>                              | <b>Possible Values</b> | <b>Definition</b>                    |
|--------------|-------------------------------------------------|------------------------|--------------------------------------|
| 1            | Lead processed for VF and pacer pulse rejection | X                      | None                                 |
|              |                                                 | 1                      | I                                    |
|              |                                                 | 2                      | II                                   |
|              |                                                 | 3                      | III                                  |
|              |                                                 | S                      | aVR                                  |
|              |                                                 | T                      | aVL                                  |
|              |                                                 | U                      | aVF                                  |
|              |                                                 | V                      | V                                    |
|              |                                                 | +                      | V+                                   |
|              |                                                 | a                      | V1                                   |
|              |                                                 | b                      | V2                                   |
|              |                                                 | c                      | V3                                   |
|              |                                                 | d                      | V4                                   |
|              |                                                 | e                      | V5                                   |
|              |                                                 | f                      | V6                                   |
|              |                                                 | A                      | dV1                                  |
|              |                                                 | B                      | dV2                                  |
|              |                                                 | C                      | dV3                                  |
|              |                                                 | D                      | dV4                                  |
|              |                                                 | E                      | dV5                                  |
|              |                                                 | F                      | dV6                                  |
| 2            | ECG filter                                      | M                      | Monitor                              |
|              |                                                 | D                      | Off                                  |
|              |                                                 | E                      | ESU                                  |
| 3            | Pacemaker detection                             | C                      | On - Artifact Rejection <Medium>     |
|              |                                                 | c                      | Off - Artifact Rejection <Medium>    |
|              |                                                 | H                      | Fusion - Artifact Rejection <Medium> |
| 4            | QRS/ARR processing                              | 2                      | ECG1 & ECG2                          |
|              |                                                 | 1                      | ECG1                                 |

| Digit | Description                             | Possible Values | Definition                                                        |
|-------|-----------------------------------------|-----------------|-------------------------------------------------------------------|
| 5     | Patient category/<br>QRS classification | <Space>         | Adult, neither lead completed learning                            |
|       |                                         | 1               | Adult, ECG1 lead completed learning                               |
|       |                                         | 2               | Adult, ECG2 lead completed learning                               |
|       |                                         | B               | Adult, ECG1&2 lead completed learning                             |
|       |                                         | n               | Neonate                                                           |
| 6     | Leads available<br>for processing       | 0               | No valid lead to process                                          |
|       |                                         | 1               | ECG1 is valid to process                                          |
|       |                                         | 2               | ECG2 is valid to process                                          |
|       |                                         | 3               | ECG1 & ECG2 are valid to process                                  |
| 7     | VT count                                | 5-F             | Value = VT count<br>(where A-F corresponds to 10-15)              |
| 8     | VT rate                                 | 0-A             | Value = (VT rate - 100)/10<br>(where A corresponds to 10)         |
| 9     | SVT count                               | 3-A             | value = SVT count<br>(where A corresponds to 10)                  |
| 10    | SVT rate                                | 0-A             | value = (SVT rate - 100)/10<br>(where A corresponds to 10)        |
| 11    | TACH count                              | 5-F             | value = TACH count<br>(where A-F corresponds to 10-15)            |
| 12    | TACH rate                               | 0-A             | value = (TACH rate - 100)/10<br>(where A corresponds to 10)       |
| 13    | BRDY rate                               | 0-F             | value = (BRDY rate - 30)/5<br>(where A-F corresponds to 10-15)    |
| 14    | PAUS rate                               | 0-5             | value = (PAUS rate - 1.0)/0.5<br>(where A-F corresponds to 10-15) |
| 15    | HR source                               | E               | ECG is HR source                                                  |
|       |                                         | P               | IBP (AP) is HR source                                             |
|       |                                         | S               | SPO2 is HR source                                                 |
| 16    | RESP mode                               | O               | RESP monitoring off                                               |
|       |                                         | M               | Manual                                                            |
|       |                                         | A               | Automatic                                                         |
|       |                                         | <Space>         | N/A                                                               |

| Digit  | Description                                  | Possible Values                 | Definition                                                                                                                          |
|--------|----------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 17     | RESP size                                    | 1-K<br><Space>                  | value = (RESP size)/5<br>(where A-K corresponds to 10-20)<br>N/A                                                                    |
| 18, 19 | Minutes since breath detector initialization | 00-99<br><Space>                | Number of minutes that have elapsed since the breath detector was initialized (where 99 corresponds to $\geq 99$ minutes)<br>N/A    |
| 20     | Not used                                     | <Space>                         | N/A                                                                                                                                 |
| 21     | Monitor model                                | A<br>B<br>I<br>J<br>K<br>T<br>V | SC 9000<br>Delta/Delta XL/Kappa/GammaX XL<br>SC 6000<br>Gamma/Gamma XL/Vista<br>Infinity ACS<br>Infinity Telemetry<br>Infinity M300 |
| 22-26  | Software version                             | xxxxx<br>(ASCII)                | First five characters of base software (for example, VA1-1)                                                                         |

## Continuous Recordings

Continuous recordings run until stopped manually and can only be generated for an individual patient. They consist of up to two waveforms which are printed the same size as those displayed.

Pressure waveforms are recorded in either *Standard* or *Overlapped* mode, depending on selected bedside monitor setting.

### Header Information

Continuous recordings contain a header, which is identical to that of timed recordings, except that the recording type data is different and there is no data for DELAY TIME.

## Reports

---

Depending on the configuration of your Infinity CentralStation, the following reports can be generated.

- Simultaneous ECG Report
- Full Disclosure (Strip, Hour, 24-Hour, Shift, Selected Strip, Graphical Trend, Patient Status, and Caliper Reports)
- Event Disclosure (Strip, Graphical Trend, Shift, and Selected Events Reports)
- VentCentral Reports

Reports are processed and printed even if you leave the respective application. Pacer marks, Pacemaker identifiers, and time change indicators are printed, where applicable.

All printed reports include the following:

### Printed Report Header

- Report Title
- Patient Name
- Patient ID
- Date of Birth
- Care Unit Label
- Bed Label
- Admit Date
- Pacer/ICD/PCD Identifier
- Report Stop Time
- Duration

### Printed Report Footer

- COMMENTS area for handwritten comments
- Page Number and Total Number of Pages (e.g. Page 1 of 3)
- Date area (for dating signature)
- Signature area

## Simultaneous ECG Report

### WARNING

**If you require a diagnostic-quality recording, you must use a Rest ECG Report.**

A simultaneous ECG Report for any patient monitored by the Infinity CentralStation can be printed from the patient's Bed View. The report includes three 4 x 2.5 second 25mm per second ECG waveforms for each lead (up to 12), one 10 second ECG waveform for lead II, and ST values for each lead. This is a Non-Diagnostic report. To request a Simultaneous ECG Report, see page 194.

## **Full Disclosure Strip Report**

The Full Disclosure strip report prints the screen-displayed waveforms. A report page may contain a 2-channel strip, each twenty seconds long with data centered around the cursor time.

To request a Full Disclosure strip report, see page 193.

## **Full Disclosure Hour Report**

The Hour Report consists of up to three user-selectable waveforms in a compressed format of one minute per row. There are 60 minutes of data per page. Each data channel produces one report page. Data is captured 30 minutes before cursor time.

To request this report, see page 193.

## **Full Disclosure 24 Hour Report**

The 24 Hour Report is similar to the Hour Report except that it prints 24 hours of data (24 pages). The end time is the cursor time and data is collected 24 hours before that time. The report supports one waveform selection from one of three channels (Channel 1 is the default).

To request this report, see page 193.

## **Event Disclosure Strip Reports**

An Event Disclosure strip report contains 20 seconds of data centered around the time of the event. It also includes event class label, time, date, and the ECG lead on which an event is detected (if applicable).

To request this report, see page 193.

## Shift Report

You can request a 2, 4, 8, 12, or 24 hour shift report from Full or Event Disclosure. From either application the cursor time represents the report 'stop time'. The report 'start time' equals 'stop time' minus the selected report length.

The shift report has four sections, described in the following table:

| Shift Report Section | Description/Contents                                                                                                                                                                                                                                                                                               | Maximum Pages            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Shift Report Cover   | <ul style="list-style-type: none"> <li>Header</li> <li>Three waveform channels (10 seconds each)</li> <li>Comments/signature/date field</li> <li>Footer</li> </ul>                                                                                                                                                 | One                      |
| Patient Status       | <ul style="list-style-type: none"> <li>Header</li> <li>HR (calculated from trend data)</li> <li>Bradycardia</li> <li>Pacemaker</li> <li>Ventricular and Supraventricular Ectopics</li> <li>S-T Deviation</li> <li>Caliper Results</li> <li>Latest Alarm Event (within alarm categories)</li> <li>Footer</li> </ul> | One                      |
| Graphical Trends     | <ul style="list-style-type: none"> <li>Header</li> <li>Trend graphs</li> <li>Footer</li> </ul>                                                                                                                                                                                                                     | Two<br>(6 trends/page)   |
| Selected Events      | <ul style="list-style-type: none"> <li>Header</li> <li>Event strips (10 seconds each) with annotations</li> <li>Footer</li> </ul>                                                                                                                                                                                  | Three<br>(4 events/page) |

### NOTE

- You must select each event on the report. Click on the printer icon so that a check mark displays on the icon (see page 216).
- You can select up to 12 events that occurred before the request time.
- The parameter values at the time of the event appear at the top of each strip.
- Event selections are saved through program resets and power losses.

## Shift Report Cover

The form is titled "2 Hour Shift Report" and contains the following sections:

- Header Section:**
  - A:** A bracket indicates the time range from "Start Time: 14:21:54 08-Aug-2005" to "Stop Time: 16:21:54 08-Aug-2005".
  - B:** Points to the "Delay: 10 Seconds" and "Speed: 25mm/Second" settings.
- Patient Information Section:**
  - C:** Points to the "Patient ID" field.
  - D:** Points to the "Date of Birth: 0-Feb-0" and "Race/CO/POD" fields.
- ECG Waveform Section:**
  - E:** A bracket indicates the duration of the three ECG waveforms (I, V2, and I), each showing 10 seconds of data (5 seconds pre-event and 5 seconds real-time).
- Comments Section:**
  - Two sections labeled "COMMENTS:" with lines for text entry.
  - Each section includes fields for "Signature:" and "Date:".
- Footer:**
  - Page 1 of 4

- A** Cursor time minus selected report length (2, 4, 8, 12, 24 hrs)
- B** Delay
- C** Recording Speed
- D** Patient information
- E** Each waveform contains 5 seconds of pre-event data and 5 seconds of real-time data.



## Patient Status

The *Patient Status* section of the **Shift Report** is a summary report focusing on a patient's arrhythmia status. A **Patient Status Report** is also

available as an independent report and can be configured to print 2, 4, 8, 12, or 24 hours of data (see page 194).

|   |                             |               |                                  |                 |             |
|---|-----------------------------|---------------|----------------------------------|-----------------|-------------|
| A | Patient:                    |               | Patient Status Report            |                 |             |
|   | Care Unit: VS               | Bed: T295, 13 | Start Time: 13:13:53 01-Aug-2005 |                 |             |
|   | Admit Date: 0-Feb-0         |               | Stop Time: 15:13:53 01-Aug-2005  |                 |             |
|   |                             |               | Duration: 2 Hours                |                 |             |
|   | Patient ID:                 |               | Date of Birth: 0-Feb-0           | Patient/CD/POD: |             |
| J | HR (Trended)                |               | Caliper                          | Reference       | Latest      |
|   | Minimum                     | 80            |                                  | 15:13:53        | 01-Aug-2005 |
|   | Maximum                     | 80            |                                  |                 |             |
|   | Average                     | 80            |                                  |                 |             |
| I | Pacemaker                   |               | IRI                              |                 |             |
|   | % Paced                     | 0             | PR                               |                 |             |
|   |                             |               | QRS                              |                 |             |
|   |                             |               | QT                               |                 |             |
|   |                             |               | QTc                              |                 |             |
| H | Bradycardia                 |               | S-T Deviation                    | Minimum         | Maximum     |
|   |                             |               | STI                              | -0.1            | 1.0         |
|   |                             |               | STII                             | -0.2            | 1.0         |
|   |                             |               | STIII                            | -0.2            | 1.0         |
|   |                             |               | STaVR                            | -0.1            | 1.0         |
|   |                             |               | STaVL                            | -0.2            | 1.0         |
|   |                             |               | STaVF                            | -0.1            | 1.0         |
|   |                             |               | STV2                             | -0.2            | 1.0         |
|   |                             |               | STV5                             | -0.3            | 1.0         |
|   |                             |               | STV6                             | -0.1            | 1.0         |
|   |                             |               | STV3                             | -0.1            | 1.0         |
|   |                             |               | STV4                             | -0.3            | 1.0         |
|   |                             |               | STV8                             | -0.2            | 1.0         |
|   |                             |               | STVM                             | -0.2            | 1.0         |
|   |                             |               | STCVM                            | 3200.0          | 1.0         |
| G | Ventricular Ectopics        |               |                                  |                 |             |
|   | PVC/min                     | 0 0           |                                  |                 |             |
|   | Ventricular Fibrillation VF | 0 0.0%        |                                  |                 |             |
|   | Ventricular Ectopics VT     | 0 0.0%        |                                  |                 |             |
|   | Accelerated IIR A/VR        | 0 0.0%        |                                  |                 |             |
|   | Couplet CPT                 | 0 0.0%        |                                  |                 |             |
|   | Run RUN                     | 0 0.0%        |                                  |                 |             |
|   | Regimen                     | 0 0.0%        |                                  |                 |             |
| F | Supraventricular Ectopics   |               |                                  |                 |             |
|   | Sinus Tachycardia TACH      | 0 0.0%        |                                  |                 |             |
|   | Supraventricular Tachy SVT  | 0 0.0%        |                                  |                 |             |
|   | Pause PAUS                  | 0 0.0%        |                                  |                 |             |
|   | Other                       | 0 0.0%        |                                  |                 |             |
| E | Latest Alarm Event          |               |                                  |                 |             |
| D | COMMENTS:                   |               |                                  |                 |             |
|   | Signature:                  |               | Date:                            | Page: of 1      |             |

A Header Information

B Caliper Results

C S-T Deviation (from ST trends)

D Comment area

E Latest Alarm Event (in alarm categories)

F Supraventricular Ectopics

G Ventricular Ectopics

H Bradycardia

I Pacemaker

J Heart Rate, calculated from trend data

Graphical Trends

In the Graphical Trends section of the Shift Report each parameter *trend* is plotted according to type.

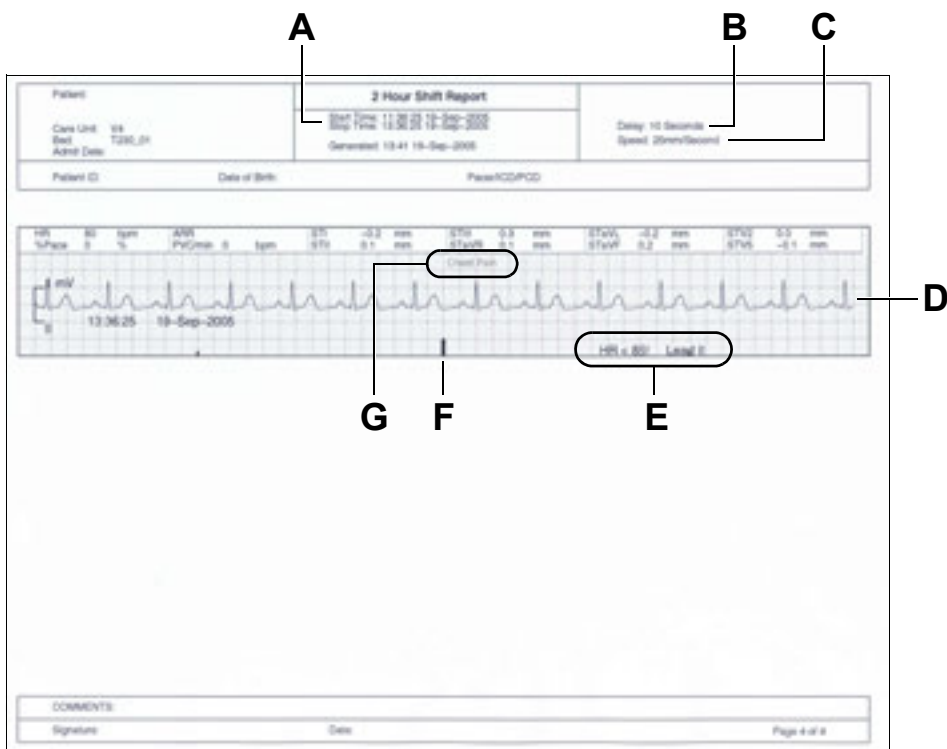
| Parameter Type                          | Appearance on Trend Graph                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Single-value, continuous such as HR     | Single, continuous line                                                                                                                            |
| Single-value, intermittent such as C.O. | Cross-hairs representing each value                                                                                                                |
| Multi-value, continuous such as ART     | Lines with a dotted line in the center indicating the mean value<br><br>e.g., The top and bottom line can represent systolic and diastolic values. |
| Multi-value, intermittent such as NIBP  | Vertical line with a blank space in the center indicating the mean value                                                                           |

This section resembles the full version Graphical Trend Report in appearance and content (see page 188) but is numbered and labeled as part of the Shift Report.

Selected Events

Depending on how many events you select (see page 222), the selected events section may consist of up to three pages (4 waveform strips each). You must select each event that will appear in this section of the shift report by clicking on the following icon so that a check mark displays on the icon.





- A** Cursor time minus selected report length (2, 4, 8, 12, 24 hrs)
- B** Delay
- C** Recording speed
- D** The waveform contains 5 seconds of pre-event data and 5 seconds of real-time data. The event marks the middle of the waveform.
- E** Event Label
- F** Event Marker
- G** Example of an annotated event

## Graphical Trend Report

The Graphical Trend Report is a multi-parameter report containing up to 15 trended parameters from bedside or telemetry patients. The report can be configured to print 2, 4, 8, 12, or 24 hours of Full or Event Disclosure data. The cursor time represents the report 'stop time'; the report 'start time' equals 'stop time' minus the report length.

You can customize the trend selection per patient (see page 196) or at system setup (see page 62). The trend scale can be a fixed scale or have AutoScale values. When you select **Enable AutoScale** (see page 196) in the **Trend Order Setup** screen, the low and high scale values are determined by an AutoScale factor to the highest and lowest parameter value over the currently selected trend time. To request a Graphical Trend Report, see page 193.

| Parameter                  | AutoScale Factor (approximate) |
|----------------------------|--------------------------------|
| HR                         | 10 bpm                         |
| PVC/min                    | 10                             |
| % Paced                    | 10%                            |
| SpO2                       | 10%                            |
| PLS                        | 10                             |
| ST                         | 1.0 mm                         |
| STVM                       | 0.10 mV                        |
| STCVM                      | 0.10 mV                        |
| Pressures (including NIBP) | 10 mmHg<br>1 kPa               |
| All others                 | 10                             |

## Selected Strip Report

The **Selected Strip Report** allows the user to select a waveform of interest within Full Disclosure and define beginning and ending cursor times (not to exceed one hour). The report can contain 8 sec-

onds to 1 hour of information (8 second default) with 8 seconds of waveform data per row. To request this report, see page 195.

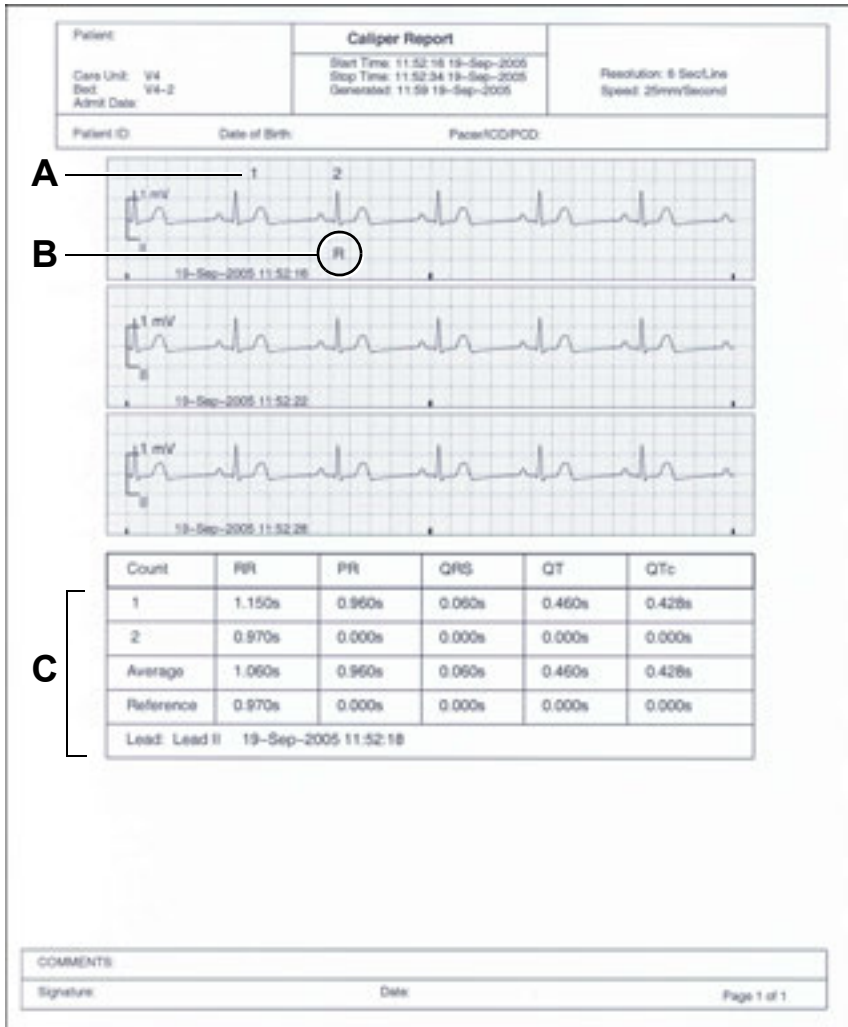
## Selected Events Report

The **Selected Events Report** may consist of up to six pages (4 event strips per page). You can select and print up to 24 events. Each waveform contains 10 seconds of data, and is annotated with *Lead Printed*, *Lead Processed*, *Event Cause*, *12 of the most recent parameter values*, *Event End Time*, and *Event Date*. If the lead processed is stored in

the Infinity CentralStation database, it will be printed on the report. If not, the top lead of the displayed Full Disclosure data will appear. To request this report, see page 195.

## ECG Caliper Report

The ECG **Caliper Report** includes an 18-second ECG waveform strip and up to ten rows of interval measurements. To request this report, see page 196.



- A** Caliper position numbers on the graph correspond to Count column numbers in the table.
- B** “R” displays under complex from which reference measurements were taken when it is in the 18-second ECG strip.
- C** Summary table from Caliper Review screen can include up to 10 rows of interval measurements.

## Ventilator Report

With the VentCentral option you can request a ventilator report for a bedside patient who is connected to a ventilator.

A ventilator report is divided into two sections:

- Parameters And Settings
- Waveforms and Loops

To request a ventilator report, see page 194.

### Parameters and settings

This portion contains the parameter values, units of measure, the settings, and the time the settings were last changed. The order of the parameters is fixed. Depending on the number of available parameters originating from the bedside monitor, this portion of the report may consist of several pages.

#### NOTE

Certain parameters, identified as measured values, are derived from other values provided by the ventilator and may not reflect the true measured values. See the operating instructions for your specific monitor for detailed information on derived measurements.

### Waveforms and loops

The waveforms and loops page contains the ventilator waveforms and loops displayed in the VentCentral Review screen and may contain up to three waveforms and up to two loops.

Each waveform consists of 7 seconds of data and begins at the time the report is generated. Each loop represents the first detected breath within the 7 seconds of data.

All waveforms are printed at 25mm/sec.

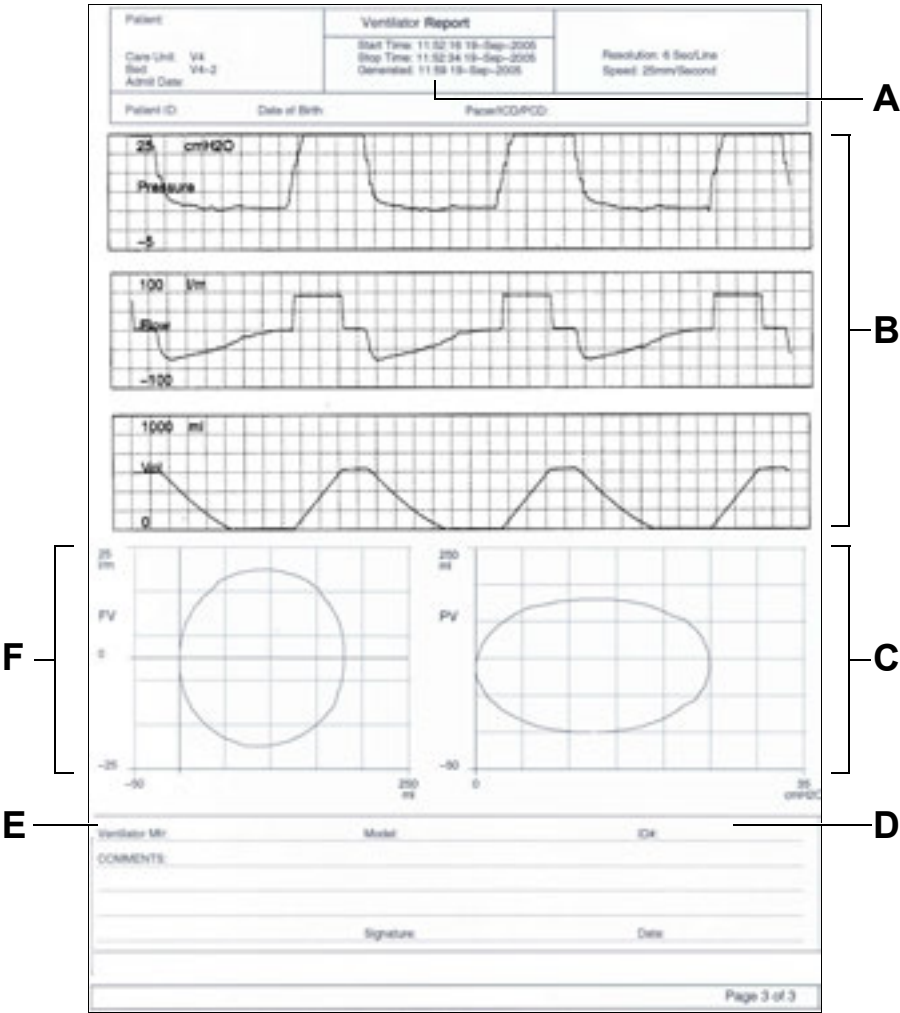
The labels and the scale of the printed waveforms and loops are identical to the displayed waveforms and loops in the VentCentral Review screen.

## Parameters and Settings

| Patient                |                        | Ventilator Report                |                                 | Respiration: 6 Sec/Line<br>Speed: 25mm/Second |  |
|------------------------|------------------------|----------------------------------|---------------------------------|-----------------------------------------------|--|
| Care Unit: V4          | Bed: V4-2              | Start Time: 11:52:16 18-Sep-2005 | Stop Time: 11:52:34 18-Sep-2005 | Generated: 11:59:19 18-Sep-2005               |  |
| Admit Date:            |                        |                                  |                                 |                                               |  |
| Patient ID:            | Date of Birth:         | Pace1001000                      |                                 |                                               |  |
| Parameter              | Units                  | Measured                         | Setting                         | Setting Time                                  |  |
| VENT MODE              |                        |                                  | VC                              | 10:58 18-Sep-2005                             |  |
| CMV FREQ               | breaths/min            |                                  | 30.0                            | 10:58 18-Sep-2005                             |  |
| SMV FREQ               | breaths/min            |                                  | 30.0                            | 10:58 18-Sep-2005                             |  |
| PEEP                   | cmH <sub>2</sub> O     | 0                                |                                 |                                               |  |
| PEEP set               | cmH <sub>2</sub> O     |                                  | 0.0                             | 10:58 18-Sep-2005                             |  |
| RRV                    | breaths/min            | 30                               |                                 |                                               |  |
| IO <sub>2</sub>        | %                      | 21                               |                                 |                                               |  |
| MAP                    | cmH <sub>2</sub> O     | 50                               |                                 |                                               |  |
| Pause                  | cmH <sub>2</sub> O     | 50                               |                                 |                                               |  |
| TVi                    | ml                     | 290                              |                                 |                                               |  |
| TVe                    | ml                     | 300                              |                                 |                                               |  |
| MVl                    | l/min                  | 20.0                             |                                 |                                               |  |
| MVe                    | l/min                  | 21.0                             |                                 |                                               |  |
| Cdyn                   | ml/cmH <sub>2</sub> O  |                                  |                                 |                                               |  |
| Raw                    | cmH <sub>2</sub> O/l/s |                                  |                                 |                                               |  |
| Insp T%                | %                      | 25                               |                                 |                                               |  |
| INSP T%                | %                      |                                  | 25.0                            | 10:58 18-Sep-2005                             |  |
| I:E I-Part             |                        | 1.0                              |                                 |                                               |  |
| I:E E-Part             |                        | 1.5                              |                                 |                                               |  |
| BAROM PRESS            | mbar                   | 1000                             |                                 |                                               |  |
| O <sub>2</sub> SUPPLY  | mbar                   | 3000                             |                                 |                                               |  |
| AIR SUPPLY             | mbar                   | 3000                             |                                 |                                               |  |
| PIP                    | cmH <sub>2</sub> O     | 100                              |                                 |                                               |  |
| TRIG SENS-PEEP         | cmH <sub>2</sub> O     |                                  | 0.0                             | 10:58 18-Sep-2005                             |  |
| IO <sub>2</sub> set    | %                      |                                  | 21.0                            | 10:58 18-Sep-2005                             |  |
| IO <sub>2</sub> set LO | %                      |                                  | 15.0                            | 10:58 18-Sep-2005                             |  |
| IO <sub>2</sub> HI LIM | %                      |                                  | 30.0                            | 10:58 18-Sep-2005                             |  |
| P HI LIM               | cmH <sub>2</sub> O     |                                  | 60                              | 10:58 18-Sep-2005                             |  |
| MVe LO LIM             | l/min                  |                                  | 5.0                             | 10:58 18-Sep-2005                             |  |
| MVe HI LIM             | l/min                  |                                  | 30.0                            | 10:58 18-Sep-2005                             |  |
| PATIENT RANGE          |                        |                                  | Adult                           | 10:58 18-Sep-2005                             |  |
| P SUPPORT              | cmH <sub>2</sub> O     |                                  | 0.0                             | 10:58 18-Sep-2005                             |  |
| COMMENTS:              |                        |                                  |                                 |                                               |  |
| Signature:             |                        | Date:                            |                                 | Page 1 of 2                                   |  |

- A** The time the report request was processed
- B** The Setting Time column displays the time the setting was updated or data collection resumed after the monitor was disconnected from the patient
- C** The Parameter column displays the parameters measured at the time the report request was processed
- D** The Measured column displays the measured parameter values. Certain parameters, identified as measured values, are derived from other values or settings provided by the ventilator and may not reflect actual values. See the operating instructions for your specific monitor for detailed information on derived measurements.

Waveforms and loops



- A** The time the report request was processed
- B** Top 3 waveforms selected in VentCentral screen
- C** Pressure Loop (represents the first breath)
- D** Area to record Ventilator Identification Number
- E** Area to record Ventilator manufacturer
- F** Flow Loop (represents the first breath)



## Requesting Recordings and Reports

### NOTE

The print queue can handle multiple report requests. You are notified when the print queue is full, and must wait until some reports print before requesting more.

| Recording Report                                                                                                                                                                                                                                                                                                                                                                                                             | Required Steps                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timed recording<br>(from Main Screen)                                                                                                                                                                                                                                                                                                                                                                                        | <p>For an individual patient:</p> <ul style="list-style-type: none"> <li>Click on the <b>REC</b> button in the patient's parameter.</li> </ul> <p>If the patient is discharged, in <i>Standby</i>, or waveforms are stopped, the <b>REC</b> button is unavailable for selection and you cannot request a timed recording.</p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>For all patients admitted to Main Screen:</p> <ul style="list-style-type: none"> <li>Press the keyboard F11 key.</li> </ul>                                                                                                                                                                                                |
| <h3>NOTE</h3> <ul style="list-style-type: none"> <li>If a recording was previously requested for a patient from Main Screen or the patient's waveforms are stopped, no recording is printed for that patient.</li> <li>If the Infinity CentralStation is operating in Dual Display mode, recordings are only printed for patients of the screen in which the mouse pointer is located at the time of the request.</li> </ul> |                                                                                                                                                                                                                                                                                                                               |
| Timed recording<br>(from Bed View)                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Click on <b>Record</b> on the menu bar.</li> </ul>                                                                                                                                                                                                                                     |
| Timed manual recording<br>(from transmitter)                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Press transmitter recording button for less than 3 seconds.</li> </ul>                                                                                                                                                                                                                 |
| Automatic alarm recording                                                                                                                                                                                                                                                                                                                                                                                                    | Automatically generated timed recordings are described in the chapters starting on page 157 (Alarms) and page 129 (Arrhythmia).                                                                                                                                                                                               |
| Continuous recording                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Click on the <b>CONT</b> button.</li> </ul> <p>If the patient is either discharged, in <i>Standby</i>, or waveforms are stopped, the <b>CONT</b> button will appear unavailable for selection and you will not be able to request a timed recording.</p>                               |
| Continuous recording<br>(from transmitter)                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Press transmitter recording button for more than 3 seconds (Infinity Telemetry only, not Infinity M300).</li> </ul>                                                                                                                                                                    |
| Print Screen                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Main Screen:</p> <ul style="list-style-type: none"> <li>Press the keyboard <b>Print Screen</b> key.</li> </ul> <p>Bed View:</p> <ol style="list-style-type: none"> <li>Click on <b>Print in the Bed View menu bar.</b></li> <li>Click on <b>Bed View.</b></li> </ol>                                                       |

| Recording Report                                                                                                                                                              | Required Steps                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOTE</b><br>If the <b>Print</b> button appears unavailable for selection, the <b>Printer Connected</b> setting is <b>Off</b> in the Configure Central menu (see page 291). |                                                                                                                                                                                                                                                                                                                                                                                   |
| Simultaneous ECG Report                                                                                                                                                       | <ol style="list-style-type: none"> <li>1 Open patient's Bed View Screen.</li> <li>2 Click on <b>Simultaneous ECG Report</b>.</li> </ol>                                                                                                                                                                                                                                           |
| Ventilator Report                                                                                                                                                             | <ol style="list-style-type: none"> <li>1 Open the <b>Ventilator Settings Review</b> Screen (see page 240).</li> <li>2 Click on <b>Report</b>.</li> </ol>                                                                                                                                                                                                                          |
| <b>Full Disclosure Reports</b>                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>– Strip Report</li> <li>– Hour Report</li> <li>– 24 Hour Report</li> </ul>                                                             | <ol style="list-style-type: none"> <li>1 From the <b>Full Disclosure</b> screen (see page 215) click on <b>Reports...</b></li> <li>2 Click on and select desired report.<br/>If you request a <b>24 Hour Report</b>, a pop-up displays to alert you that this report will keep the printer busy. You must confirm your request by clicking on <b>OK</b> in the pop-up.</li> </ol> |
| <b>NOTE</b><br>You can change the cursor time by clicking on it.                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>– Batch Shift Reports</li> </ul>                                                                                                       | <ol style="list-style-type: none"> <li>1 Open the <b>Setup Central</b> screen (see page 60).</li> <li>2 Click on <b>Batch Shift Reports</b>.<br/>The report length selections display.</li> <li>3 Select a report length (2, 4, 8, 12, or 24 Hour).</li> <li>4 Click on <b>Accept</b> to save the setting.</li> <li>5 Press <b>F9</b> on your keyboard.</li> </ol>                |
| <ul style="list-style-type: none"> <li>– Shift Report</li> <li>– Patient Status Report</li> </ul>                                                                             | <ol style="list-style-type: none"> <li>1 From the <b>Full Disclosure</b> screen (see page 215) click on <b>Reports....</b><br/>The <b>Reports</b> menu displays.</li> <li>2 Click on and select the desired report.</li> <li>3 Select a report length from the menu (2, 4, 8, 12, or 24 Hour).</li> </ol>                                                                         |

| Recording Report                                                                                                                                                                                                                                                                                                                                                                                | Required Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>– Selected Strip Report</li> </ul>                                                                                                                                                                                                                                                                                                                       | <ol style="list-style-type: none"> <li>1 From the <b>Full Disclosure</b> screen (see page 215) click on <b>Reports....</b></li> <li>2 Select a report length from an entry on the menu (1, 5 10, or 60 minutes, and Selected).<br/>The menu item ‘Selected’ is unavailable for selection until a second cursor is defined.</li> <li>3 For a <i>Selected</i> report length</li> <li>4 Right-click on the waveform to set a second cursor. The second cursor displays with dotted lines and can represent either the <i>Stop</i> or <i>Start Time</i> depending on its location relative to the first cursor.<br/>A confirmation pop-up appears that shows <b>Start Time</b>, <b>Stop Time</b>, and <b>Total Pages</b>.</li> <li>5 Click on <b>Continue</b> or <b>Cancel</b> in the pop-up.</li> </ol> |
| <p><b>NOTE</b></p> <ul style="list-style-type: none"> <li>– You cannot set the cursors so that the report is longer than 60 minutes or less than 2 seconds. A pop-up will display that cancels the selected period.</li> <li>– The minimum report length is 8 seconds. Also, the report length will automatically adjust so that each row will contain a full 8 seconds of ECG data.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recording Report                  | <h3>Required Steps</h3>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Graphical Trend Report            | <p>On a <b>Graphical Trend Report</b>, which is also included in a <b>Shift Report</b>, you can print a report with the system-assigned trend settings (see page 62) or you can configure the report to display particular trends for the individual patient as follows:</p> <ol style="list-style-type: none"> <li>From the <b>Full Disclosure</b> screen (see page 215) click on <b>Reports...</b> From the <b>Reports</b> menu click on <b>Trend Report Setup</b>. The <b>Trend Order Setup</b> screen displays.</li> </ol>  <ol style="list-style-type: none"> <li>Select the trends you want to display and move them to the <b>Display List</b> window. The trends typically display on a fixed scale, but you can change the display by enabling AutoScale.</li> <li>If you want to change the display scale to AutoScale, first click on the parameter you want to autoscale, and then click on the selection box next to <b>Enable AutoScale</b>.</li> <li>Select a number of <b>Trends Per Page</b> for the report.</li> <li>Click on <b>Accept</b> to keep the trend order setup or on <b>Cancel</b>.</li> </ol> |
| Caliper Report                    | <ol style="list-style-type: none"> <li>Open the <b>Caliper Review</b> screen (see page 231).</li> <li>Click on <b>Print Report</b>.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <h3>Event Disclosure Reports</h3> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Event Disclosure Strip Report     | <ol style="list-style-type: none"> <li>From the <b>Event Disclosure</b> screen (see page 215) click on the button, <b>Reports...</b></li> <li>Click on <b>Strip Report</b>.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| <b>Recording Report</b>                 | <b>Required Steps</b>                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event Disclosure Shift Report           | <b>1</b> From the <b>Event Disclosure</b> screen (see page 215) click on the button, <b>Reports....</b>                                                                                                                                                                                                                          |
| Event Disclosure Graphical Trend Report | <b>2</b> Click on and select desired report.<br><b>3</b> Select a report length from the menu (2, 4, 8, 12, or 24 hr).                                                                                                                                                                                                           |
| Selected Events Report                  | <b>1</b> From the <b>Event Disclosure</b> screen (see page 215) click on the button, <b>Reports....</b><br><b>2</b> Click on <b>Selected Events Report</b> .<br><b>3</b> Select the events for the report. A pop-up displays the number of events selected.<br><b>4</b> Click on <b>Accept</b> to continue or on <b>Cancel</b> . |

## Canceling a Recording

| <b>Recording Type</b>  | <i>To cancel in Main Screen:</i>                        | <i>To cancel in Bed View:</i>              | <i>To cancel at the recorder:</i> |
|------------------------|---------------------------------------------------------|--------------------------------------------|-----------------------------------|
| Manual timed recording | Click the patient's <b>REC</b> key to deselect it       | Click the <b>RECORD</b> key to deselect it | Press the <b>STOP</b> key         |
| Alarm recording        | Not possible                                            | Not possible                               | Press the <b>STOP</b> key         |
| ARR event recording    | Not possible                                            | Not possible                               | Press the <b>STOP</b> key         |
| Continuous recording   | Click the patient's <b>CONT REC</b> key to deselect it. | Not possible                               | Press the <b>STOP</b> key         |

## Actions that Affect Recordings

| <b>Action</b>                          | <b>Effect on Pending Recordings</b> | <b>Effect on Recordings in Progress</b> |
|----------------------------------------|-------------------------------------|-----------------------------------------|
| Patient Discharge                      | Pending recordings are discarded    | Recording is canceled                   |
| Edit patient's demographic data        | No effect                           | No effect                               |
| Patient monitoring put in Standby mode | No effect                           | Recording is canceled                   |
| Restore factory defaults               | Pending recordings are discarded    | Recording is canceled                   |

| Action                                                                                                                      | Effect on Pending Recordings                                                                                        | Effect on Recordings in Progress                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configuration change of Main Screen resulting in removal of a patient                                                       | Patient's pending recordings are canceled                                                                           | Recording is canceled                                                                                                                                                               |
| Change in scale settings or parameter order (parameters might become available/unavailable due to restored/missing signals) | No effect                                                                                                           | Recordings will continue but waveform and parameter data for any removed parameter appear as blanks. Added parameters are <i>not</i> included on a recording currently in progress. |
| Recorder assignment change                                                                                                  | No effect. Additional recordings are printed on the new recorder.                                                   | No effect on recordings.                                                                                                                                                            |
| Network time/date change                                                                                                    | No effect on active or pending recordings. All new and continuous recordings in progress reflect the new date/time. | Continuous recordings reflect the change immediately (as will all future timed recordings).                                                                                         |
| R 50 recorder becomes unavailable, runs out of paper, or jams                                                               | No effect                                                                                                           | Recording is not lost, but routed to another assigned recorder, or stored for printing when the recorder becomes available.                                                         |
| Remote device requesting a recording of a local patient goes offline                                                        | Deletes all recordings associated with this device                                                                  | Recording is canceled                                                                                                                                                               |
| Patient disconnect for transport                                                                                            | Cancels all pending recordings for that bed                                                                         | Recording is canceled                                                                                                                                                               |

## Recording Status Messages

| Recorder Status Messages                                                                                                                                                                                                                           |          |                                                                                                                                      |                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Status Message                                                                                                                                                                                                                                     | Tone     | Description                                                                                                                          | Action                                                                          |
| <ul style="list-style-type: none"> <li>– &lt;xxxx&gt; Recording Request Accepted<sup>1</sup></li> <li>– Cont. Recording Now Timed (for Gamma/Gamma XL /Vista monitors only)</li> <li>– &lt;xxxx&gt; Report Request Accepted<sup>2</sup></li> </ul> | None     | Request is accepted, but stored until a printing device is available.                                                                | None                                                                            |
| <ul style="list-style-type: none"> <li>– &lt;xxxx&gt; Recording Started<sup>1</sup></li> <li>– &lt;xxxx&gt; Report Request Started<sup>2</sup></li> </ul>                                                                                          | None     | Request is being printed.                                                                                                            | None                                                                            |
| <ul style="list-style-type: none"> <li>– &lt;xxxx&gt; Recording Finished<sup>1</sup></li> <li>– &lt;xxxx&gt; Report Request Finished<sup>2</sup></li> </ul>                                                                                        | None     | Printout is finished or is stored.                                                                                                   | None                                                                            |
| <xxxx> Report Request Canceled - Queue full <sup>2</sup>                                                                                                                                                                                           | None     | Report is canceled due to a full print queue.                                                                                        | None                                                                            |
| <xxx> Disconnected <sup>3</sup>                                                                                                                                                                                                                    | Advisory | Recorder(s) not connected to the network.                                                                                            | Connect the recorder.                                                           |
| <xxxx> Recording Interrupted <sup>1</sup>                                                                                                                                                                                                          | None     | Recording stalled because recorder is disconnected or out of paper, its door is open, or an error occurred during data transmission. | Contact the Hospital Bio-medical Engineering Dept.                              |
| <xxx> Door Open <sup>3</sup>                                                                                                                                                                                                                       | Advisory | Recorder door open                                                                                                                   | Close door                                                                      |
| <xxx> Out Of Paper <sup>3</sup>                                                                                                                                                                                                                    | Advisory | Recorder is out of paper                                                                                                             | Load paper                                                                      |
| <xxx> Failure <sup>3</sup>                                                                                                                                                                                                                         | Serious  | Recorder failed to operate                                                                                                           | Take recorder out of service and contact service personnel.                     |
| <xxx> Offline <sup>3</sup>                                                                                                                                                                                                                         | Serious  | Recorder is offline. Recorder communication with the network failed.                                                                 | Contact the Hospital Bio-medical Engineering Dept. to check network connection. |
| <xxx> Duplicate Address <sup>3</sup>                                                                                                                                                                                                               | Serious  | Recorder detected duplicate address.                                                                                                 | Contact the Hospital Bio-medical Engineering Dept.                              |

| Recorder Status Messages                                                                                                                            |           |                                                                                                                                                   |                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Status Message                                                                                                                                      | Tone      | Description                                                                                                                                       | Action                                                                                     |
| Recorder Not Assigned                                                                                                                               | Attention | No recorder assigned.                                                                                                                             | Assign a recorder (see page 63).                                                           |
| Recording Request Not Accepted                                                                                                                      | Attention | Bedside monitor's print queue is full, or telemetry patient's recording is requested within 5 seconds of request for a higher-priority recording. | Wait until some of the pending recordings are printed before requesting another recording. |
| Recording Status Unknown - Connection Failed                                                                                                        | Attention | Recording status unknown due to failed bed connection (telemetry or bedside).                                                                     | Contact the Hospital Bio-medical Engineering Dept.                                         |
| Cursor out of Range                                                                                                                                 | Attention | More than 60 minutes of data is defined by Selected Strip Report cursors.                                                                         | Set cursors for 8 seconds to 60 minutes of FD data.                                        |
| <sup>1</sup> <xxxx> represents recording title.<br><sup>2</sup> <xxxx> represents report title.<br><sup>3</sup> <xxx> represents device/host label. |           |                                                                                                                                                   |                                                                                            |

| Printer Status Messages                 |           |                                                                                          |                                                                          |
|-----------------------------------------|-----------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Status Message                          | Tone      | Description                                                                              | Action                                                                   |
| Print Request Finished                  | None      | Request is complete                                                                      | None                                                                     |
| Printer Not Configured                  | Attention | No printer is configured at the Infinity CentralStation.                                 | Configure the printer (see page 291).                                    |
| Print Request Not Accepted - Queue Full | Attention | Print screen request was rejected because the print queue handles one request at a time. | Wait until previous screens complete printing before requesting another. |
| Printer/Communication Error             | Attention | Communication to network is faulty.                                                      | Consult the Hospital Bio-medical Engineering Dept.                       |
| Printer Selection Failed                | Attention | Selected printer is not set up properly.                                                 | Consult the Hospital Bio-medical Engineering Dept.                       |



# Trends

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## Overview

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Trend information is available at the Infinity Central-Station for remote viewing of bedside and telemetry devices.

## Accessing a Patient's Trends

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To access a patient's trends:

- 1 Open the patient's Bed View screen.
- 2 Click on **Review**.

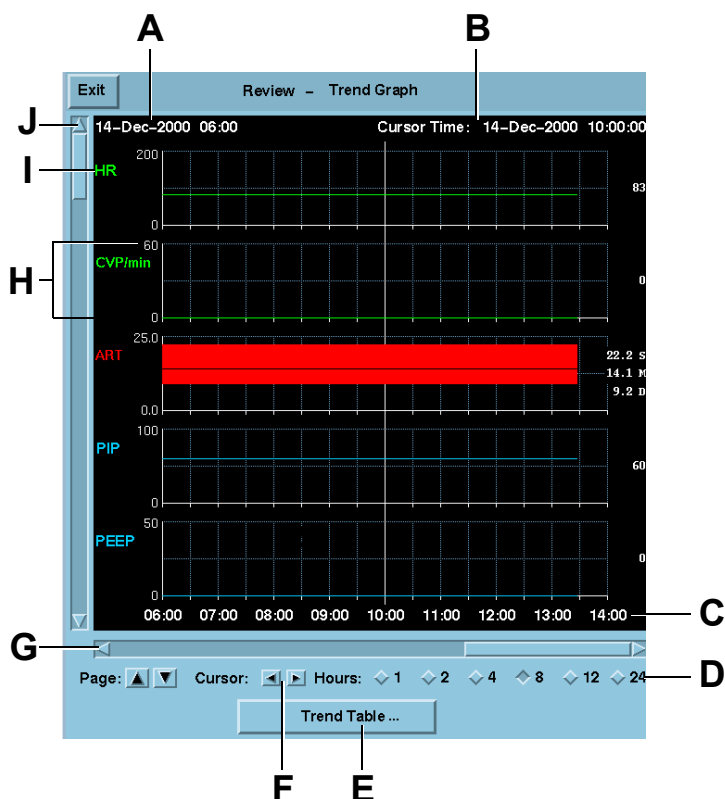
- 3 Click on **Trend Graphs...** or **Trend Table....**

When you open the Trend Graphs screen, data is centered around the cursor time.

## Trend Graphs

---

Trend graphs show stored trend data for each parameter. Each trend page can accommodate up to five parameter graphs that may be blank if the parameter is not selected for trending. Trends are updated automatically, with the most recent data entering continuously on the right. When you click on the trended parameter label, the scale toggles between *fixed* and *autoscale* values (see page 188).



- A** Start date/time of Trend data
- B** Time and date corresponding to current cursor position
- C** Time labels
- D** Radio buttons to select graph time period
- E** Opens Trend Table
- F** Cursor scrolling keys
- G** Horizontal arrows scroll time periods
- H** Interactive scale permits selection of:
  - autoscale or fixed scale (telemetry patients)
  - autoscale or bed scale (bedside patients)
- I** Parameter label (click on Parameter Label to autoscale)
- J** Vertical arrows scroll one graph at a time

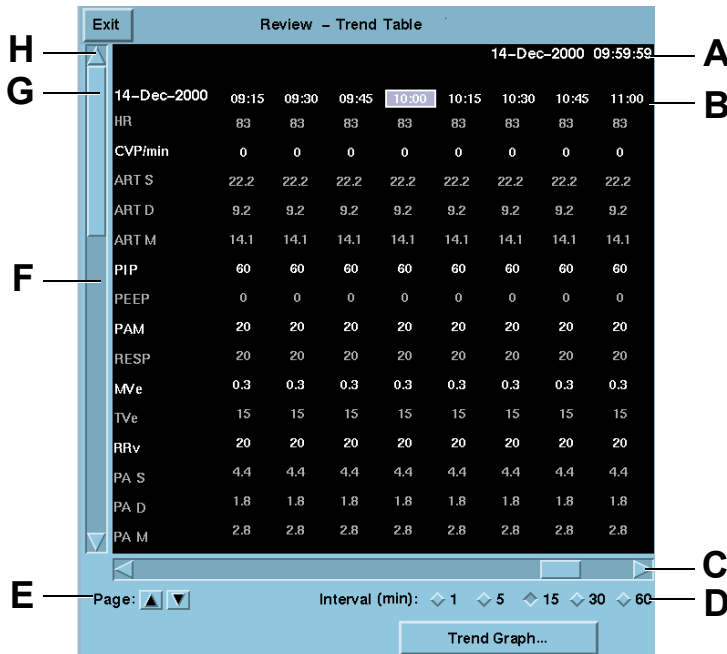
## Parameter Trend Scale Defaults for Telemetry Patients

| Parameter        | Scale: Upper/Lower                     |
|------------------|----------------------------------------|
| HR               | 0 - 200 bpm                            |
| PVC/min          | 0 - 60                                 |
| % Paced          | 0 - 100%                               |
| SPO <sub>2</sub> | 50 - 100%                              |
| PLS              | 0 - 200 bpm                            |
| STI              | +/-5 mm / +/- .5 mV                    |
| STII             | +/-5 mm / +/- .5 mV                    |
| STIII            | +/-5 mm / +/- .5 mV                    |
| ST aVR           | +/-5 mm / +/- .5 mV                    |
| ST aVL           | +/-5 mm / +/- .5 mV                    |
| ST aVF           | +/-5 mm / +/- .5 mV                    |
| STV1/dV1/STV+    | +/-5 mm / +/- .5 mV                    |
| STV2/dV2         | +/-5 mm / +/- .5 mV                    |
| STV3/dV3         | +/-5 mm / +/- .5 mV                    |
| STV4/dV4         | +/-5 mm / +/- .5 mV                    |
| STV5/dV5         | +/-5 mm / +/- .5 mV                    |
| STV6/dV6         | +/-5 mm / +/- .5 mV                    |
| STVM             | +10.0 mm to 0.0 mm or 1.0 mV to 0.0 mV |
| STCVM            | +10.0 mm to 0.0 mm or 1.0 mV to 0.0 mV |
| NBP              | 0 to 300 mmHg                          |

## Trend Table

Each trend table row shows a parameter's values; each column shows a trend data set for all parameters at that time. If a parameter is assigned to trending but cannot be trended, the row lists the

parameter label but values are blank. Regardless of the selected interval, a trend table displays up to 8 data columns. A trend table may also represent special events (see page 207).



- A Date and time of cursor position
- B Time stamps
- C Horizontal arrows scroll time columns
- D Radio buttons for selecting time period
- E Permits page scrolling
- F Parameter labels
- G Start date/time of trend data
- H Vertical arrows scroll rows of data

- Trend values appear in groups of associated parameter rows.
- Discrete events such as NPB or CO measurements cause a set of trend data to be stored for all parameters.
- Such data appears as a separate trend column with a green time stamp heading. This is also true when a mini-calc computation is initiated on an Delta/Delta XL/Kappa/GammaX XL monitor.
- The trend table updates automatically whenever an interval elapses. The most recent column is at the right of the screen.

# Trend Cursor

On a trend graph, the cursor appears as a full-screen white vertical line (see arrow in the following illustration). On a trend table the cursor is the highlighted column time (circled below). Cursor time displays in the upper-right area of the screen.

Trend Graph



Trend Table

| 14-Dec-2000 | 09:15 | 09:30 | 09:45 | 10:00 | 10:15 | 10:30 | 10:45 | 11:00 |
|-------------|-------|-------|-------|-------|-------|-------|-------|-------|
| HR          | 83    | 83    | 83    | 83    | 83    | 83    | 83    | 83    |
| CVP/min     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| ART S       | 22.2  | 22.2  | 22.2  | 22.2  | 22.2  | 22.2  | 22.2  | 22.2  |
| ART D       | 9.2   | 9.2   | 9.2   | 9.2   | 9.2   | 9.2   | 9.2   | 9.2   |
| ART M       | 14.1  | 14.1  | 14.1  | 14.1  | 14.1  | 14.1  | 14.1  | 14.1  |
| PIP         | 60    | 60    | 60    | 60    | 60    | 60    | 60    | 60    |
| PEEP        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| PAM         | 20    | 20    | 20    | 20    | 20    | 20    | 20    | 20    |
| RESP        | 20    | 20    | 20    | 20    | 20    | 20    | 20    | 20    |
| MVe         | 0.3   | 0.3   | 0.3   | 0.3   | 0.3   | 0.3   | 0.3   | 0.3   |
| TVe         | 15    | 15    | 15    | 15    | 15    | 15    | 15    | 15    |
| RRv         | 20    | 20    | 20    | 20    | 20    | 20    | 20    | 20    |
| PA S        | 4.4   | 4.4   | 4.4   | 4.4   | 4.4   | 4.4   | 4.4   | 4.4   |
| PA D        | 1.8   | 1.8   | 1.8   | 1.8   | 1.8   | 1.8   | 1.8   | 1.8   |
| PA M        | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   |

- When using the time anchor function between Full Disclosure, Event Disclosure and trends, the cursor time will match one of the trend table column time stamps. The column time is highlighted and appears in the center of the trend table screen.
- If the cursor time does not exactly match one of the columns in the trend table, cursor time is displayed but no column is highlighted. Click on any column time to highlight it.
- If the cursor time is not within the time span of the current trend page, the cursor time displays but the cursor does not. Scroll to the desired data to display the cursor on the current trend.
- If the cursor time is older than any of the available trend data, the oldest trend data is displayed without the cursor.
- 24 hours of trend graph data is displayed. One hour of tabular trend data is displayed.

## Representation of Special Conditions

| Condition                                                       | Representation in Trend Table                                                                               |                                                                 | Representation in Trend Graphs                                    |                                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
|                                                                 | Telemetry Patient                                                                                           | Bedside Patient                                                 | Telemetry Patient                                                 | Bedside Patient                                                   |
| Discrete trend samples such as those associated with NPB and CO | None                                                                                                        | Column with a green time stamp                                  | None                                                              | Vertical line with a gap in the center indicating the mean value  |
| A trend event stored at Delta/Delta XL/Kappa bedside monitor    | None                                                                                                        | Column with a green time stamp                                  | None                                                              | Not shown in Trend graphs                                         |
| Physiological conditions such as an asystole or an apnea        | <ul style="list-style-type: none"> <li>– ASY = Asystole</li> <li>– VF = Ventricular Fibrillation</li> </ul> | ASY = Asystole<br>FIB = Ventricular Fibrillation<br>APN = Apnea | – Parameter value = 0                                             | Parameter value = 0<br>Parameter value = 0<br>Parameter value = 0 |
| No parameter could be derived                                   | * * *                                                                                                       | * * *                                                           | Blanks in the graphs                                              | Blanks in the graphs                                              |
| Lead-Off condition                                              | *L*                                                                                                         | ***                                                             | *L*                                                               | ***                                                               |
| Out-of-range values                                             | +++ (high) or --- (low)                                                                                     | +++ (high) or --- (low)                                         | Blanks (move the cursor over the blanks to display actual values) | Blanks (move the cursor over the blanks to display actual values) |
| Artifact                                                        | *A*                                                                                                         | ***                                                             | *A*                                                               | ***                                                               |
| Lead wires, electrodes are unplugged                            | *U*                                                                                                         | ***                                                             | *U*                                                               | ***                                                               |
| Hardware failure                                                | *F*                                                                                                         | ***                                                             | *F*                                                               | ***                                                               |
| Interference                                                    | *I*                                                                                                         | None                                                            | Blanks in the graphs                                              | shaded                                                            |
| No signal                                                       | *N*                                                                                                         | Blank spaces                                                    | Blanks in the graphs                                              | Blanks in the graphs                                              |
| Time change at bedside monitor                                  | None                                                                                                        | Yellow line between trend columns                               | None                                                              | Blank spaces in the graph                                         |

| Condition                                                                          | Representation in Trend Table                            |                                                                                                                                             | Representation in Trend Graphs                             |                                                            |
|------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
|                                                                                    | Telemetry Patient                                        | Bedside Patient                                                                                                                             | Telemetry Patient                                          | Bedside Patient                                            |
| A trend event has been stored by initiating a mini-calc computation at the bedside | None                                                     | Column designated with this symbol  and a green time stamp | None                                                       | not shown in graphs                                        |
| Change in units of measure                                                         | New units take effect during the next patient admission. | None                                                                                                                                        | New units take effect during the next patient admission.   | None                                                       |
| Power loss in server                                                               | Blank spaces are stored in place of trend values         | Blank spaces are stored in place of trend values                                                                                            | Blank spaces are stored in place of trend values           | Blank spaces are stored in place of trend values           |
| Telemetry receiver goes offline                                                    | Blank spaces instead of trend values                     | None                                                                                                                                        | Blank spaces instead of trend values                       | None                                                       |
| Relearning of ST complexes                                                         | LRN                                                      | None                                                                                                                                        | Vertical dotted line that extends through the entire graph | Vertical dotted line that extends through the entire graph |
| Changing of ST ISO points and ST measuring points                                  | CHG                                                      | None                                                                                                                                        | Vertical dotted line that extends through the entire graph | Vertical dotted line that extends through the entire graph |



## Full/Event Disclosure

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|                                                          |     |
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## Overview

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The Infinity CentralStation has the ability to store Full Disclosure waveforms continuously for a minimum of 2 hours to a maximum of 72 hours, depending on the configuration. The number of waveforms stored in Full Disclosure also depends upon the configuration. Event Disclosure is storage of

arrhythmia, alarm events, or manually triggered events. Depending on the license option, storage capacity for Full and Event Disclosure ranges from 2 to 72 hours, depending on the configuration, and up to 1000 events per patient.

## Full/Event Disclosure Census Operations

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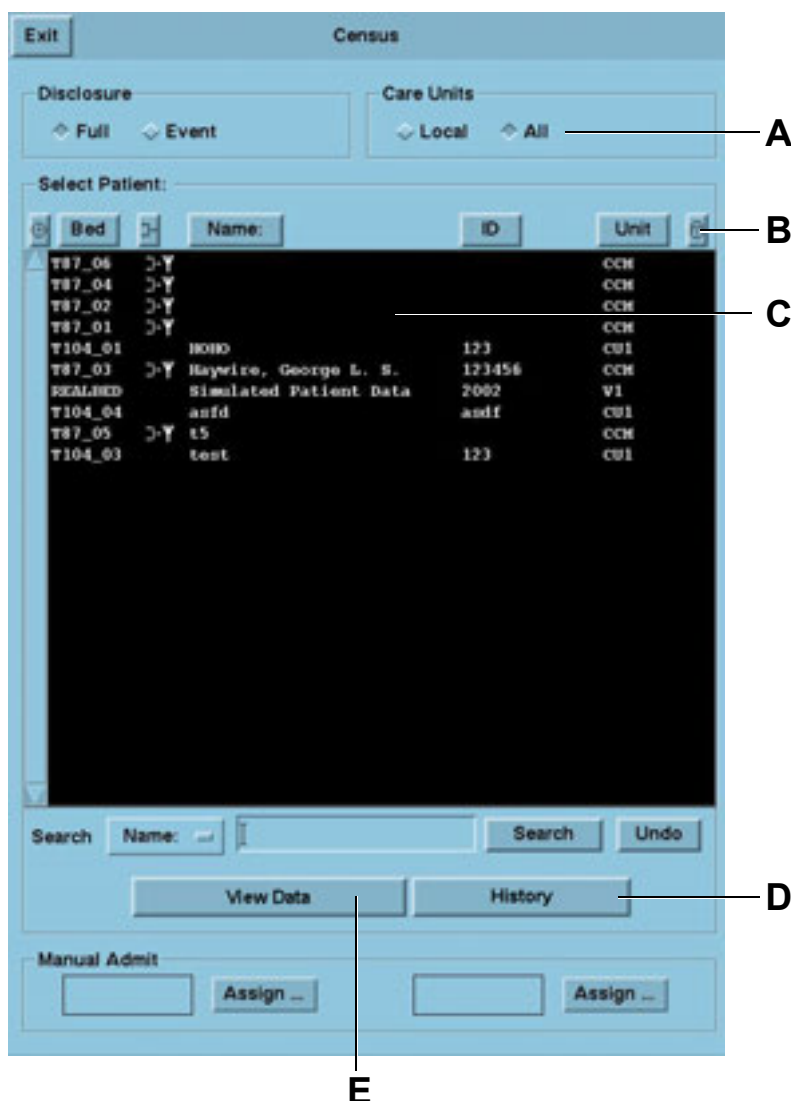
### Accessing the Census Screen

To access the Census screen from the Bed View:

- 1 Click on **Review**.
- 2 Click on **Full Disclosure...** or **Event Disclosure...**
- 3 Click on **Census** in the menu bar.

To access the Census screen from the Main Screen:

- 1 Click on **View**.
- 2 Click on **Full Disclosure...** or **Event Disclosure...**



- A** Local - all patients with Full/Event Disclosure records on local Infinity CentralStation  
All - all patients with Full/Event Disclosure records on the Infinity Network
- B** Click on heading to sort columns
- C** Patient's Census screen
- D** Opens the selected patient's History window
- E** Opens the Disclosure Review Screen

| Screen                                                                                                        | Selection                                                                                        | Description                                                                                                       |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Disclosure                                                                                                    | Full                                                                                             | Full disclosure data                                                                                              |
|                                                                                                               | Event                                                                                            | Event disclosure data                                                                                             |
| Care Units                                                                                                    | Local                                                                                            | Patients with disclosure records being stored in the database of the Infinity CentralStation currently being used |
|                                                                                                               | All                                                                                              | Patients with disclosure records accessible on the Infinity Network                                               |
| Select Patient:<br>Available patient records are listed and can be sorted by clicking on one of the headings. | Record Lock<br> | Denotes a “locked” record. Up to 16 records may be locked at one time.                                            |
|                                                                                                               |                 | Time Disclosure record was acquired<br>Sort order places most recent record first.                                |
|                                                                                                               | Bed                                                                                              | Infinity Network bed label                                                                                        |
|                                                                                                               |                 | Denotes <i>Active/Inactive</i> status<br>When the symbol displays, disclosure data collection is active.          |
|                                                                                                               | Name                                                                                             | Patient Name                                                                                                      |
|                                                                                                               | ID                                                                                               | Patient ID Number                                                                                                 |
|                                                                                                               | Care Unit                                                                                        | Care Unit Label                                                                                                   |
| Search                                                                                                        | – Name                                                                                           | Selects field that will be searched for keyword entered in the text entry box                                     |
|                                                                                                               | – ID                                                                                             |                                                                                                                   |
|                                                                                                               | – Bed                                                                                            |                                                                                                                   |
|                                                                                                               | text box                                                                                         | Text entry box for keyword search criteria (20-character max.)                                                    |
|                                                                                                               | Search                                                                                           | Activates search function for information entered into text box                                                   |
|                                                                                                               | Undo                                                                                             | Returns Census Screen                                                                                             |
| View Data                                                                                                     | Opens selected patient’s disclosure review screen (Full or Event)                                |                                                                                                                   |
| History                                                                                                       | Opens selected patient’s disclosure history screen                                               |                                                                                                                   |
| Manual Admit                                                                                                  | Assign...                                                                                        | Manually admit up to 2 additional patients to Disclosure data collection (see page 214)                           |

## Admitting a Patient to Full/Event Disclosure

Full/Event Disclosure storage begins automatically when a patient is admitted at the Main Screen of the Infinity CentralStation. There are two additional Manual Admit records for patients within a Monitoring Unit, but they may not be added at the Main Screen.

### NOTE

A Monitoring Unit is a logical group of beds that provide patient monitoring services such as alarm annunciation, recordings, and remote control to its members.

### Data Collection Rules

A patient remains admitted to the Infinity CentralStation and data storage continues for as long as the patient remains on the same monitor on the Infinity network.

If the patient leaves the Monitoring Unit and returns with the same bedside monitor (Pick and Go) or to the same Infinity CentralStation, data storage resumes automatically when the patient is reconnected to the Infinity Network. If the patient's monitor/Infinity CentralStation is different, the new care unit data is linked to the previous data set and is available on the patient's Full or Event Disclosure Review screen.

"Inactive" records are included in the Census Screen patient list and have no "active data collection" symbol (see page 212). Stored waveforms for patients who are removed from Main Screen are saved and sorted according to time spent, and have an inactive status. As soon as the storage capacity is reached, the oldest data is replaced by the most recent. Each inactive record deletes automatically after 72 hours, except when the record is locked (see page 212).

### NOTE

- If the bedside monitor is disconnected for Pick and Go and reconnected to the network, Disclosure storage resumes automatically.
- Whenever you disconnect a bedside monitor from the network, small gaps will appear in the Full Disclosure waveforms.

Once the Full/Event Disclosure capacity is reached (4, 8, 12, 16, or 32 patients, depending on server option) new patients can be admitted as patients are discharged and slots become available.

The Infinity CentralStation database permits storage of "active" and "inactive" patient records. The total number of "inactive" records available is calculated using the following formula:

$$64 - (n + 2)$$

with "n" equal to the number of active patients.

## Admitting a Patient on the Census screen

You can admit two additional patients to Full/Event Disclosure.

To admit a patient:

- 1 Open the Census screen (see page 210).
- 2 Click on the button labeled **Assign...** in the **Manual Admit** section of the **Census** screen.

The *Manual Assign Bed* pop-up displays.

- 3 Double-click on the patient you wish to admit or  
Highlight the patient you wish to admit and click on **Accept**.

## Accessing Disclosure History

---

To access disclosure history:

- 1 Open the Census screen (see page 210).
- 2 Highlight the desired record in the Census screen patient list (see page 211).

- 3 Click on **History**.

The *Patient Care History* pop-up displays.

- 4 Select the record you wish to view.
- 5 Click on **View Data**.

## Locking a Full/Event Disclosure Record

You can lock up to 16 disclosure records in each Infinity CentralStation database. When you **lock** the record, data will not be automatically deleted. When all 16 locks are used, the button is unavailable for selection.

To Lock/Unlock a Record:

### NOTE

If the clinical password function is enabled on the **Biomed-Configure Central** screen (see page 291), you will need this password to unlock a record.

- 1 Access the *Patient Care History* pop-up.
- 2 Select the record you wish to lock or unlock.
- 3 Click on the **Lock/Unlock** toggle button.

## Accessing the Full/Event Disclosure Review Screen

---

To access from the Main Screen:

- 1 Open the Census Screen (see page 210).
- 2 Highlight a record in the Census screen patient list (page 211) to activate the **View Data** button.
- 3 Click on **View Data** to display the Full/Event Disclosure Review screen.

To access from Bed View:

- 1 Click on **Review**.
- 2 Click on **Full Disclosure...** or **Event Disclosure...**

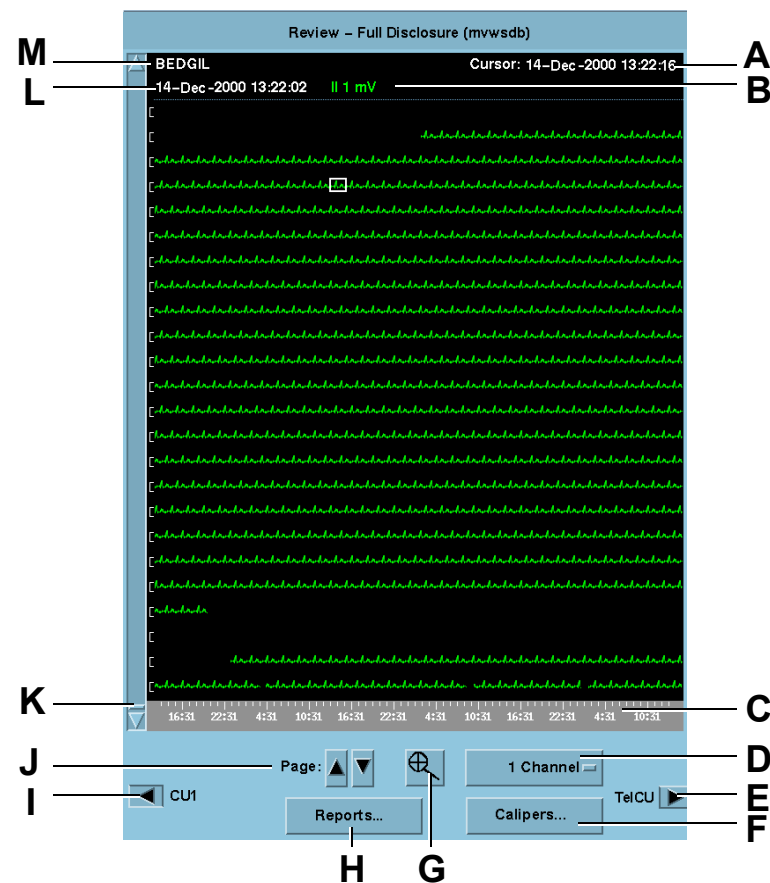
The Full Disclosure screen displays in compressed screen (zoom out) format.

All time changes that occur at the Infinity Central-Station are identified on the waveform by a yellow tick mark. Appended data is marked with a white tick mark, which represents the time the data was appended.

### NOTE

When moving between Disclosure screens and other screens (such as Trends) near a system time change, time stamps may be slightly different on the two screens. This occurs because the Infinity CentralStation and the bedside monitors have different methods of annotating time. These time differences do not affect the displayed data which *correlates exactly in both applications* to an internal time stamp.

# Full Disclosure Review Screen



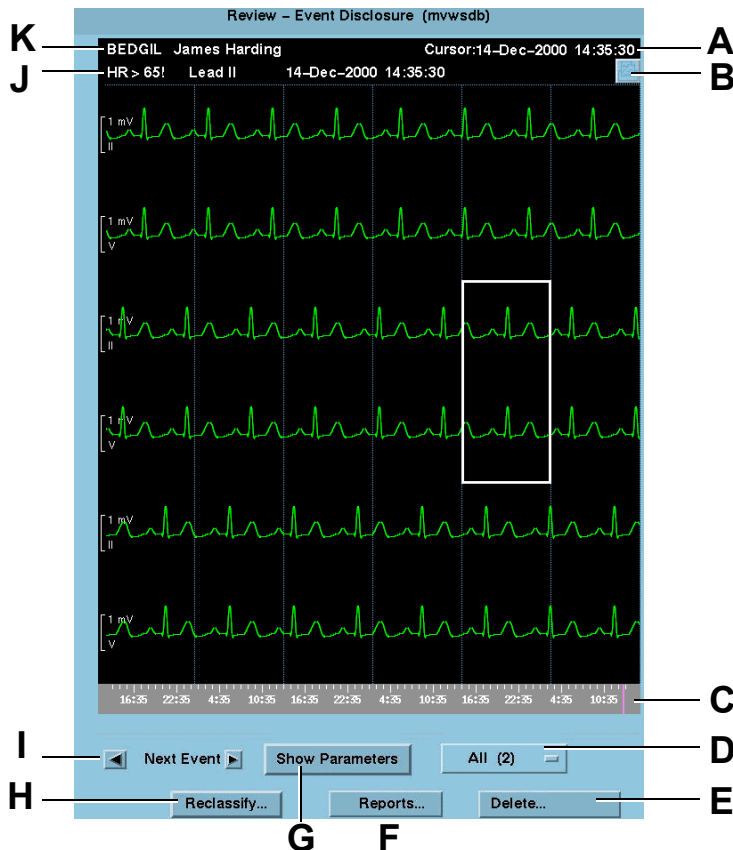
- A** Time/date at the cursor position
- B** Gain setting and lead label
- C** Time Line reflects Full/Event Disclosure Storage Option (see page 209). When navigating through care units the time line changes to reflect the storage time of the selected care unit
- D** Select number of waveforms to display
- E** Scroll through care unit data when electronic patient transfer or manual transfer. When there is no *previous* or *next* care unit data, the arrow is unavailable for selection and no care unit label displays.
- F** Open ECG Caliper Tool Screen
- G** Select viewing mode (Zoom In or Zoom Out)
- H** Generate Full Disclosure reports
- I** (Same definition as label E.)
- J** Navigate by page
- K** Vertical scroll bar
- L** Time/date of oldest displayed data
- M** Patient identification.

All time changes that occur at the Infinity Central-Station are identified on the waveform by a yellow tick mark. Appended data is marked with a white tick mark, which represents the time the data was appended.



## Event Disclosure Review Screen

When you open the **Review-Event Disclosure** screen, the most recent event information displays along with its associated waveform (if the waveform is being stored).



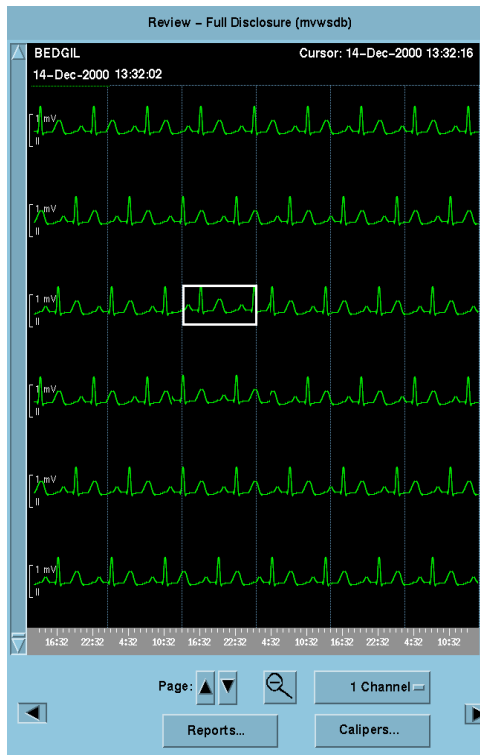
- A** Time/date at the cursor position
- B** To select an event for the shift report, click on the symbol. When an event is annotated, the comment displays.
- C** Time line with tick marks for all stored events. Prominent tick mark identifies displayed event
- D** Event class
- E** Deletes event classes
- F** Opens Reports submenu
- G** Displays parameter values at time of the event (see page 224)
- H** Permits reclassification of event
- I** Scrolls to next/previous event on time line. To set up event classes, see (see page 225).
- J** Cause, lead, time, and date of stored event
- K** Patient identification.

# Full/Event Disclosure Review Screen Functions

You can view a patient’s Disclosure data (Full or Event) in **Zoom Out** (compressed) or **Zoom In** (expanded) format.

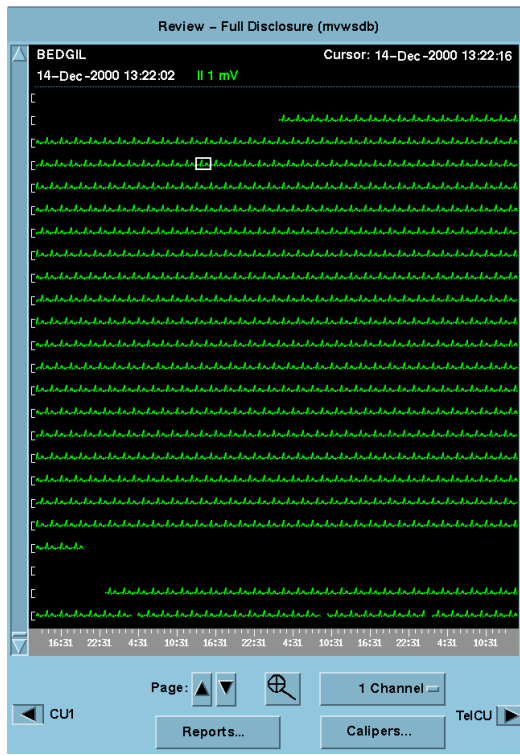
## Zoom In

For 1 channel approximately 36 seconds of data displays on a page.



## Zoom Out

For 1 channel approximately 9.5 minutes of data displays on a page.



The following lists the functions available on the Disclosure Review screens.

To navigate through and view stored Full/Event Disclosure data for multiple care units, see page 216.

### Full Disclosure

- To generate reports, see page 193.
- To select how many channels of data to view, see page 216.
- To select gain settings, see page 219.
- To select which waveforms to store, see page 220.
- To access the ECG caliper tool, see page 227.
- To configure the trend order setup for Graphical Trend Report, see page 196.

### Event Disclosure

- To view parameter values, see page 224.
- To delete events (an individual event or an entire event class), see page 224.
- To annotate events, see page 222.
- To rename or reclassify events, see page 225.

| Function                             | Steps                                                                                                                                                                                                                                                                                                                                                                                                                       | Notes                                                                            |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| To select which waveforms to display | <ol style="list-style-type: none"> <li><b>1</b> Click on <b>Options</b> on the <b>Disclosure Review screen</b> menu bar.</li> <li><b>2</b> Click on <b>Display Options...</b></li> <li><b>3</b> Click on the channel's <i>Parameter</i> button. A list of available settings appears.</li> <li><b>4</b> Click on the desired setting.</li> <li><b>5</b> Repeat steps 3 and 4 for each channel.</li> </ol>                   | The time at the cursor appears in white text on the <b>Patient Setup</b> screen. |
| To set the display gain              | <ol style="list-style-type: none"> <li><b>1</b> Click on <b>Options</b> on the <b>Disclosure Review screen</b> menu bar.</li> <li><b>2</b> Click on <b>Display Options...</b></li> <li><b>3</b> Click on the channel's <i>Gain/Scale</i> button. A pop-up of available settings appears (see page 221).</li> <li><b>4</b> Click on the desired setting.</li> <li><b>5</b> Repeat steps 3 and 4 for each channel.</li> </ol> | None                                                                             |

| Function                                                                                                                                                                                                                                                                                                                                                | Steps                                                                                                                                                                                                                                                                            | Notes                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To turn the pacer mark ON/OFF                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Click on <b>Pacer Mark</b> toggle button on the bottom of the waveform selection screen.</li> </ul>                                                                                                                                       | Pacer marks for bedside patients only display at the Infinity CentralStation if they are available at the bedside monitor.                                                                                       |
| To select which waveforms to store                                                                                                                                                                                                                                                                                                                      | <ol style="list-style-type: none"> <li>Click on <b>Options</b> on the <b>Disclosure Review screen</b> menu bar.</li> <li>Click on <b>Storage Options...</b></li> <li>Click on each channel's 'Parameter' button.</li> <li>Click on the desired setting in the pop-up.</li> </ol> | Channel defaults:<br>1 - Lead II<br>2 - Lead V<br>3 - ART<br>4 - SpO2                                                                                                                                            |
| Enable/Disable Automatic waveform tracking                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Click on the <b>Auto Track</b> On/OFF toggle button near the bottom of the Waveform Storage screen.</li> </ul>                                                                                                                            | When <b>AutoTrack</b> is <b>ON</b> , the first four waveforms displayed at the bedside are stored automatically.<br><br>When <b>AutoTrack</b> is <b>OFF</b> , you must manually select which waveforms to store. |
| <b>NOTE</b><br>When the locked option for the expanded waveform is installed, AutoTracking is always enabled (and it cannot be disabled).                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                  |
| <b>WARNING</b><br><b>If you change leads at the bedside monitor when AutoTracking is set to OFF at the Infinity CentralStation, you must also change the lead selection at the Infinity CentralStation. Otherwise, the bedside waveforms may no longer correspond to the Infinity CentralStation waveform storage settings and data may be missing.</b> |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>Click on <b>Accept</b> to save changes, <b>Undo</b> to revert to the original settings, or <b>Cancel</b> to exit without saving changes.</li> </ul>                                                                                                                                                              |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                  |

**Available Gain Settings**

| <b>Parameter</b>               | <b>Available Settings</b>                                                                                        | <b>Default Size</b>                                                                                                             |
|--------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| ECG                            | 0.25, 0.5, 1, 2, 4, and 8 mV                                                                                     | 1 mV                                                                                                                            |
| ART/IBP, LV, GP1, GP2, P1a-P3d | 0 to: 20, 40, 50, 100, 125, 150, 175, 200, 225, 250, 300 mm Hg<br>0 to: 8, 12, 16, 20, 24, 32, 40 kPa            | Adult<br>0 to 200 mm Hg/0 to 24 kPa<br><br>Neonatal<br>0 to 100 mm Hg/0 - 16 kPa<br><br>Pediatric<br>0 to 150 mm Hg/0 to 20 kPa |
| PA, RV                         | 0 to: 20, 40, 50 100, 125, 150 mm Hg<br>0 to: 4, 6, 8, 12, 16, 20 kPa                                            | 0 to 50 mm Hg/0 to 8 kPa                                                                                                        |
| LA, RA, CVP                    | -5 to: 5, 15, 20, 25, 40, 50, 100, 150, 200, 250, 300 mm Hg<br>-1 to: 2, 4, 5, 6, 8 kPa                          | -5 to 20 mm Hg/ -1 to 5 kPa                                                                                                     |
| ICP                            | -30 to 30:<br>0 to: 20, 40, 50, 100, 150, 200, 250, 300 mm Hg<br><br>-4 to 4:<br>0 to: 1, 2, 3, 4, 8, 16 kPa     | 0 to 20 mm Hg/0 - 4 kPa                                                                                                         |
| SpO <sub>2</sub>               | 10, 20, 30, 40, 50, 60, 70, 80, 90, 100%                                                                         | 40%                                                                                                                             |
| etCO <sub>2</sub>              | 0 to: 40, 60, 80 mm Hg<br>0 to: 5, 8, 12 kPa<br>0 to: 5, 8, 12%                                                  | 0 to 40 mm Hg/0 to 5 kPa<br>0 to 5%<br>0 to 12%/0 to 12 kPa (MIB acquired)                                                      |
| Ventilator flow (MIB)          | -5 to 5 L/min<br>-10 to 10 L/min<br>-20 to 20 L/min<br>-50 to 50 L/min<br>-100 to 100 L/min<br>-200 to 200 L/min | Adult:<br>-100 to 100 L/min<br><br>Neonatal:<br>-20 to 20 L/min<br><br>Pediatric:<br>-50 to 50 L/min                            |
| Ventilator pressure (MIB)      | -5 to 25 cmH <sub>2</sub> O<br>-10 to 50 cmH <sub>2</sub> O<br>-20 to 120 cmH <sub>2</sub> O                     | -5 to 25 cmH <sub>2</sub> O                                                                                                     |

| Parameter                                                               | Available Settings                     | Default Size |
|-------------------------------------------------------------------------|----------------------------------------|--------------|
| Agent (MGM):<br>Halothane, Isoflurane, Enflurane, Seflurane, Desflurane | 0-1%, 0-2%, 0-3%, 0-5%,<br>0-0%, 0-20% | 0-3%         |
| O <sub>2</sub> (MGM)                                                    | 20 - 50%, 20-100%                      | 20-100%      |
| Resp                                                                    | 5-100%                                 | 40%          |

## Annotating Events

On the **Event Disclosure Review** Screen you can annotate clinically significant events. For example, you can add a remark to an event such as *Chest Pain Decrease* after administering medications. A maximum of 100 annotations can be stored per patient. If you attempt to store a comment once the maximum capacity has been reached, a pop-up appears alerting you that the storage capacity is full. If this happens, you must delete some comments to make room for new ones.

A list of frequently used terms is available to select from to annotate an event.

To annotate an event:

- 1 Access the patient's Event Disclosure Review screen (see page 215).
- 2 Scroll to the desired event either by clicking within the Event window or by using the ◀ **Next Event** ▶ button.
- 3 Click the right mouse button to activate the annotate pop-up list.
- 4 Either click on a term within the pop-up or type in a new comment (maximum of 25 characters).
- 5 Click on **Annotate** to add the comment to the event database or on **Exit** to leave the pop-up without storing an annotation. After an annotation is assigned to an event, it appears at the top of the waveform of the Even Disclosure Review screen.

## Creating Your Own Annotations List

The annotations list contains a predefined set of terms that you cannot alter. However, you can add your own terms to the list (maximum capacity = 100 terms).

To create your own annotations list:

- 1 Open the patient's Event Disclosure Review screen (see page 215).
- 2 Click on the right mouse button to activate the Notes pop-up.

- 3 Type in the note and click on the + button to *add* the entry to the top of the list; or highlight the list item to *delete* and press the – button.

### NOTE

You cannot delete a predefined entry from the annotation list; you can only delete terms that you have added manually.

## Viewing Events

To view and navigate through stored events:

- Click on the time line of the Event Review screen and click on the ◀ **Next Event** ▶ arrows.

To view a class of events:

- 1 Click on the *event class* option button to display an event class list with the number of events for each.

**All** includes all stored events and is the default event class that displays when the screen opens.

### NOTE

Only event classes that have an event in the last 28, 48, or 72 hours (depending on the configured option) are listed unless the disclosure record was locked.

- 2 Click on the desired event class.

## Deleting Events

You can either delete individual events or entire event categories.

To delete an individual event:

- 1 Open the **Event Review** screen (see page 215).
- 2 Select the event you wish to delete.
- 3 Click on **Delete...**
- 4 Click on **Event** to delete the currently displayed event (the previous event is displayed; if no previous events exist, the *next* event is displayed instead).

To delete an entire event class:

- 1 Click on the *Event Class* button to display a list.
- 2 Click on the Event Class you wish to delete.
- 3 Click on **Delete...** to delete the entire Event Class. This activates a confirmation pop-up.
- 4 Click on **Yes** in the pop-up to delete all of the events for the selected class or on **No** to exit the pop-up without deleting any events.

## Viewing Parameter Values

To see parameter values stored at the time of an event:

- 1 Open the **Event Review** screen (see page 215).
- 2 Click on **Show Parameters**.

Arrow buttons may appear on either side of the parameter screen pop-up depending on the number of stored parameters for the events. Use these arrows to scroll through the list.



## Renaming/Reclassifying Events

You can rename any stored arrhythmia event using labels from a preconfigured selection menu, as follows:

- 1 Open the **Event Review** screen (see page 215) and select the event you wish to rename.
- 2 Click on **Reclassify**. The *Reclassify Event* pop-up displays.

The pop-up list includes all current Infinity arrhythmia class labels and ten additional labels with a preceding asterisk. These additional labels can either be the defaults or user-configured event names (see page 293).

- 3 Select the desired event label and click on **Accept**.  
A confirmation pop-up displays which asks you to confirm that you want to reclassify the **Current Event** with your selection.
- 4 Click on **Continue** in the confirmation pop-up to accept the change or **Cancel** to keep the current label.

The new label for the event displays on the Event Disclosure Review screen. If an event has a label that is user-configured, it is grouped in the event class, "other" (see page 223).

You can only reclassify events if they are in one of the following arrhythmia categories.

|      |       |     |       |
|------|-------|-----|-------|
| ASY  | VF    | VT  | RUN   |
| AIVR | CPT   | BGM | TACH  |
| BRDY | PAUSE | SVT | OTHER |

Otherwise, the **Reclassify** button is unavailable for selection on the Event Review Screen.

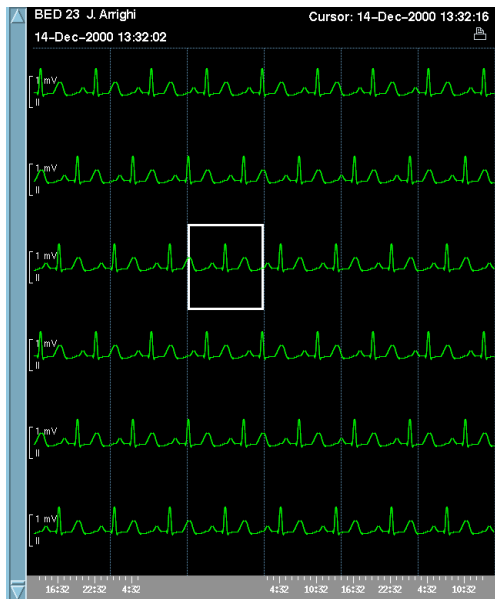
Disclosure Review Screen Cursor

When you open a Disclosure Review screen data is centered around the cursor time, which appears at the top right of the screen.

Changing the cursor time

- Click the mouse anywhere on the displayed waveforms to change the cursor time.

1 Channel cursor



NOTE

Cursor time does not change when you scroll through the data or select a time on the time line.

The cursor height varies with the number of channels selected.

2 Channel cursor



## ECG Caliper Tool

---

An ECG caliper tool lets you obtain, calculate, store, and review time-based interval measurements, and averages using Full Disclosure.

Before you can use the ECG caliper tool, the Full Disclosure **Channel 1** setting must be set to display ECG data (see page 219).

To access the ECG Caliper Screen:

- 1 Open the **Full Disclosure Review** screen (see page 215).
- 2 Position cursor over the ECG area you wish to measure.
- 3 Click on **Calipers...** A submenu displays.
- 4 Click on the desired function:

### Measure

- Executes **Caliper Measure** screen which uses 18 seconds of complete Full Disclosure ECG data following cursor time.

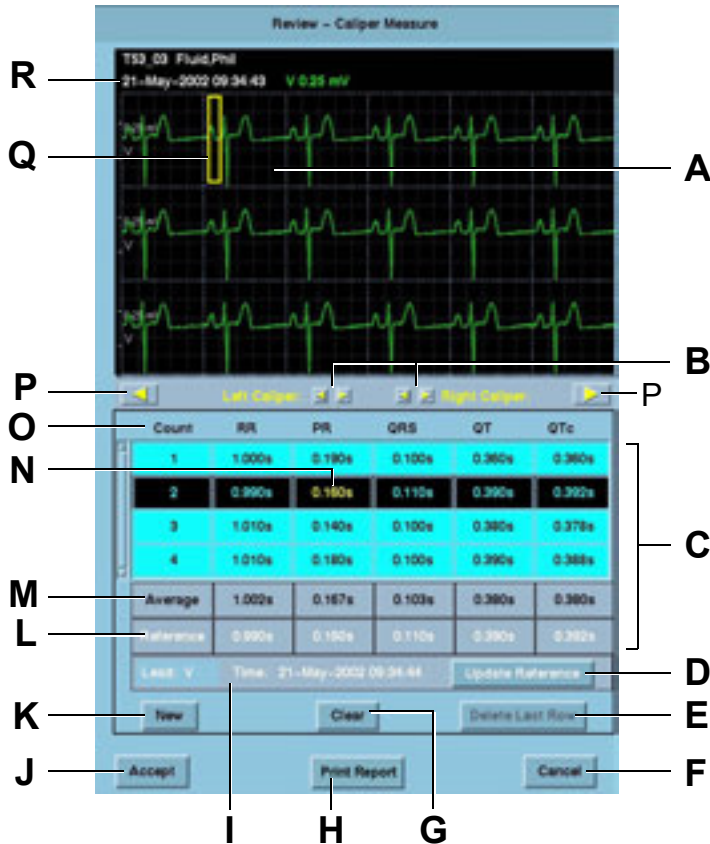
### Review

- Displays the **Caliper Review** screen from which you can view stored measurements.

### NOTE

The **Review** button is unavailable for selection when no measurements are stored for the patient.

## Caliper Measure Screen



- A** Waveform area: Right-click mouse to move right caliper anywhere on the waveform.
- B** Controls to adjust the caliper
- C** Summary Table
- D** Updates reference
- E** To delete the last row of measurements
- F** To cancel measurements
- G** To reset the selected measurement to zero
- H** To generates a printed report of current measurements (see page 189)
- I** Last saved reference lead, date, and time
- J** To accept/save measurements, calculations, and averages

- K** To generate a new row of caliper measurements
- L** Reference values (white text)
- M** Interval averages
- N** Current measurement (yellow text)
- O** Interval measurements and QTc headings
- P** **Interval Advance Arrows** rotate or move calipers to the next common interval in the strip. In order for the calipers to move, you must measure the first RR interval in the strip.
- Q** Left click mouse to move left caliper anywhere on the waveform.
- R** Cursor date and time

## Waveform Area

Cursor time marks the first second of an 18-second ECG strip that displays at the top of the **Caliper Measure** screen. If less than 18 seconds of data follows the cursor in the FD review screen, the last full 18 seconds of data will display.

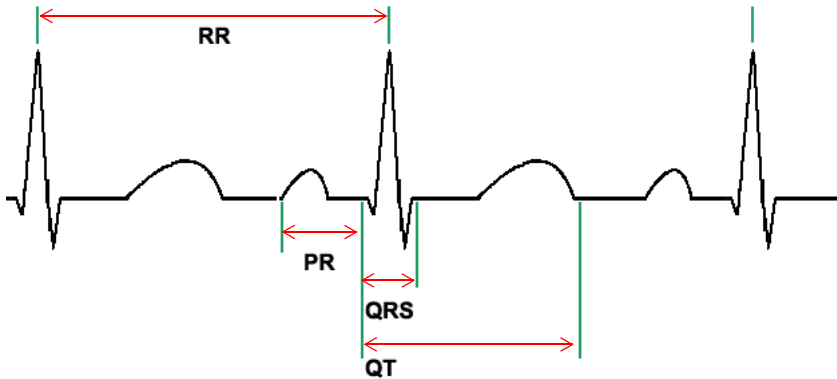
Initially the calipers appear at the beginning of the waveform but can be set by clicking the mouse buttons.

## Summary Table

The summary table of the **Caliper Measure** screen displays interval measurements taken (see page 230), calculated averages, and saved reference information (see page 231).

It includes up to ten **Count** rows that contain RR, PR, QRS, and QT interval measurement entries, and calculates QTc for each corresponding RR and QT entry.

## Interval Measurement Calculation



|         |                                          |
|---------|------------------------------------------|
| RR avg  | Sum of RR intervals/quantity of samples  |
| PR avg  | Sum of PR intervals/quantity of samples  |
| QRS avg | Sum of QRS intervals/quantity of samples |
| QT avg  | Sum of QT intervals/quantity of samples  |
| QTc     | QT/square root of RR                     |
| QTc avg | Sum of QTc values/quantity of samples    |

## Taking Interval Measurements

When the **Caliper Measure** screen first opens there is one highlighted row of measurements with zero values. As soon as one row of measurements is taken, these values display as **Average** measurements. All rows, except those containing no measurements, are used to calculate averages.

Your first measurement must be the RR interval in order to determine the QT interval. After you measure the first RR interval, measure the PR, QRS and QT intervals.

To measure an interval:

- 1 Move the cursor to the RR interval measurement box and click. The active measurement displays in yellow text.
- 2 Position the cursors over a complex in the ECG strip.
- 3 Right-click the mouse button to set the right caliper.
- 4 Left-click the mouse button to set the left caliper.

### NOTE

It is not possible to set negative measurement values.

- 5 Move the cursor to the next interval measurement box and click.
- 6 Repeat steps 2 through 5 for each desired measurement interval of that complex.
- 7 Once you measure an RR interval between complexes, you can use the **Interval Advance Arrows** (see page 228) to *march* or rotate the cursors the distance of the RR interval along the strip.

After all your measurements are complete you can save the data for review or future report generation or you can print an ECG Caliper Report immediately.

- To save the measurements click on **ACCEPT**.
- To print an ECG Caliper report immediately click on **Print Report**.

If you **Cancel** the **Caliper Measurement** screen without saving data, a pop-up displays. To save measurements not previously stored, click **Yes** in the pop-up.

## Adding/Removing/Changing Measurement Rows

The **Caliper Measure** screen *Summary Table* holds up to ten rows of interval measurements that are used to determine averages.

To add a row of interval measurements:

- 1 Click on **New**. A row of interval measurements with zero values is added and is highlighted.
- 2 Complete interval measurements for the complex (see page 230).
- 3 Repeat steps 1 and 2 for each new row. You can have up to ten **Count** rows.

To remove the last row of interval measurements:

- 1 Click on any interval measurement in the last measurement row to select and highlight the row.
- 2 Click on **Delete Last Row** to remove the row from the summary table.

The **Delete Last Row** button is unavailable for selection until more than one row of measurements exists and the last row of measurements is highlighted.

To reset a row of interval measurements:

- 1 Click on any interval measurement in a row to select and highlight the row.
- 2 Click on **Clear** to reset all measurements to zero.

## Saving References

At the bottom of the summary table saved reference measurements appear in the row labeled **Reference** and display in white text. **Reference Time:** shows the last saved reference date and time.

Reference measurements are displayed and stored with all associated caliper interval measurements until a new reference is selected.

To save a reference:

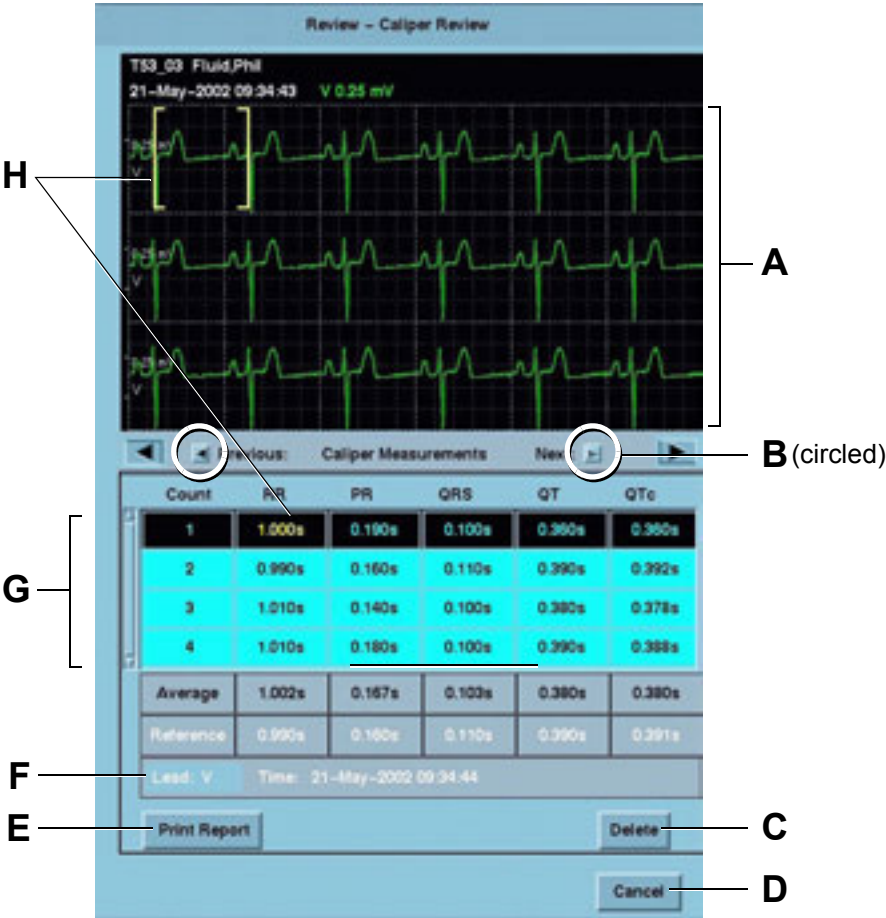
- 1 Click on any interval measurement in a row to select and highlight the row.
- 2 Select **Update Reference** to save these reference measurements.
- 3 When you save a new reference and reference values already exist, a pop-up displays requesting confirmation.
- 4 Select **YES** in the pop-up to update the reference.

## Caliper Review Screen

The **Caliper Review** screen lists all saved caliper measurements in its summary table. It can contain up to 72 hours of saved measurements.

From the **Caliper Review** screen you can:

- Delete saved measurements
- Generate a Caliper Report (see page 189)



- A Waveform area
- B Scroll through the saved measurements. (Arrows are unavailable for selection at the beginning and end of stored data.)
- C Delete the saved measurements
- D Exit the screen
- E Generate a Caliper Report (see page 189)
- F Last saved reference lead, date, and time
- G Summary table
- H Calipers display selected measurement (yellow text)

When the **Caliper Review** screen opens the latest saved interval measurements, averages, and associated references display along with the ECG waveform. Whenever you click on an interval measurement in the summary table, the calipers display the measured interval on the actual complex in the waveform area.

To open the **Caliper Review** screen, see page 227.



### Deleting interval measurements

To delete interval measurements:

- 1 Open the **Caliper Review** Screen (page see page 227).
- 2 Click on **Delete**.  
A pop-up displays. To remove **all** of the measurements in the summary table, click on **Yes** in the pop-up.

## Full and Event Disclosure Reports

---

The following reports are generated from the **Full Disclosure Review** screen:

- Strip reports
- Shift reports
- Graphical Trend reports
- Selected Events reports
- One-hour reports
- 24-hour reports
- Selected Strip reports
- Patient Status reports
- Caliper reports

The following reports are generated from the **Event Disclosure Review** screen:

- Strip reports
- Shift reports
- Graphical Trend reports
- Selected Events reports

Full Disclosure and Event Disclosure reports are printed on the laser printer even if you exit the screen.

## Exporting Full/Event Disclosure Data

---

### Exporting Full Disclosure Data to a Third-Party Device

When exporting data, a minimum of a patient ID must be entered in the ADT screen.

The **Export Waveforms** feature exports up to 25 hours of four continuous waveforms to a compatible third party device (contact your local Dräger representative for a list of compatible third party devices).

To Export Waveforms from Full Disclosure:

- 1 Open the patient's **Full Disclosure** Screen (see page 215).
- 2 Click on **Reports...**  
A pop-up menu displays.
- 3 Click on **Export Waveforms**.

#### NOTE

- If the **Export Waveforms** selection is not listed, the destination IP address is not configured for the device. Consult the Hospital Biomedical Engineering Dept.
- The waveform export feature supports the transfer of one file at a time, during which time the Export Waveforms selection is unavailable for selection.

The **Export Strip** feature exports up to 4 waveforms each containing 18 seconds of data in which the cursor time marks the center of the exported waveform segment.

To Export Full Disclosure Data:

- 1 Open the patient's **Full Disclosure** Review Screen (see page 215).
- 2 Click on the cursor to set the *end time* of the data to be exported.
- 3 Click on **Reports**.  
A pop-up menu displays.
- 4 Click on **Export Strip**.

#### NOTE

If the **Export Strip** selection does not appear, the Infinity MegaCare IP address is not configured. Consult the hospital biomedical engineering department.

### Exporting Events from Event Disclosure

The **Export Event** feature exports 18 seconds of the selected event's waveform data in which the event is the middle of the waveform segment, the event's cause string (e.g. ASY), and the parameter values at the time of the event.

To export the Event Disclosure Data:

- 1 Open the patient's **Event Disclosure**. Review screen (see page 215).
- 2 Access the desired event

- 3 Click on **Reports...**  
A pop-up menu displays.
- 4 Click on **Export Event**.

#### NOTE

If the **Export Event** button does not appear, the Infinity MegaCare IP address is not configured. Consult the hospital biomedical engineering department.

## Special Conditions

Various situations during patient monitoring may affect Full/Event Disclosure data collection.

| Special Condition                                                                                                            | Effect                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The monitoring device is being powered up.                                                                                   | <ul style="list-style-type: none"> <li>– Full Disclosure data storage resumes accordingly.</li> <li>– No Events are stored until the monitor is operational.</li> </ul>                                                                         |
| The monitoring device is powered down.                                                                                       | The Full/Event Disclosure data is stored.                                                                                                                                                                                                       |
| CPS/IDS is offline, has failed, or is powered down.                                                                          | <ul style="list-style-type: none"> <li>– Blank spaces appear in the Full Disclosure data for the duration of the interruption.</li> <li>– No events are stored during the interruption.</li> </ul>                                              |
| The patient monitor is in <b>Standby</b> mode.                                                                               | <ul style="list-style-type: none"> <li>– Blank spaces appear in the Full Disclosure data for the mode duration.</li> <li>– No events are stored for the mode duration.</li> </ul>                                                               |
| The <b>Standby</b> mode is canceled.                                                                                         | Storage of Full/Event Disclosure data resumes.                                                                                                                                                                                                  |
| The patient is discharged at the bedside.                                                                                    | Full/Event Disclosure data remains until manually deleted at the Infinity CentralStation.                                                                                                                                                       |
| The patient demographics are changed.                                                                                        | Full/Event Disclosure data is stored under the edited demographics.                                                                                                                                                                             |
| The bedside monitor is disconnected.                                                                                         | <ul style="list-style-type: none"> <li>– The Full Disclosure database shows blanks for the duration of the interruption. A message describes the absence of data.</li> <li>– No events are stored while the monitor is disconnected.</li> </ul> |
| The pod/cartridge or a signal source such as a transducer is connected.                                                      | Full Disclosure data collection begins for any displayed parameter that was selected for the Full Disclosure application.                                                                                                                       |
| The pod/cartridge or a signal source is disconnected.                                                                        | Blanks will appear in the Full Disclosure display of the parameter for the duration of the interruption.                                                                                                                                        |
| Network date and/or time changes.                                                                                            | Up to 20 system time changes are stored. The time stamps on the waveforms and event calls mark the time the data was actually stored.                                                                                                           |
| The patient leads are changed at the bedside monitor while Full/Event Disclosure automatic waveform tracking is <b>OFF</b> . | Data may be missing since the leads selected for display on the bedside monitor may no longer correspond to the Infinity CentralStation storage settings.                                                                                       |

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# VentCentral Option

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|                                                   |     |                                              |     |
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## Overview

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With the VentCentral option you can review a patient's ventilator information at the Infinity CentralStation.

### NOTE

- Certain parameters, identified as measured values, are derived from other values or settings provided by the ventilator and may not reflect actual values. See the operating instructions for your specific device for detailed information on derived measurements.
- The VentCentral option supports only modular monitors with VF and later software. If VentCentral is intended for use with an MIB device, the MIB must also have VF (or later) software.

### WARNING

**Ventilator data should be used for informational purposes only. Refer to the primary monitoring device before making therapeutic or diagnostic decisions. Always verify alarm condition at the actual device.**

The displayed parameters and settings may originate from bedside monitor-connected devices, such as the following:

### NOTE

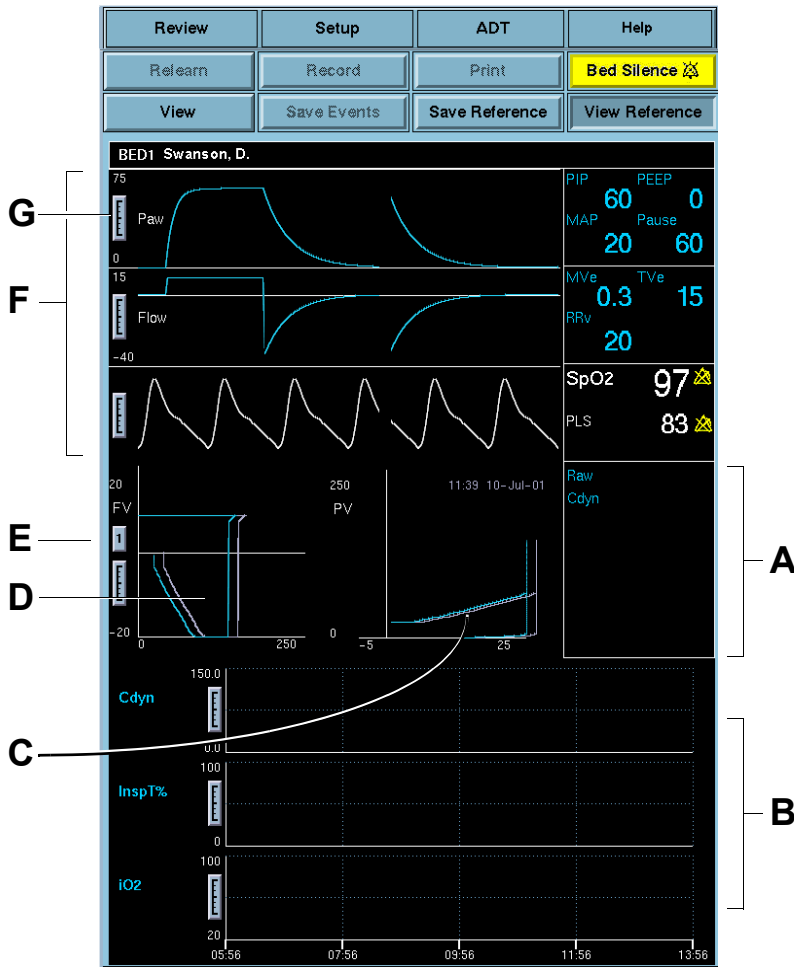
You cannot change any MIB-connected device settings from the Infinity CentralStation.

- Medical Information Bus (MIB)-connected Ventilators and anesthesia machines (see the tables starting on page 248.)
- Infinity hemodynamic bedside monitors (parameters only)
- etCO<sub>2</sub>/Respiratory Mechanics Pod directly connected to a bedside monitor (parameters only)

You may interact with the VentCentral option via three Ventilator screens:

- *Ventilator screen*
- *Ventilator Settings Review screen*
- Ventilator Setup screen

## The Ventilator Screen



- A** Parameter box with fixed set of parameters. If a parameter is not available, the area is blank.
- B** 8-hour trend of user-selected parameters
- C** Pressure vs. Volume Loop
- D** Flow vs. Volume Loop
- E** Select "1" for single loop and "4" for multi-loop display. During multi-loop configuration, four successive loops (of different shades) show. All loops erase and the sequence restarts after the fourth loop is complete.
- F** Display of 3 waveform channels
- G** Waveform adjustment scale

## Accessing the Ventilator Screen

To access the Ventilator screen:

- 1 Click on the patient's Main Screen parameter box. For information on accessing a remote patient, see page 109.
- 2 Click on **View**.
- 3 Click on **Ventilator**.

### Scaling

If the VentCentral waveforms, loops, or trends appear clipped, you can change the display size by clicking on the scale at the left edge of the waveform area, or to the left of the trends.

## Waveform Area in Ventilator Screen

The Ventilator screen allows you to select up to three waveforms for display and prioritize their order (see page 245). If a waveform is not available from the bedside or the MIB-connected device, the next waveform in the priority list is used to fill the three waveform areas. If less than three waveforms are available, waveform areas are blank.

### Parameter Box

The parameter boxes to the right of the waveform area may consist of MIB and/or Infinity parameters and cannot be changed. These parameter boxes are independent of the bedside monitor and do not flash when a parameter is in alarm. MIB parameters do not support units, limits, and/or alarm-off icons. For Infinity parameters a crossed bell icon appears next to the parameter label when the alarm function is turned off.

## Loops Area in Ventilator Screen

The real-time ventilator loops in the Ventilator screen are synchronous with the associated breath. Flow Volume is shown on the left and Pressure Volume on the right. Loops are drawn with the same display scale as the associated waveforms.

### Displaying Loops

You can save and display a **Reference Loop** with the actual loop display. The color of the reference loop is magenta. The Ventilator screen can display a **Single Loop** or **Multi-loop** draw.



### Loop Parameter Box

The parameter box next to the loop area contains dynamic compliance, dynamic resistance, and arterial blood gas values.

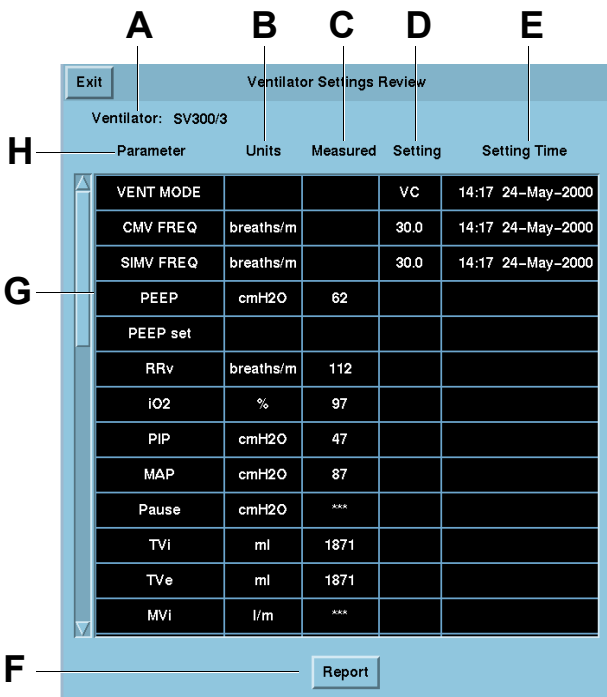
If blood gas values are not available, the labels do not appear. The following table displays the parameter order and ranges.

| Parameter        | Units                      | Measurement Range                |
|------------------|----------------------------|----------------------------------|
| pH               | pH, nmol/L                 | 6.6 - 7.8                        |
| HCO <sub>3</sub> | mEq/l, mmol/L              | 1-200 mEq/l                      |
| Hct (Hematocrit) | %                          | 15-75%                           |
| tHb (Hemoglobin) | g/dl, g/100ml, g/L, mmol/L | 5-25 g/dl                        |
| PCO <sub>2</sub> | mmHg (kPa)                 | 10-200 mmHg (1.333 - 26.66 kPa)  |
| PO <sub>2</sub>  | mmHg (kPa)                 | 10 - 700 mmHg (1.33 - 93.31 kPa) |

### Trend Area in Ventilator Screen

Three trend graphs on the **Ventilator** screen show the most recent 8 hours of trend data for user-selected parameters (see page 246). If a selected parameter is not available from the bedside/MIB device, the next parameter from the **Trends Setup** screen is automatically displayed. If less than three parameters are available, the **Ventilator** screen parameter graph is blank.

## The Ventilator Settings Review Screen



- A** The Ventilator field displays the ventilator type.
- B** The Units column displays the selected units of measure at the MIB device or ventilator.
- C** The Measured column shows the most recently updated measurement value. Certain parameters, identified as measured values or settings, are derived from the values provided by the ventilator and may not reflect the actual values.
- D** The Setting column displays the most recent setting of the ventilator/MIB device.
- E** The Setting Time column displays the time and date a setting was modified.
- F** The Report button generates a Ventilator report (see page 171).
- G** The scroll bar provides access to additional parameters.
- H** The Parameter column displays the parameter or setting labels.

## Accessing the Ventilator Settings Review Screen

To review the Ventilator settings:

- 1 Click on the patient's parameter box.
- 2 Click on **Review**.
- 3 Click on **Ventilator Settings Review**.

## Parameter and Settings Order

The **Ventilator Settings Review** screen displays settings and supported parameters sent by the connected MIB device, Infinity bedside monitor, or compatible bedside monitor pod.

Each **Ventilator Settings Review** screen parameter and setting displays in a fixed order. If a setting or parameter is not available, the corresponding field on the **Ventilator Settings** screen is blank. Additional settings and parameters not assigned to the fixed order are at the end of data received from the MIB-connected device. Parameters/settings sent to Infinity monitors for various MIB-compliant devices are listed in tables starting on page 248.

### NOTE

Ventilator data displayed in VentCentral will vary by ventilator model version.

### NOTE

The Infinity CentralStation does not support Acoustic Evoked Potential (AEP) and Somatosensory Evoked Potential (SEP) functions that are enabled at a bedside monitor.

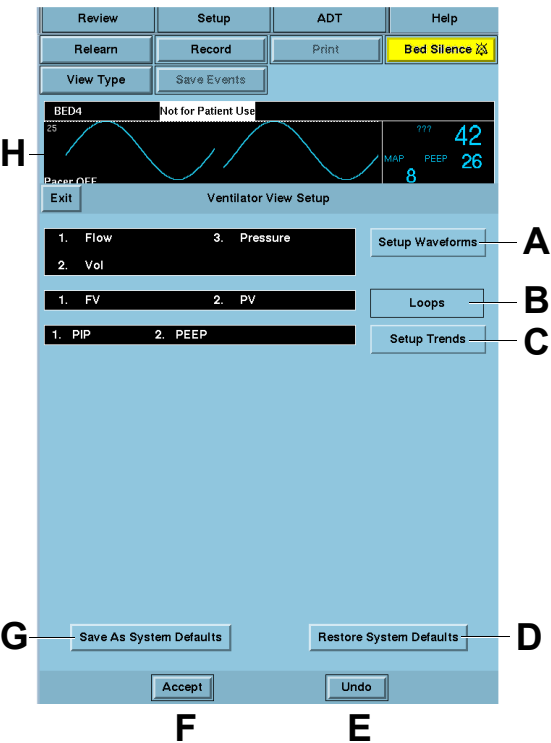
## Special Conditions

The following table displays special conditions affecting the Ventilator Settings Review screen.

| Condition                                                                                                             | Effect                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Disconnected parameter hardware                                                                                       | The VentCentral application is not available and cannot be accessed for the respective patient.   |
| Reset of MIB- device, bedside monitor, or Infinity CentralStation                                                     | The <b>Ventilator Settings</b> screen time stamp reflects the first setting following the reset.  |
| Bedside monitor PICK AND GO                                                                                           | The <b>Ventilator Settings</b> screen time stamp reflects the monitor's reconnect to the network. |
| First connection of MIB-compliant ventilator, etCO <sub>2</sub> /Respiratory Mechanics Pod, or Delta/Delta XL monitor | Time stamp reflects the first setting received after the device is connected.                     |
| Blood gas values stored for more than 30 minutes                                                                      | Values are removed from the VentCentral screen.                                                   |
| Device sends more digits than the Ventilator Settings screen field can accommodate                                    | Instead of the parameter values, overflow characters display.                                     |

# Ventilator View Setup Screen

The **Ventilator View Setup** screen permits selecting an individualized display for a specific patient.



- A** Opens the Waveforms Setup screen
- B** Fixed - Always shows Flow vs. Volume and Pressure vs. Volume
- C** Opens the Trends Setup screen
- D** Restores system defaults
- E** Reverts to previous settings
- F** Accepts new settings/changes
- G** Saves current setup as the system default for all patients
- H** Top ventilation waveform

## Accessing the Ventilator View Setup Screen

To access the Ventilator View Setup screen:

- 1 Click on the patient's Main Screen parameter box.
- 2 Click on **Setup**.
- 3 Click on **Ventilator View Setup**.

To access a remote patient, see page 109.

## Selecting Waveforms for Display

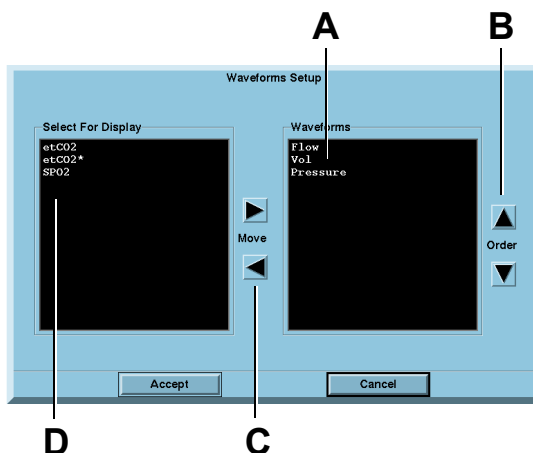
To set up the display priority of ventilator parameters:

### NOTE

If a selected waveform is not available from the bedside monitor, the next waveform is selected to fill available Infinity CentralStation waveform channels.

- 1 Open the **Ventilator View Setup** screen (see page 244).
- 2 Click on **Setup Waveforms**.

The **Waveforms Setup** pop-up displays.



- A Parameters selected for display
- B Arrow buttons allow you to change the priority order of selected waveforms
- C Arrow buttons allow you to move selected parameters from one window to the other
- D Parameters available for selection

- 3 Select the desired display configuration.

The left/right arrows move desired waveforms from the display window to the *Waveforms* window. The up/down arrows move parameters in a desired sort order

### NOTE

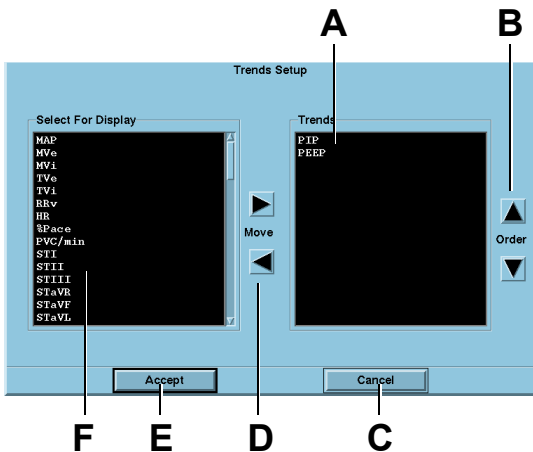
You may select several parameters at once before you move them from one window to the other.

- 4 Click on **Accept** to save or **Cancel** to keep prior settings.
- 5 Click on **Save as System Defaults** in the **Ventilator View Setup** screen to retain this setup as the system default.

## Selecting Waveforms for the Trend Display

To select and prioritize ventilator parameters for trend display:

- 1 Open the **Ventilator View Setup** screen (see page 244).
- 2 Click on **Setup Trends**. The **Trends Order Setup** pop-up displays.



- A Trends selected for display
- B Arrow buttons allow you to change the priority order of the selected trends
- C Cancel current setup
- D Arrow buttons allow you to move selected parameters from one window to the other
- E Accepts current setup
- F Parameters available for selection

- 3 Select the desired display configuration.

The left/right arrows move the desired parameters from the display window to the *Trend* window. The up/down arrows move parameters in a desired sort order.

### NOTE

You can select several parameters at once to move them from one window to the other.

- 4 Click on **Accept** to save or **Cancel** to keep prior settings.
- 5 Click on **Save as System Defaults** in the **Ventilator View Setup** screen to keep this setup as the system default.

## MIB Ventilator Alarms

---

**MIB-connected ventilators:** Orange alarm messages appear in the lower right corner of the top-most Main Screen waveform channel and are available without the VentCentral option enabled.

### WARNING

- **Ventilation alarms function independently of the bedside monitor.**
- **The bedside monitor does not annunciate ventilation alarms.**
- **The connected ventilator allows no remote control.**

In the VentCentral screens alarm messages appear on the right side of the information area. When an MIB-connected ventilator is active at the Infinity CentralStation and the signal is lost, the Infinity CentralStation sounds a one-time alarm and displays the message, “MIB Disconnected.”

### NOTE

MIB alarm messages will be sent to and displayed at the Infinity CentralStation when you put a patient's bedside monitor in *Standby* mode.

Parameter boxes in the VentCentral screens that originate at MIB-connected ventilators do not flash when they are in alarm.

Alarm messages that originate at MIB-connected ventilators are displayed for as long as the ventilator sends them. If multiple alarm conditions occur simultaneously, the messages rotate.

Alarms that originate from the etCO<sub>2</sub>/Respiratory Mechanics Pod can be silenced remotely when the corresponding remote silence function is enabled.

## Ventilator Report

---

For detailed information on ventilator reports and how to request them, see page 190.

## VentCentral Messages

---

| Message                      | Description                                                                                                                                             | Action                                  |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Trend data not available     | When you admit a new patient to Main Screen, it takes a moment for the trend data to be compiled. Within 1 minute the database will contain trend data. | Exit and re-enter the trend application |
| No trend parameters selected | Trend parameters have not been selected.                                                                                                                | Select trend parameters (see page 246)  |
| Serious Alarm at <%s>        | There is an MIB alarm at the corresponding bed.                                                                                                         | Check the patient.                      |
| MIB Disconnected             | MIB cable is disconnected.                                                                                                                              | Re-connect MIB cable                    |

## Ventilator and Device Messages/Labels

---

Tables on the following pages list messages, labels, and settings sent by various MIB-connected devices to the Infinity bedside monitors for display at the Infinity CentralStation.

### CAUTION

The following tables list possible labels and messages for identification purposes. For complete information about message interpretation and device operation, please refer to the equipment documentation.



## Dräger Cato Anesthesia System

(as of Evita 2 Dura/4 Software version 4.22;  
as of Evita XL version 6.12)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Cato Anesthesia Systems |                             |
|--------------------------------|-----------------------------|
| VentCentral Parameters         | Description                 |
| iHAL                           | Agent inspiratory iHAL      |
| iENF                           | Agent inspiratory iENF      |
| iISO                           | Agent inspiratory iISO      |
| iDES                           | Agent inspiratory iDES      |
| iSEV                           | Agent inspiratory iSEV      |
| etHAL                          | Agent expiratory            |
| etENF                          | Agent expiratory            |
| etISO                          | Agent expiratory            |
| etDES                          | Agent expiratory            |
| etSEV                          | Agent expiratory            |
| MAP                            | Mean Breathing pressure     |
| Pause                          | Plateau pressure            |
| PEEP                           | PEEP                        |
| PIP                            | PIP                         |
| TVe                            | Expiratory Tidal Volume     |
| MVe                            | Minute Volume               |
| RRv                            | Breathing Frequency         |
| iCO <sub>2</sub>               | insp. CO <sub>2</sub>       |
| etCO <sub>2</sub>              | End Tidal CO <sub>2</sub>   |
| iO <sub>2</sub>                | insp. O <sub>2</sub>        |
| etO <sub>2</sub>               | End Tidal O <sub>2</sub>    |
| iN <sub>2</sub> O              | insp. N <sub>2</sub> O      |
| etN <sub>2</sub> O             | Expiratory N <sub>2</sub> O |

| <b>Dräger Cato Anesthesia Systems</b> |                    |
|---------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>          | <b>Description</b> |
| FLOW                                  | Flow               |
| PAW                                   | Airway Pressure    |

## Dräger Cicero Series Anesthesia Systems

| <b>Dräger Cicero Series Anesthesia Systems</b> |                              |
|------------------------------------------------|------------------------------|
| <b>VentCentral Parameters</b>                  | <b>Description</b>           |
| iHAL                                           | Agent inspiratory iHAL       |
| iENF                                           | Agent inspiratory iENF       |
| iISO                                           | Agent inspiratory iISO       |
| iDES                                           | Agent inspiratory iDES       |
| iSEV                                           | Agent inspiratory iSEV       |
| eO2                                            | Tidal volume, expired breath |
| iN2O                                           | insp. N2O                    |
| iO2                                            | Insp. O2                     |
| MAP                                            | Mean Breathing pressure      |
| MVe                                            | Minute Volume                |
| Pause                                          | Plateau pressure             |
| PEEP                                           | PEEP                         |
| PIP                                            | PIP                          |
| RRc                                            | Resp. Rate CO2               |
| RRv                                            | Breathing Frequency          |
| TVe                                            | Expiratory Tidal Volume      |
| etHAL                                          | Agent expiratory             |
| etENF                                          | Agent expiratory             |
| etISO                                          | Agent expiratory             |
| etDES                                          | Agent expiratory             |
| etSEV                                          | Agent expiratory             |

| <b>Dräger Cicero Series Anesthesia Systems</b> |                    |
|------------------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>                   | <b>Description</b> |
| FLOW                                           | Flow               |
| PAW                                            | Airway Pressure    |

## Dräger Julian Anesthesia System

As of version 2.02

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| <b>Dräger Julian Anesthesia System</b> |                         |
|----------------------------------------|-------------------------|
| <b>VentCentral Parameters</b>          | <b>Description</b>      |
| iHAL                                   | Agent inspiratory iHAL  |
| iENF                                   | Agent inspiratory iENF  |
| iISO                                   | Agent inspiratory iISO  |
| iDES                                   | Agent inspiratory iDES  |
| iSEV                                   | Agent inspiratory iSEV  |
| etHAL                                  | Agent expiratory        |
| etENF                                  | Agent expiratory        |
| etISO                                  | Agent expiratory        |
| etDES                                  | Agent expiratory        |
| etSEV                                  | Agent expiratory        |
| MAP                                    | Mean Breathing pressure |
| Pause                                  | Plateau pressure        |
| PEEP                                   | PEEP                    |
| PIP                                    | PIP                     |
| TVe                                    | Expiratory Tidal Volume |
| MVe                                    | Minute Volume           |
| RRv                                    | Breathing Frequency     |
| RRc*                                   | Resp. Rate CO2          |
| iCO2                                   | insp. CO2               |

| Dräger Julian Anesthesia System |                |
|---------------------------------|----------------|
| VentCentral Parameters          | Description    |
| etCO2                           | End Tidal CO2  |
| iN2O                            | insp. N2O      |
| etN2O                           | Expiratory N2O |
| I:E I-Part                      |                |
| I:E E-Part                      |                |

| Dräger Julian Anesthesia System |                 |
|---------------------------------|-----------------|
| VentCentral Waveforms           | Description     |
| FLOW                            | Flow            |
| PAW                             | Airway Pressure |

| Dräger Julian Anesthesia System |                 |
|---------------------------------|-----------------|
| Ventilator Settings             | Description     |
|                                 | Ventilator Type |
| I:E I-Part                      | I:E I-Part      |

## Dräger Primus Anesthesia System

As of release 2.03

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Primus Anesthesia System |                        |
|---------------------------------|------------------------|
| VentCentral Parameters          | Description            |
| Cdyn                            | Compliance             |
| iHAL                            | Agent inspiratory iHAL |
| iENF                            | Agent inspiratory iENF |
| iISO                            | Agent inspiratory iISO |
| iDES                            | Agent inspiratory iDES |

| <b>Dräger Primus Anesthesia System</b> |                         |
|----------------------------------------|-------------------------|
| <b>VentCentral Parameters</b>          | <b>Description</b>      |
| iSEV                                   | Agent inspiratory iSEV  |
| etHAL                                  | Agent expiratory        |
| etENF                                  | Agent expiratory        |
| etISO                                  | Agent expiratory        |
| etDES                                  | Agent expiratory        |
| etSEV                                  | Agent expiratory        |
| Barom pressure                         | Barom pressure          |
| MAP                                    | Mean Breathing pressure |
| Pause                                  | Plateau pressure        |
| PEEP                                   | PEEP                    |
| PIP                                    | PIP                     |
| TVe                                    | Expiratory Tidal Volume |
| MVe                                    | Minute Volume           |
| TV lk                                  | MV leak                 |
| RRv                                    | Breathing Frequency     |
| RRc*                                   | Resp. Rate CO2          |
| iCO2                                   | insp. CO2               |
| etCO2                                  | endtidales CO2          |
| iO2                                    | insp. O2                |
| iN2O                                   | insp. N2O               |
| etN2O                                  | expiratorisches N2O     |
| I:E I-Part                             | I:E I-Part              |
| I:E E-Part                             | I:E E-Part              |
| I:E                                    | I:E                     |

| <b>Dräger Primus Anesthesia System</b> |                                  |
|----------------------------------------|----------------------------------|
| <b>VentCentral Waveforms</b>           | <b>Description</b>               |
| FLOW                                   | Flow                             |
| PV Loop                                | Flow Loop (derived from 01H)     |
| PAW                                    | Airway Pressure                  |
| FV Loop                                | Pressure Loop (derived from 00H) |

| <b>Dräger Primus Anesthesia System</b> |                    |
|----------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>           | <b>Description</b> |
| CO2                                    | CO2                |

| <b>Dräger Primus Anesthesia System</b> |                    |
|----------------------------------------|--------------------|
| <b>Ventilator Modes</b>                | <b>Description</b> |
| VC                                     | Volume Mode        |
| SIMV                                   | Volume Mode + Sync |
| PC                                     | Pressure Mode      |
| PS                                     | Pressure Support   |
| Fresh gas ext.                         | Fresh Gas external |
| Man/Spon                               | Man/Spon           |
| Standby                                | Standby            |

| <b>Dräger Primus Anesthesia System</b> |                     |
|----------------------------------------|---------------------|
| <b>Ventilator Settings</b>             | <b>Description</b>  |
|                                        | Ventilator Type     |
| iO2 set                                | Insp. O2            |
| IPPV Freq                              | IPPV Freq           |
| I:E I-Part                             | I:E I-Part          |
| Ti set                                 | Insp. Time          |
| TVi set                                | Insp. Tidal Volume  |
| PIP set                                | Max./Insp. Pressure |
| Int PEEP set                           | int PEEP            |
| P Support                              | P Support           |
| FLOW Trig set                          | Flow Trigger        |
| ASB RAMP set                           | ASB Ramp            |

## Dräger Narkomed 6000/64000 System

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Narkomed 6000/64000 System |                                  |
|-----------------------------------|----------------------------------|
| VentCentral Parameters            | Description                      |
| iHAL                              | Agent inspiratory iHAL           |
| iENF                              | Agent inspiratory iENF           |
| iISO                              | Agent inspiratory iISO           |
| iDES                              | Agent inspiratory iDES           |
| iSEV                              | Agent inspiratory iSEV           |
| etHAL                             | Agent expiratory                 |
| etENF                             | Agent expiratory                 |
| etISO                             | Agent expiratory                 |
| etDES                             | Agent expiratory                 |
| etSEV                             | Agent expiratory                 |
| MAP                               | Mean Breathing pressure          |
| PEEP                              | PEEP                             |
| PIP                               | PIP                              |
| TVe                               | Expiratory Tidal Volume          |
| MVe                               | Minute Volume                    |
| RRv                               | Breathing Frequency              |
| RRc                               | Resp. Rate CO <sub>2</sub>       |
| iCO <sub>2</sub>                  | insp. CO <sub>2</sub>            |
| etCO <sub>2</sub>                 | endtidales CO <sub>2</sub>       |
| iO <sub>2</sub>                   | insp. O <sub>2</sub>             |
| iN <sub>2</sub> O                 | insp. N <sub>2</sub> O           |
| etN <sub>2</sub> O                | expiratorisches N <sub>2</sub> O |

| Dräger Narkomed 6000/64000 System |             |
|-----------------------------------|-------------|
| VentCentral Waveforms             | Description |
| FLOW                              | Flow        |

| <b>Dräger Narkomed 6000/64000 System</b> |                    |
|------------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>             | <b>Description</b> |
| PAW                                      | Airway Pressure    |

## Dräger Fabius Tiro / GS Anesthesia Systems

(as of version 3.11)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| <b>Dräger Fabius Tiro / GS Anesthesia Systems</b> |                         |
|---------------------------------------------------|-------------------------|
| <b>VentCentral Parameters</b>                     | <b>Description</b>      |
| iENF                                              | Agent inspiratory iENF  |
| iISO                                              | Agent inspiratory iISO  |
| iDES                                              | Agent inspiratory iDES  |
| iSEV                                              | Agent inspiratory iSEV  |
| etHAL                                             | Agent expiratory        |
| etENF                                             | Agent expiratory        |
| etISO                                             | Agent expiratory        |
| etDES                                             | Agent expiratory        |
| etSEV                                             | Agent expiratory        |
| MAP                                               | Mean Breathing pressure |
| Pause                                             | Plateau pressure        |
| PEEP                                              | PEEP                    |
| PIP                                               | PIP                     |
| TVe                                               | Expiratory Tidal Volume |
| MVe                                               | Minute Volume           |
| RRv                                               | Breathing Frequency     |
| RRc*                                              | Resp. Rate CO2          |
| iCO2                                              | insp. CO2               |
| etCO2                                             | End tidal CO2           |



| <b>Dräger Fabius Tiro / GS Anesthesia Systems</b> |                           |
|---------------------------------------------------|---------------------------|
| <b>VentCentral Parameters</b>                     | <b>Description</b>        |
| iO2                                               | insp. O2                  |
| etO2                                              | End tidal O2              |
| iN2O                                              | insp. N2O                 |
| etN2O                                             | Expiratory N2O            |
| Air Flow                                          | Air flow (Fabius GS only) |
| N2O Flow                                          | Nitrous Oxide flow        |
| O2 Flow                                           | Oxygen Flow               |

| <b>Dräger Fabius Tiro / GS Anesthesia Systems</b> |                                  |
|---------------------------------------------------|----------------------------------|
| <b>VentCentral Waveforms</b>                      | <b>Description</b>               |
| FLOW                                              | Flow                             |
| Flow Loop                                         | Flow Loop (derived from 01H)     |
| PAW                                               | Airway Pressure                  |
| Pressure Loop                                     | Pressure Loop (derived from 00H) |
| Volume                                            | Volume                           |
| CO2                                               | CO2                              |
| FLOW                                              | Flow                             |

| <b>Dräger Fabius Tiro / GS Anesthesia Systems</b> |                    |
|---------------------------------------------------|--------------------|
| <b>Ventilator Modes</b>                           | <b>Description</b> |
| SIMV                                              | SIMV               |
| PC                                                | PC                 |
| VC                                                | VC                 |
| PS                                                | PS                 |
| Standby                                           |                    |
| Man/Spon                                          | Man/Spont          |
| PC+PS                                             | PS                 |

| <b>Dräger Fabius Tiro / GS Anesthesia Systems</b> |                     |
|---------------------------------------------------|---------------------|
| <b>Ventilator Settings</b>                        | <b>Description</b>  |
| IppV Freq                                         |                     |
| I:E E-Part                                        | I:E E-part          |
| I:E I-Part                                        | I:E I-part          |
| Pt/It                                             |                     |
| iFLOW set                                         | Max. insp. Flow     |
| TVi set                                           | Insp. Tidal Volume  |
| PIP set                                           | Max./Insp. Pressure |
| Insp pressure                                     |                     |
| int PEEP set                                      | int.PEEP            |
| p Support                                         | P Sup               |
| FLOW Trig set                                     | Flow Trigger        |

## Dräger Fabius CE Anesthesia System

(as of Version 4)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| <b>Dräger Fabius CE Anesthesia System</b> |                         |
|-------------------------------------------|-------------------------|
| <b>VentCentral Parameters</b>             | <b>Description</b>      |
| MAP                                       | Mean Breathing pressure |
| PEEP                                      | PEEP                    |
| PIP                                       | PIP                     |
| TVe                                       | Expiratory Tidal Volume |
| MVe                                       | Minute Volume           |
| RRv                                       | Breathing Frequency     |
| iO2                                       | insp. O2                |

| <b>Dräger Fabius CE Anesthesia System</b> |                    |
|-------------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>              | <b>Description</b> |
| FLOW                                      | Flow               |
| PAW                                       | Airway Pressure    |

| <b>Dräger Fabius CE Anesthesia System</b> |                    |
|-------------------------------------------|--------------------|
| <b>Ventilator Modes</b>                   | <b>Description</b> |
| IPPV                                      | IPPV               |
| Man/Spont                                 |                    |

| <b>Dräger Fabius CE Anesthesia System</b> |                    |
|-------------------------------------------|--------------------|
| <b>Ventilator Settings</b>                | <b>Description</b> |
|                                           | Ventilator Type    |
| TVi set                                   | Insp. Tidal Volume |
| I:E E-Part                                | I:E E-Part         |
| I:E I-Part                                | I:E Insp           |
| IPPV Freq                                 | IppV Freq          |
| PEEP set                                  | int peep           |
| PIP                                       | Pmax               |
| Pt/It                                     | Pauset/Itime       |

## Dräger Zeus Anesthesia System

(as of Version 4.02)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| <b>Dräger Zeus Anesthesia System</b> |                        |
|--------------------------------------|------------------------|
| <b>VentCentral Parameters</b>        | <b>Description</b>     |
| Raw                                  | Resistance             |
| iHAL                                 | Agent inspiratory iHAL |

| <b>Dräger Zeus Anesthesia System</b> |                           |
|--------------------------------------|---------------------------|
| <b>VentCentral Parameters</b>        | <b>Description</b>        |
| iENF                                 | Agent inspiratory iENF    |
| iISO                                 | Agent inspiratory iISO    |
| iDES                                 | Agent inspiratory iDES    |
| iSEV                                 | Agent inspiratory iSEV    |
| etHAL                                | Agent expiratory          |
| etENF                                | Agent expiratory          |
| etISO                                | Agent expiratory          |
| etDES                                | Agent expiratory          |
| etSEV                                | Agent expiratory          |
| Barom pressure                       | Barom pressure            |
| MAP                                  | Mean Breathing pressure   |
| Pause                                | Plateau pressure          |
| PEEP                                 | PEEP                      |
| PIP                                  | PIP                       |
| MVe                                  | Minute Volume             |
| TVe                                  | Expiratory Tidal Volume   |
| RRv                                  | Breathing Frequency       |
| Mve s                                | Spontaneous Minute Volume |
| Tvs                                  |                           |
| Leakage                              |                           |
| RRc*                                 | Resp. Rate CO2            |
| iCO2                                 | insp. CO2                 |
| etCO2                                | End Tidal CO2             |
| etO2                                 | End Tidal O2              |
| iO2                                  | insp. O2                  |
| iN2O                                 | insp. N2O                 |
| etN2O                                | Expiratory N2O            |
| O2 supply pressure                   | O2 supply pressure        |
| Air Flow                             | Air Flow                  |
| N2O Flow                             | N2O Flow                  |
| O2 Flow                              | O2 Flow                   |
| MvM                                  | Mandatory MV              |

| <b>Dräger Zeus Anesthesia System</b> |                              |
|--------------------------------------|------------------------------|
| <b>VentCentral Parameters</b>        | <b>Description</b>           |
| RRm                                  | Respiratory rate (mandatory) |

| <b>Dräger Zeus Anesthesia System</b> |                    |
|--------------------------------------|--------------------|
| <b>VentCentral Waveforms</b>         | <b>Description</b> |
| PAW                                  | Airway Pressure    |
| CO2                                  | CO2                |

| <b>Dräger Zeus Anesthesia System</b> |                      |
|--------------------------------------|----------------------|
| <b>Ventilator Modes</b>              | <b>Description</b>   |
| VC                                   | Volume               |
| VC                                   | Volume / Autoflow    |
| PC+PS                                | Volume / AutoFlow+PS |
| PC                                   | Pressure             |
| SIMV                                 | Pressure +PS         |
| PS                                   | CPAP+PS              |
| PS                                   | CPAP                 |
| Man/Spon                             | Man/Spon             |
| Standby                              | Standby              |

| <b>Dräger Zeus Anesthesia System</b> |                    |
|--------------------------------------|--------------------|
| <b>Ventilator Settings</b>           | <b>Description</b> |
|                                      | Ventilator Type    |
| iO2 set                              | Insp. O2           |
| PIP set                              | Pmax               |
| TVi set                              | Insp. Tidal Volume |
| TV set                               | Insp. Time         |
| Teset                                | Exp. Time          |
| IppV Freq                            |                    |
| PEEP set                             | PEEP               |
| FLOW Trig set                        | Flow Trigger       |
| Tslope                               | ASB Ramp set       |

## Dräger Babylog 8000 Ventilator

(as of version 5.01)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Babylog 8000 Ventilator |                                    |
|--------------------------------|------------------------------------|
| VentCentral Parameters         | Description                        |
| Cdyn                           | Compliance                         |
| RAW                            | Resistance                         |
| MAP                            | Mean Breathing Pressure            |
| PEEP                           | PEEP Airway Pressure               |
| PIP                            | Peak Airway Pressure               |
| Gas Trans Coeff                | Gas Transport Coefficient          |
| TV i m                         | Inspiratory mandatory Tidal Volume |
| TVhf                           | High frequency Tidal Volume        |
| TVe                            | Tidal Volume                       |
| TVLeak                         | Leakage                            |
| MV s%                          | Spontaneous Fraction Minute Volume |
| MVe                            | Respiratory Minute Volume          |
| RRv                            | Respiratory Rate                   |
| iO2                            | Insp. O <sub>2</sub>               |

| Dräger Babylog 8000 Ventilator |                                  |
|--------------------------------|----------------------------------|
| VentCentral Waveforms          | Description                      |
| FLOW                           | Flow                             |
| Flow Loop                      | Flow Loop (derived from 01H)     |
| PAW                            | Airway Pressure                  |
| Pressure Loop                  | Pressure Loop (derived from 00H) |

| <b>Dräger Babylog 8000 Ventilator</b> |                                 |
|---------------------------------------|---------------------------------|
| <b>Ventilator Alarms</b>              | <b>Description</b>              |
| %O2 ERR                               | Insp. O2 Measurement inoperable |
| VENTILATOR ALARM                      | Flow Measurement inoperable     |

| <b>Dräger Babylog 8000 Ventilator</b> |                    |
|---------------------------------------|--------------------|
| <b>Ventilator Modes</b>               | <b>Description</b> |
| IPPV                                  | MODE IPPV          |
| SIMV                                  | MODE SIMV          |
| CPAP                                  | MODE CPAP          |
| basic mode                            | MODE VIVE          |
| PS                                    | MODE PSV           |

| <b>Dräger Babylog 8000 Ventilator</b> |                              |
|---------------------------------------|------------------------------|
| <b>Ventilator Settings</b>            | <b>Description</b>           |
|                                       | Ventilator Type              |
| iO2 set                               | Insp. O2                     |
| iFLOW                                 | Inspiratory Flow             |
| FLOW                                  | Expiratory Flow              |
| Ti set                                | Inspiratory Time             |
| Te set                                | Expiratory Time              |
| I:E I-Part                            | I:E Insp                     |
| I:E E-Part                            | I:E Exp                      |
| SIMV FREQ                             | Frequency IMV (SIMV)         |
| PEEP set                              | PEEP/CPAP                    |
| APNEA TIME                            | Apnea Time                   |
| PIP set                               | Maximum Inspiratory Pressure |
| TRIG VOL                              | Trigger Volume               |
| MV HI LIM                             | High Minute Volume Limit     |
| MV LO LIM                             | Low Minute Volume Limit      |
| MV ALM DLY                            | Minute Volume Alarm Delay    |
| HFV FREQ                              | Frequency HFV                |
| HFV AMPL                              | Amplitude HFV                |
| Tvi set                               | set tidal volume             |

## Dräger Evita 2 Ventilator

| Dräger Evita 2 Ventilator  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm Messages             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Displayed in...                                                                                                                                                                                                                                                                      |
| Paw > HI LIM               | Paw exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Main Screen - lower right corner of the top waveform channel                                                                                                                                                                                                                         |
| AW -TEMP>HI LIM            | AW -TEMP exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                      |
| Ventilator Alarm           | <ul style="list-style-type: none"> <li>– Airway temperature measurement unsuccessful</li> <li>– Check expiration - valve</li> <li>– Check flow sensor</li> <li>– CO<sub>2</sub> device failure</li> <li>– CO<sub>2</sub> not calibrated</li> <li>– CO<sub>2</sub> sensor disconnected or malfunctioning</li> <li>– Failure to cycle</li> <li>– Gas mixer inoperable</li> <li>– Pressure measurement unsuccessful</li> <li>– Problems with respirator</li> <li>– Respirator synchronization inoperable</li> <li>– Device temperature is too high.</li> <li>– Volume measurement unsuccessful</li> <li>– Volume is inconsistent.</li> <li>– Airway temperature sensor alarm</li> </ul> | <p>Bed View - information area, just below menu buttons</p> <p>Alarm causes that are labelled as 'Ventilator Alarm' refer to non-clinical device conditions. All other alarm messages refer to clinical events that violate a setting which can be configured at the ventilator.</p> |
| APNEA                      | Apnea alarm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      |
| ASB > 4 SEC                | ASB is more than 4 sec                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                      |
| Patient Disconnect         | Patient may be disconnected from ventilator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      |
| etCO <sub>2</sub> < LO LIM | etCO <sub>2</sub> below lower alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                      |
| etCO <sub>2</sub> > HI LIM | etCO <sub>2</sub> exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      |
| AIR SUPPLY?                | Gas supply alarm, air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                      |
| O <sub>2</sub> SUPPLY?     | Gas supply alarm, O <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                      |
| iO <sub>2</sub> < LO LIM   | iO <sub>2</sub> less than lower alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      |
| iO <sub>2</sub> > HI LIM   | iO <sub>2</sub> exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                      |
| % O <sub>2</sub> ERR       | iO <sub>2</sub> measurement unsuccessful                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                      |
| MVe < LO LIM               | MVe is below lower alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                      |
| MVe > HI LIM               | MVe exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                      |
| PEEP > HI LIM              | PEEP exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                      |
| RR > HI LIM                | RR exceeds upper alarm limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                      |



| <b>Dräger Evita 2 Ventilator</b> |                                               |                                                        |
|----------------------------------|-----------------------------------------------|--------------------------------------------------------|
| <b>Parameter Labels</b>          | <b>Description</b>                            | <b>Displayed in...</b>                                 |
| Paw MIN                          | Minimum airway pressure                       | Ventilator Settings<br>Review screen<br>(see page 243) |
| AW-Temp                          | Airway temperature                            |                                                        |
| VCO2                             | CO2 production, minute volume                 |                                                        |
| Cdyn                             | Dynamic compliance                            |                                                        |
| TVd aw%                          | Dead space                                    |                                                        |
| TVd aw                           | Dead space, airway                            |                                                        |
| etCO2                            | end-tidal CO2                                 |                                                        |
| I:E I-Part                       | I:E ratio (inspiratory component)             |                                                        |
| iO2                              | Inspired O2                                   |                                                        |
| MAP                              | Mean airway pressure                          |                                                        |
| MVe                              | Expired minute volume                         |                                                        |
| MVe s                            | Spontaneous expired minute volume             |                                                        |
| Occlusion Press                  | Occlusion pressure                            |                                                        |
| Pause                            | Pause pressure                                |                                                        |
| PEEP                             | Peak end expiratory airway pressure           |                                                        |
| iPEEP                            | Intrinsic peak end expiratory airway pressure |                                                        |
| PIP                              | Peak inspired airway pressure                 |                                                        |
| Raw                              | Dynamic resistance                            |                                                        |
| RRv                              | Pediatric respiratory rate - vol/flow         |                                                        |
| RRs                              | Spontaneous respiratory rate                  |                                                        |
| TVe                              | Tidal volume expired breath                   |                                                        |
| Trapped VOL                      | Trapped volume                                |                                                        |

| Dräger Evita 2 Ventilator |                                     |                                                        |
|---------------------------|-------------------------------------|--------------------------------------------------------|
| Selected Settings         | Description                         | Displayed in...                                        |
| APNEA TIME                | Apnea time                          | Ventilator Settings<br>Review screen<br>(see page 243) |
| ASB RAMP set              | Assisted spontaneous breathing ramp |                                                        |
| ASB set                   | Assisted spontaneous breathing      |                                                        |
| BIPAP HI P set            | BIPAP high pressure limit           |                                                        |
| BIPAP HI T set            | BIPAP high time limit               |                                                        |
| BIPAP LO P set            | BIPAP low pressure limit            |                                                        |
| BIPAP LO T set            | BIPAP low time limit                |                                                        |
| FLOW Trig set             | Flow trigger setting                |                                                        |
| SIMV FREQ                 | Frequency, IMV - SIMV               |                                                        |
| IPPV FREQ                 | Frequency IPPV                      |                                                        |
| I:E E-Part                | I:E ratio - expiratory component    |                                                        |
| I:E I-Part                | I:E ratio - inspiratory component   |                                                        |
| iO2 set                   | Inspired O2 setting                 |                                                        |
| MAX IFLOW set             | Maximum inspired flow               |                                                        |
| PIP set                   | Peak inspired airway pressure       |                                                        |
| PEEP set                  | PEEP setting                        |                                                        |
| INT PEEP set              | Intermittent PEEP setting           |                                                        |
| TACHY LENGTH              | Tachypnea duration                  |                                                        |
| TACHY FREQ                | Tachypnea frequency                 |                                                        |
| TVi set                   | Inspired tidal volume setting       |                                                        |
| TRIG PRESS                | Trigger pressure                    |                                                        |
| Vent:                     | Evita 2                             |                                                        |

## Dräger Evita 2 Dura/4/XL Ventilator

(as of Evita 2 Dura/4 Software version 4.22;  
as of Evita XL version 6.12)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Evita 2 Dura/4/XL Ventilator |                    |
|-------------------------------------|--------------------|
| VentCentral Parameters              | Description        |
| AW-TEMP                             | Airway Temperature |
| Cdyn                                | Compliance         |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                                               |
|--------------------------------------------|-----------------------------------------------|
| <b>VentCentral Parameters</b>              | <b>Description</b>                            |
| etCO <sub>2</sub>                          | Endtidal CO <sub>2</sub> (etCO <sub>2</sub> ) |
| I:E I-Part                                 | I:E Insp                                      |
| I:E E-Part                                 | I:E Exp                                       |
| iO <sub>2</sub>                            | Insp. O <sub>2</sub>                          |
| iPEEP                                      | Intrinsic PEEP                                |
| MAP                                        | Mean Airway Pressure                          |
| MVe                                        | Minute Volume                                 |
| MVe s                                      | Spont Minute Volume                           |
| Neg iForce                                 | Negative Inspiratory Force                    |
| Occlusion Press                            | Occlusion Pressure                            |
| Pause                                      | Plateau Airway Pressure                       |
| PEEP                                       | PEEP Airway Pressure                          |
| PIP                                        | Peak Airway Pressure                          |
| Raw                                        | Resistance                                    |
| RRs                                        | Spont Breathing Frequency                     |
| RRv                                        | Breathing Frequency                           |
| Trapped Vol                                | Trapped Volume                                |
| TVd aw                                     | Dead Space                                    |
| TVd aw%                                    | Relative Dead Space                           |
| TVe                                        | Expiratory Tidal Volume                       |
| VCO <sub>2</sub>                           | CO <sub>2</sub> -Production (VC02)            |
| VTasb                                      | Inspiratory spont. Support volume             |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                                  |
|--------------------------------------------|----------------------------------|
| <b>VentCentral Waveforms</b>               | <b>Description</b>               |
| FLOW                                       | Flow                             |
| Flow Loop                                  | Flow Loop (derived from 01H)     |
| PAW                                        | Airway Pressure                  |
| Pressure Loop                              | Pressure Loop (derived from 00H) |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                                        |
|--------------------------------------------|----------------------------------------|
| <b>VentCentral Alarm Messages</b>          | <b>Description</b>                     |
| % O2 ERR                                   | Insp. O2 Measurement inoperable        |
| iO2 > HI LIM                               | iO2 more than high limit               |
| iO2 < LO LIM                               | iO2 less than low limit                |
| AIR SUPPLY?                                | Check Air Supply                       |
| APNEA                                      | Apnoe detected by Evita                |
| ASB > 4 SEC                                | Assisted Spontaneous Breathing > 4 sec |
| AW-TEMP > HI LIM                           | Airway Temperature > High Limit        |
| Ventilator Alarm                           | Airway Temperature Measurement inop    |
| Ventilator Alarm                           | Safety mode                            |
| Battery                                    | Battery low or malfunction             |
| Ventilator Alarm                           | Clean CO2 Sensor (Window occluded)     |
| Ventilator Alarm                           | CO2 Device Failure                     |
| Ventilator Alarm                           | CO2 Sensor disconnected or fault       |
| Ventilator Alarm                           | CO2 Zero-Calibration requested         |
| Ventilator Alarm                           | Fail to Cycle                          |
| etCO2 > HI LIM                             | Entidal CO2 > high Limit               |
| etCO2 < LO LIM                             | Entidal CO2 < low Limit                |
| Ventilator Alarm                           | Check Expiration Valve                 |
| Ventilator Alarm                           | Check Flow Sensor                      |
| O2 SUPPLY?                                 | O2 Supply Press low                    |
| MVe > HI LIM                               | Minute Volume > high Limit             |
| Mve < LO LIM                               | Minute Volume < low Limit              |
| Ventilator Alarm                           | Gas Mixer inoperable (Alarm)           |
| Ventilator Alarm                           | Neonatal Flow Sensor not at Y-piece    |
| Ventilator Alarm                           | Neo. Volume Measurement inoperable     |
| Patient Disconnect                         | Disconnection Ventilator               |
| Paw > HI LIM                               | Airway Pressure > high limit           |
| PEEP > HI LIM                              | PEEP > high Limit                      |
| Ventilator Alarm                           | Pressure Measurement inoperable        |
| RR > HI LIM                                | Respiratory Rate > high Limit          |
| Ventilator Alarm                           | Respirator Synchronization inoperable  |
| Tve > HI LIM                               | Tidal Volume > high Limit              |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                               |
|--------------------------------------------|-------------------------------|
| <b>VentCentral Alarm Messages</b>          | <b>Description</b>            |
| Ventilator Alarm                           | Volume Measurement inoperable |
| Ventilator Alarm                           | Evita Standby                 |
| Ventilator Alarm                           | Loss of Data                  |
| Ventilator Alarm                           | PEEP Valve inoperable         |
| Standby                                    | Ventilator STANDBY            |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                           |
|--------------------------------------------|---------------------------|
| <b>Ventilator Modes</b>                    | <b>Description</b>        |
| IPPV                                       | MODE IPPV                 |
| IPPV                                       | MODE IPPV/ASSIST          |
| SIMV                                       | MODE SIMV                 |
| SIMV                                       | MODE SIMV/ASB             |
| BIPAP                                      | MODE BIPAP                |
| BIPAP                                      | MODE BIPAP/ASB            |
| SIMV/Auto                                  | MODE SIMV/AutoFlow        |
| SIMV                                       | MODE SIMV/ASB/AutoFlow    |
| IPPV/Auto                                  | MODE IPPV/AutoFlow        |
| IPPV                                       | MODE IPPV/ASSIST/AutoFlow |
| PRVC                                       | MODE APRV                 |
| MMV                                        | MODE MMV                  |
| MMV                                        | MODE MMV/ASB              |
| MMV                                        | MODE MMV/AutoFlow         |
| MMV                                        | MODE MMV/ASB/AutoFlow     |
| CPAP                                       | MODE CPAP                 |
| CPAP                                       | MODE CPAP/ASB             |
| CPAP                                       | MODE CPAP/PPS             |
| BIPAP                                      | MODE BIPAP/ASSIST         |
| Standby                                    | Ventilator STANDBY        |

| <b>Dräger Evita 2 Dura/4/XL Ventilator</b> |                      |
|--------------------------------------------|----------------------|
| <b>Ventilator Settings</b>                 | <b>Description</b>   |
| <Evita>                                    | Ventilator Type      |
| Adult                                      | MODE Adults          |
| Pedi                                       | MODE Pediatrics      |
| Neonate                                    | MODE Neonates        |
| iO2 set                                    | Insp. O2             |
| iFLOW set                                  | Max. insp. Flow      |
| TVi set                                    | Insp. Tidal Volume   |
| Ti set                                     | Insp. Time           |
| I:E I-Part                                 | I:E Insp             |
| I:E E-Part                                 | I:E Exp              |
| SIMV FREQ                                  | Frequency            |
| PEEP set                                   | PEEP                 |
| INT PEEP set                               | Interm. PEEP         |
| APRV Lo Prsr                               | APRV Low Pressure    |
| APRV Hi Prsr                               | APRV High Pressure   |
| APRV Lo Time                               | APRV Low Time        |
| APRV Hi Time                               | APRV High Time       |
| APNEA TIME                                 | Apnea Time           |
| ASB set                                    | ASB                  |
| PIP set                                    | Max./Insp. Pressure  |
| TACHY FREQ                                 | Tachyapnea Frequency |
| FLOW Trig set                              | Flow Trigger         |
| ASB RAMP set                               | ASB Ramp             |
| FLOW ASSIST                                | Flow Assist          |
| VOL ASSIST                                 | Volume Assist        |
| Tdisconnect                                | TDeconnect           |

## Dräger Savina Ventilator

(as of version 3.01)

### NOTE

Not all products listed in this section are available for all countries. For details, contact your sales representative.

| Dräger Savina Ventilator |                           |
|--------------------------|---------------------------|
| VentCentral Parameters   | Description               |
| AW-TEMP                  | Airway Temperature        |
| Cdyn                     | Compliance                |
| I:E I-Part               | I:E Insp                  |
| I:E E-Part               | I:E Exp                   |
| Ti                       | Inspiratory Time          |
| iO2                      | Insp. O2                  |
| MAP                      | Mean Airway Pressure      |
| MVe                      | Minute Volume             |
| Mve s                    | Spont Minute Volume       |
| Max iFlow                | Flow Peak                 |
| Pause                    | Plateau Airway Pressure   |
| PEEP                     | PEEP Airway Pressure      |
| PIP                      | Peak Airway Pressure      |
| Raw                      | Resistance                |
| RRs                      | Spont Breathing Frequency |
| RRv                      | Breathing Frequency       |
| TVe                      | Expiratory Tidal Volume   |

| Dräger Savina Ventilator |                 |
|--------------------------|-----------------|
| VentCentral Waveforms    | Description     |
| FLOW                     | Flow            |
| Flow Loop                | Flow Loop       |
| PAW                      | Airway Pressure |
| Pressure Loop            | Pressure Loop   |

| <b>Dräger Savina Ventilator</b> |                                              |
|---------------------------------|----------------------------------------------|
| <b>Ventilator Alarms</b>        | <b>Description</b>                           |
| % O2 ERR                        | Insp. O2 Measurement inoperable              |
| iO2 > HI LIM                    | iO2 higher than high limit                   |
| iO2 < LO LIM                    | iO2 less than low limit                      |
| ASB > 4 SEC                     | Assisted Spontaneous Breathing exceeds 4 sec |
| APNEA                           | Apnoe detected by Savina                     |
| AW-TEMP > HI LIM                | Airway Temperature > high Limit              |
| Ventilator Alarm                | Airway Temperature Measurement inop          |
| Battery                         | Battery low or malfunction                   |
| Battery                         | Battery low                                  |
| Ventilator Alarm                | Check Expiration Valve                       |
| Ventilator Alarm                | Check Cooling                                |
| Ventilator Alarm                | Check Flow Sensor                            |
| O2 SUPPLY?                      | O2 Supply Press low                          |
| Mve > HI LIM                    | Minute Volume > high Limit                   |
| Mve < LO LIM                    | Minute Volume < low Limit                    |
| Paw > HI LIM                    | Airway Pressure > high                       |
| Patient Disconnect              | Disconnection Ventilator                     |
| PEEP > HI LIM                   | PEEP > high Limit                            |
| Ventilator Alarm                | PEEP Valve inoperable                        |
| Ventilator Alarm                | Pressure Measurement inoperable              |
| RR > HI LIM                     | Respiratory Rate > high Limit                |
| Ventilator Alarm                | Savina Standby                               |
| TVe > HI LIM                    | Tidal Volume > high Limit                    |
| Ventilator Alarm                | Volume Measurement inoperable                |

| <b>Dräger Savina Ventilator</b> |                    |
|---------------------------------|--------------------|
| <b>Ventilator Modes</b>         | <b>Description</b> |
| IPPV                            | MODE IPPV          |
| IPPV                            | MODE IPPV/ASSIST   |
| SIMV                            | MODE SIMV          |
| SIMV                            | MODE SIMV/ASB      |
| BIPAP                           | MODE BIPAP         |



| <b>Dräger Savina Ventilator</b> |                           |
|---------------------------------|---------------------------|
| <b>Ventilator Modes</b>         | <b>Description</b>        |
| BIPAP                           | MODE BIPAP/ASB            |
| SIMV/Auto                       | MODE SIMV/AutoFlow        |
| SIMV                            | MODE SIMV/ASB/AutoFlow    |
| IPPV/Auto                       | MODE IPPV/AutoFlow        |
| IPPV                            | MODE IPPV/ASSIST/AutoFlow |
| CPAP                            | MODE CPAP                 |
| CPAP                            | MODE CPAP/ASB             |
| Standby                         | Ventilator STANDBY        |

| <b>Dräger Savina Ventilator</b> |                                     |
|---------------------------------|-------------------------------------|
| <b>Ventilator Settings</b>      | <b>Description</b>                  |
|                                 | Ventilator Type                     |
| iO2 set                         | Insp. O2                            |
| TVi set                         | Insp. Tidal Volume                  |
| Ti set                          | Insp. Time                          |
| I:E I-Part                      | I:E Insp                            |
| I:E E-Part                      | I:E Exp                             |
| SIMV FREQ                       | Frequency                           |
| PEEP set                        | PEEP                                |
| INT PEEP set                    | Interm. PEEP                        |
| APNEA TIME                      | Apnea Time                          |
| ASB set                         | ASB                                 |
| PIP set                         | Max./Insp. Pressure                 |
| FLOW Trig set                   | Flow Trigger                        |
| FREQbackup                      | Backup Frequency                    |
| TVbackup                        | Backup Tidal Volume                 |
| Insp Press                      | Insp. Pressure (P <sub>insp</sub> ) |
| TDeconnect                      | TDeconnect                          |
| Flow Accel                      | Flow Acceleration                   |

## Ohmeda Modulus CD Anesthesia System

| Ohmeda Modulus CD Anesthesia System |                                |                                                        |
|-------------------------------------|--------------------------------|--------------------------------------------------------|
| Parameter                           | Description                    | Displayed in...                                        |
| MAP                                 | Mean Airway Pressure           | Ventilator Settings<br>Review screen<br>(see page 243) |
| MVe                                 | Minute Volume, expired         |                                                        |
| Pause                               | Pause Pressure                 |                                                        |
| PIP                                 | Peak inspired airway pressure  |                                                        |
| RRv                                 | Respiratory rate (volume/flow) |                                                        |
| TVe                                 | Tidal volume, expired breath   |                                                        |
| Selected Settings                   |                                |                                                        |
| iO2 set                             | iO2 setting                    | Ventilator Settings<br>Review screen<br>(see page 243) |

## Puritan-Bennett 7200/840 Ventilators

| Puritan-Bennett 7200/840 Ventilators |                                  |                                                                                                                                                                                                                   |
|--------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ventilator Alarms                    | Description                      | Displayed in...                                                                                                                                                                                                   |
| Paw > HI LIM                         | Paw exceeds upper alarm limit    | Main Screen - lower right corner of the top waveform channel                                                                                                                                                      |
| Paw < LO LIM                         | Paw fell below lower alarm limit |                                                                                                                                                                                                                   |
| APNEA                                | Apnea event                      | Bed View - information area, just below menu buttons                                                                                                                                                              |
| AIR SUPPLY                           | Gas air supply pressure          |                                                                                                                                                                                                                   |
| O2 SUPPLY                            | Gas O2 supply pressure           | Alarm causes that are labelled as 'Ventilator Alarm' refer to non-clinical device conditions. All other alarm messages refer to clinical events that violate a setting which can be configured at the ventilator. |
| Ventilator Alarm                     | I:E Alarm                        |                                                                                                                                                                                                                   |
| Ventilator Alarm (P-B 7200 only)     | Leakage, system                  |                                                                                                                                                                                                                   |
| MVe < LO LIM                         | MVe is below lower alarm limit   |                                                                                                                                                                                                                   |
| PEEP < LO LIM (P-B 7200 only)        | PEEP is below lower alarm limit  |                                                                                                                                                                                                                   |
| RR > HI LIM                          | RR exceeds upper alarm limit     |                                                                                                                                                                                                                   |
| TVe < LO LIM                         | Tidal volume is low              |                                                                                                                                                                                                                   |

| <b>Puritan-Bennett 7200/840 Ventilators</b> |                                   |                                                        |
|---------------------------------------------|-----------------------------------|--------------------------------------------------------|
| <b>VentCentral Parameter</b>                | <b>Description</b>                | <b>Displayed in...</b>                                 |
| Cdyn<br>(P-B 7200 only)                     | Dynamic compliance                | Ventilator Settings<br>Review screen<br>(see page 243) |
| Cs<br>(P-B 7200 only)                       | Static compliance                 |                                                        |
| I:E E-Part                                  | I:E ratio (expiratory component)  |                                                        |
| I:E I-Part<br>(P-B 840 only)                | I:E ratio (inspiratory component) |                                                        |
| MAP                                         | Mean airway pressure              |                                                        |
| MVe                                         | Expired minute volume             |                                                        |
| MVe s                                       | Spontaneous minute volume         |                                                        |
| Pause                                       | Pause pressure                    |                                                        |
| PIP                                         | Peak inspired airway pressure     |                                                        |
| Raw                                         | Dynamic resistance                |                                                        |
| Rs                                          | Static resistance                 |                                                        |
| RRv                                         | Respiratory rate                  |                                                        |
| TVe                                         | Tidal volume expired breath       |                                                        |
| VC                                          | Vital capacity                    |                                                        |

| <b>Puritan-Bennett 7200/840 Ventilators</b> |                               |                                                        |
|---------------------------------------------|-------------------------------|--------------------------------------------------------|
| <b>Ventilator Alarms</b>                    | <b>Description</b>            | <b>Displayed in...</b>                                 |
| FLOW WAVE set                               | Flow waveform setting         | Ventilator Settings<br>Review screen<br>(see page 243) |
| FREQ set                                    | Frequency                     |                                                        |
| iO2 set                                     | Inspired O2 setting           |                                                        |
| 100%O2                                      | Inspired O2% setting          |                                                        |
| NEBUL set                                   | Nebulizer setting             |                                                        |
| PIF set                                     | Peak inspiratory flow setting |                                                        |
| PEEP set                                    | PEEP setting                  |                                                        |
| Plateau set                                 | Plateau setting               |                                                        |
| SIGH ON                                     | Sigh enabled                  |                                                        |
| TVi set                                     | Tidal volume                  |                                                        |
| TRIG PRESS                                  | Trigger pressure              |                                                        |
| VENT MODE                                   | Selected ventilation mode     |                                                        |
| VENT:                                       | Selected Ventilator type      |                                                        |

| <b>Puritan-Bennett 7200/840 Ventilators</b> |                    |                                     |
|---------------------------------------------|--------------------|-------------------------------------|
| <b>VentCentral Waveforms<sup>1</sup></b>    |                    |                                     |
| <b>Parameter</b>                            | <b>Description</b> | <b>Displayed in...</b>              |
| Paw                                         | Airway Pressure    | Ventilator screen<br>(see page 239) |
| Flow                                        | Flow (Insp/exp)    |                                     |

<sup>1</sup> 7200 model only

## Taema Horus Ventilator

| <b>Taema Horus Ventilator</b> |                                             |                                                                                                                                                                                                                   |
|-------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ventilator Alarms</b>      | <b>Description</b>                          | <b>Displayed in...</b>                                                                                                                                                                                            |
| Paw > HI LIM                  | Paw exceeds upper alarm limit               | Main Screen - lower right corner of the top waveform channel                                                                                                                                                      |
| APNEA                         | Apnea event                                 |                                                                                                                                                                                                                   |
| Expiration Valve?             | Check expiration - valve                    |                                                                                                                                                                                                                   |
| Patient Disconnect            | Patient may be disconnected from ventilator | Bed View - information area, just below menu buttons                                                                                                                                                              |
| I:E Alarm                     | I:E error                                   |                                                                                                                                                                                                                   |
| iO2 < LO LIM                  | iO2 is below lower alarm limit              | Alarm causes that are labelled as 'Ventilator Alarm' refer to non-clinical device conditions. All other alarm messages refer to clinical events that violate a setting which can be configured at the ventilator. |
| iO2 > HI LIM                  | iO2 exceeds upper alarm limit               |                                                                                                                                                                                                                   |
| MVe < LO LIM                  | MVe is below lower alarm limit              |                                                                                                                                                                                                                   |
| MVe > HI LIM                  | MVe exceeds upper alarm limit               |                                                                                                                                                                                                                   |
| PEEP > HI LIM                 | PEEP exceeds upper alarm limit              |                                                                                                                                                                                                                   |
| Flow Sensor?                  | Check flow sensor                           |                                                                                                                                                                                                                   |
| AIR SUPPLY?                   | Gas supply alarm, air                       |                                                                                                                                                                                                                   |
| O2 SUPPLY?                    | Gas supply alarm, O2                        |                                                                                                                                                                                                                   |

| <b>Taema Horus Ventilator</b> |                               |                                                        |
|-------------------------------|-------------------------------|--------------------------------------------------------|
| <b>VentCentral Parameter</b>  | <b>Description</b>            | <b>Displayed in...</b>                                 |
| Cdyn                          | Dynamic compliance            | Ventilator Settings<br>Review screen<br>(see page 243) |
| MAP                           | Mean airway pressure          |                                                        |
| MVe                           | Expired minute volume         |                                                        |
| MVi                           | Inspired minute volume        |                                                        |
| Pause                         | Pause pressure                |                                                        |
| PIP                           | Peak inspired airway pressure |                                                        |
| PEEP                          | Peak end expiratory pressure  |                                                        |
| RRv                           | Respiratory rate              |                                                        |
| TVe                           | Tidal volume, expired         |                                                        |
| TVi                           | Tidal volume, inspired        |                                                        |
| iO2                           | Inspired O2                   |                                                        |

| <b>Taema Horus Ventilator</b> |                           |                                                        |
|-------------------------------|---------------------------|--------------------------------------------------------|
| <b>Ventilator Settings</b>    | <b>Description</b>        | <b>Displayed in...</b>                                 |
| FREQ set                      | Frequency                 | Ventilator Settings<br>Review screen<br>(see page 243) |
| PEEP set                      | PEEP setting              |                                                        |
| TVi set                       | Tidal volume              |                                                        |
| TRIG PRESS                    | Trigger pressure          |                                                        |
| VENT MODE                     | Selected ventilation mode |                                                        |
| VENT:                         | Selected ventilator type  |                                                        |

| <b>Taema Horus Ventilator</b> |                    |                                     |
|-------------------------------|--------------------|-------------------------------------|
| <b>VentCentral Waveforms</b>  |                    |                                     |
| <b>Parameter</b>              | <b>Description</b> | <b>Displayed in...</b>              |
| Paw                           | Airway Pressure    | Ventilator screen<br>(see page 239) |
| Flow                          | Flow (Insp/exp)    |                                     |

## Hamilton Galileo Ventilator

| Hamilton Galileo Ventilator |                                                                                                                                                                                                             |                                                                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ventilator Messages         | Description                                                                                                                                                                                                 | Displayed in...                                                                                                                                                                                                   |
| Paw > HI LIM                | Paw exceeds upper alarm limit                                                                                                                                                                               | Main Screen - lower right corner of the top waveform channel                                                                                                                                                      |
| Ventilator Alarm            | <ul style="list-style-type: none"> <li>Minute volume exceeds upper alarm limit</li> <li>Failure to cycle</li> <li>Gas supply alarm</li> <li>PEEP valve inoperable</li> <li>Ventilator disconnect</li> </ul> |                                                                                                                                                                                                                   |
| APNEA                       | Apnea event                                                                                                                                                                                                 | Bed View - information area, just below menu buttons                                                                                                                                                              |
| %O2 Error                   | %O2 measurement error                                                                                                                                                                                       |                                                                                                                                                                                                                   |
| MVe < LO LIM                | MVe is below lower alarm limit                                                                                                                                                                              |                                                                                                                                                                                                                   |
| RR > HI LIM                 | RR exceeds upper alarm limit                                                                                                                                                                                |                                                                                                                                                                                                                   |
|                             |                                                                                                                                                                                                             | Alarm causes that are labelled as 'Ventilator Alarm' refer to non-clinical device conditions. All other alarm messages refer to clinical events that violate a setting which can be configured at the ventilator. |

| Hamilton Galileo Ventilator |                                       |                                                  |
|-----------------------------|---------------------------------------|--------------------------------------------------|
| Ventilator Parameter        | Description                           | Displayed in...                                  |
| Cdyn                        | Dynamic compliance                    | Ventilator Settings Review screen (see page 243) |
| I:E E-Part                  | I:E ratio (expiratory component)      |                                                  |
| I:E I-Part                  | I:E ratio (inspiratory component)     |                                                  |
| iO2                         | Inspired O2                           |                                                  |
| MAP                         | Mean airway pressure                  |                                                  |
| MVe                         | Expired minute volume                 |                                                  |
| Pause                       | Pause pressure                        |                                                  |
| PIP                         | Peak inspired airway pressure         |                                                  |
| PEEP                        | Peak end expiratory airway pressure   |                                                  |
| Raw                         | Dynamic resistance                    |                                                  |
| RRv                         | Pediatric respiratory rate - vol/flow |                                                  |
| RRs                         | Spontaneous respiratory rate          |                                                  |
| TVe                         | Tidal volume, expired                 |                                                  |
| TVi                         | Tidal volume, inspired                |                                                  |
| iO2                         | Inspired O2                           |                                                  |

| <b>Hamilton Galileo Ventilator</b> |                                  |                                                  |
|------------------------------------|----------------------------------|--------------------------------------------------|
| <b>Ventilator Settings</b>         | <b>Description</b>               | <b>Displayed in...</b>                           |
| iO2 set                            | Inspired O2 setting              | Ventilator Settings Review screen (see page 243) |
| INSP t%                            | I:E ratio, inspiratory component |                                                  |
| PAUSE t%                           | I:E ratio, expiratory component  |                                                  |
| PEEP set                           | PEEP setting                     |                                                  |
| TVi set                            | Inspired tidal volume setting    |                                                  |
| TRIG PRESS                         | Trigger pressure                 |                                                  |
| Vent:                              | Selected Ventilator type         |                                                  |

### Maquet SV 300/SV 300A/Servo<sup>i</sup> Ventilator

| <b>Maquet SV 300/SV 300A/Servo<sup>i</sup> Ventilator</b> |                                               |                                                              |
|-----------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|
| <b>Ventilator Messages</b>                                | <b>Description</b>                            | <b>Displayed in...</b>                                       |
| Paw > HI LIM                                              | An apnea detected                             | Main Screen - lower right corner of the top waveform channel |
| APNEA                                                     | Battery alarm                                 |                                                              |
| BATTERY                                                   | CMV potentiometer error                       | Bed View - information area, just below menu buttons         |
| CMV potentiometer                                         | Gas supply alarm                              |                                                              |
| GAS SUPPLY?                                               | High continuous pressure                      |                                                              |
| HI CONT PRESS                                             | iO2 is below lower limit                      |                                                              |
| iO2 < LO LIM                                              | iO2 exceeds upper limit                       |                                                              |
| iO2 > HI LIM                                              | Mains failure                                 |                                                              |
| MAINS FAIL                                                | MVe is too high/too low                       |                                                              |
| MVe > HI / LO LIM                                         | Mode switch error                             |                                                              |
| VENT ERR                                                  | O2 cell disconnect, or O2 potentiometer error |                                                              |
| % O2 ERR                                                  | Over range or Range switch error              |                                                              |
| DEVICE ERR                                                | An apnea detected                             |                                                              |

| <b>Maquet SV 300/SV 300A/Servo<sup>i</sup> Ventilator</b> |                         |                                                  |
|-----------------------------------------------------------|-------------------------|--------------------------------------------------|
| <b>Ventilator Settings</b>                                | <b>Description</b>      | <b>Displayed in...</b>                           |
| BAROM PRESS                                               | Barometric pressure     | Ventilator Settings Review screen (see page 243) |
| AIR SUPPLY                                                | Gas air supply pressure |                                                  |
| O2 SUPPLY                                                 | Gas O2 supply pressure  |                                                  |

| <b>Maquet SV 300/SV 300A/Servo<sup>i</sup> Ventilator</b> |                                                      |                                                        |
|-----------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|
| <b>Ventilator Settings</b>                                | <b>Description</b>                                   | <b>Displayed in...</b>                                 |
| <sup>1</sup> I:E I-Part                                   | I:E ratio (inspiratory component)                    | Ventilator Settings<br>Review screen<br>(see page 243) |
| <sup>1,2</sup> INSP t%                                    | I:E ratio (inspiratory component)%                   |                                                        |
| iO <sub>2</sub>                                           | Inspired O <sub>2</sub>                              |                                                        |
| MAP cmH <sub>2</sub> O                                    | Mean airway pressure                                 |                                                        |
| MVe I                                                     | Expired minute volume                                |                                                        |
| MVi I                                                     | Inspired minute volume                               |                                                        |
| Pause                                                     | Pause pressure                                       |                                                        |
| PEEP                                                      | Peak end expiratory pressure                         |                                                        |
| PIP                                                       | Peak inspired pressure                               |                                                        |
| RRv                                                       | Respiratory rate                                     |                                                        |
| TVe                                                       | Tidal volume expired breath                          |                                                        |
| TVi                                                       | Tidal volume inspired breath                         |                                                        |
| CMV FREQ                                                  | Frequency of CMV setting                             |                                                        |
| SIMV FREQ                                                 | Frequency of SIMV setting                            |                                                        |
| INSP t%                                                   | I:E ratio - inspiratory component                    |                                                        |
| I RISE%                                                   | I:E ratio - inspiratory rise time component          |                                                        |
| PAUSE t%                                                  | I:E ratio - pause component                          |                                                        |
| iO <sub>2</sub> set                                       | iO <sub>2</sub> setting                              |                                                        |
| iO <sub>2</sub> set LO                                    | Lower alarm limit of iO <sub>2</sub>                 |                                                        |
| iO <sub>2</sub> HI LIM                                    | Upper alarm limit of iO <sub>2</sub>                 |                                                        |
| MVe LO LIM                                                | Lower alarm limit of MVe                             |                                                        |
| MVe HI LIM                                                | Upper alarm limit of MVe                             |                                                        |
| PATIENT RANGE                                             | Patient type<br>(Adult= 4; Pediatric=15; Neonate=16) |                                                        |
| PEEP set                                                  | PEEP setting                                         |                                                        |
| P CTL > PEEP                                              | Pressure control level above PEEP                    |                                                        |
| P HI LIM                                                  | Upper pressure alarm limit                           |                                                        |
| P SUPPORT                                                 | Pressure support level above PEEP                    |                                                        |
| TRIG SENS > PEEP                                          | Trigger sensitivity level below PEEP                 |                                                        |
| VENT MODE <sup>3</sup>                                    | Ventilation mode setting                             |                                                        |
| SV 300, SV 300A,<br>Servo <sup>i</sup>                    | Selected Ventilator type                             |                                                        |
| VOL set                                                   | Volume setting                                       |                                                        |



| Maquet SV 300/SV 300A/Servo <sup>i</sup> Ventilator                                                                                                                                                                                                                                              |                 |                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------|
| Ventilator Settings                                                                                                                                                                                                                                                                              | Description     | Displayed in...                 |
| Paw                                                                                                                                                                                                                                                                                              | Airway pressure | Ventilator Settings             |
| Flow                                                                                                                                                                                                                                                                                             | Flow (insp/exp) | Review screen<br>(see page 243) |
| <sup>1</sup> Certain parameters, identified as measured values, are derived from other values or settings provided by the ventilator and may not reflect actual values. See the operating instructions for your specific monitor or ventilator for detailed information on derived measurements. |                 |                                 |
| <sup>2</sup> Display of I:E value may be affected by mode of ventilation.                                                                                                                                                                                                                        |                 |                                 |
| <sup>3</sup> Mode value may be blanked during certain conditions and modes of ventilation.                                                                                                                                                                                                       |                 |                                 |

| <b>Maquet SV 300/SV 300A/Servo<sup>i</sup> Ventilator</b> |                                        |
|-----------------------------------------------------------|----------------------------------------|
| <b>Ventilation Mode Display (Servo<sup>i</sup>)</b>       |                                        |
| <b>Ventilator Mode</b>                                    | <b>Mode at Infinity CentralStation</b> |
| SIMV (PRVC)                                               | SIMV (PRVC)                            |

### Maquet SV 900 Ventilator

| <b>Maquet SV 900 Ventilator</b> |                               |                                                        |
|---------------------------------|-------------------------------|--------------------------------------------------------|
| <b>Ventilator Parameter</b>     | <b>Description</b>            | <b>Displayed in...</b>                                 |
| RRv                             | Respiratory rate              | Ventilator Settings<br>Review screen<br>(see page 243) |
| TVe                             | Tidal volume, expired         |                                                        |
| Tvi                             | Tidal volume, inspired        |                                                        |
| MVi                             | Minute volume, inspired       |                                                        |
| MVe                             | Minute volume, expired        |                                                        |
| PIP                             | Peak inspired airway pressure |                                                        |
| MAP                             | Mean airway pressure          |                                                        |
| Pause                           | Pause pressure                |                                                        |
| PEEP                            | Peak End Expiratory Pressure  |                                                        |
| iO <sub>2</sub>                 | Inspired O <sub>2</sub>       |                                                        |

## Viasys Bear 1000 Adult Ventilator

| Viasys Bear 1000 Adult Ventilator |                                            |                                                              |
|-----------------------------------|--------------------------------------------|--------------------------------------------------------------|
| Alarm Messages                    | Description                                | Displayed in...                                              |
| Paw > HI LIM                      | Paw exceeds upper alarm limit              | Main Screen - lower right corner of the top waveform channel |
| Paw < LO LIM                      | Paw is below lower limit                   |                                                              |
| GAS SUPPLY?                       | Gas supply alarm                           | Bed View - information area, just below menu buttons         |
| RR > HI LIM                       | Respiratory rate exceeds upper alarm limit |                                                              |
| RR < LO LIM                       | Respiratory rate is below lower limit      |                                                              |
| MVe < LO LIM                      | Minute volume is below lower limit         |                                                              |
| MVe > HI LIM                      | Minute volume exceeds upper alarm limit    |                                                              |
| I:E Alarm                         | I:E Limit (I:E Error)                      |                                                              |
| Cycle Failed                      | Fail to cycle                              |                                                              |

| Viasys Bear 1000 Adult Ventilator |                                |                     |
|-----------------------------------|--------------------------------|---------------------|
| Parameter                         | Description                    | Displayed in...     |
| PIP                               | Peak inspired airway pressure  | Ventilator Settings |
| MAP                               | Mean airway pressure           | Review screen       |
| %MMV                              | % Mandatory minute ventilation | (see page 243)      |
| TVe                               | Tidal volume, expired          |                     |
| Ti                                | Inspiratory time               |                     |
| MVe                               | Minute volume, expired         |                     |
| MVe s                             | Spontaneous minute volume      |                     |
| RRv                               | Respiratory rate               |                     |
| Pause                             | Pause (plateau) pressure       |                     |
| RRs                               | Respiratory rate, spontaneous  |                     |

| <b>Viasys Bear 1000 Adult Ventilator</b> |                                               |                                                  |
|------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| <b>Settings</b>                          | <b>Description</b>                            | <b>Displayed in...</b>                           |
| Ti set                                   | Inspiratory time                              | Ventilator Settings Review screen (see page 243) |
| iO2 set                                  | Inspired O2 setting                           |                                                  |
| MAX iFLOW set                            | Maximum inspiratory flow                      |                                                  |
| TVi set                                  | Tidal volume, inspired                        |                                                  |
| Pause t                                  | I:E ratio (Pause component) Inspiratory pause |                                                  |
| Pslope                                   | Pressure slope                                |                                                  |
| TRIG PRESS                               | Pressure trigger sensitivity                  |                                                  |
| FLOW Trig set                            | Flow trigger                                  |                                                  |
| Paw set                                  | Inspiratory pressure                          |                                                  |
| MMV                                      | Mandatory minute ventilation                  |                                                  |
| Freq set                                 | Breath rate                                   |                                                  |
| PSupport                                 | Pressure support                              |                                                  |

| <b>Viasys Bear 1000 Adult Ventilator</b> |                                        |
|------------------------------------------|----------------------------------------|
| <b>Ventilation Mode Display</b>          |                                        |
| <b>Ventilation Mode at Device</b>        | <b>Mode at Infinity CentralStation</b> |
| Assist Control                           | Assist Control                         |
| Pressure Control                         | PCV                                    |
| SIMV/CPAP/PSV                            | SIMV/CPAP/PSV                          |
| PC-SIMV/CPAP/PSV                         | PC-SIMV/CPAP/PSV                       |

| <b>Viasys Bear 1000 Adult Ventilator</b> |                    |                                  |
|------------------------------------------|--------------------|----------------------------------|
| <b>Waveforms</b>                         |                    |                                  |
| <b>Parameter</b>                         | <b>Description</b> | <b>Displayed in...</b>           |
| Paw                                      | Airway Pressure    | Ventilator screen (see page 239) |
| Flow                                     | Flow (Insp/exp)    |                                  |
| Loops                                    | Ventilator loops   |                                  |

## Viasys BearCub Infant Ventilator

| Viasys BearCub Infant Ventilator |                                                   |                                                              |
|----------------------------------|---------------------------------------------------|--------------------------------------------------------------|
| Alarm Messages                   | Description                                       | Displayed in...                                              |
| Paw > HI LIM                     | Paw exceeds upper alarm limit                     | Main Screen - lower right corner of the top waveform channel |
| Paw < LO LIM                     | Paw is below lower limit                          |                                                              |
| GAS SUPPLY?                      | Gas supply alarm                                  | Bed View - information area, just below menu buttons         |
| RR > HI LIM                      | Respiratory rate exceeds upper alarm limit        |                                                              |
| PEEP < LO LIM                    | Peak End Expiratory Pressure is below lower limit |                                                              |
| MVe < LO LIM                     | Minute volume is below lower limit                |                                                              |
| APNEA                            | Apnea                                             |                                                              |
| Battery                          | Low battery supply                                |                                                              |
| Cycle Failed                     | Fail to cycle                                     |                                                              |

| Viasys BearCub Infant Ventilator |                               |                     |
|----------------------------------|-------------------------------|---------------------|
| Parameter                        | Description                   | Displayed in...     |
| PIP                              | Peak inspired airway pressure | Ventilator Settings |
| PEEP                             | Peak End Expiratory Pressure  | Review screen       |
| Te                               | Expiratory time               | (see page 243)      |
| TVe                              | Tidal volume, expired         |                     |
| Ti                               | Inspiratory time              |                     |
| MVe                              | Minute volume, expired        |                     |
| RRv                              | Respiratory rate              |                     |
| TVi                              | Tidal volume, inspired        |                     |
| MAP                              | Mean Airway Pressure          |                     |
| O2 Supply Prsr                   | O2 Supply pressure            |                     |

| Viasys BearCub Infant Ventilator |                          |                     |
|----------------------------------|--------------------------|---------------------|
| Settings                         | Description              | Displayed in...     |
| Ti set                           | Inspiratory time         | Ventilator Settings |
| Freq set                         | Breath rate              | Review screen       |
| Vlimit                           | Volume limit setting     | (see page 243)      |
| FLOW Trig set                    | Flow trigger setting     |                     |
| iFLOW set                        | Inspiratory flow setting |                     |

| <b>Viasys BearCub Infant Ventilator</b> |                                        |
|-----------------------------------------|----------------------------------------|
| <b>Ventilation Mode Display</b>         |                                        |
| <b>Ventilation Mode at Device</b>       | <b>Mode at Infinity CentralStation</b> |
| Assist Control                          | Assist Control                         |
| Pressure Control                        | PCV                                    |
| SIMV/CPAP/PSV                           | SIMV/CPAP/PSV                          |
| PC-SIMV/CPAP/PSV                        | PC-SIMV/CPAP/PSV                       |

| <b>Viasys BearCub Infant Ventilator</b> |                    |                                     |
|-----------------------------------------|--------------------|-------------------------------------|
| <b>Waveforms</b>                        |                    |                                     |
| <b>Parameter</b>                        | <b>Description</b> | <b>Displayed in...</b>              |
| Paw                                     | Airway Pressure    | Ventilator screen<br>(see page 239) |
| Flow                                    | Flow (Insp/exp)    |                                     |
| Loops                                   | Ventilator loops   |                                     |

## Baxter Vigilance/Vigilance II Cardiac Output Monitor

| Baxter Vigilance/Vigilance II Cardiac Output Monitor                             |                                    |                                                        |
|----------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------|
| Parameter                                                                        | Description                        | Displayed in...                                        |
| BT                                                                               | Blood Temperature                  | Ventilator Settings<br>Review screen<br>(see page 243) |
| CCO                                                                              | Continuous Cardiac Output          |                                                        |
| CCI                                                                              | Continuous Cardiac Output Index    |                                                        |
| ICO                                                                              | Intermittent Cardiac Output        |                                                        |
| ICI                                                                              | Intermittent Cardiac Output Index  |                                                        |
| SaO2                                                                             | Arterial Oxygen Saturation         |                                                        |
| SvO2                                                                             | Venous Oxygen Saturation           |                                                        |
| SVR                                                                              | Systemic Vascular Resistance       |                                                        |
| SVRI                                                                             | Systemic Vascular Resistance Index |                                                        |
| VO2                                                                              | O2 Consumption                     |                                                        |
| DO2                                                                              | O2 Delivery                        |                                                        |
| <i>The following parameters are available from the Baxter Vigilance II only.</i> |                                    |                                                        |
| SV                                                                               | Stroke Volume                      |                                                        |
| SVI                                                                              | Stroke Volume Index                |                                                        |
| EDV                                                                              | End Diastolic Volume               |                                                        |
| EDVI                                                                             | End Diastolic Volume Index         |                                                        |
| ESV                                                                              | End Systolic Volume                |                                                        |
| ESVI                                                                             | End Systolic Volume Index          |                                                        |
| EF                                                                               | Ejection Fraction                  |                                                        |

**etCO<sub>2</sub>/Respiratory Mechanics Pod**

| <b>etCO<sub>2</sub>/Respiratory Mechanics Pod</b> |                                                 |                                                        |
|---------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|
| <b>Parameter</b>                                  | <b>Description</b>                              | <b>Displayed in...</b>                                 |
| PIP                                               | Peak inspired pressure                          | Ventilator Settings<br>Review screen<br>(see page 243) |
| PEEP                                              | Peak end expiratory pressure                    |                                                        |
| MAP                                               | Mean arterial pressure                          |                                                        |
| TVd aw                                            | Dead space, airway                              |                                                        |
| MVe s/m                                           | Expiratory minute volume spontaneous/mechanical |                                                        |
| MVe                                               | Expiratory minute volume                        |                                                        |
| RR s/m                                            | Respiratory rate, spontaneous/mechanical        |                                                        |
| RRv                                               | Respiratory rate                                |                                                        |
| Cdyn                                              | Dynamic compliance                              |                                                        |
| C20/Cdyn                                          | Dynamic compliance of the last 20% breath       |                                                        |
| Raw e                                             | Dynamic resistance                              |                                                        |
| PEF                                               | Peak Expiratory Flow                            |                                                        |
| TValv                                             | Alveolar tidal volume                           |                                                        |
| VCO <sub>2</sub>                                  | CO <sub>2</sub> production, minute volume       |                                                        |
| InspT %                                           | Percent inspired time                           |                                                        |
| MValv                                             | Alveolar mixed minute volume                    |                                                        |
| MValvs                                            | Alveolar spontaneous minute volume              |                                                        |
| Pause                                             | Pause pressure                                  |                                                        |
| PeCO <sub>2</sub>                                 | Mixed expired CO <sub>2</sub>                   |                                                        |
| PEF                                               | Peak expiratory flow                            |                                                        |
| RAW                                               | Dynamic airway resistance                       |                                                        |
| RAW i                                             | Inspired airway resistance                      |                                                        |
| RSBI                                              | Rapid shallow breathing index                   |                                                        |
| Ti                                                | Inspiratory time                                |                                                        |
| T e                                               | Expired tidal volume                            |                                                        |
| TV lk                                             | ET tube leakage                                 |                                                        |
| TV CO <sub>2</sub>                                | CO <sub>2</sub> tidal volume                    |                                                        |

| etCO2/Respiratory Mechanics Pod |                             |                                     |
|---------------------------------|-----------------------------|-------------------------------------|
| Parameter                       | Description                 | Displayed in...                     |
| Waveforms                       |                             | Ventilator Screen<br>(see page 239) |
| PAW                             | Airway Pressure             |                                     |
| Flow                            | Flow (Insp/exp)             |                                     |
| Pressure Loop                   | Represents the first breath |                                     |
| Flow Loop                       | Represents the first breath |                                     |



## Biomed Functions

---

|                                                             |     |
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## Biomed Screen Selections

---

The Biomed functions are password-protected and intended only for the hospital's biomedical personnel or Dräger service representatives.

| Selection                   | Description                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Show Devices...             | Allows your Biomed to view the status of network devices within the monitoring unit of the Infinity CentralStation                                               |
| Configure Central...        | Provides access to setup features                                                                                                                                |
| Configure Locked Options... | Allows your Biomed to enable available options                                                                                                                   |
| System Console...           | Activates a window from where you can perform remote diagnostics, shut down the Infinity CentralStation, select Rest ECG report formats, etc.                    |
| Diagnostic Log              | Displays a log of 1000 hardware and software error conditions                                                                                                    |
| Clinical Events Log         | Displays a log of 1000 clinically significant events, such as alarm detection, Alarm Silence requests, recording status, and remote control of bedside monitors. |
| Configure Telemetry...      | Accesses Telemetry setup functions (see page 299).                                                                                                               |

### Accessing the Biomed Screens

To access the Biomed screens:

- 1 Click on **Biomed** in the Main Screen menu bar.
- 2 Click on the desired menu selection.
- 3 Enter the Biomed password in the password pop-up.
- 4 Click on **Accept** in the password pop-up.

The password pop-up window displays.

### Show Devices Screen

---

The Biomed - Show Devices screen lists status information for all network devices assigned to the monitoring unit of the specific Infinity CentralStation. To open the Biomed - Show Devices screen, see page 290.

| Selection                                               | Description                                                            |
|---------------------------------------------------------|------------------------------------------------------------------------|
| Show Central Status                                     | Opens a pop-up with status information for the Infinity CentralStation |
| Bedsides, Recorders, Centrals, CPS's, All radio buttons | Radio buttons for listing device-specific information                  |
| Page arrow buttons                                      | Scrolls pages of information                                           |

## Configure Central Screen

The following table displays the Biomed - Configure Central Screen selections.

| Selection                                                                                                                                             | Description                                                                                                                                                                                                                    | Available Settings                                                         | Default |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------|
| Hospital Name                                                                                                                                         | Defines hospital name                                                                                                                                                                                                          | 25 character max.                                                          | –       |
| Language                                                                                                                                              | Displays the language that was set during installation                                                                                                                                                                         | –                                                                          | –       |
| Biomed Password                                                                                                                                       | Sets or modifies Biomed password                                                                                                                                                                                               | 8 character max.                                                           | –       |
| Clinical Password ON/OFF                                                                                                                              | Enables/disables clinical password                                                                                                                                                                                             | ON/OFF                                                                     | ON      |
| Clinical Password                                                                                                                                     | Sets or modifies clinical password                                                                                                                                                                                             | 8 character max.                                                           | ICS     |
| Central Layouts Password                                                                                                                              | Enables/disables password protection.<br>If enabled, screen is password protected.                                                                                                                                             | ON / OFF                                                                   | OFF     |
| Printer Connected                                                                                                                                     | Sets the printer connection                                                                                                                                                                                                    | – OFF<br>– Network<br>(Central Station is connected to a network printer.) | OFF     |
| Network Printer Address                                                                                                                               | Displays network printer IP address configured during installation (for viewing only)                                                                                                                                          | –                                                                          | –       |
| Cabrera                                                                                                                                               | Permits activation/deactivation of the Hexaxial lead configuration<br><br>When you turn Cabrera on, the leads are displayed in the Hexaxial display format.<br><br>If you turn Cabrera off, the leads appear in standard mode. | ON / OFF                                                                   | OFF     |
| Event Class Editor                                                                                                                                    | Permits editing of user-selectable event classification labels (see page 293)                                                                                                                                                  | Edit / Hide                                                                | Hide    |
| Patient Confidentiality                                                                                                                               | Determines whether a patient's name appears on Infinity CentralStation displayed screens                                                                                                                                       | ON / OFF                                                                   | OFF     |
| Restore Default Settings                                                                                                                              | Restores system default settings                                                                                                                                                                                               | –                                                                          | –       |
| <b>NOTE</b><br>Whenever default settings are restored, the Infinity CentralStation must be restarted.<br>An entry is made to the clinical events log. |                                                                                                                                                                                                                                |                                                                            |         |

| <b>Selection</b>      | <b>Description</b>                                                              | <b>Available Settings</b> | <b>Default</b> |
|-----------------------|---------------------------------------------------------------------------------|---------------------------|----------------|
| Hospital Label        | Identifies the hospital label for the network                                   | 7 character max.          | –              |
| Monitoring Unit Label | Identifies the monitoring unit to which the Infinity CentralStation is assigned | 7 character max.          | –              |
| Care Unit Label       | Identifies the care unit to which the Infinity CentralStation is assigned       | 7 character max.          | –              |
| Host Label            | View only                                                                       | –                         | –              |

## Editing Event Class Names

To edit the event class names:

- 1 Open the **Biomed - Configure Central** screen (see page 290).
- 2 Click on the toggle button next to **Event Class Editor** so that it displays **Edit**.

A pop-up displays with ten default labels:

| Class Name | Description         |
|------------|---------------------|
| AFIB       | Atrial Fibrillation |
| ATAC       | Atrial Tachycardia  |
| BBB        | Bundle branch block |
| PNC        | Pacer not captured  |
| PNP        | Pacer not pacing    |
| TRIG       | Trigeminy           |
| MTI        | Multifocal PVC      |
| RONT       | R-on-T              |
| IRR        | Irregular HR        |
| NVTA       | Non-sustained VTAC  |

- 3 After you make the desired changes, click on **Accept** to save changes, or **Undo**.

## Modifying the Configure Central selections

To modify the configuration of the central screen:

- 1 Open the **Biomed - Configure Central** screen (see page 290).
- 2 Select a button or text entry box to set and click.
- 3 After entering information and modifying settings click on **Accept** to save changes, or **Undo**.

## Configure Locked Options Screen

---

This screen lists software that must be purchased separately and can only be enabled with a unique password for each Infinity CentralStation. For a list of available software options, contact your local Dräger representative.

### Enabling Available Options

To enable available options:

- 1 Open the **Configure Locked Options** screen (see page 290).
- 2 Click on one of the available menu choices.

- 3 Enter the password for the option you wish to enable.

- 4 Click on **Accept**.

After an option is enabled, a pop-up displays the following message:

*The Infinity CentralStation must be restarted to allow the locked options to take effect.*

- 5 Restart the Infinity CentralStation.

Whenever you lock or enable an option, a message is stored in the clinical events log.

## Raid Option

---

With the Infinity CentralStation RAID Option, RAID 1 mirroring provides disclosure disk redundancy by writing all full/event disclosure data to two identical hard drives. With this option configured, all disclosure disk data is written to a second, identical hard drive in the background.

### CAUTION

In the event of a RAID (Redundant Array of Independent Disks) failure, no tone sounds, and no message appears. However, an LED on the hard drive that failed will light.

## System Console

---

This Biomed menu selection activates a console window with a prompt from which you can:

- Run remote diagnostics for all network devices
- Shut down or restart the Infinity CentralStation
- Review an alarm history log for telemetry patient channels consisting of the last 5000 events (with time and date of the alarm occurrences)
- Select Rest ECG report formats

For information about opening the **System Console**, see page 290.

## Rest ECG Reports

---

Rest ECG reports are available in several formats which you can customize. As soon as the data arrives from the bedside monitor, the Infinity CentralStation analyzes it and generates a printed report.

An entry is made in the Clinical Events log whenever the Infinity CentralStation issues a report. This entry indicates whether or not the report was processed successfully (see page 298 for information about viewing logs).

If multiple printing requests exist, the Rest ECG report is queued until the assigned printer becomes available.

### **WARNING**

**Diagnosis based on interpretation of 12-lead monitoring results should only be done by qualified personnel. Prior to final interpretation and diagnosis, qualified physicians should review suggested diagnostic statements and all other available information.**

## Available Report Formats

The format of a Rest ECG report determines the layout and the type of information included in the report. For more detailed information on generating ECG reports, please refer to the bedside monitor documentation.

| Report Type                            | Resolution                             | Diagnostic Statements Y/N | Pages | Orientation        |
|----------------------------------------|----------------------------------------|---------------------------|-------|--------------------|
| 6x2 waves, 2.5 s                       | 50 mm/s                                | Y                         | 1     | Landscape          |
| 6x2 waves, 2.5 s                       | 50 mm/s                                | N                         | 1     | Landscape          |
| 6x2 waves, 2.5 s                       | 50 mm/s                                | Y                         | 1     | Landscape, Cabrera |
| 6x2 waves, 2.5 s                       | 50 mm/s                                | N                         | 1     | Landscape, Cabrera |
| Complexes and 3 waves (V1, II, V5) 7 s | Complexes: 50mm/s<br>Waveforms: 25mm/s | Y                         | 1     | Portrait, Standard |
| 12 waveforms, each 5 s                 | 50 mm/s                                | N                         | 2     | Cabrera            |
| 12 waveforms, each 5 s                 | 50 mm/s                                | N                         | 2     | Landscape          |
| 3x4 waveforms, each 2.5 s              | 25 mm/s                                | N                         | 1     | Landscape          |
| 3x4 waveforms, each 2.5 s              | 25 mm/s                                | Y                         | 1     | Landscape          |
| 6x2 waveforms, each 5 s                | 25 mm/s                                | Y                         | 1     | Landscape          |
| 6x2 waveforms, each 5 s                | 25 mm/s                                | N                         | 1     | Landscape          |
| 12 waveforms, each 10s                 | 25 mm/s                                | Y                         | 2     | Landscape          |
| 12 waveforms, each 10s                 | 25 mm/s                                | N                         | 2     | Landscape          |



## Accessing the Rest ECG Report Selection Menu

To access the Rest ECG Report Selection Menu:

- 1 Open the **System Console** (see page 290).
- 2 Type the following at the system prompt:

**rekgSelect**

A selection list appears, as shown in the following example:

```

                                Biomed – System Console
$ rekgSelect

      ---- Rest ECG Report Selection Menu ----

1. 6x2 waves @ 2.5 secs, with diagnostics
2. 6x2 waves @ 2.5 secs, without diagnostics
3. 6x2 waves @ 2.5 secs, with diagnostics, cabrera format
4. 6x2 waves @ 2.5 secs, without diagnostics, cabrera format
5. Complexes & 3 waves @ 7 secs, matrix diags
6. 12 waves @ 5.0 secs, cabrera format (2 pages)
7. 12 waves @ 5.0 secs (2 pages)
8. 3x4 waves @ 2.5 secs, without diagnostics
9. 3x4 waves @ 2.5 secs, with diagnostics
10. 6x2 waves @ 5.0 secs, with diagnostics
11. 6x2 waves @ 5.0 secs, without diagnostics
12. 12 waves @ 10.0 secs, with diagnostics (2 pages)
13. 12 waves @ 10.0 secs (2 pages)
14. 1 copy at printer
15. 2 copies at printer
99. Set or change hospital name

Select from 1 ~ 15, 99 or press <Enter> for the default (9)
Enter 0 or q to quit.
Select: █

```

- 3 Enter the number before the format or function you wish to select (1 - 15, 99) or press **<Enter>** for the default (9).

To change the hospital name, first make your selection, enter the new information, and then press **<Enter>** on the keyboard.

## Viewing Logs

---

The password-protected diagnostic and clinical events logs support recording hardware and software conditions and clinical events. The *Diagnostic log* contains a record of up to 1000 hardware and software events and are helpful in determining system status. The *Clinical Events log* contains information on up to 1000 events of clinical significance.

Typical clinical events log entries may include:

- Transitions in/out of local alarm silence
- Infinity CentralStation Alarm State Change (e.g., Audible Alarm Annunciation Change from active to silenced or vice versa)
- Alarm Limit Change from the Infinity CentralStation
- Arrhythmia Setup Change from the Infinity CentralStation
- Bedside State Change (e.g., online, offline)
- CPS/IDS State Change (e.g., online, offline, duplicate IP address detected)
- Main Screen Layout Change
- Edits of Patient Demographics
- Alarm Volume Change
- Clinical Password Modification
- Language Change
- Restoration of Factory Default Settings
- Recording Status (other than cancellation of continuous or timed recordings)
- Recorder State Change (e.g., Recorder offline, Recorder Failure, CPS/IDS Offline)
- Time and Date Change
- Locked Options Change
- A request for SpO<sub>2</sub> with no additional SpO<sub>2</sub> licenses available
- Infinity CentralStation detection of incompatible Infinity M300 software

Log entries appear as separate lines and are displayed in chronological order with the most recent entry at the top. Each entry includes date, time, event class, event code, and event description.

To open the **Diagnostic** or **Clinical Events** logs, see page 290.

## Copying Logs

To copy logs to a USB device:

- 1 Plug the USB storage media into one of the USB connectors on the Infinity CentralStation.

### NOTE

Both the Diagnostic **and** Clinical Events logs are copied during a copy request.

- 2 Access the log you wish to copy (see page 298).
- 3 Click on **Copy Logs To Disk**.

Logs are copied as individual files. During the copying process, the Infinity CentralStation is fully operational. However, the **Copy Logs To Disk** button remains unavailable for selection until the process completes. If the operation fails, a status message displays and an advisory tone sounds.

## Telemetry System Set Up

---

The following Biomed setup functions pertain exclusively to the Infinity M300. Before you can access these functions, the Infinity M300 option must be enabled (see page 294).

### Infinity M300 Set Up

#### Accessing the Infinity M300 Telemetry Setup Screens

To access the Infinity M300 Telemetry Setup screens:

- 1 Click on **Biomed** in the Main Screen menu bar.
- 2 Click on **Configure Telemetry...**
- 3 Click on **Telemetry Devices** or **M300 Setup**.
- 4 Enter the Biomed password in the password pop-up.
- 5 Click on **Accept** in the pop-up.  
The selected screen displays.

You must use the Biomed password to access these screens (see Installation Instructions, included with this Instructions for Use in product packaging).

#### Biomed - Infinity M300 Setup Screen

##### NOTE

You can only configure the Infinity M300 Alarm Paused key, STAFF ALERT key, Record key, and speaker for an individual patient if the Per Patient option (see page 303) is selected during setup.

### Infinity M300 Devices Screen

The **Telemetry Devices** screen permits the Biomed to configure up to 255 devices to use with the Infinity CentralStation.

To configure the telemetry devices:

- 1 Open the **Telemetry Devices** screen.
- 2 Click on the **New Device** button.
- 3 Enter the **IP address** in a row of the table for each available device.
- 4 Assign a unique **Telemetry ID** (4 character maximum).
- 5 Repeat steps 2 and 3 for each device.
- 6 When you have added all of the devices, click on the **Accept** button.

Once you assign the *Telemetry IDs*, those devices are available for selection from the **Telemetry ID** section of the **Admit** screen (see page 121).

| Selection      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Settings                                                                                                                                               | Default      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Alarm Pause    | <p>Determines how the Infinity M300 <b>Alarm Pause</b> key can be used.</p> <p>If you want to be able to individually set this function for each patient, you must select the <i>Per Patient</i> option (see page 303).</p> <p>If you select <i>All Off Alarm Paused</i> key will be disabled for all Infinity M300 devices in the system.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>– All On</li> <li>– All Off</li> <li>– Per Patient</li> </ul>                                                   | All Off      |
| Record         | <p>Determines how the Infinity M300 <b>Record</b> key can be used.</p> <p>If you want to be able to individually set this function for each patient, you must select the <i>Per Patient</i> option (see page 303).</p> <p>If you want to generate manual timed recordings when you press the Infinity M300 <b>Record</b> key, you must select <i>Record All</i> or <i>Rec/Store All</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>– All Off</li> <li>– Record All</li> <li>– Store All</li> <li>– Rec/Store All</li> <li>– Per Patient</li> </ul> | Record All   |
| Staff Alert    | <p>Determines how the Infinity M300 STAFF ALERT key can be used.</p> <p>If you want to be able to individually set this function for each patient, you must select the <i>Per Patient</i> option (see page 303).</p> <p>If you want to generate a STAFF ALERT alarm at the Infinity Central Station when you press the Infinity M300 <b>STAFF ALERT</b> key, you must select All On.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>– All On</li> <li>– All Off</li> <li>– Per Patient</li> </ul>                                                   | All On       |
| Volume Control | <p>Determines if the Infinity M300 speaker can be enabled.</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p><b>WARNING</b></p> <p>The Infinity M300 speaker is intended for use only in the patient vicinity. It is not intended for primary alarm annunciation. Use the Infinity CentralStation speaker for primary alarm annunciation.</p> </div> <div style="width: 45%;"> <p><b>NOTE</b></p> <p>When the Infinity M300 <b>Volume Control</b> is set to <b>All Disabled</b>, the Volume screen is not available on the M300. The M300 speaker volume is fixed to whatever is set on the <b>Biomed\Telemetry Setup\Speaker Volume</b> setting.</p> <p>Unless you select the <i>Per Patient</i> option, you cannot individually set this function for each patient (see page 303).</p> </div> </div> | <ul style="list-style-type: none"> <li>– All Enabled</li> <li>– All Disabled</li> <li>– Per Patient</li> </ul>                                         | All Disabled |

| Selection              | Description                                                                     | Settings                                                              | Default |
|------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------|
| Speaker Volume         | Determines volume of Infinity M300 audible alarm.                               | – OFF<br>– 10 - 100%                                                  | OFF     |
| Display Shut Off Timer | Determines how long the Infinity M300 display remains on with no user activity. | – 1 min<br>– 2 min<br>– 3 min<br>– 4 min<br>– 5 min                   | 1 min   |
| ECG Notch Filter       | Assigns Infinity M300 ECG Notch filter setting.                                 | – 50 Hz<br>– 60 Hz                                                    | 60 Hz   |
| Régulation             | Sets French Homologation mode.                                                  | – Default<br>– France                                                 | Default |
| Accept/Undo            | Enables you to save or discard the changes made since the last save.            | – Accept saves any changes<br>– Undo reverts to the previous settings |         |

## Infinity Telemetry Set Up

### Accessing the Infinity Telemetry Setup Screens

To access the Infinity Telemetry Setup screens:

- 1 Click on **Biomed** in the Main Screen menu bar.
- 2 Click on **Configure Telemetry...**
- 3 Click on one of the following:
  - Receiver Setup...
  - Transmitter Setup...
  - RF Diagnostics...
- 4 Enter the Biomed password in the password pop-up.
- 5 Click on **Accept** in the pop-up.

Once you access a Telemetry Setup screen with the Biomed password, you can open all other Telemetry Setup screens without having to re-enter the Biomed password.

### Infinity Telemetry Receiver Setup

The **Receiver Setup** screen is further protected by a special password, which is only accessible to authorized service personnel. For detailed information on the various setup functions of this screen, please consult the accompanying service documentation.

To open the **Receiver Setup** screen, see page 299.

| Selection                                                                          | Description                                                                                 |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Country                                                                            | Permits selection of country from option list (Password protected. Contact Dräger Service). |
| ECG Notch Filter                                                                   | Displays list of possible <b>ECG Notch Filter</b> selections                                |
| Regulation                                                                         | Permits selection of possible <b>Regulation</b> configurations                              |
| <b>Receiver Details</b>                                                            |                                                                                             |
| S/W Version                                                                        | Displays Infinity CentralStation software version                                           |
| ID Label                                                                           | Displays receiver identification name assigned at installation                              |
| Ethernet Address                                                                   | Displays telemetry receiver IP address configured during installation                       |
| Antenna System                                                                     | Displays <b>Antenna System</b> type                                                         |
| Channel Spacing                                                                    | Permits channel selections for selected country                                             |
| Receiver Type                                                                      | Permits selection of UHF or VHF                                                             |
| Frequency Range                                                                    | Graphic display of <b>Channel Spacing</b> range                                             |
| – Click on <b>Accept</b> to save changes or <b>Undo</b> to keep original settings. |                                                                                             |

## Configuring the Transmitter Buttons

In the **Transmitter Setup** screen, you can configure the transmitter **Recording** and the **Staff Alert** buttons.



Recording button



Staff Alert button

These settings determine whether or not a recording or a staff alert can be initiated from the transmitter.

To configure the transmitter buttons:

- 1 Open the **Transmitter Setup** screen (see page 299).

The top of the screen contains the **Transmitter Record** and **Transmitter Staff Alert** buttons.

- 2 Click on the **Transmitter Record** or **Transmitter Staff Alert** button and scroll to a setting

The following table displays the Transmitter Record and Staff Alert button selections.

| Selection   | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All On      | This selection activates the transmitter <b>Staff Alert</b> and <b>Recording</b> buttons for <b>all</b> telemetry channels. A timed recording is initiated if you press the recording button, and if you press the staff alert button, a serious alarm is generated at the Infinity CentralStation. Also, an alarm message appears in the patient's corresponding Main Screen and Bed View waveform areas. |
| All Off     | The transmitter staff alert button and recording button are deactivated for <b>all</b> telemetry channels. You cannot generate any recordings or staff alert calls from any transmitter.                                                                                                                                                                                                                   |
| Per Patient | This setting allows you the greatest flexibility because the transmitter buttons can be enabled or disabled for each patient <i>individually</i> in the Transmitter Setup menu (see page 53).                                                                                                                                                                                                              |

- 3 Click on **Accept** to save the setting, or the **Undo** button to keep the previous setting.

## Radio Frequency Diagnostics

To open the **RF Diagnostics** screen, see “Infinity Telemetry Set Up” on page 301.

### Available Information on the RF Diagnostics Screen

The **RF Diagnostics** screen displays the frequency and I.D. for active Infinity Telemetry patients that currently have a transmitter assigned to the receiver channel. The screen also provides the information outlined in the following table. This information is updated every minute even as you are viewing it. Please consult the Service documentation for further information.:

| Information                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                | Possible Values                         |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Description                                        | This selection activates the transmitter <b>Staff Alert</b> and <b>Recording</b> buttons for <b>all</b> telemetry channels. A timed recording is initiated if you press the recording button, and if you press the staff alert button, a serious alarm is generated at the Infinity CentralStation. Also, an alarm message appears in the patient's corresponding Main Screen and Bed View waveform areas. | 1 to 4; 1 to 8; 1 to 12; 1 to 16; blank |
| Frequency                                          | The transmitter staff alert button and recording button are deactivated for <b>all</b> telemetry channels. You cannot generate any recordings or staff alert calls from any transmitter.                                                                                                                                                                                                                   | abc.defg; blank                         |
| I.D.                                               | This setting allows you the greatest flexibility because the transmitter buttons can be enabled or disabled for each patient <i>individually</i> in the Transmitter Setup menu (see page 53).                                                                                                                                                                                                              | 1 to 255; blank                         |
| RSSI-dBm                                           | Receiver's signal strength indicator for each channel (averaged over 1 minute).                                                                                                                                                                                                                                                                                                                            | 0 to -125 dBm; blank                    |
| 1 min,<br>5 min,<br>10 min,<br>1 Hour,<br>24 Hours | Percentage of good ECG signal samples received over the specified interval.                                                                                                                                                                                                                                                                                                                                | 0 to 100%                               |



## Patient Preparation

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|                                           |     |
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| <b>Attaching the Pulse Oximeter</b> ..... | 308 |

## General

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Careful skin preparation and proper electrode placement ensure strong signals with minimal artifact. In case of a technical alarm (e.g., a detached lead), re-prepare the patient according to the following recommendations.

### **WARNING**

**The Infinity M300 or Infinity Telemetry device should be placed to reduce the possibility of:**

- **the device falling on or injuring patient**
- **any accessory cables accidentally entangling patient's neck.**

Follow hospital protocol for preparing the patient's skin. To ensure a good quality signal, change electrodes every 24 to 48 hours (or in accordance with manufacturer's instructions). Electrodes may have to be changed more frequently under the following conditions:

- ECG signal degradation
- Excessive patient perspiration
- Patient skin irritation

A wide selection of reusable and single-use electrodes is available. Select the best electrode for the monitoring situation. Dräger recommends Ag/AgCl single-use electrodes. If you are using pre-gelled electrodes, verify that there is enough gel in the electrode gel-filled area. Never use disposable electrodes after their expiration date or when the gel has dried out.

Choose electrode sites in the configuration that will provide the best ECG based on the patient's underlying cardiac condition and the number of leads you wish to monitor. (P- and T-wave amplitudes should be no more than one third of the QRS amplitude.) Select flat, non-muscular sites to maximize electrode contact and minimize muscle artifact. Avoid joints or bony protrusions. Consider the following special conditions when selecting sites for electrode placement:

- **Surgery** — Keep the electrodes as far from the surgical site as possible.
- **Burn Patients** — Use sterile electrodes. Clean the equipment thoroughly. Follow hospital infection control procedures.

## ECG Electrode Placement

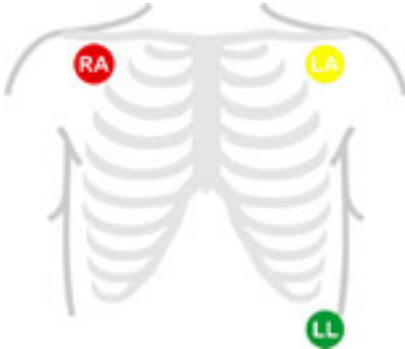
---

### **WARNING**

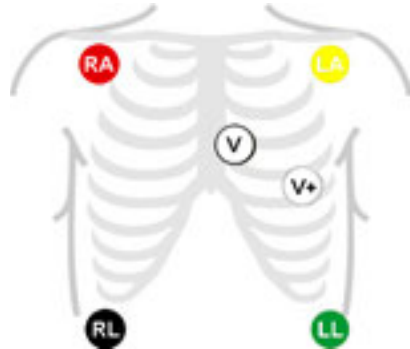
**When placing electrodes, ensure all connecting wires are placed safely and cannot injure the patient. Serious injury can occur if the lead wires are allowed to encircle the neck or extremities.**

The following illustrations show typical ECG electrode placement for 3-, 5-, and 6-wire lead sets.

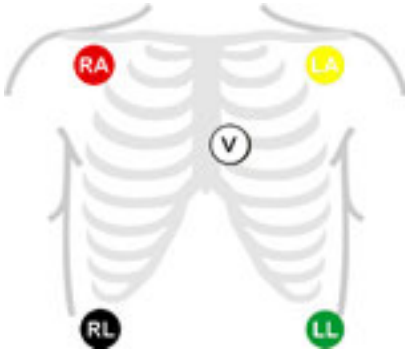
**3-Wire Standard**



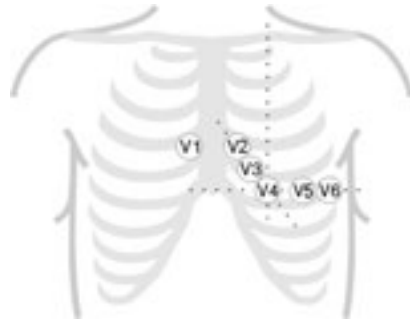
**6-Wire Standard**



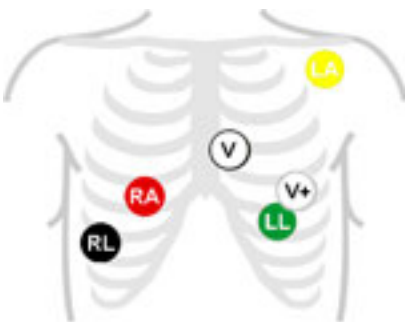
**5-Wire Standard**



**Chest and Lead Standard**



**Paced**



V+ is only used with 6-wire monitoring.

Dräger recommends that you do not use this modified placement for TruST monitoring.

## Lead Wire Color Codes

Lead wire color codes designated by the IEC and the AHA/AAMI appear in the table that follows:

| ECG Lead | IEC            | AHA/AAMI (US)  |
|----------|----------------|----------------|
| LA       | Yellow         | Black          |
| LL       | Green          | Red            |
| RA       | Red            | White          |
| RL       | Black          | Green          |
| V        | White          | Brown          |
| V+       | Gray and White | Gray and Brown |

## Attaching the Pulse Oximeter

The accuracy of SpO<sub>2</sub> monitoring depends largely on the strength and quality of the SpO<sub>2</sub> signal.

If a finger is used as a monitoring site, remove any nail polish. Cut the patient's finger nails, if necessary, for better sensor placement. Use only approved sensors and apply them per the sensor manufacturer's recommendation (contact your local Dräger representative for a list of approved sensors). Ambient light can interfere with pulse oximetry measurements if the sensor is not properly attached, causing erratic measurement or missing values. Ensure proper sensor placement and cover the sensor with opaque material if interference due to ambient light is suspected.

To attach the pulse oximeter to the patient:

- 1 Select the sensor type and size best suited for your patient.
- 2 If the sensor is reusable, clean it before and after each patient use.
- 3 Position the sensor correctly and attach it to your patient.
- 4 Connect the sensor to the patient cable.
- 5 Inspect the sensor application site frequently. If the sensor is too tight it may damage the tissue and impede blood flow. If the sensor is damaged, do not use it.

### CAUTION

Read the instructions provided with the sensor for optimal application techniques and for safety information

## Technical Specifications

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### NOTE

- Specifications are subject to change.
- The telemetry system complies with the Radio Equipment and Telecommunications Terminal Equipment Directive (1999/5/EC).

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## Infinity CentralStation

### Site of Operation

The site of operation of all equipment must meet environmental requirements as specified in this section.

For related information, refer to "General WARNINGS and CAUTIONS" on page 16.

The Infinity CentralStation, Telemetry System, and Infinity M300 are not intended for home care use or the following hospital environments:

- Hyperbaric Chambers,
- CT areas,
- Environments containing MRI equipment.

| Infinity CentralStation              |                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Display Specifications</b>        |                                                                                                                                                               |
| Display Size                         | 432 mm (17 in.), 533 mm (21 in.) diagonal, CRT or TFT                                                                                                         |
| Resolution                           | 1280 x 1024 pixels (native resolution)                                                                                                                        |
| <b>User Controls</b>                 |                                                                                                                                                               |
| Input Device Controls                | PS/2-compatible keyboard and PS/2-compatible optical mouse included in country-specific kit. Optional Dräger Medical supplied touch screen is also supported. |
| <b>Central Processing Unit (CPU)</b> |                                                                                                                                                               |
| Processor                            | Minimum 3.20 GHz processor                                                                                                                                    |
| Storage                              | <ul style="list-style-type: none"> <li>– 1 GB RAM, USB, DVD/CDRW</li> <li>– 2 x 73 GB hard drives</li> </ul>                                                  |
| Disk Array                           | SCSI Raid 1 in locked drive bay (optional)                                                                                                                    |
| Software Updates                     | CD-ROM                                                                                                                                                        |
| Connections                          | <ul style="list-style-type: none"> <li>– 2 Asynchronous RS-232 serial ports</li> <li>– 4 USB ports (2 front, 2 rear)</li> <li>– 2 LAN connections</li> </ul>  |
| Network Connectivity                 | <ul style="list-style-type: none"> <li>– Infinity Network</li> <li>– Infinity TruST™ Telemetry Network</li> </ul>                                             |
| Video Output                         | Dual Head PCI graphic board, 1280 x 1024 @ 75 Hz                                                                                                              |
| Audio Output                         | Internal speaker standard. Dräger Medical supplied external speakers also required                                                                            |
| Alarm Grades                         | Advisory, Serious, Life-threatening (Audible and visual indications)                                                                                          |
| Patients per CPU                     | <ul style="list-style-type: none"> <li>– 16 patients using one display</li> <li>– 32 patients using two displays</li> </ul>                                   |
| <b>Electrical Specifications</b>     |                                                                                                                                                               |
| Power consumption                    | 115V / 230V 4.0A/2.0A                                                                                                                                         |

| <b>Infinity CentralStation</b>      |                                                                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BTU Output                          | Up to 460W                                                                                                                                                                                                                                                                     |
| <b>Environmental Specifications</b> |                                                                                                                                                                                                                                                                                |
| Temperature Range:                  | Operating: 0 to 45° C (32 to 113° F)<br>Storage: -20 to 50° C (-4 to 122° F)                                                                                                                                                                                                   |
| Altitude                            | Operating: 3500 m (11000 feet)<br>Storage: 10600 m (35000 feet)                                                                                                                                                                                                                |
| Mode of operation                   | Continuous                                                                                                                                                                                                                                                                     |
| Acoustic Noise                      | < 46 dBA                                                                                                                                                                                                                                                                       |
| <b>Physical Specifications</b>      |                                                                                                                                                                                                                                                                                |
| Size (H x W x D)                    | 426.7 x 218.4 x 508.0 mm (16.8 x 8.6 x 20.0 in.)                                                                                                                                                                                                                               |
| Weight                              | 12.7 kg (28 lbs.)                                                                                                                                                                                                                                                              |
| Chassis Color                       | White                                                                                                                                                                                                                                                                          |
| <b>Standards/Compliance</b>         |                                                                                                                                                                                                                                                                                |
|                                     | FCC Class B CISPR II UL 60950-1 and EN60601-1-2 EMC for medical equipment. UL/CSA/CE Certification as required per Original Equipment Manufacturers (OEM). Infinity CentralStation is CE marked in accordance with the requirements of the 93/42/EEC Medical Device Directive. |

## Touch Screen Display

| Touch Screen Display         |                                                                               |
|------------------------------|-------------------------------------------------------------------------------|
| Type                         | 482.6 mm (19") TFT active matrix LCD, anti-glare, hard-coated                 |
| Active Display Area (h x w)  | 301 x 376 mm (11.9 x 14.8 in.)                                                |
| Native Resolution            | 1280 x 1024 (1600 x 1200 maximum resolution)                                  |
| Response Time                | 20 ms                                                                         |
| Viewing Angle                | 85° to 85°                                                                    |
| Base Stand                   | Tilt, swivel, height, and pivot                                               |
| Electrical Specifications    |                                                                               |
| Power Consumption            | < 60W                                                                         |
| Power Input                  | 100-250 Vac, 1.4A-0.7A, 60/50Hz, universal                                    |
| Power Output                 | 12VDC, 5A                                                                     |
| Environmental Specifications |                                                                               |
| Temperature Range:           | Operating: 10 to 40° C (50 to 104° F)<br>Storage: -20 to 60° C (-4 to 140° F) |
| Physical Specifications      |                                                                               |
| Size (H x W x D)             | 467 x 432 x 251 mm (18.3 x 17 x 9.8 in.)                                      |
| Weight                       | 9.7 kg (21.3 lbs.)                                                            |
| VESA Mount                   | 100 x 100 mm (3.9 x 3.9 in.)                                                  |
| Chassis Color                | Grey                                                                          |
| Standards/Compliance         |                                                                               |
|                              | FCC, CE, CUL, TUV/GS, VCC, TCO                                                |



## Uninterruptible Power Supply

### CAUTION

Dräger recommends the use of an Uninterruptible Power Manager in conjunction with the Infinity CentralStation at all times. Without an Uninterruptible Power Manager, power losses may cause the Infinity CentralStation to reboot and electrical anomalies may cause unwanted operation.

| Uninterruptible Power Supply (620VA, 120V, 220V) |                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------|
| Connections                                      | Infinity CentralStation, Infinity TruST Telemetry Receiver                   |
| Audio output                                     | < 45 dB at 1 m (3 ft.)                                                       |
| <b>Physical Specifications</b>                   |                                                                              |
| Size (H x W x D)                                 | 168 x 119 x 368 mm (6.61 x 4.68 x 14.48 in.)                                 |
| Weight                                           | 12.27 kg (27.05 lbs.)                                                        |
| <b>Electrical Specifications</b>                 |                                                                              |
| Waveform type                                    | Stepped approximation to a sine wave                                         |
| Input voltage                                    | 115, 220, 230, 240 VAC $\pm$ 20% (nominal, user-selectable)                  |
| Input frequency (Hz)                             | 50/60 Hz $\pm$ 3% (auto sensing)                                             |
| Surge energy rating                              | 320 joules                                                                   |
| Battery recharge                                 | 5.50 hours                                                                   |
| Backup time                                      | 15.7/5.5 minutes (half/full loaded)                                          |
| Internal batteries                               | Maintenance free lead-acid battery with suspended electrolyte<br>- leakproof |
| Indicators                                       | LED for replace battery, overload, on battery                                |
| <b>Environmental Specifications</b>              |                                                                              |
| Temperature Range:                               | Operating: 0 to 40° C (32 to 104° F)<br>Storage: -15 to 45° C (5 to 113° F)  |
| Altitude                                         | Operating: 3000 m (10000 feet)<br>Storage: 3000 m (10000 feet)               |
| <b>Standards/Compliance</b>                      |                                                                              |
| CE, GOST, VCCI, VDE                              |                                                                              |

| Laser Printer                                                                                                                                    |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Black and white 1200 x 1200 dpi laser printers, 115V/220V sold separately.<br>Contact Dräger Medical to obtain list of supported laser printers. |  |

| Strip Recorder                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infinity R50-N Network Strip Recorder provides two-channel strip recordings of automatically-captured arrhythmia or alarm events, or manually initiated recordings. |

## Uninterruptible Power Supply (Japan)

### CAUTION

Dräger recommends the use of an Uninterruptible Power Manager in conjunction with the Infinity CentralStation at all times. Without an Uninterruptible Power Manager power losses may cause the Infinity CentralStation to reboot and electrical anomalies may cause unwanted operation.

| Uninterruptible Power Supply (750VA Japan) |                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------|
| Connections                                | Infinity CentralStation, Infinity TruST Telemetry Receiver                  |
| Audio output                               | < 55 dB                                                                     |
| <b>Physical Specifications</b>             |                                                                             |
| Size (H x W x D)                           | 157 x 137 x 358 mm (6.18 x 5.39 x 14.09 in.)                                |
| Weight                                     | 13.18 kg (29.05 lbs.)                                                       |
| <b>Electrical Specifications</b>           |                                                                             |
| Waveform type                              | Sine wave                                                                   |
| Input voltage                              | 160-285 V                                                                   |
| Input frequency (Hz)                       | 50/60 Hz $\pm$ 3% (auto sensing)                                            |
| Surge energy rating                        | 340 joules                                                                  |
| Battery recharge                           | 3 hours                                                                     |
| Backup time                                | 4.8/16.4 minutes (half/full loaded)                                         |
| Internal batteries                         | Maintenance free lead-acid battery with suspended electrolyte - leakproof   |
| Indicators                                 | LED for replace battery, overload, on battery                               |
| <b>Environmental Specifications</b>        |                                                                             |
| Temperature Range:                         | Operating: 0 to 40° C (32 to 104° F)<br>Storage: -15 to 45° C (5 to 113° F) |
| Altitude                                   | Operating: 3000 m (10000 feet)<br>Storage: 3000 m (10000 feet)              |
| <b>Standards/Compliance</b>                |                                                                             |
|                                            | C-tick, CE, EN50091                                                         |

## Infinity M300

| Infinity M300                       |                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physical Specifications</b>      |                                                                                                                                                                                                                                                                                           |
| Size, H x W x D                     | 142.2 x 76.2 x 30.5 mm (5.6 x 3 x 1.2 in)                                                                                                                                                                                                                                                 |
| Weight                              | 276.4 g (9.75 oz) with battery                                                                                                                                                                                                                                                            |
| Cooling                             | Convection                                                                                                                                                                                                                                                                                |
| Connections                         | ECG, Communication port for SpO <sub>2</sub> , or Programming Cable, Bedside Charger, Central Charger                                                                                                                                                                                     |
| <b>Environmental Specifications</b> |                                                                                                                                                                                                                                                                                           |
| Atmospheric Pressure                | Operating: 64.7 to 106.0 kPa<br>Storage: 50.0 to 106.0 kPa                                                                                                                                                                                                                                |
| Temperature                         | Operating: 0 to 40° C (32 to 104° F)<br>Storage: -20 to 60° C (-4 to 140° F)                                                                                                                                                                                                              |
| Humidity (non condensing)           | Operating: 10% to 85%<br>Storage: 10% to 85%                                                                                                                                                                                                                                              |
| Free Fall                           | IEC 60068-2-32, Procedure 1<br>Height of fall: 1 m<br>Number of falls: 1 on each of six surfaces                                                                                                                                                                                          |
| Protection against ingress of water | IPX7, temporary immersion                                                                                                                                                                                                                                                                 |
| <b>Electrical Characteristics</b>   |                                                                                                                                                                                                                                                                                           |
| Power Source                        | Rechargeable 3.75 V lithium ion battery, available from Dräger                                                                                                                                                                                                                            |
| Protection Class                    | Internally powered                                                                                                                                                                                                                                                                        |
| Mode of operation                   | Continuous                                                                                                                                                                                                                                                                                |
| Battery operating time              | ECG only: 17 to 19 hours<br>ECG + continuous SpO <sub>2</sub> : 14 to 16 hours<br>Operation time varies according to use of display, alarm alerts, and wireless environment (roaming)                                                                                                     |
| Battery recharging time             | Using Bedside Charger to 100%, approximately:<br>0 to 25% = 2 hours<br>0 to 50% = 4 hours<br>0 to 75% = 6 hours<br>0 to 100% = 8 hours<br><br>Using Central Charger to 100%, approximately:<br>0 to 25% = 40 minutes<br>0 to 50% = 1.5 hours<br>0 to 75% = 2 hours<br>0 to 100% = 3 hours |

| <b>Infinity M300</b>                          |                                                                                                                                                                                                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Interface</b>                         |                                                                                                                                                                                                                                                           |
| Controls                                      | 6 function keys: alarm pause, view screen, staff alert, record/mark event, up/down scroll                                                                                                                                                                 |
| Alarms                                        | Audible and visible alarm indication (user controlled)<br>3 severity levels: Life threatening, Serious, Advisory                                                                                                                                          |
| <b>Display</b>                                |                                                                                                                                                                                                                                                           |
| Size/viewing area                             | 5.08 x 5.08 cm (2 x 2 in) diagonal LCD                                                                                                                                                                                                                    |
| Resolution                                    | 220 x 176 pixels                                                                                                                                                                                                                                          |
| <b>Communications</b>                         |                                                                                                                                                                                                                                                           |
| Network                                       | IEEE 802.11b                                                                                                                                                                                                                                              |
| Radio Power Output                            | 30 mW maximum                                                                                                                                                                                                                                             |
| <b>ECG</b>                                    |                                                                                                                                                                                                                                                           |
| Available leads                               | 3-wire: I, II, III<br>5-wire: I, II, III, aVL, aVR, aVF, V<br>6-wire: I, II, III, aVL, aVR, aVF, V, V+<br>6-wire with Infinity TruST 12-lead: I, II, III, aVL, aVR, aVF, dV1, V2, dV3, dV4, V5, dV6                                                       |
| Leads Analyzed                                | Any two user-selected ECG leads (simultaneously), or any one user-selected ECG lead                                                                                                                                                                       |
| Measurement Range                             | 15 to 300 bpm                                                                                                                                                                                                                                             |
| Accuracy                                      | ± 2 bpm or ± 1%, whichever is greater                                                                                                                                                                                                                     |
| Display update period                         | 2 seconds nominal                                                                                                                                                                                                                                         |
| QRS detection                                 | Amplitude: 0.5 - 5.0 mV<br>Duration: 40 - 120 ms                                                                                                                                                                                                          |
| HR Level Alarm Adjustment Range               | 20 to 300 bpm                                                                                                                                                                                                                                             |
| Frequency Response                            | 0.5 - 40 Hz: ± 3 dB                                                                                                                                                                                                                                       |
| Sweep Speed                                   | 25 mm/sec ± 10% (accuracy only guaranteed for 17" displays)                                                                                                                                                                                               |
| Detected Events/Rhythms                       | Asystole, Ventricular Fibrillation, Ventricular Tachycardia, Sinus Bradycardia, Ventricular Run, Accelerated Idioventricular Rhythm, Supraventricular Tachycardia, Ventricular Couplet, Ventricular Bigeminy, Sinus Tachycardia, Pause, Artifact, PVC/min |
| Degree of protection against electrical shock | Type CF                                                                                                                                                                                                                                                   |
| Defibrillation Protection                     | In accordance per IEC 60601-2-27                                                                                                                                                                                                                          |
| Electrosurgery and Cautery                    | Not intended for use during ESU                                                                                                                                                                                                                           |

| Infinity M300                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pacer detection:</b>                                                                                                                                                                                                                                                                                        | Leads I and (II or III)                                                                                                                                                                                |
| <b>NOTE</b><br>The Infinity M300 device contains a magnet which generates an extremely low static magnetic field of approximately 2 gauss at 12.7 mm (0.5 in.) distance. Please refer to the manufacturer's Instructions for Use of any third party medical devices in the patient vicinity for compatibility. |                                                                                                                                                                                                        |
| Amplitude                                                                                                                                                                                                                                                                                                      | ±2 to ±900 mV                                                                                                                                                                                          |
| Width (dp)                                                                                                                                                                                                                                                                                                     | 0.1 to 2.0 ms                                                                                                                                                                                          |
| <b>ST Segment Analysis</b>                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                        |
| Leads analyzed                                                                                                                                                                                                                                                                                                 | 3-wire: I, II, or III<br>5-wire: I, II, III, aVL, aVR, aVF, V<br>6-wire: I, II, III, aVL, aVR, aVF, V, V+<br>6-wire with Infinity TruST 12-lead: I, II, III, aVL, aVR, aVF, dV1, V2, dV3, dV4, V5, dV6 |
| Default ISO                                                                                                                                                                                                                                                                                                    | Default: -28 msec                                                                                                                                                                                      |
| ST Measurement Points                                                                                                                                                                                                                                                                                          | Default: +80 msec                                                                                                                                                                                      |
| ST Complex                                                                                                                                                                                                                                                                                                     | Length: 900 msec (250 samples)<br>Frequency response: 0.05 to 40 Hz                                                                                                                                    |
| Update Interval                                                                                                                                                                                                                                                                                                | 15 seconds                                                                                                                                                                                             |
| ST Measurement and Alarm Limit Ranges                                                                                                                                                                                                                                                                          | -15.0 to 15.0 mm, -1.5 to 1.5 mV                                                                                                                                                                       |
| ST Measurement Resolution                                                                                                                                                                                                                                                                                      | 0.1 mm (0.01 mV)                                                                                                                                                                                       |
| <b>Pulse Oximetry (optional)</b>                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                        |
| Parameter Display                                                                                                                                                                                                                                                                                              | Percentage of functional (oxygen-saturated) hemoglobin Saturation (%SpO <sub>2</sub> ), pulse rate                                                                                                     |
| Measurement method                                                                                                                                                                                                                                                                                             | Absorption-spectrophotometry                                                                                                                                                                           |
| Measurement and Display range                                                                                                                                                                                                                                                                                  | SpO <sub>2</sub> : 1 - 100%<br>Pulse rate: 30-250 bpm                                                                                                                                                  |
| Calibration range                                                                                                                                                                                                                                                                                              | 70-100%                                                                                                                                                                                                |
| Display update period                                                                                                                                                                                                                                                                                          | 2 seconds nominal                                                                                                                                                                                      |
| Maximum hold from previous update                                                                                                                                                                                                                                                                              | 30 seconds (in the event of artifact or other error)                                                                                                                                                   |

| Infinity M300 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accuracy:     | <p>SpO<sub>2</sub>: 0 to 69% not specified<br/>70 to 100% sensor-specific as follows:</p> <p><u>Masimo® LNOP® Sensors</u><br/>SpO<sub>2</sub> Accuracy<sup>1, 2, 3, 4</sup>:<br/>LNOP adt, LNOP Pdt, LNOP neo,<br/>LNOP DCI, LNOP TC-I, LNOP DCIP,<br/>LNOP YI ±3%<br/>Pulse Rate Accuracy: ±3 bpm or ±3%<br/>(whichever is greater)<br/>Low Perfusion Accuracy, SpO<sub>2</sub>: ±2%<br/>Low Perfusion Accuracy, Pulse Rate: ±3 bpm or ±3%<br/>(whichever is greater)</p> <p><u>Masimo® LNCS® Sensors</u><br/>SpO<sub>2</sub> Accuracy<sup>1, 2, 3, 4</sup>:<br/>LNCS DCI®, LNCS DCIP, LNCS Adtx,<br/>LNCS Pdtx, LNCS Inf ±2%<br/>Pulse Rate Accuracy: ±3 bpm or ±3%<br/>(whichever is greater)<br/>Low Perfusion Accuracy, SpO<sub>2</sub>: ±2%<br/>Low Perfusion Accuracy, Pulse Rate: ±3 bpm or ±3%<br/>(whichever is greater)</p> <p><u>Nellcor® Sensors</u><br/>SpO<sub>2</sub> Accuracy<sup>1, 2, 3, 4</sup>:<br/>OxiMAX® MAX-A, OxiMAX MAX-AL,<br/>OxiMAX MAX-P, DS100A ±3%</p> <p><u>Dräger Sensors</u><br/>SpO<sub>2</sub> Accuracy<sup>1, 2, 3, 4</sup>:<br/>MS16444 Disposable Foam Pedi,<br/>MS16445 Disposable Foam Adt,<br/>MS16449 Disposable Vinyl Adt,<br/>MS16448 Disposable Vinyl Pedi,<br/>MS13235 Reusable Sensor ±2%</p> |

## Infinity M300

### NOTE

<sup>1</sup> Since pulse oximeter measurements are statistically distributed, only about two-thirds of those measurements can be expected to fall within  $\pm 1 A_{rms}$  of the value measured by a co-oximeter.

<sup>2</sup> These accuracies have been validated using blood samples obtained from healthy adult volunteers during induced hypoxia studies in the range of 70-100% SpO<sub>2</sub> against a laboratory co-oximeter and ECG monitor.

<sup>3</sup> SpO<sub>2</sub> accuracies are expressed as  $\pm$  "X" digits between indicated saturation levels. Accuracy of the SpO<sub>2</sub> measurement is specified within 1  $A_{rms}$  of the value measured by a co-oximeter.

<sup>4</sup> The pulse rate accuracy has been validated on healthy adult volunteers during induced hypoxia studies in the range of 70-100% SpO<sub>2</sub> against a laboratory co-oximeter and ECG monitor. This variation equals  $\pm 1 A_{rms}$  of the pulse rate value measured by the ECG monitor.

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| Nominal wavelength: | Nellcor: Red: 660 nm IR: 910 nm<br>Masimo: Red: 660 nm IR: 905 nm |
|---------------------|-------------------------------------------------------------------|

|                                          |            |
|------------------------------------------|------------|
| SpO <sub>2</sub> Alarm Adjustment Range: | 20 to 100% |
|------------------------------------------|------------|

|                                    |               |
|------------------------------------|---------------|
| Pulse Rate Alarm Adjustment Range: | 30 to 240 bpm |
|------------------------------------|---------------|

|        |                                                                                                                                                      |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power: | Nellcor: Red: 3 mW (max.) IR: 4 mW (max.)<br>Masimo: Red: 0.9 mW (max.) IR: 0.9 mW (max.)<br><br>LED drive is current limited by hardware mechanisms |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|

### Standards/Compliance

IEC 60601-1:1988 + A1:1991 + A2:1995,  
IEC 60601-1-2:2001,  
IEC 60601-2-27:2005,  
IEC 60601-2-49:2001,  
ANSI/AAMI EC13:2002(R)2007.



## Infinity M300 Bedside Charger

| Infinity M300 Bedside Charger       |                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Physical Specifications</b>      |                                                                                                  |
| Size, H x W x D                     | 45.72 x 162.56 x 99.06 mm (1.8 x 6.4 x 3.9 in)                                                   |
| Weight                              | 224.0 g (7.9 oz)                                                                                 |
| Cooling                             | Convection                                                                                       |
| Connections                         | One (1) Infinity M300                                                                            |
| <b>Electrical Specifications</b>    |                                                                                                  |
| Input Voltage                       | 92 - 264 VAC, 50/60 Hz (±5%)                                                                     |
| Protection Class                    | Class 2                                                                                          |
| Mode of Operation                   | Continuous                                                                                       |
| <b>Environmental Requirements</b>   |                                                                                                  |
| Temperature                         | Operating: 0 to 40° C (32 to 104° F)<br>Storage: -20 to 60° C (-4 to 140° F)                     |
| Humidity (non condensing)           | Operating: 10% to 85%<br>Storage: 10% to 85%                                                     |
| Atmospheric Pressure                | Operating: 64.7 kPa to 106.0 kPa<br>Storage: 50.0 kPa to 106.0 kPa                               |
| Free Fall                           | Height of fall: 1 m<br>Number of falls: 1 on each of six surfaces<br>IEC 60068-2-32, Procedure 1 |
| Protection against ingress of water | IPX4, splashing water                                                                            |
| <b>Standards/Compliance</b>         |                                                                                                  |
| IEC 60601-1 and IEC 60601-1-2       |                                                                                                  |

## Infinity M300 Central Charger

| Infinity M300 Central Charger       |                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------|
| <b>Physical Specifications</b>      |                                                                                  |
| Size, H x W x D                     | 520.7 x 215.9 x 190.5 mm (20.5 x 8.5 x 7.5 in)                                   |
| Weight                              | 6.5 kg (14.4 lbs)                                                                |
| Cooling                             | Convection                                                                       |
| Connections                         | Up to ten (10) Infinity M300 devices                                             |
| <b>Electrical Specifications</b>    |                                                                                  |
| Input Voltage                       | 92 - 264 VAC                                                                     |
| Input frequency (Hz)                | 50/60 Hz                                                                         |
| Protection Class                    | Class 1                                                                          |
| Mode of Operation                   | Continuous                                                                       |
| <b>Environmental Requirements</b>   |                                                                                  |
| Atmospheric Pressure                | Operating: 70 to 106 kPa<br>Storage: 50 to 106 kPa                               |
| Temperature                         | Operating: 10 to +45° C (50 to 113° F)<br>Storage: -40 to +70° C (-40 to 158° F) |
| Humidity (non-condensing)           | Operating: 10% to 95%<br>Storage: 10% to 95%                                     |
| Protection against ingress of water | IPX1 (dripping) per IEC 60529                                                    |
| <b>Standards/Compliance</b>         |                                                                                  |
| IEC 60601-1 and IEC 60601-1-2       |                                                                                  |

## Infinity TruST Telemetry System

| <b>TruST Telemetry Receiver</b>        |                                                       |
|----------------------------------------|-------------------------------------------------------|
| Network Connectivity                   | Infinity Network; Infinity TruST Telemetry Network    |
| Antenna Type                           | Dual conversion super heterodyne                      |
| Modulation                             | Gaussian Minimum Shift Keying (GMSK)                  |
| Channels                               | 4 channels per card, maximum 16 channels              |
| Frequencies                            | UHF 400 - 460 MHz, UHF 608 - 614 MHz, VHF 174-216 MHz |
| Channel Spacing                        | Varies by country, 25, 40 or 50 kHz                   |
| <b>Electrical Specifications</b>       |                                                       |
| Power Consumption                      | 2A @ 100 VAC, 1A @ 240 VAC                            |
| Input Voltage                          | 100/120/220/230-240 VAC selectable +/- 10%            |
| Mains Frequency                        | 50/60 Hz. (nominal)                                   |
| Indicators                             | LED indicator for power                               |
| <b>Environmental Conditions</b>        |                                                       |
| Cooling                                | Fan circulation                                       |
| Operation Temperature                  | 10 to 40° C (50 to 104° F)                            |
| Non-Operating Conditions               | -20 to 60° C (-4 to 140° F)                           |
| <b>Physical Attributes</b>             |                                                       |
| Chassis Color                          | White                                                 |
| Weight                                 | 16 kg (35 lbs)                                        |
| H x W x D                              | 191 x 432 x 508 mm (7.5 x 17.0 x 20.0 in.)            |
| <b>Regulatory/Standards Compliance</b> |                                                       |
| Compliances                            | – IEC 60601-1 and IEC 60601-1-2                       |

| <b>TruST Telemetry Transmitter</b>                                                                                                                                                                                                                                                                                               |                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Lead Type                                                                                                                                                                                                                                                                                                                        | 3-wire, 5-wire, 6-wire                                                                                                          |
| Indicators                                                                                                                                                                                                                                                                                                                       | LEDs indicating lead off and low battery                                                                                        |
| Leads-off Sensors                                                                                                                                                                                                                                                                                                                | One for each lead: RA, LA, LL, V, V+                                                                                            |
| Lead-off Threshold                                                                                                                                                                                                                                                                                                               | 12 M $\Omega$ , $\pm$ 25% for RA, LA, LL, V, V+<br>2.7 M $\Omega$ , $\pm$ 25% for RL                                            |
| Common Mode Rejection                                                                                                                                                                                                                                                                                                            | Meets CMRR test specification in AAMI EC 13-2002 4.2.9.10                                                                       |
| Controls                                                                                                                                                                                                                                                                                                                         | Staff alert, timed and continuous recording start, self-test                                                                    |
| Pacemaker Detection                                                                                                                                                                                                                                                                                                              | 0.1 to 2 ms, 2.5 to 700 mV (positive or negative)<br>Independent detectors for Lead I and II                                    |
| <b>NOTE</b><br>The Infinity Telemetry transmitter device contains a magnet which generates an extremely low static magnetic field of approximately 11 gauss at 12.7 mm (0.5 in.) distance. Please refer to the manufacturer's Instructions for Use of any third party medical devices in the patient vicinity for compatibility. |                                                                                                                                 |
| Defibrillation Protection                                                                                                                                                                                                                                                                                                        | Will withstand 360 joules                                                                                                       |
| <b>Electrical Specifications</b>                                                                                                                                                                                                                                                                                                 |                                                                                                                                 |
| Protection against electrical shock                                                                                                                                                                                                                                                                                              | Class B                                                                                                                         |
| Electrosurgery and Cautery                                                                                                                                                                                                                                                                                                       | Not intended for use during ESU                                                                                                 |
| Battery                                                                                                                                                                                                                                                                                                                          | – 9 V alkaline: 2 days<br>– 9 V lithium: 4 days                                                                                 |
| <b>Environmental Specifications</b>                                                                                                                                                                                                                                                                                              |                                                                                                                                 |
| Operation Temperature                                                                                                                                                                                                                                                                                                            | 10 to 40° C (50 to 104° F)                                                                                                      |
| Non-Operating Conditions                                                                                                                                                                                                                                                                                                         | -25 to 60° C (-13 to 140° F)                                                                                                    |
| RF Power Output                                                                                                                                                                                                                                                                                                                  | 1 MW, -3 to + 4 dBm                                                                                                             |
| Frequency Response                                                                                                                                                                                                                                                                                                               | 0.5 - 40 Hz: $\pm$ 3 dB                                                                                                         |
| Water Resistance                                                                                                                                                                                                                                                                                                                 | Electronics protected from damage due to accidental immersion in up to 1 m of water for 30 minutes. Meets EN 60529 rating IPX7. |
| Drop and Free fall                                                                                                                                                                                                                                                                                                               | Withstands 20 drops from 1.5 m onto a tiled concrete surface with no degradation of performance                                 |

| <b>TruST Telemetry Transmitter</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical Resistance                | Resistance tested for isopropyl alcohol 70%, sodium hypochlorite (bleach) 2.5% and the following commercial disinfectants:<br><br>Buraton 10F 1%, Cetylcode-G® Concentrate, Diluent Concentrate Cidex Plus™ 28 Day Solution, Cidex OPA Solution, Cidex OPA Solution High Level Disinfectant, Cidex™ Activated Dialdehyde Solution, Compliance, Enzol (Cidezyme), Incidin® Plus 1%, Incidin® Plus 2%, Lysoformin 3000 1%, Metricide® 28 Long-Life Activated, Dialdehyde Solution, Sporidicin Sterilizing and Disinfecting Solution |
| Connections                        | <ul style="list-style-type: none"> <li>– ECG Analog cable for bedside display of Lead II scaled to 1 mV ± 15% to transmitter's ECG input leads</li> <li>– Vital Connection cable between Infinity CentralStation and Infinity monitor X8 port for data acquisition of SpO2 and/or NIBP</li> <li>– Infinity Telemetry Transmitter Programming Port for tuning each transmitter electronically over its frequency range;</li> <li>– MicroO2+ pulse oximeter.</li> </ul>                                                             |
| <b>Physical Specifications</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Size, H x W x D                    | 116 x 65 x 25 mm (4.56 x 2.56 x 1 in.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Weight                             | 160 grams (5.7 oz.) with lithium battery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| <b>Telemetry Vital Signs Measuring Capabilities</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |            |        |                              |        |                                  |                                    |                                      |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|--------|------------------------------|--------|----------------------------------|------------------------------------|--------------------------------------|
| <b>ECG</b>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |            |        |                              |        |                                  |                                    |                                      |
| Available leads                                     | <table> <tr> <td>3-wire</td><td>I, II, III</td></tr> <tr> <td>5-wire</td><td>I, II, III, aVR, aVF, aVL, V</td></tr> <tr> <td>6-wire</td><td>I, II, III, aVR, aVF, aVL, V, V+</td></tr> <tr> <td>6-wire<br/>(Infinity TruST 12-lead)</td><td>V2, V5, dV1, dV3, dV4, dV6 (Default)</td></tr> </table> <p>For TruST 12-lead, any other combination of V1-V6 can be selected; the chest lead selection determines the constructed leads (dV1 - dV6).</p> | 3-wire | I, II, III | 5-wire | I, II, III, aVR, aVF, aVL, V | 6-wire | I, II, III, aVR, aVF, aVL, V, V+ | 6-wire<br>(Infinity TruST 12-lead) | V2, V5, dV1, dV3, dV4, dV6 (Default) |
| 3-wire                                              | I, II, III                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |            |        |                              |        |                                  |                                    |                                      |
| 5-wire                                              | I, II, III, aVR, aVF, aVL, V                                                                                                                                                                                                                                                                                                                                                                                                                         |        |            |        |                              |        |                                  |                                    |                                      |
| 6-wire                                              | I, II, III, aVR, aVF, aVL, V, V+                                                                                                                                                                                                                                                                                                                                                                                                                     |        |            |        |                              |        |                                  |                                    |                                      |
| 6-wire<br>(Infinity TruST 12-lead)                  | V2, V5, dV1, dV3, dV4, dV6 (Default)                                                                                                                                                                                                                                                                                                                                                                                                                 |        |            |        |                              |        |                                  |                                    |                                      |
| Measuring range                                     | 15 - 300 bpm                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |            |        |                              |        |                                  |                                    |                                      |
| Accuracy                                            | ± 5 bpm or ± 5% of rate (whichever is greater)                                                                                                                                                                                                                                                                                                                                                                                                       |        |            |        |                              |        |                                  |                                    |                                      |
| QRS detection range<br>amplitude                    | <i>Amplitude:</i> 0.5 to 5.0 mV<br><i>Duration:</i> 40 to 120 ms                                                                                                                                                                                                                                                                                                                                                                                     |        |            |        |                              |        |                                  |                                    |                                      |
| 1 mV calibration pulse                              | 1 mV ± 5%, all leads                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |            |        |                              |        |                                  |                                    |                                      |

| Telemetry Vital Signs Measuring Capabilities |                                                                                                                                                                                                                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Arrhythmia Processing</b>                 |                                                                                                                                                                                                                        |
| Algorithm                                    | ACE (Arrhythmia Classification Expert)                                                                                                                                                                                 |
| Leads analyzed                               | Any two user-selected ECG leads (simultaneously), or any one user-selected ECG lead                                                                                                                                    |
| Detected events/rhythms                      | Asystole, ventricular fibrillation, ventricular tachycardia, sinus bradycardia, pause, ventricular run, sinus tachycardia, accelerated idioventricular rhythm, couplet, bigeminy, paced artifact, PVC/min,% paced, SVT |
| Functions                                    | User-selectable functions: alarm on/off, record on/off, store on/off, assignment of alarm grade for each event category.                                                                                               |
| <b>Trends</b>                                |                                                                                                                                                                                                                        |
| Parameters                                   | HR,% paced, PVC/min ST (I, II, III, aVL, aVR, aVF, dV1, V2 (V), dV3, dV4, V5 (V+), dV6), STCM, STVCM, SpO2, pulse rate (PLS)                                                                                           |
| Storage time                                 | 72 hours                                                                                                                                                                                                               |
| Trend Graphs                                 | 1, 2, 4, 8, 12, 24 hours (72 hours max)                                                                                                                                                                                |
| Trend Tables                                 | 1, 5, 15, 30, 60 minutes                                                                                                                                                                                               |
| <b>ST Segment Analysis</b>                   |                                                                                                                                                                                                                        |
| Leads analyzed                               | Simultaneous analysis of leads I, II, III, aVL, aVR, aVF, dV1, V2 (V), dV3, dV4, V5 (V+), dV6.                                                                                                                         |
| Sampling rate for averaged ECG               | 250 samples                                                                                                                                                                                                            |
| Resolution                                   | $\pm 10 \mu\text{V}$                                                                                                                                                                                                   |
| Alarm handling                               | Positive and negative S-T deviations                                                                                                                                                                                   |
| Isoelectric point                            | Default is set at 28 ms before QRS onset; user-selectable in 4 ms increments between QRS peak to beginning of averaged ECG segment.                                                                                    |
| ST measurements                              | The value is updated every 15 seconds from an average of all normal, non-paced beats.                                                                                                                                  |
| Trends                                       | Both graphical and tabular format for each ECG lead                                                                                                                                                                    |
| <b>General Display Capabilities</b>          |                                                                                                                                                                                                                        |
| Telemetry channels                           | Up to 16                                                                                                                                                                                                               |
| Display screens                              | 1 or 2                                                                                                                                                                                                                 |
| Screen formats                               | Refer to Infinity CentralStation Instructions for Use.                                                                                                                                                                 |
| Sweep Speeds                                 | 25 or 50 mm/s                                                                                                                                                                                                          |
| Channel amplitudes                           | 0.25; 0.5; 1; 2; 4; 8 mV/cm                                                                                                                                                                                            |
| Display mode                                 | Fixed Waveform (erase bar) mode                                                                                                                                                                                        |
| X-Y resolution                               | 1024 x 1280 pixels with ECG waveform enhancements                                                                                                                                                                      |

| Telemetry Vital Signs Measuring Capabilities |                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------|
| <b>Recordings</b>                            |                                                                                |
| Types                                        | R50 producing 50 mm wide strips with thermal array print head<br>Laser Printer |
| Number                                       | 2                                                                              |

| Transmitter Programming Port                                                |                                             |
|-----------------------------------------------------------------------------|---------------------------------------------|
| Used to tune transmitters to selected frequency and to select ECG wire mode |                                             |
| Connections                                                                 | Infinity CentralStation                     |
| H x W X D                                                                   | 44.0 x 96.5 x 96.4 mm (1.7 x 3.8 x 3.8 in.) |
| Weight                                                                      | 450 g (1 lb.)                               |

| Infinity Telemetry ECG Analog Output Cable                                  |                       |
|-----------------------------------------------------------------------------|-----------------------|
| Used to tune transmitters to selected frequency and to select ECG wire mode |                       |
| Connections                                                                 | Infinity MultiMed Pod |

## Electromagnetic Compatibility

The Infinity M300 and the Infinity Telemetry System have been designed and tested for compliance with current regulatory standards as to its capacity to limit electromagnetic emissions (EMI), and also as to its ability to block the effects of EMI from external sources. The Infinity M300 and the Infinity Telemetry System are designed to comply with the EMI/EMC standard, EN60601-1-2.

The separation distances are written with regard to the Infinity CentralStation with Infinity Telemetry, TruST Transmitters, and Infinity M300. The numbers provided will not guarantee faultless operation. This information may not be applicable to other medical electrical equipment, and older equipment may be particularly susceptible to interference.

### Reducing EMI

To reduce possible problems caused by electromagnetic interference, Dräger recommends the following:

- Use only Dräger-approved, accessories, otherwise the correct functioning of the device may be compromised. Contact your local Dräger representative for a list of approved accessories.
- Ensure that other products used in areas where patient monitoring and/or life-support products are used comply to accepted emissions standards (EN55011).
- To avoid interfering with device operation, do not operate devices (monitors, MPods, MCables, and accessories) close to equipment that emits microwave or other high-frequency emissions.

- Strictly limit access to portable radio-frequency sources, e.g., cellular phones and radio transmitters. Be aware that portable phones may periodically transmit even when in *Standby* mode.
- Maintain good cable management. Try not to route cable over electrical equipment. Do not intertwine cables.
- Be sure all electrical maintenance is performed by trained personnel.

### General Notes

Medical electrical equipment needs special precautions regarding electromagnetic compatibility (EMC) and needs to be installed and put into service according to the EMC information provided in this manual.

Portable and mobile radio frequency (RF) communications equipment can affect medical electrical equipment. The TruST telemetry receiver may be sensitive to RF interference and could possibly be interfered with by other equipment even if the equipment complies with CISPR11.

Cables and accessories not specified within the instructions for use are not authorized. Using other cables and/or accessories may adversely impact safety, performance and electromagnetic compatibility (increased emission and decreased immunity).

The equipment should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is inevitable, the equipment should be observed to verify normal operation in the configuration in which it will be used.

When using wireless networking, be aware that the system operates at 2.4 GHz range. Other equipment, even if compliant with CISPR emission requirements, could interfere with reception of wireless data. When selecting new wireless systems (e.g. cell phones, pager systems, cordless phones etc.) for use in installations where wireless networking is used, care should always be used to insure that operating frequencies are compatible. For

example, selecting cordless phones that operate at 2.4 GHz will likely cause difficulty with the phones and networking components.

### NOTE

Detailed radio frequency characteristics: 2412-2472 MHZ, Direct-sequence spread spectrum (DSSS) IEEE 802.11b compatible, limited to 100 mW. Applicable to both access points and client adaptors. When used with 802.15.1 wireless, the device will transmit with the following characteristics: 2400-2485 MHz, Frequency Hopping Spread Spectrum (FHSS), limited to 2.5 mW. See the documentation that accompanies the wireless products for further details.

### NOTE

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna,
- Increase the separation between the equipment and receiver,
- Connect the equipment into an outlet on a different circuit from that to which the receiver is connected,
- Consult the dealer or an experienced radio/TV technician for help.

Changes or modifications to this equipment not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.



Low level signals such as ECG are potentially susceptible to interference from electromagnetic energy. While the equipment meets the testing described below, it is not a guarantee of perfect operation, the 'quieter' the electrical environment the better. In general, increasing the distance between electrical devices decreases the likelihood of interference.

**NOTE**

The Infinity CentralStation with Infinity Telemetry, TruST transmitters and Infinity M300 are intended for use in the electromagnetic environments specified below. The user of this equipment should assure that is used in such an environment.

| Electromagnetic Emissions                                                                                                                                                                                                                                                                                                                    |                            |                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emissions                                                                                                                                                                                                                                                                                                                                    | Compliance according to... | Electromagnetic environment                                                                                                                                                                                                                                           |
| RF emissions (CISPR 11)<br>Infinity CentralStation<br>Telemetry Receiver<br>and Infinity M300                                                                                                                                                                                                                                                | Group 1                    | The Infinity CentralStation and receiver use RF energy only for internal function. Therefore, their RF emissions are very low and are not likely to cause any interference in nearby electronic equipment (refer to the “ <b>NOTE</b> ” at the bottom of this table). |
| RF emissions (CISPR 11)<br>Transmitter                                                                                                                                                                                                                                                                                                       | Group 2                    | The Infinity M300 and TruST Telemetry Transmitter must emit electromagnetic energy in order to perform its intended function. Nearby electronic equipment may be affected.                                                                                            |
| CISPR Emissions<br>Classification                                                                                                                                                                                                                                                                                                            | Class B                    | The equipment is suitable for use in all establishments including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.                                             |
| Harmonic emissions<br>(IEC 61000-3-2)                                                                                                                                                                                                                                                                                                        | Class A                    |                                                                                                                                                                                                                                                                       |
| Voltage fluctuations /<br>flicker (IEC 61000-3-3)                                                                                                                                                                                                                                                                                            | Complies                   |                                                                                                                                                                                                                                                                       |
| <b>NOTE</b><br>The Infinity M300 (and TruST) emit electromagnetic energy in order to communicate with the Infinity Network (and the Telemetry receiver). Nearby electronic equipment may be affected. Radio frequency characteristics are specified above. See the documentation that accompanies the wireless products for further details. |                            |                                                                                                                                                                                                                                                                       |

| Electromagnetic Immunity                                                      |                                                           |                                   |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immunity against...                                                           | IEC 60601-1-2 test level                                  | Compliance level (of this device) | Electromagnetic environment                                                                                                                                                                                                                                                                                                   |
| Electrostatic discharge, ESD (IEC 61000-4-2)                                  | <i>Contact discharge:</i><br>$\pm 6 \text{ kV}$           | $\pm 6 \text{ kV}$                | Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be kept at levels to reduce electrostatic charge to suitable levels.                                                                                                                             |
|                                                                               | <i>Air discharge:</i><br>$\pm 8 \text{ kV}$               | $\pm 8 \text{ kV}$                |                                                                                                                                                                                                                                                                                                                               |
| Electrical fast transients / bursts (IEC 61000-4-4)                           | <i>Power supply lines:</i><br>$\pm 2 \text{ kV}$          | $\pm 2 \text{ kV}$                | Mains power quality should be that of a typical commercial or hospital environment.                                                                                                                                                                                                                                           |
|                                                                               | <i>Longer input / output lines:</i><br>$\pm 1 \text{ kV}$ | $\pm 1 \text{ kV}$                |                                                                                                                                                                                                                                                                                                                               |
| Surges on AC mains lines (IEC 61000-4-5)                                      | <i>Common mode:</i><br>$\pm 2 \text{ kV}$                 | $\pm 2 \text{ kV}$                | Mains power quality should be that of a typical commercial or hospital environment.                                                                                                                                                                                                                                           |
|                                                                               | <i>Differential mode:</i><br>$\pm 1 \text{ kV}$           | $\pm 1 \text{ kV}$                |                                                                                                                                                                                                                                                                                                                               |
| Power frequency magnetic field 50/60 Hz (IEC 61000-4-8)                       | 3 A/m                                                     | 3 A/m                             | Equipment which emits high levels of power line magnetic fields (in excess of 3A/m) should be kept at a distance to reduce the likelihood of interference.                                                                                                                                                                    |
| Voltage dips and short interruptions on AC mains input lines (IEC 61000-4-11) | Dip >95%,<br>0.5 periods                                  | >95%,<br>0.5 periods              | Mains power should be that of a typical commercial or hospital environment. If user requires continued operation during power mains interruptions insure that batteries are installed and charged. Insure that battery life exceeds longest anticipated power outages or provide and additional uninterruptible power source. |
|                                                                               | Dip 60%,<br>5 periods                                     | 60%,<br>5 periods                 |                                                                                                                                                                                                                                                                                                                               |
|                                                                               | Dip 30%,<br>25 periods                                    | 30%,<br>25 periods                |                                                                                                                                                                                                                                                                                                                               |
|                                                                               | Dip >95%,<br>5 seconds                                    | >95%,<br>5 seconds                |                                                                                                                                                                                                                                                                                                                               |

| Electromagnetic Immunity                                 |                          |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------|--------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immunity against...                                      | IEC 60601-1-2 test level | Compliance level (of this device) | Electromagnetic environment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Conducted RF<br>RF coupled into lines<br>(IEC 61000-4-6) | 150 kHz – 80 MHz:        | 3 V/m                             | Portable and mobile RF communications equipment should be used no closer to any part of the monitor, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter as below.<br><br>Recommended separation distance:<br><br>$d=1.2\sqrt{P}$ 150 kHz to 80 MHz<br><br>$d=1.2\sqrt{P}$ 80 MHz to 800 MHz<br><br>$d=2.3\sqrt{P}$ 800 MHz to 2.5 GHz<br><br><i>where 'P' is the maximum output power rating of the transmitter in watts according to the transmitter manufacturer and 'd' is the recommended separation distance in metres.</i><br><br>Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey <sup>1</sup> , should be less than the compliance level in each frequency range <sup>2</sup> .<br><br>Interference may occur in the vicinity of equipment marked with the following wireless symbol:<br><br> |
| Radiated RF<br>(IEC 61000-4-3)                           | 80 MHz – 2.5 GHz         | 3 V/m                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

<sup>1</sup>Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the equipment is used exceeds the applicable RF compliance level above, the equipment should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the equipment.

<sup>2</sup>If the frequency range is 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

| Recommended separation distances between portable and mobile RF communications equipment and the equipment                                                                                                                                                                                                                     |                                                                    |                                     |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------|---------------------------------------|
| Rated maximum output power of transmitter (watts)                                                                                                                                                                                                                                                                              | Separation distance according to frequency of transmitter (meters) |                                     |                                       |
|                                                                                                                                                                                                                                                                                                                                | 150 kHz to 80 MHz<br>$d=1.2\sqrt{P}$                               | 80 MHz to 800MHz<br>$d=1.2\sqrt{P}$ | 800 MHz to 2.5 GHz<br>$d=2.3\sqrt{P}$ |
| 0.01                                                                                                                                                                                                                                                                                                                           | 0.12                                                               | 0.12                                | 0.23                                  |
| 0.1                                                                                                                                                                                                                                                                                                                            | 0.38                                                               | 0.38                                | 0.73                                  |
| 1                                                                                                                                                                                                                                                                                                                              | 1.2                                                                | 1.2                                 | 2.3                                   |
| 10                                                                                                                                                                                                                                                                                                                             | 3.8                                                                | 3.8                                 | 7.3                                   |
| 100                                                                                                                                                                                                                                                                                                                            | 12                                                                 | 12                                  | 23                                    |
| For transmitters rated at a maximum output power not listed, the recommended separation distance 'd' (in meters) can be estimated using the equation applicable to the frequency of the transmitter, where 'P' is the maximum output power rating of the transmitter (in watts) according to the transmitter manufacturer.     |                                                                    |                                     |                                       |
| <b>NOTE</b> <ul style="list-style-type: none"> <li>– At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.</li> <li>– These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.</li> </ul> |                                                                    |                                     |                                       |

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# Cleaning

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Oximeter . . . . . 337

Reusable Sensor . . . . . 337

**CAUTION**  
Do not sterilize any components.

## Cleaning the Infinity CentralStation

---

### CAUTION

Do not immerse the CPU or its peripherals.

Do not pour or spray cleaning solution directly on equipment surfaces.

Do not allow fluids to contact electrical connectors or drip into ventilation openings.

During normal use the display, CPU, accessories, and peripherals may become soiled. Clean these components according to the original manufacturer's *Instructions For Use* or hospital protocol for computer equipment and peripherals.

## Cleaning and Disinfecting Infinity Telemetry and Infinity M300

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### CAUTION

Do not use disinfectants other than those recommended by Dräger. The use of non-approved disinfectants can damage the device.

Do not use disinfectants containing phenol. Do not use strong aromatic ketone, ether, or ester solvents, or sharp tools and abrasives. These will damage the device.

Do not gas sterilize, steam autoclave, immerse, or rinse the device in fluids. If you accidentally spill liquid on the device, remove the battery and contact the Hospital Biomedical Engineering Dept. regarding the continued safety of the unit before placing it back in operation.

### Recommended Cleaning Agents

Dräger recommends the following cleaning agents for Infinity Telemetry and Infinity M300.

- Diluted isopropyl alcohol
- Sodium hypochlorite 2.5%
- Green tinctured soap

### Patient Cables

#### WARNING

**Disposable accessories (such as disposable electrodes) are for single-use only. Do not reuse disposable accessories.**

To clean and disinfect the telemetry devices:

- 1 Clean the device with gauze moistened in soapy water.
- 2 Thoroughly dry with a lint-free cloth.
- 3 Disinfect the device with gauze moistened in a cleaning agent.
- 4 Thoroughly dry with a lint-free cloth.

- Clean the patient cables with a gauze pad moistened with a soap solution.
- Dry thoroughly with a lint-free cloth.
- To disinfect patient cables, wipe the cables with a gauze moistened with diluted alcohol.
- Dry thoroughly with a lint-free cloth.



## Reusable ECG Electrodes

- Clean grabber-wire clips regularly with a toothbrush.
- Remove any gel residue from the electrode by brushing it off under running water.
- Clean the electrodes with a gauze moistened with a soap solution.
- Dry thoroughly with a lint-free cloth.
- Disinfect the electrodes with a gauze moistened with diluted alcohol.
- Dry thoroughly with a lint-free cloth.

## Reusable SpO2 Sensor

See the cleaning instructions and recommendations provided with the sensor.

## Cleaning the MicrO2/MicrO2+<sup>®</sup> Oximeter

---

### Oximeter

To clean the oximeter:

- 1 Turn the oximeter off.
- 2 Clean the oximeter with gauze moistened in soapy water.
- 3 Thoroughly dry with a lint-free cloth.
- 4 Disinfect the oximeter with gauze moistened in diluted isopropyl alcohol (70%), or sodium hypochlorite (2.5%).
- 5 Thoroughly dry with a lint-free cloth.

### Reusable Sensor

To clean a reusable sensor:

- 1 Unplug the sensor from the oximeter.

#### NOTE

Disposable sensors are intended for single-use only and should not be reprocessed.

- 2 Clean reusable sensors according to the instructions supplied with the sensor

#### CAUTION

Do not use disinfectants containing phenol. Do not use strong aromatic ketone, ether, or ester solvents, or sharp tools and abrasives. These will damage the oximeter

Do not use disinfectants other than those recommended. The use of non-approved disinfectants can damage the oximeter.

Do not gas sterilize, steam autoclave, immerse, or rinse the oximeter and its reusable sensor in fluids. If you accidentally spill liquid on the device, remove the battery and contact the Hospital Bio-medical Engineering Dept. regarding the continued safety of the unit before placing it back in operation.

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